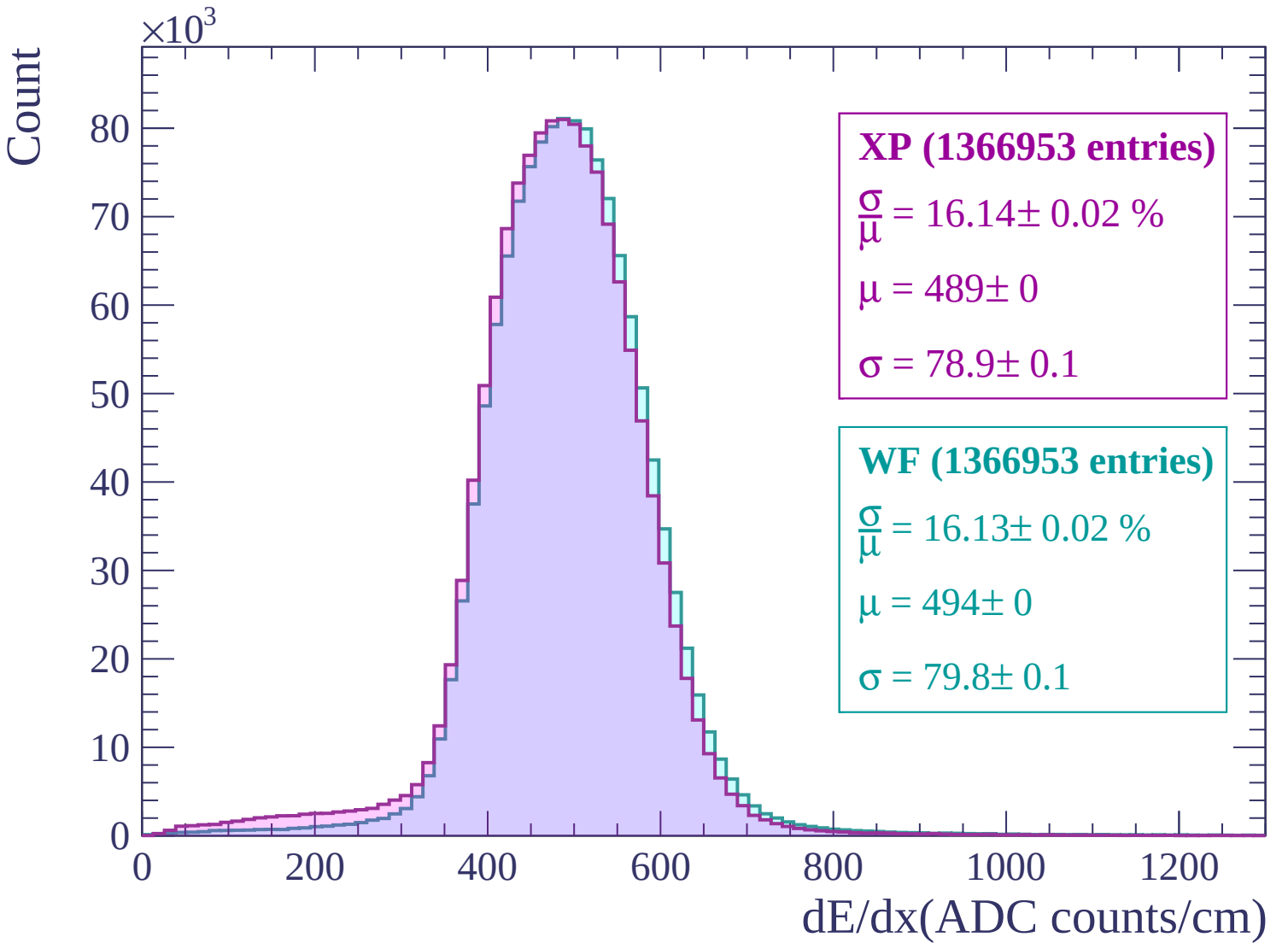
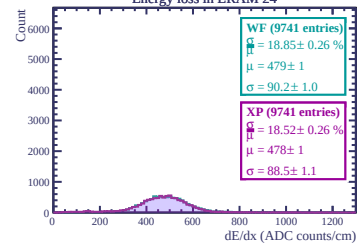
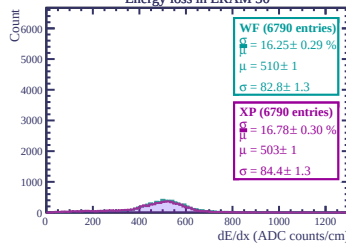




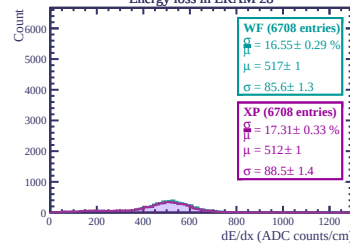
Energy loss in ERAM 14



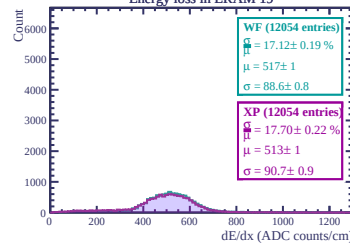
Energy loss in ERAM 30



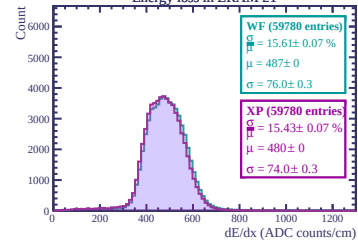
Energy loss in ERAM 30



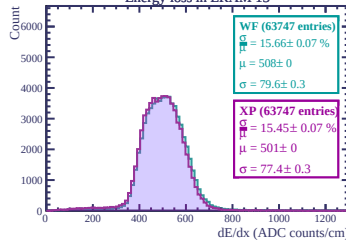
Energy loss in ERAM 30



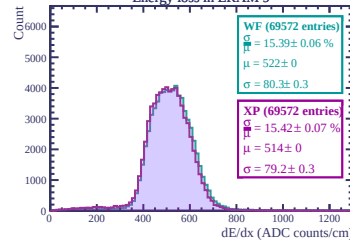
Energy loss in ERAM 21



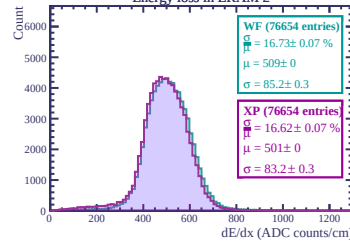
Energy loss in ERAM 13



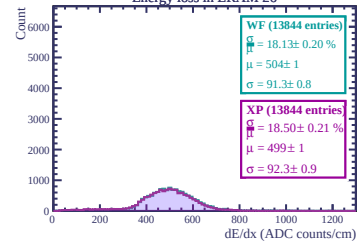
Energy loss in ERAM 9



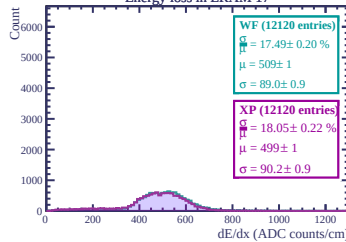
Energy loss in ERAM 2



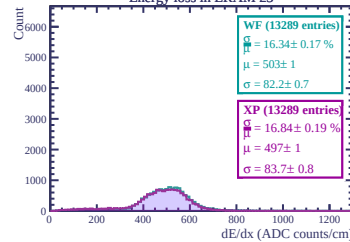
Energy loss in ERAM 26



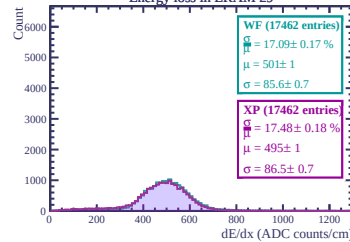
Energy loss in ERAM 17



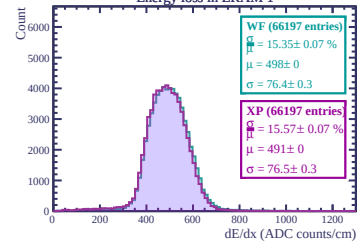
Energy loss in ERAM 23



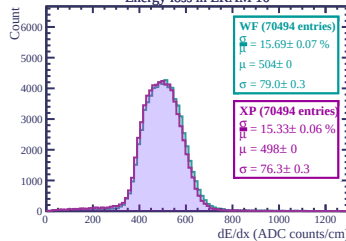
Energy loss in ERAM 29



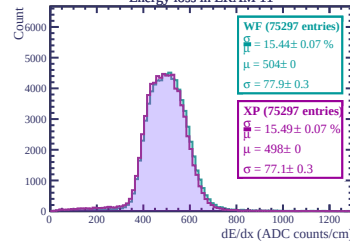
Energy loss in ERAM 1



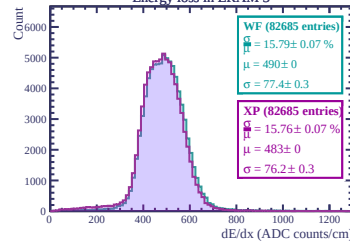
Energy loss in ERAM 10



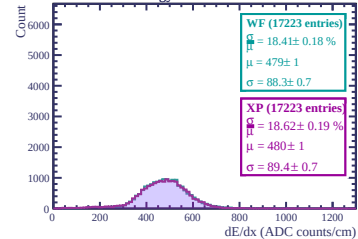
Energy loss in ERAM 11



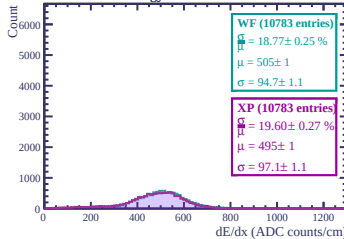
Energy loss in ERAM 3



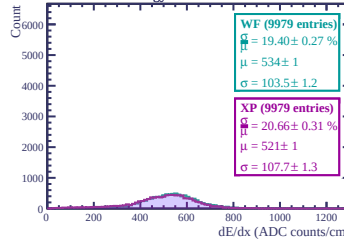
Energy loss in ERAM 17



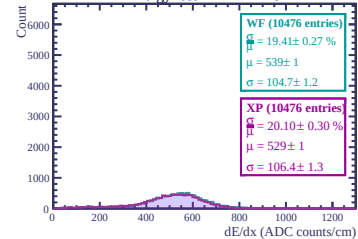
Energy loss in ERAM 16



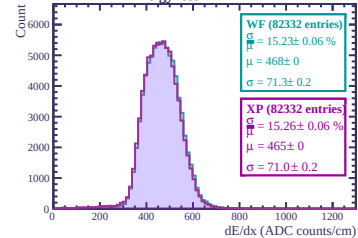
Energy loss in ERAM 15



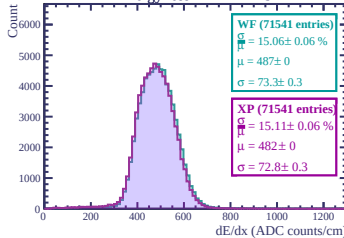
Energy loss in ERAM 14



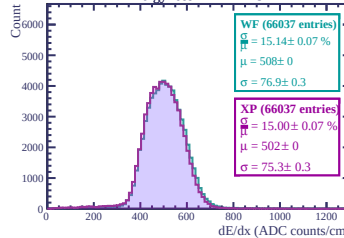
Energy loss in ERAM 42



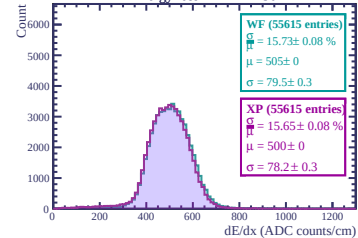
Energy loss in ERAM 45



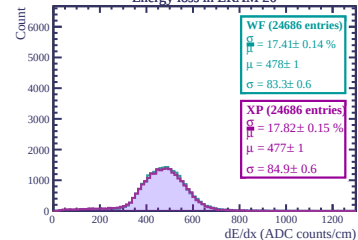
Energy loss in ERAM 37



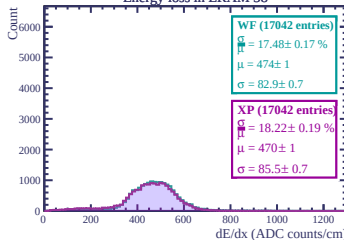
Energy loss in ERAM 36



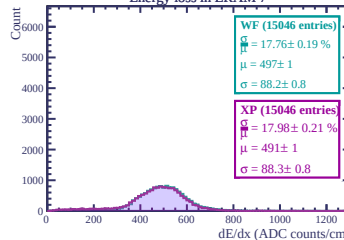
Energy loss in ERAM 20



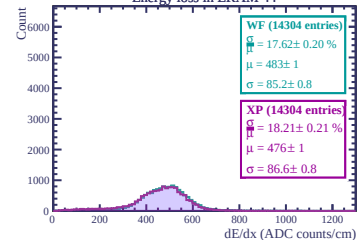
Energy loss in ERAM 38



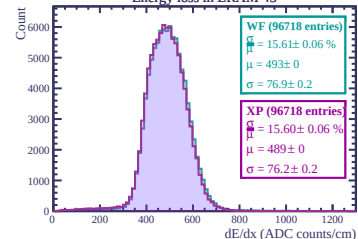
Energy loss in ERAM 7



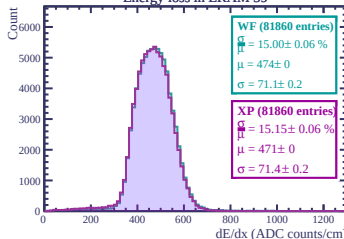
Energy loss in ERAM 44



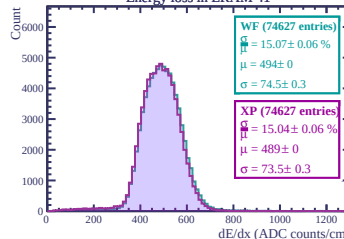
Energy loss in ERAM 43



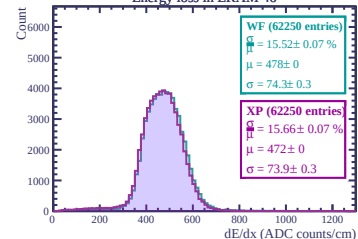
Energy loss in ERAM 39

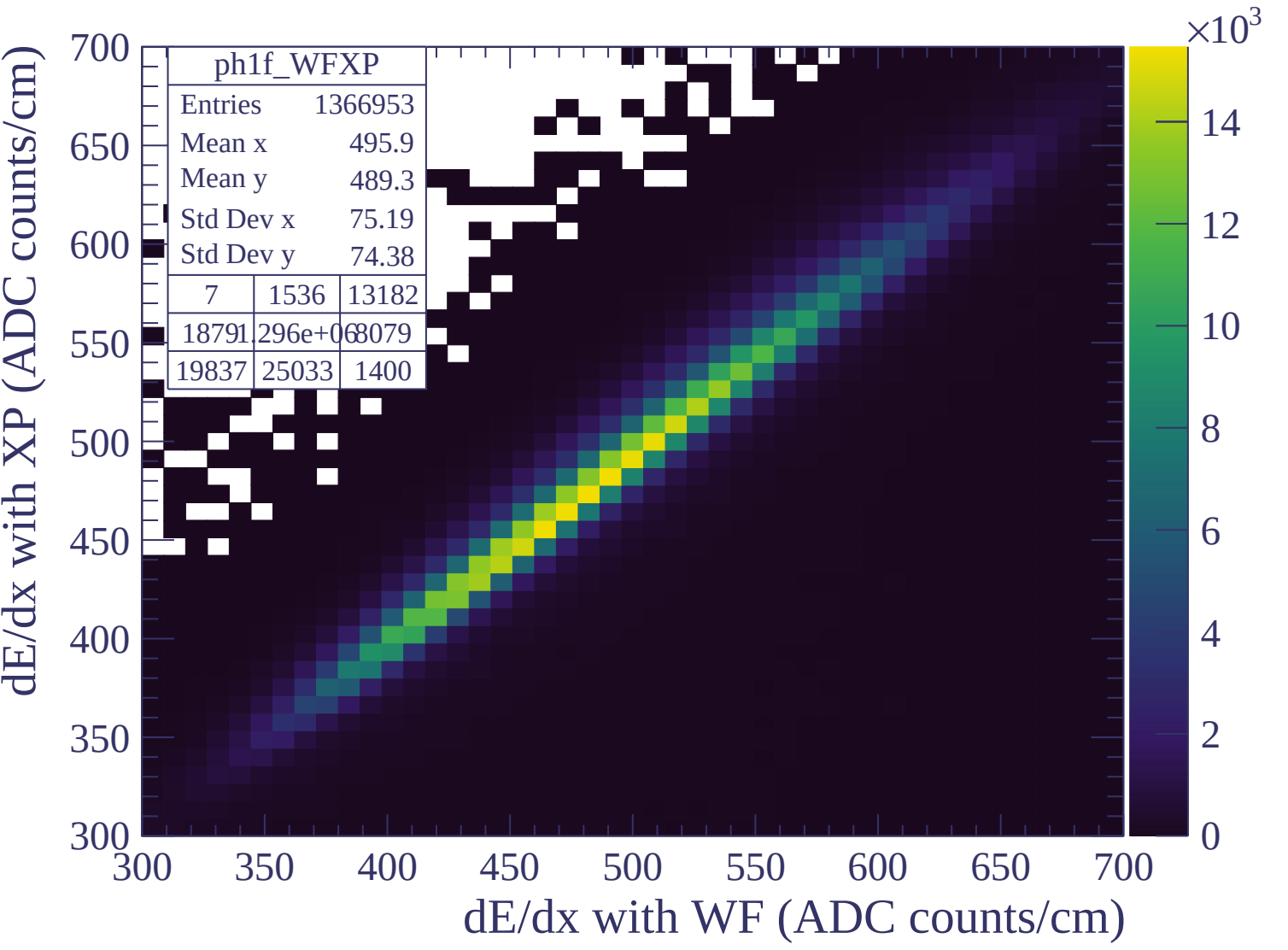


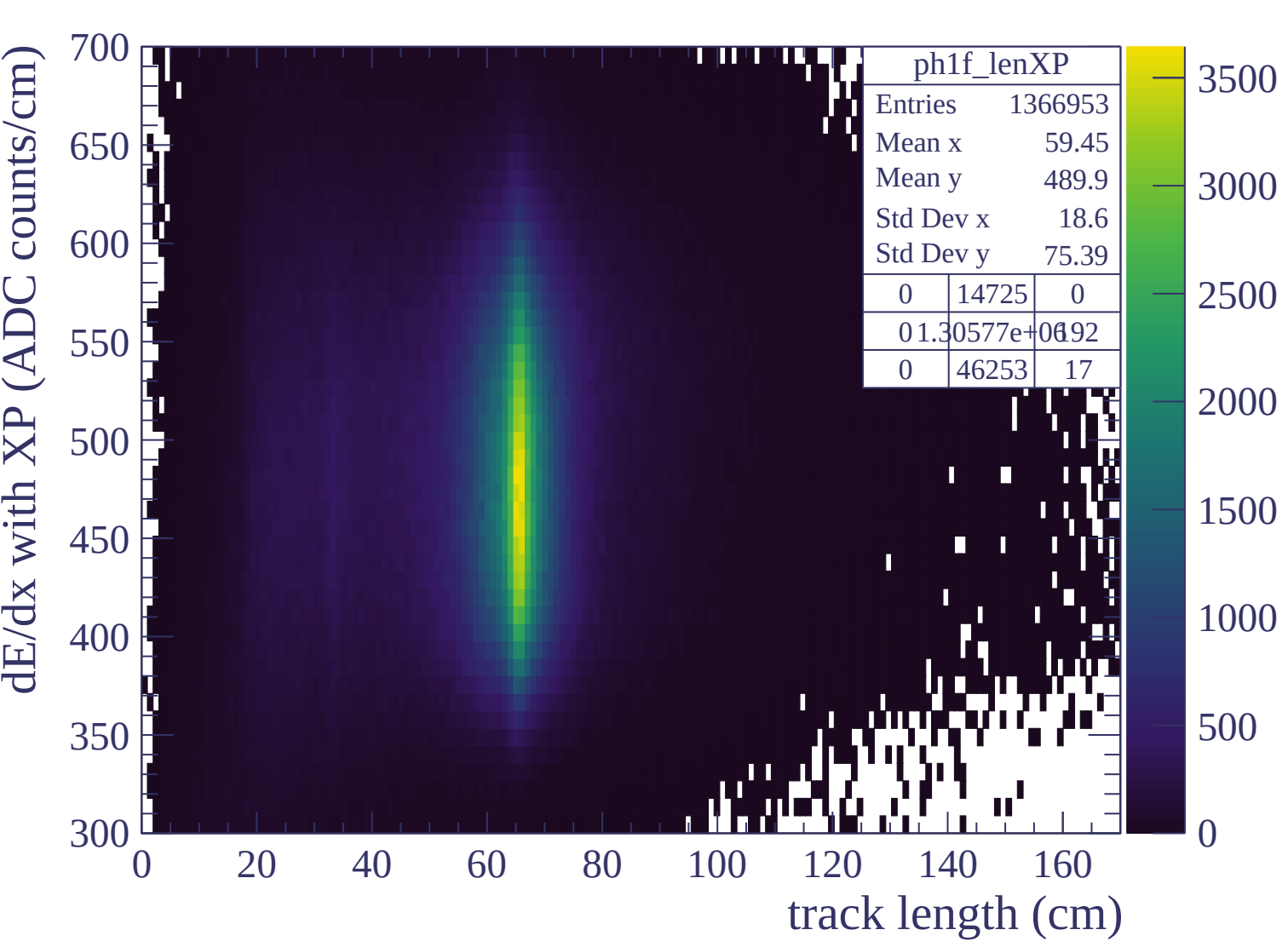
Energy loss in ERAM 41

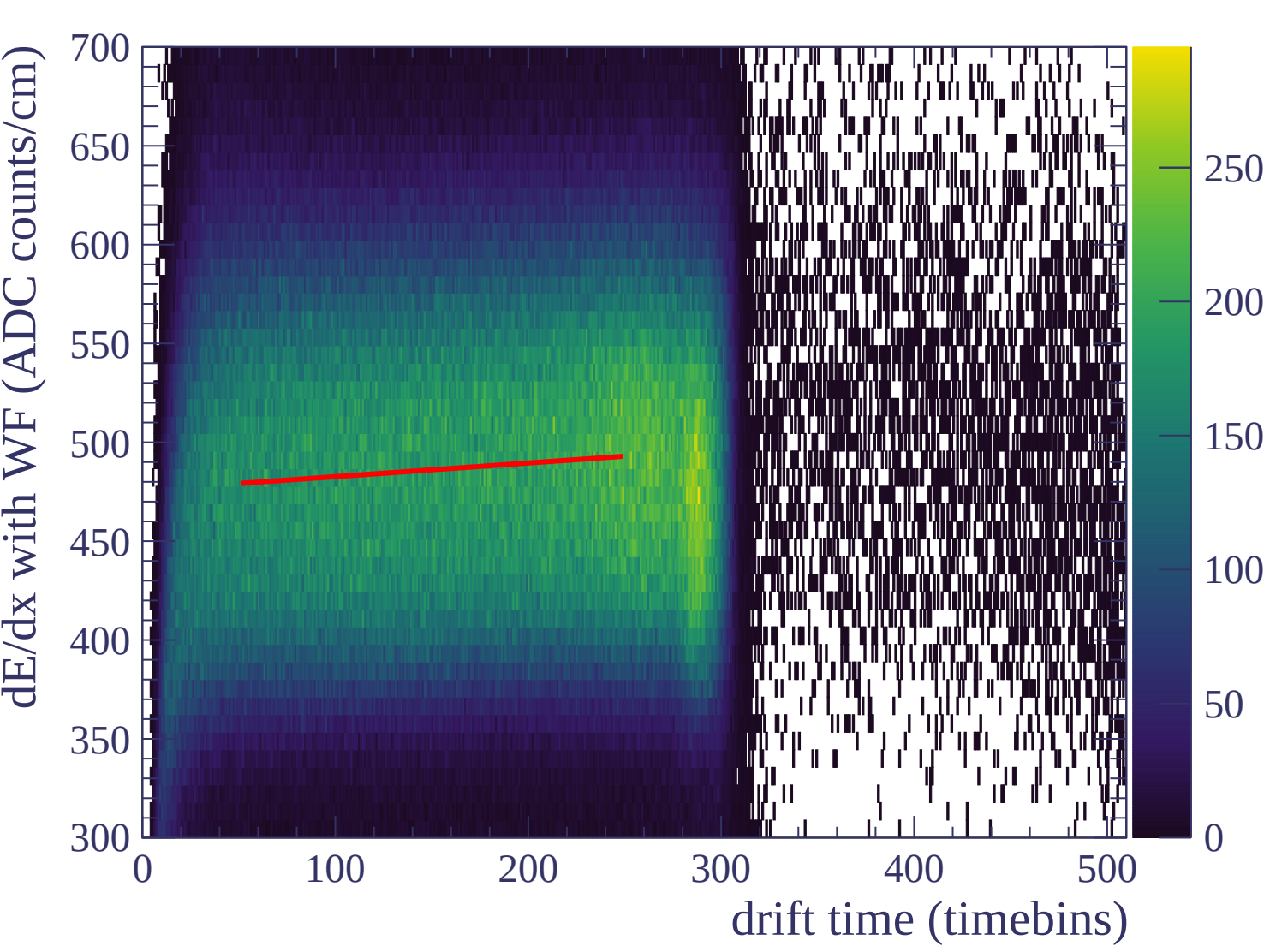


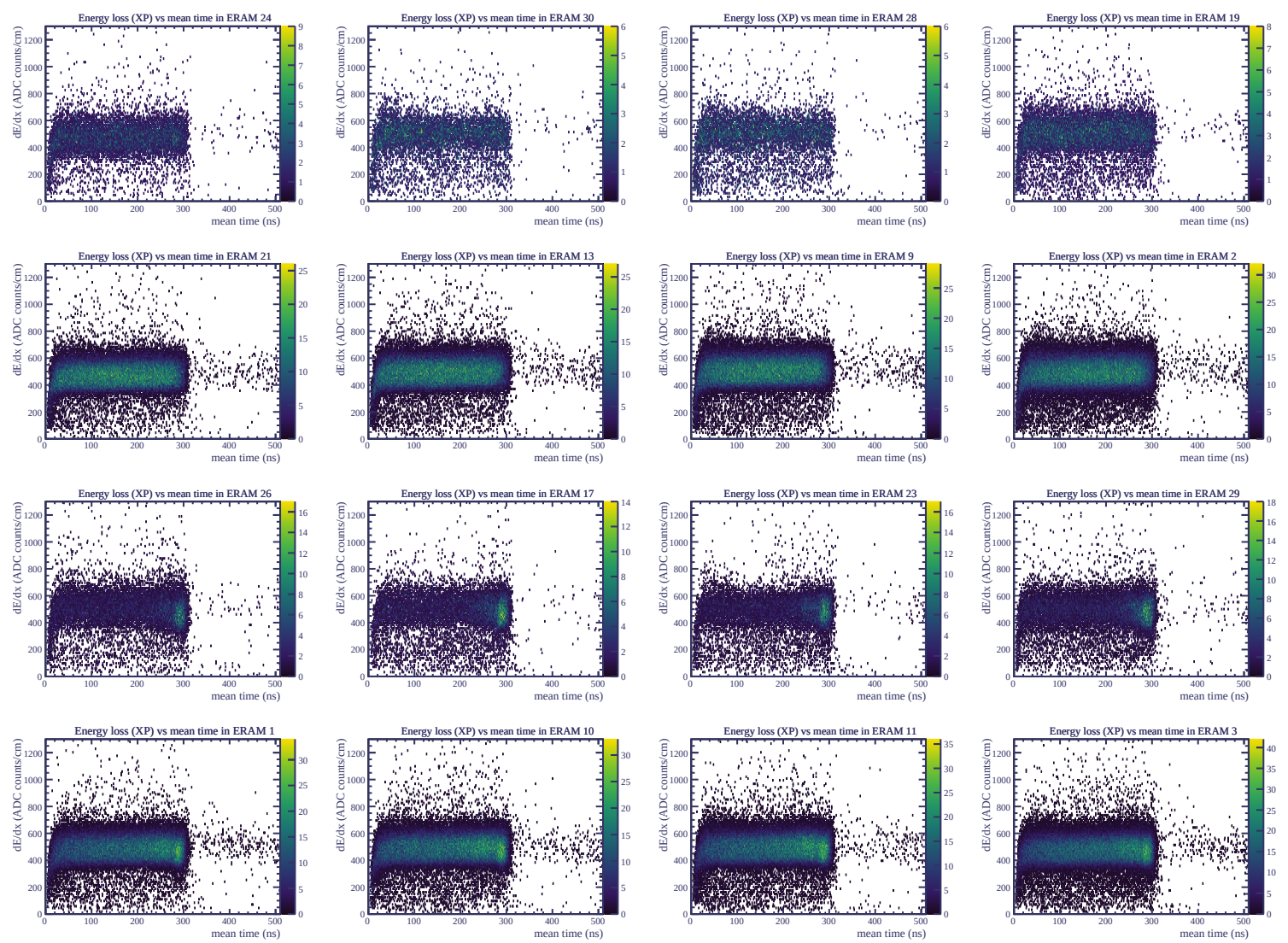
Energy loss in ERAM 46

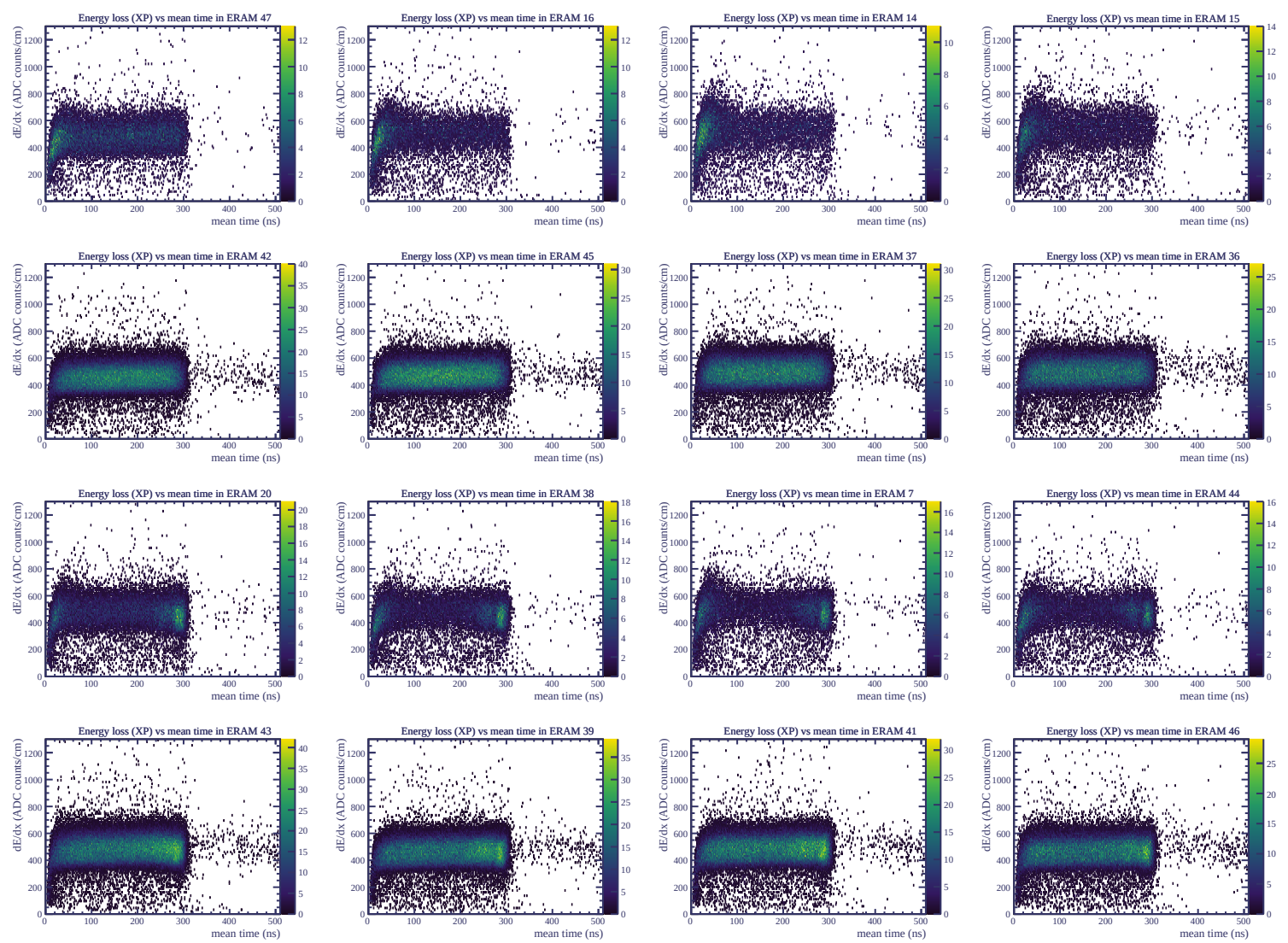




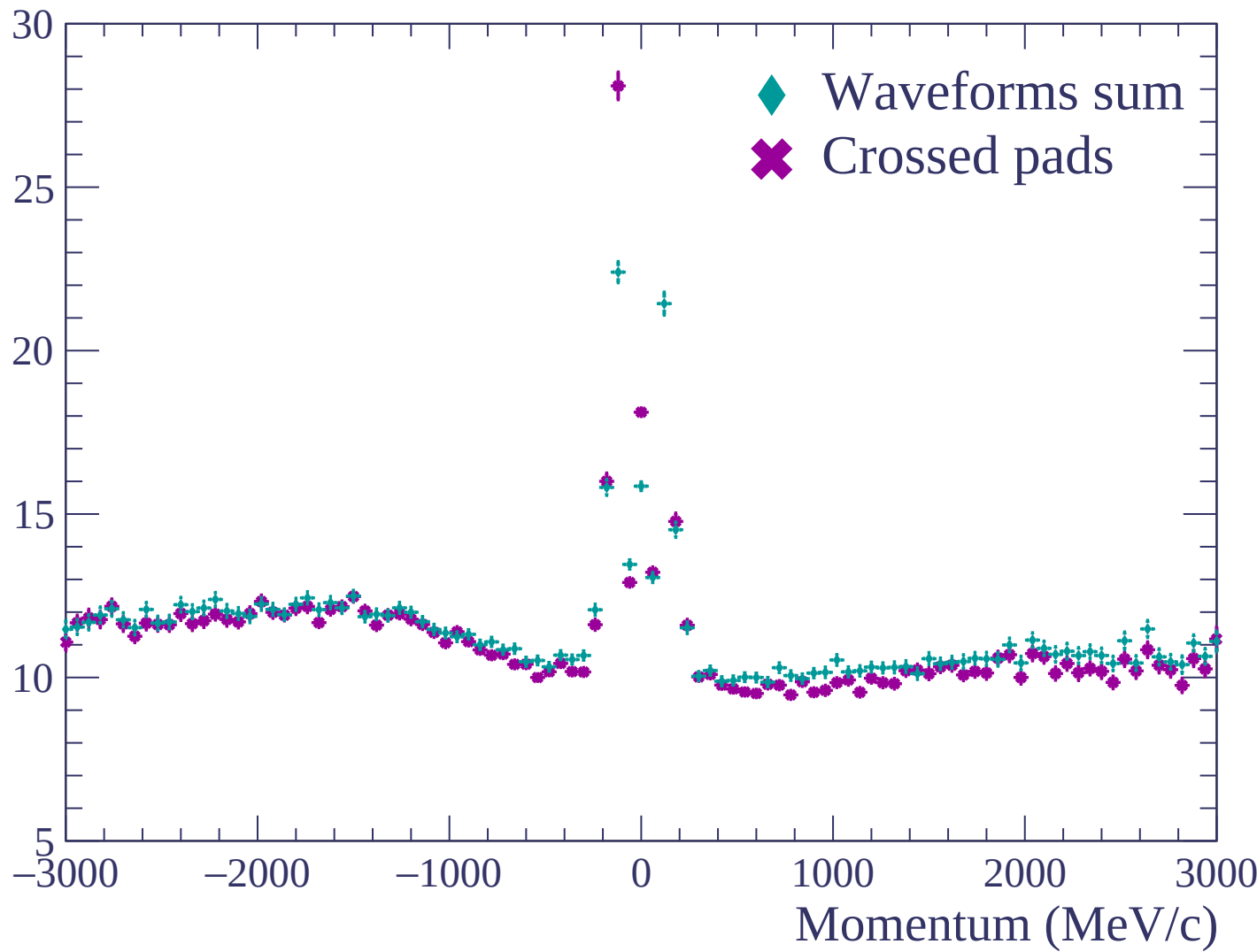


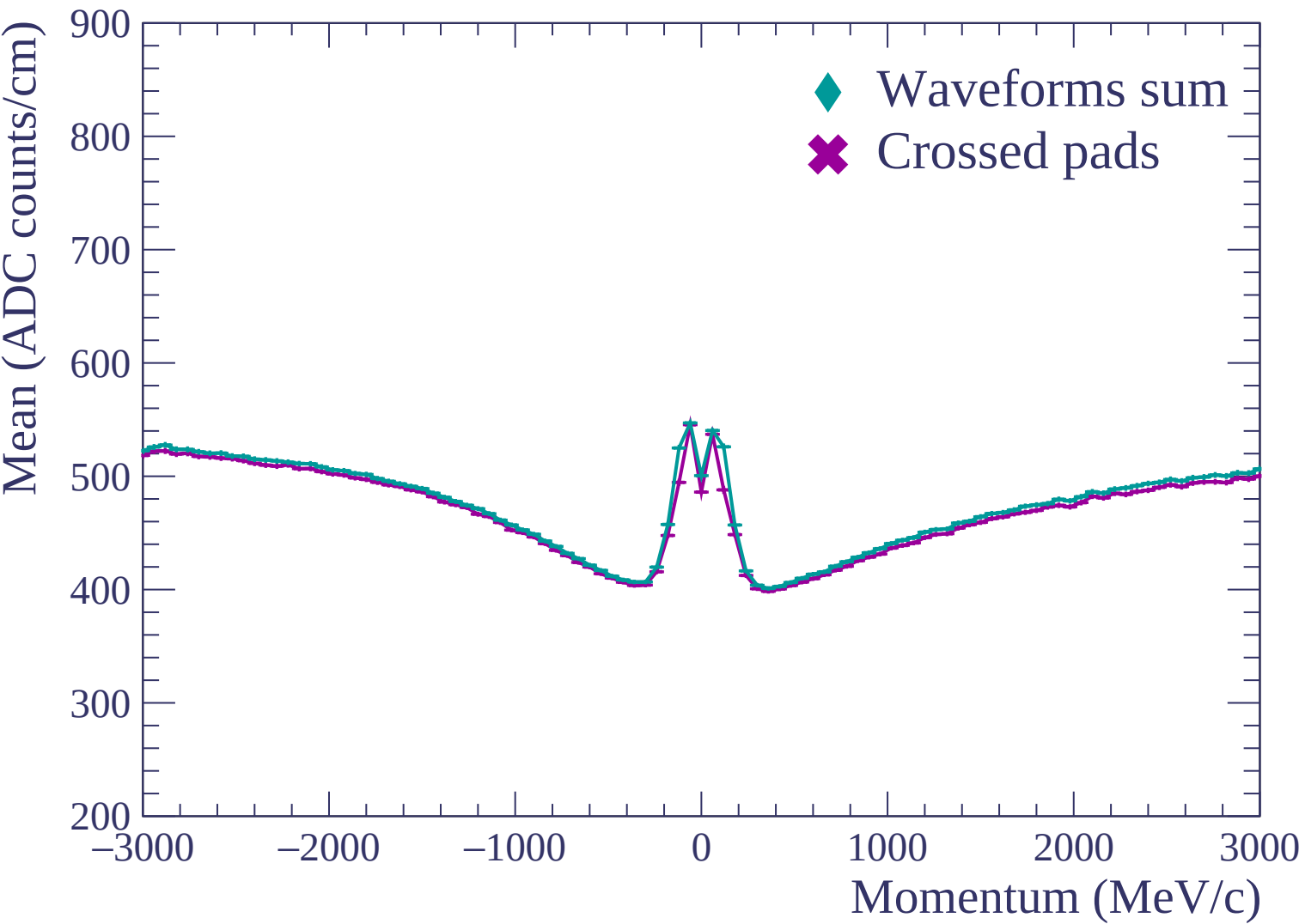


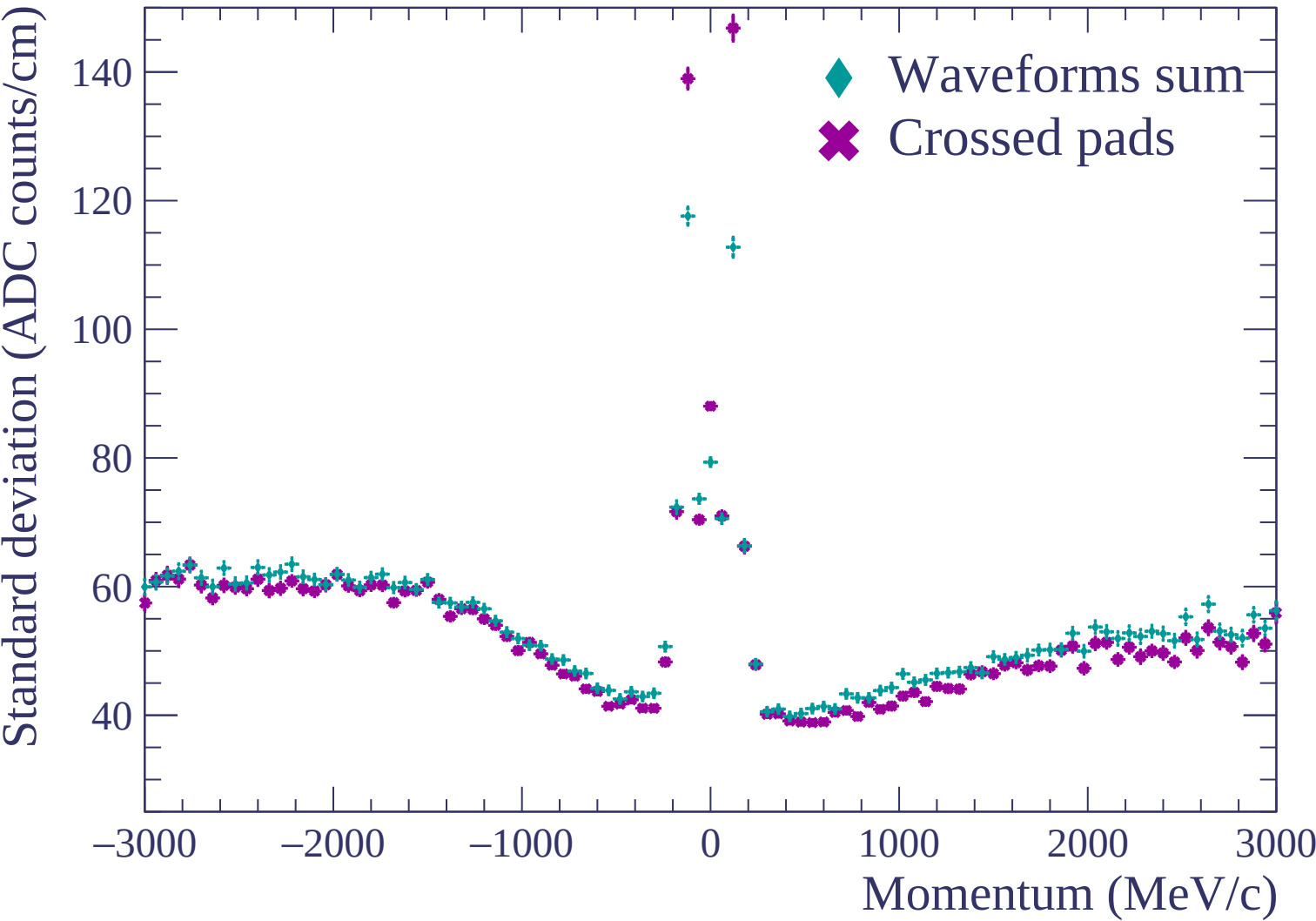


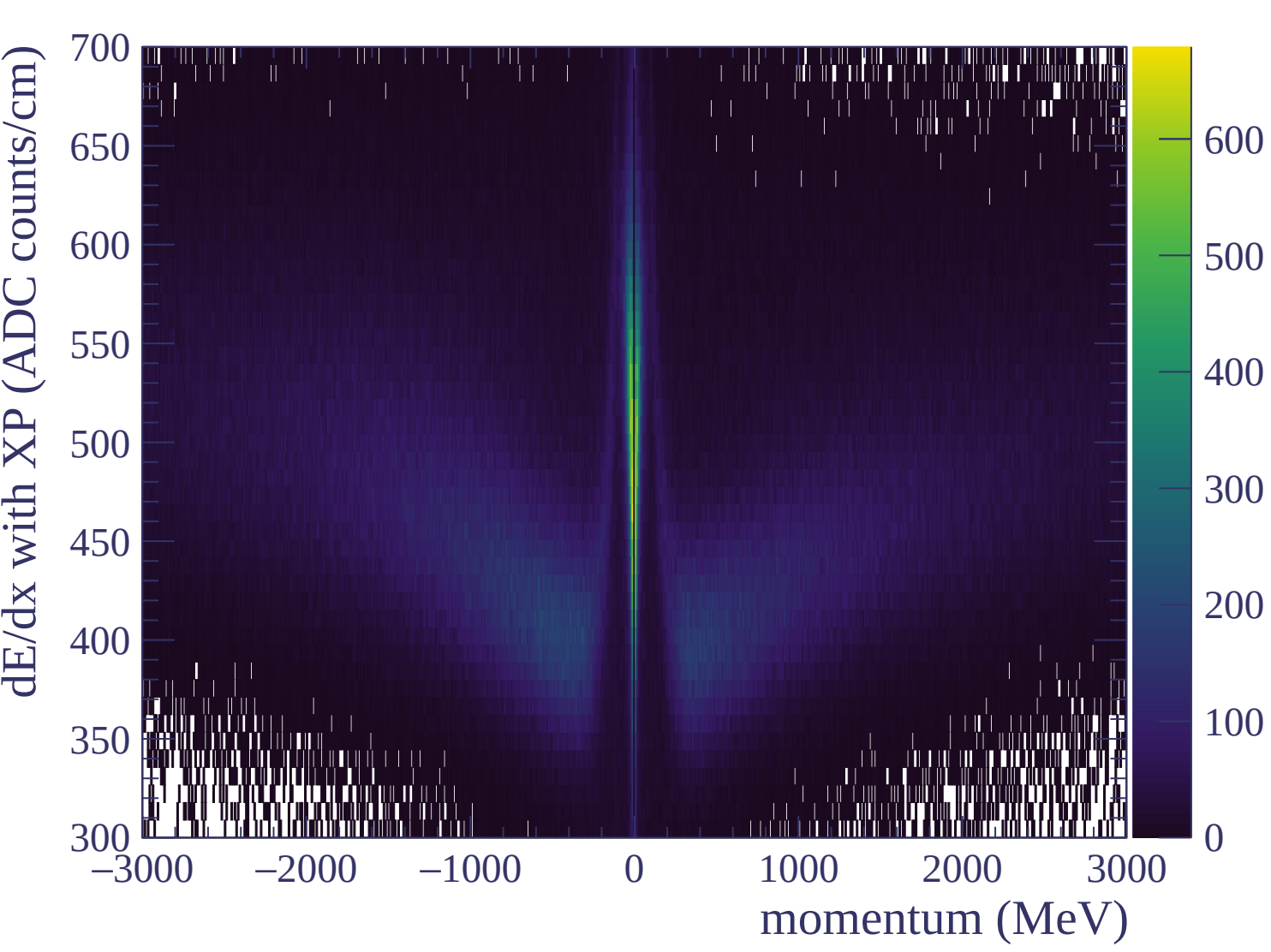


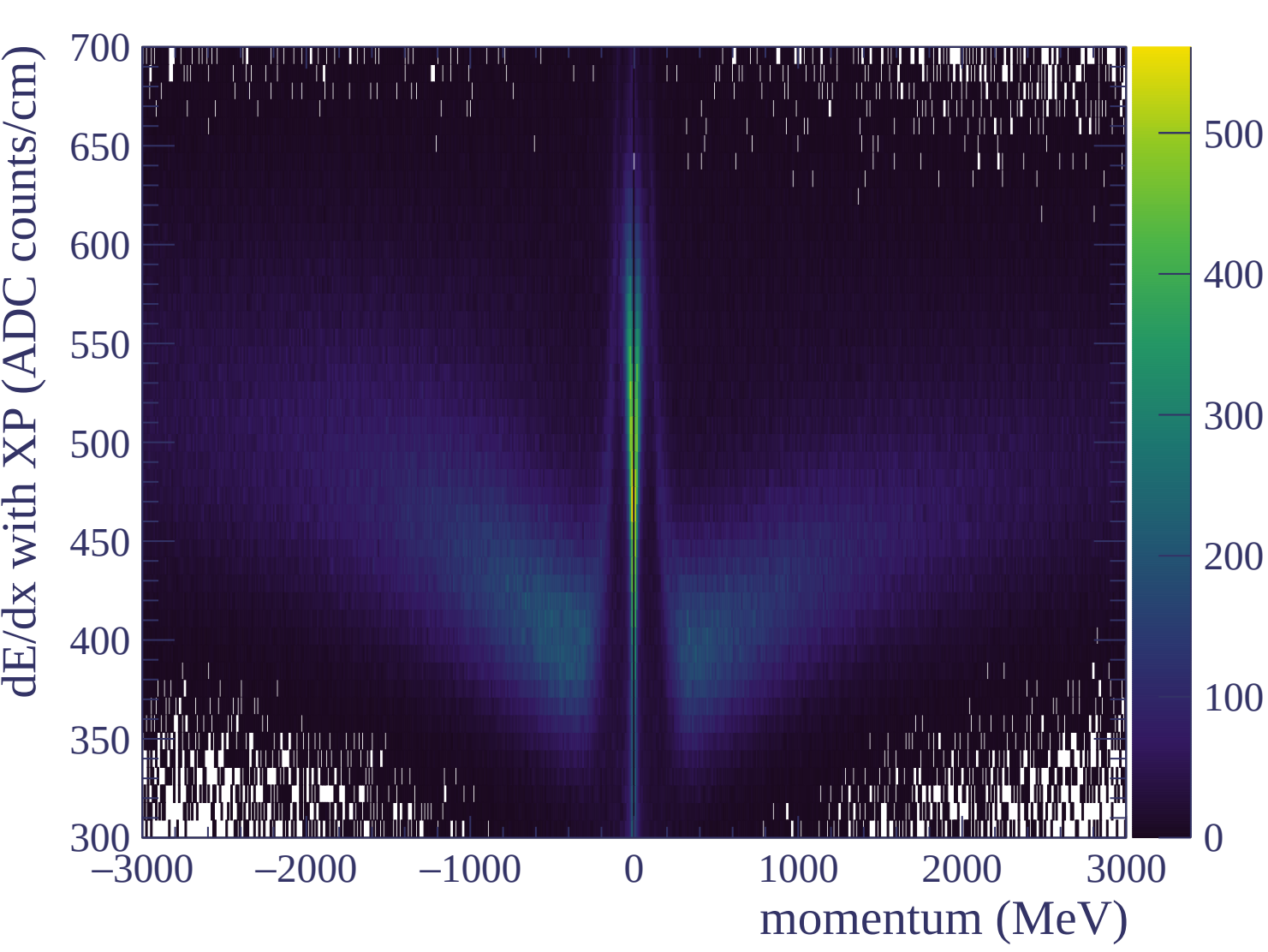
Resolution (%)

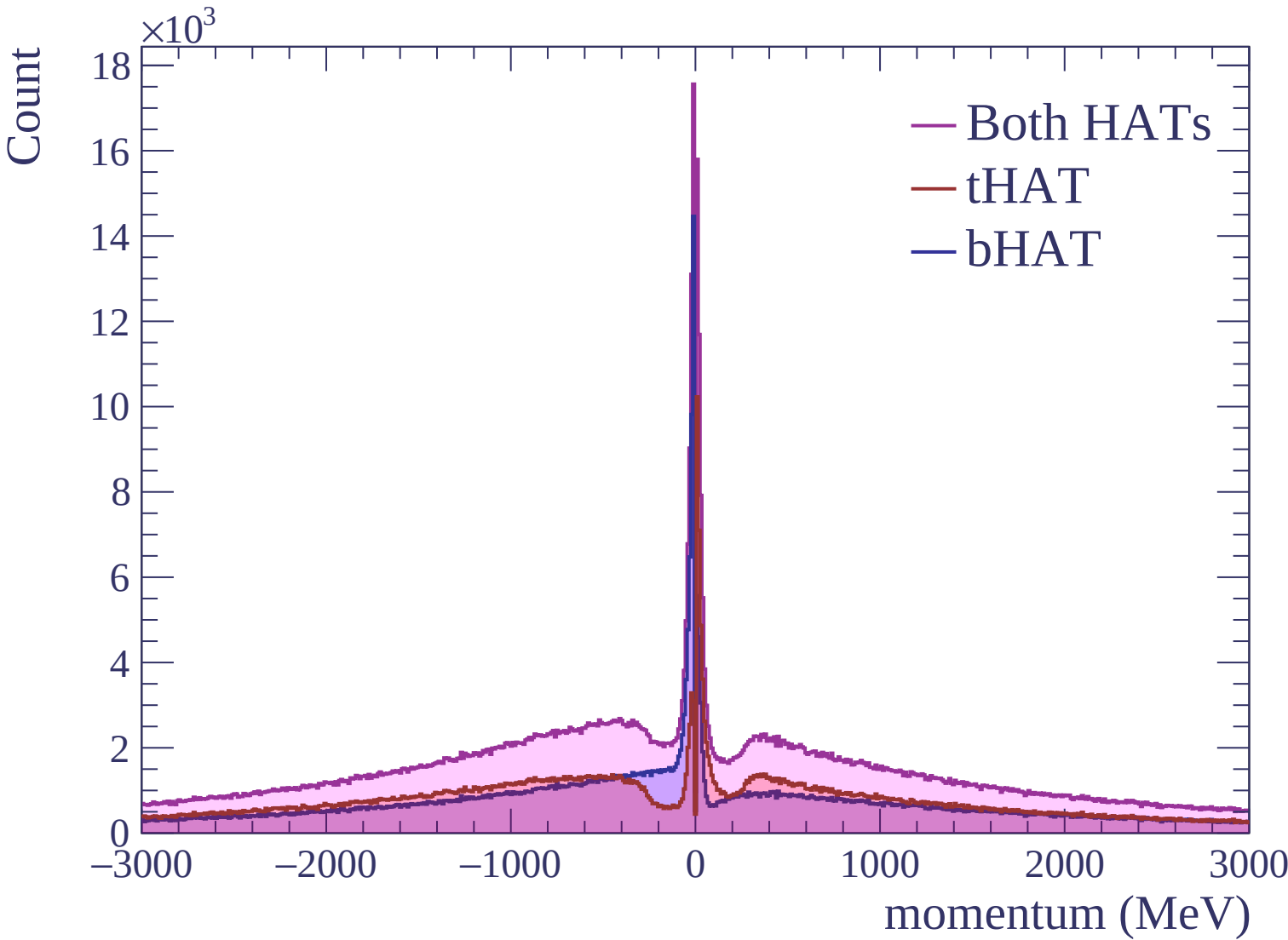






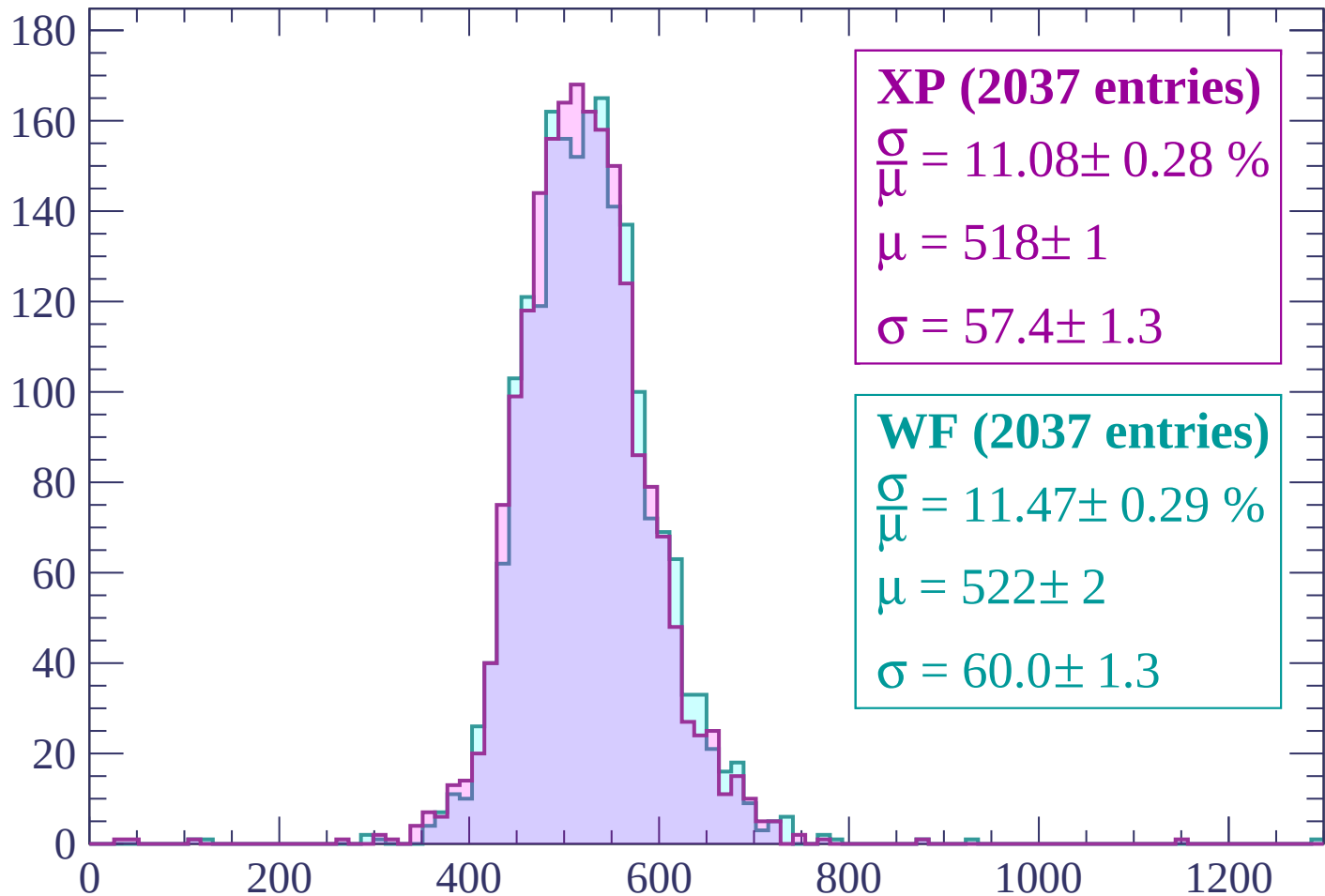






Energy loss | $-3000 < p < -2940$

Count



XP (2037 entries)

$$\frac{\sigma}{\mu} = 11.08 \pm 0.28 \%$$

$$\mu = 518 \pm 1$$

$$\sigma = 57.4 \pm 1.3$$

WF (2037 entries)

$$\frac{\sigma}{\mu} = 11.47 \pm 0.29 \%$$

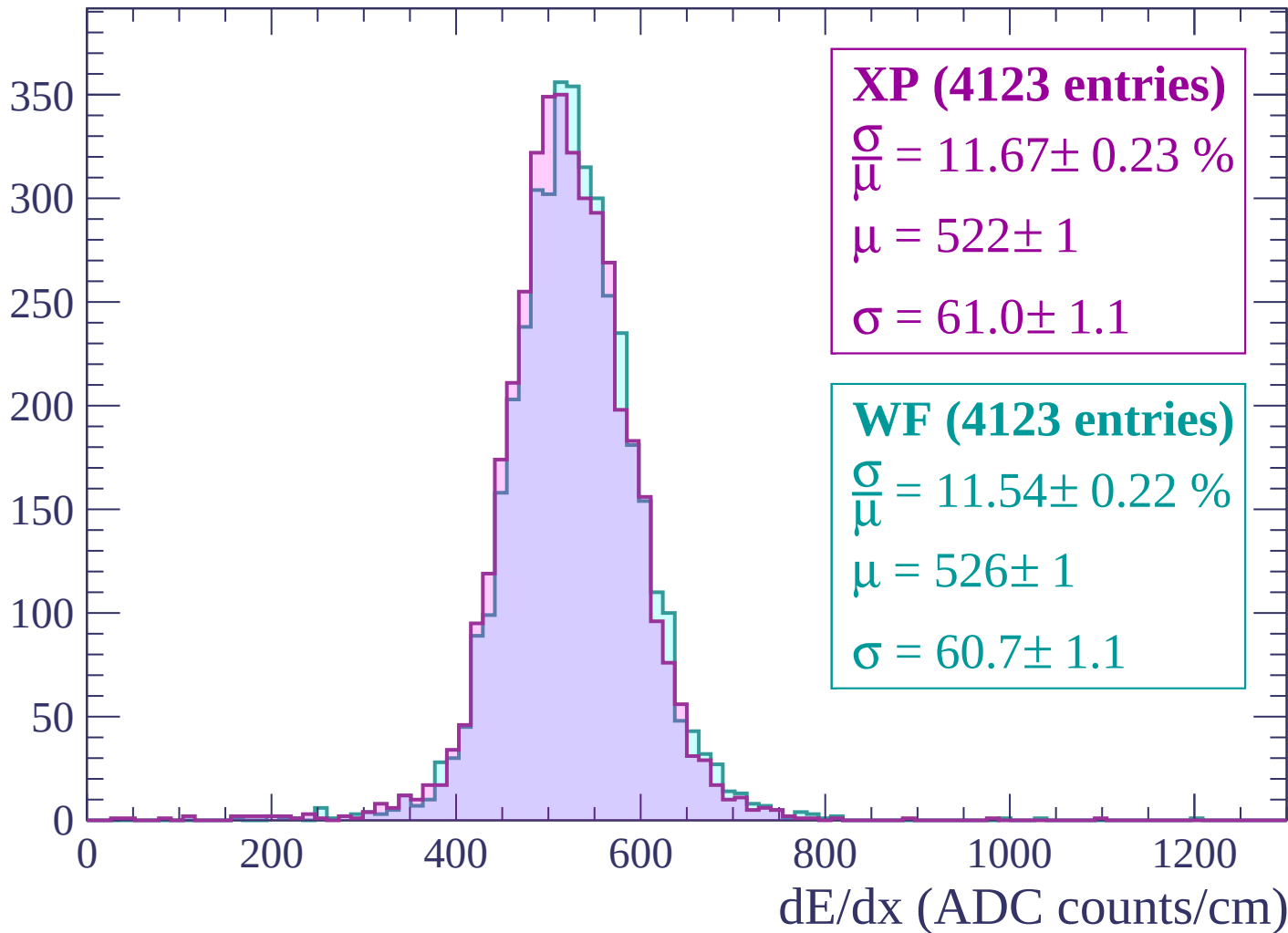
$$\mu = 522 \pm 2$$

$$\sigma = 60.0 \pm 1.3$$

dE/dx (ADC counts/cm)

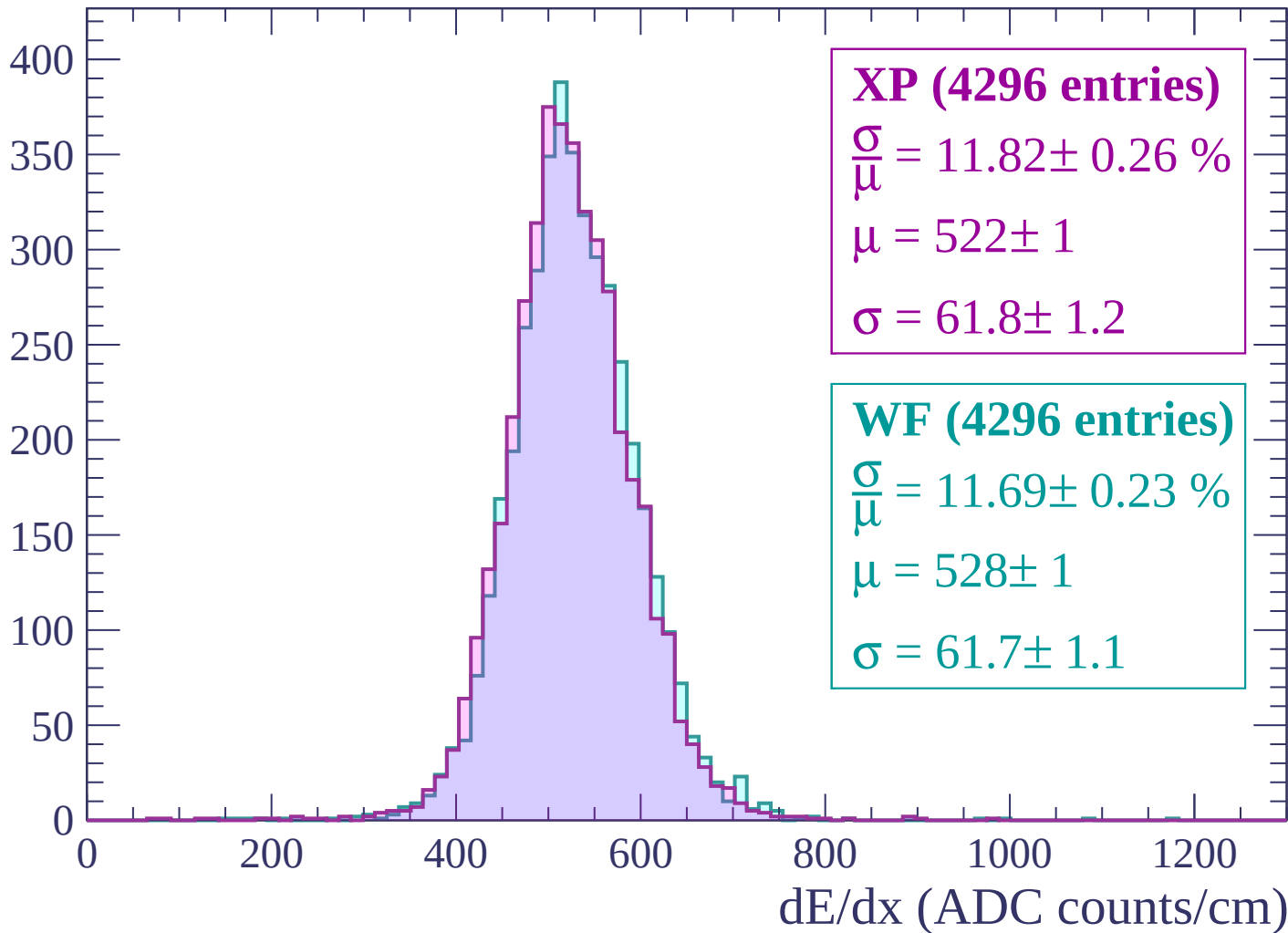
Energy loss | $-2940 < p < -2880$

Count



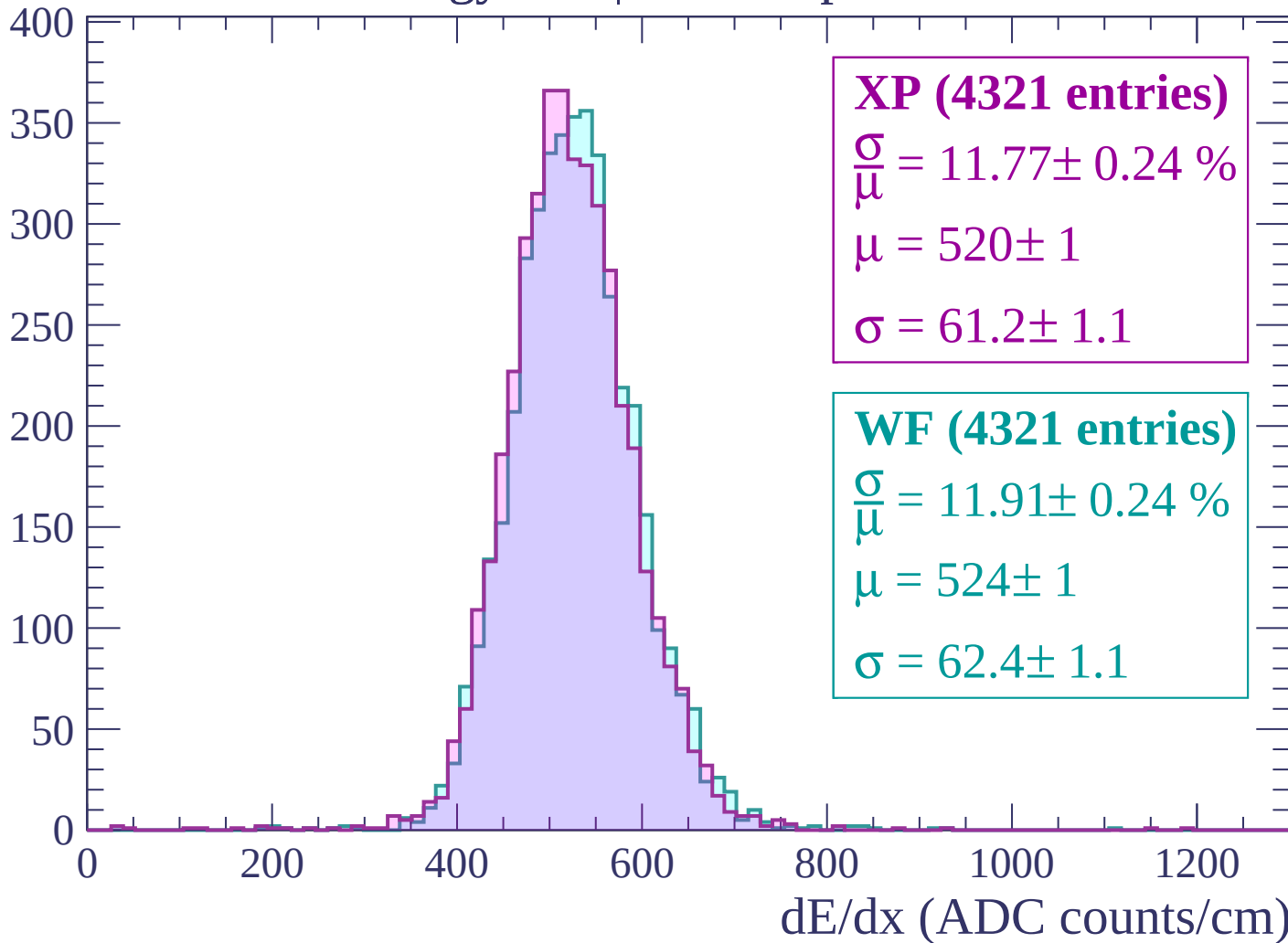
Energy loss | $-2880 < p < -2820$

Count



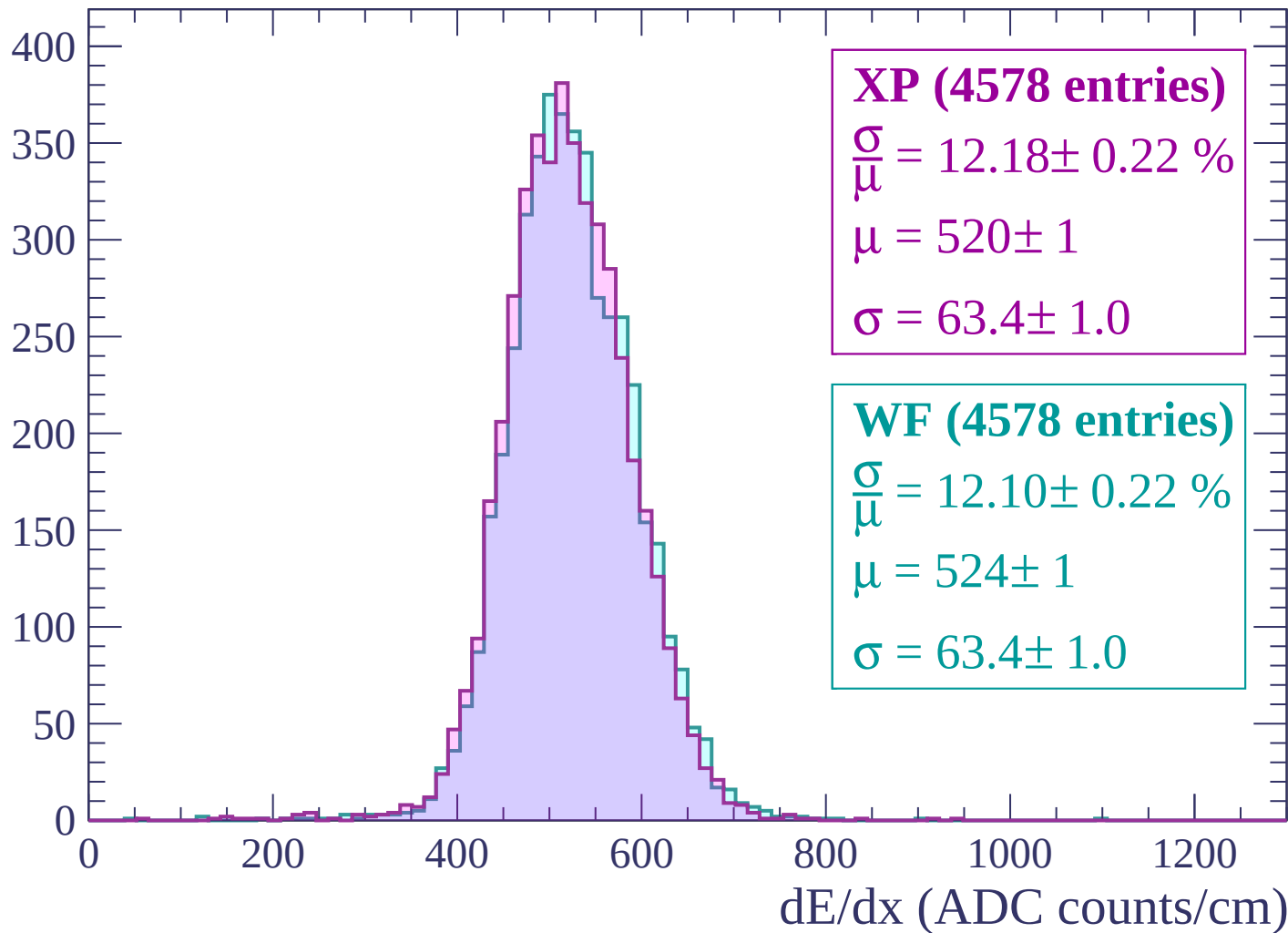
Energy loss | $-2820 < p < -2760$

Count



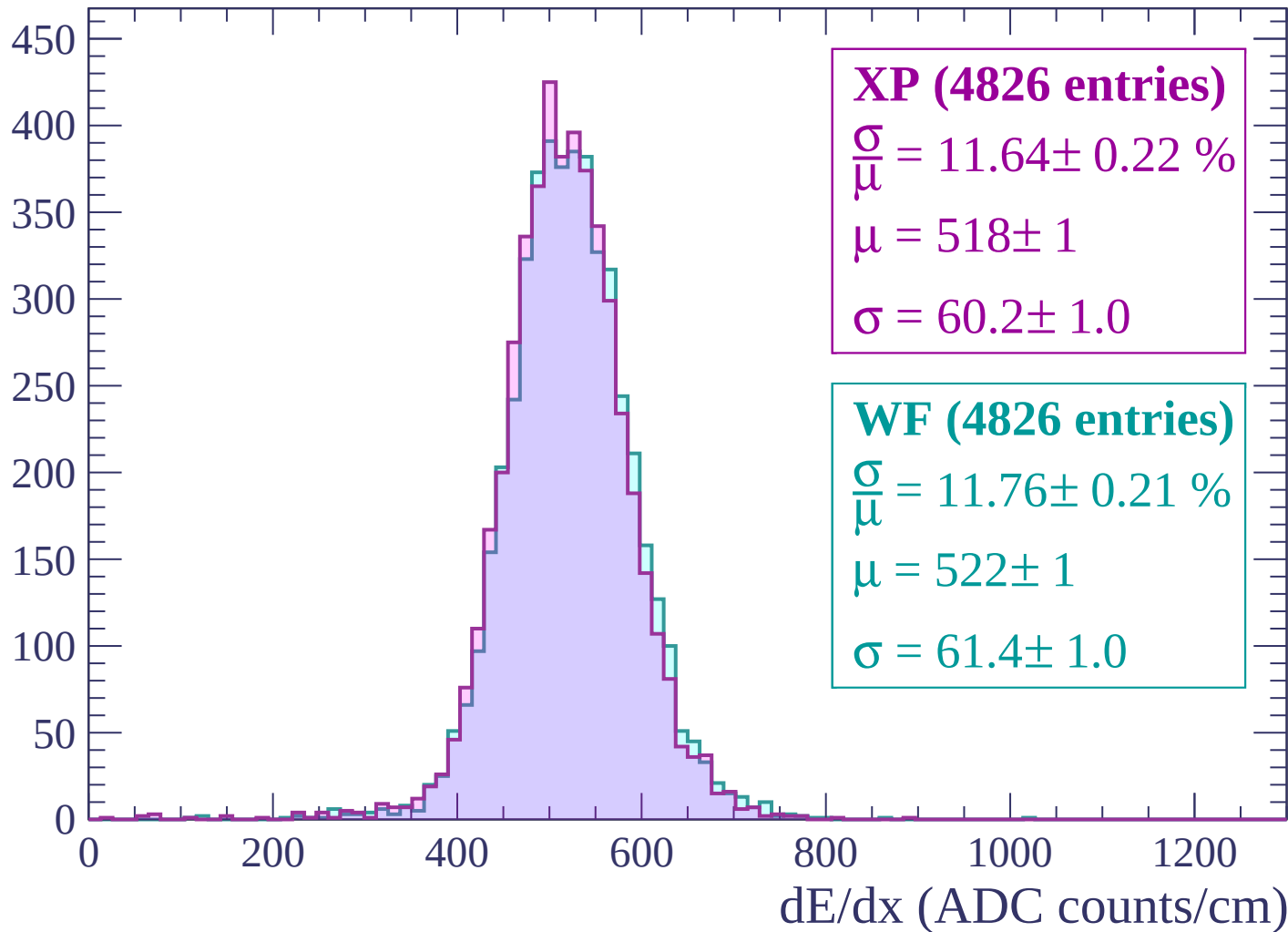
Energy loss | -2760 < p < -2700

Count



Energy loss | -2700 < p < -2640

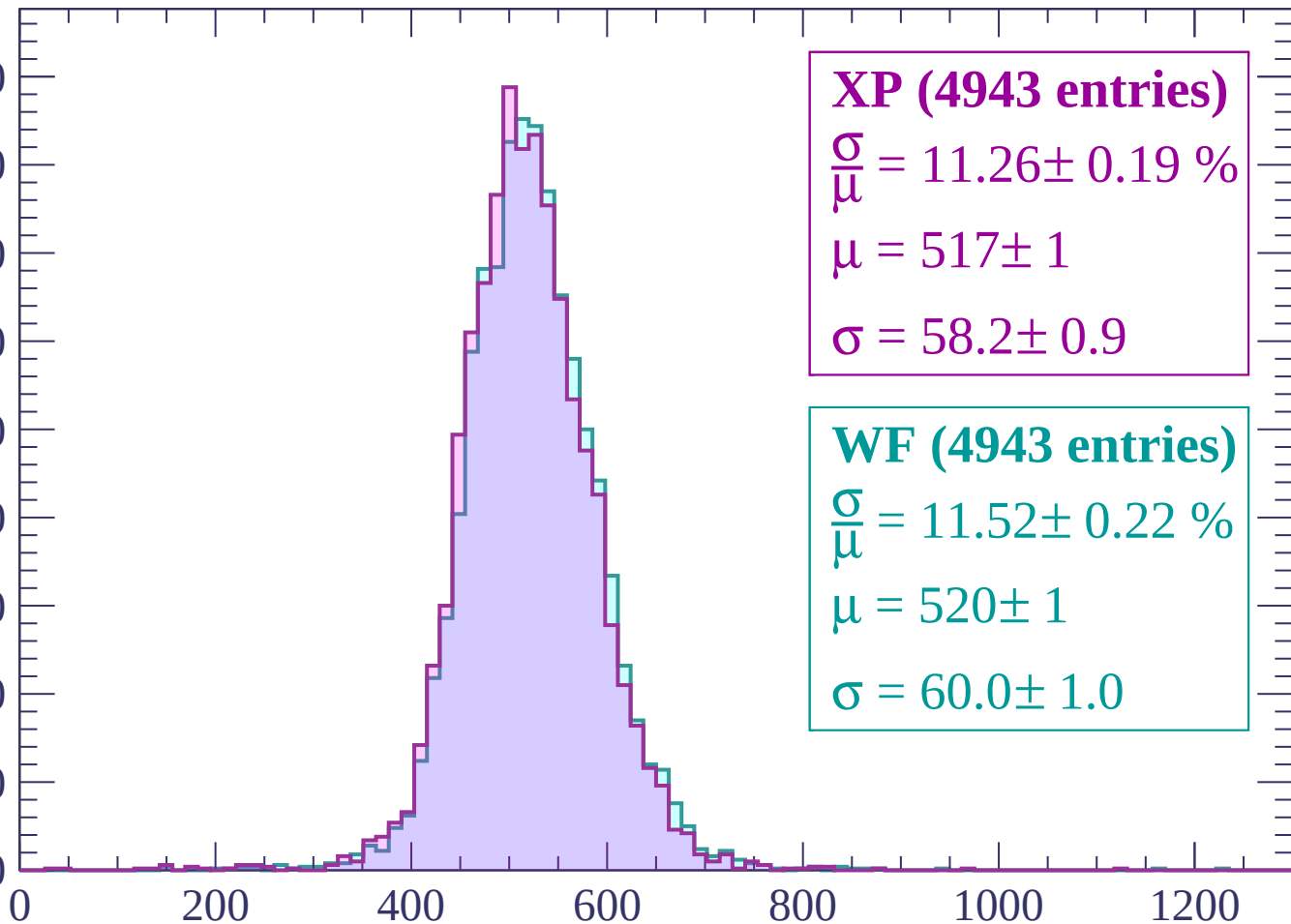
Count



Energy loss | $-2640 < p < -2580$

Count

450
400
350
300
250
200
150
100
50
0



XP (4943 entries)

$\frac{\sigma}{\mu} = 11.26 \pm 0.19 \%$

$\mu = 517 \pm 1$

$\sigma = 58.2 \pm 0.9$

WF (4943 entries)

$\frac{\sigma}{\mu} = 11.52 \pm 0.22 \%$

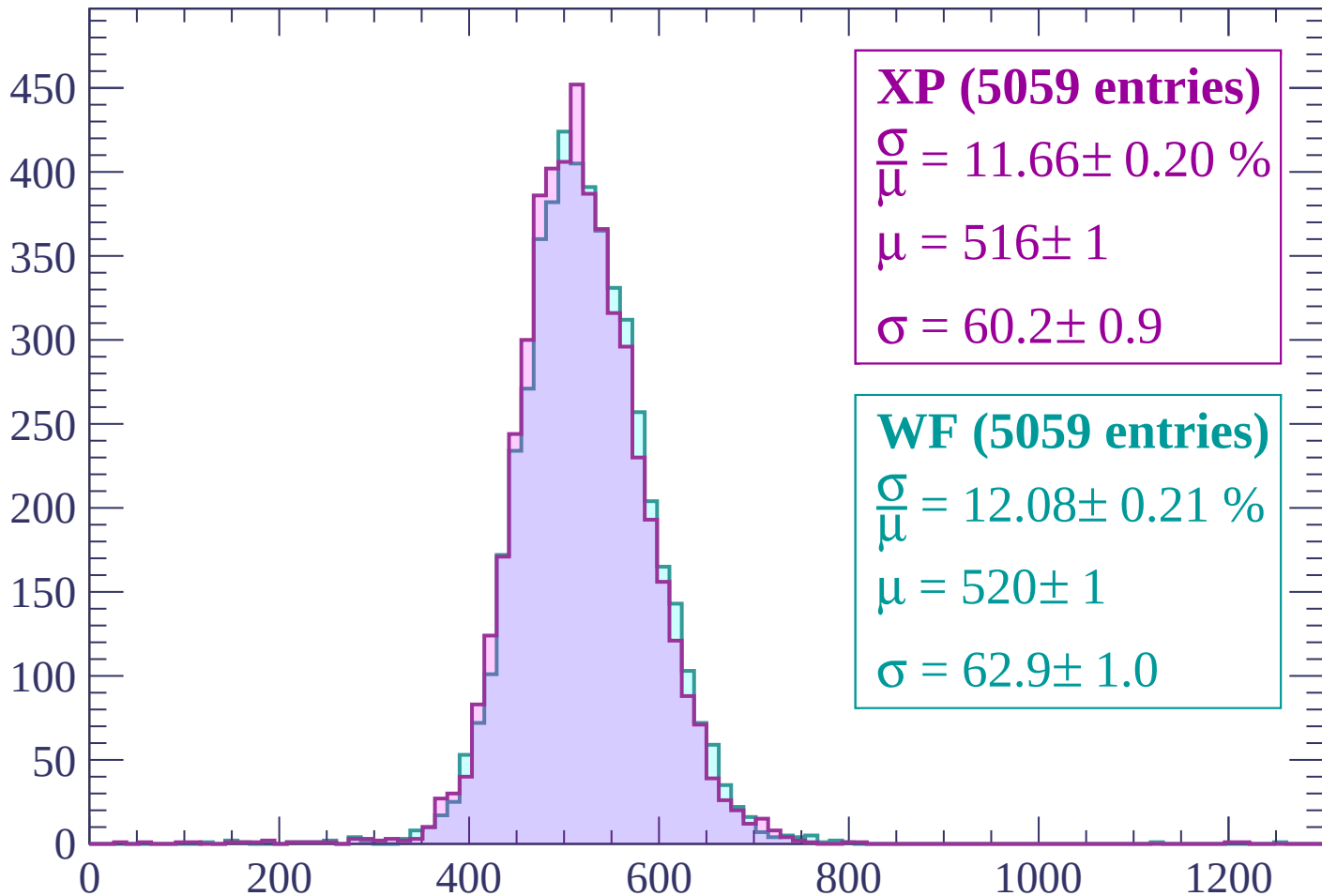
$\mu = 520 \pm 1$

$\sigma = 60.0 \pm 1.0$

dE/dx (ADC counts/cm)

Energy loss | $-2580 < p < -2520$

Count



XP (5059 entries)

$$\frac{\sigma}{\mu} = 11.66 \pm 0.20 \%$$

$$\mu = 516 \pm 1$$

$$\sigma = 60.2 \pm 0.9$$

WF (5059 entries)

$$\frac{\sigma}{\mu} = 12.08 \pm 0.21 \%$$

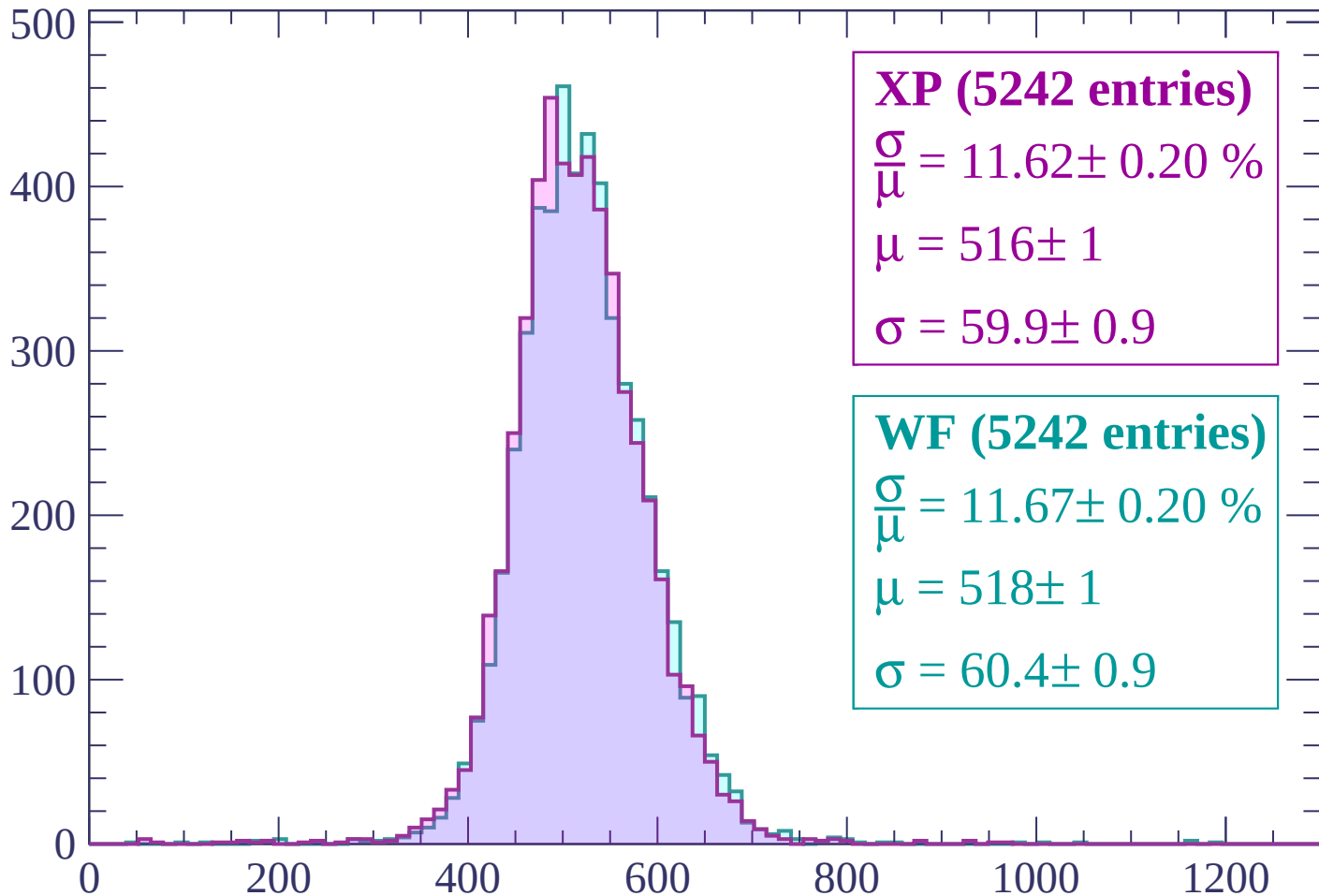
$$\mu = 520 \pm 1$$

$$\sigma = 62.9 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | -2520 < p < -2460

Count



XP (5242 entries)

$$\frac{\sigma}{\mu} = 11.62 \pm 0.20 \%$$

$$\mu = 516 \pm 1$$

$$\sigma = 59.9 \pm 0.9$$

WF (5242 entries)

$$\frac{\sigma}{\mu} = 11.67 \pm 0.20 \%$$

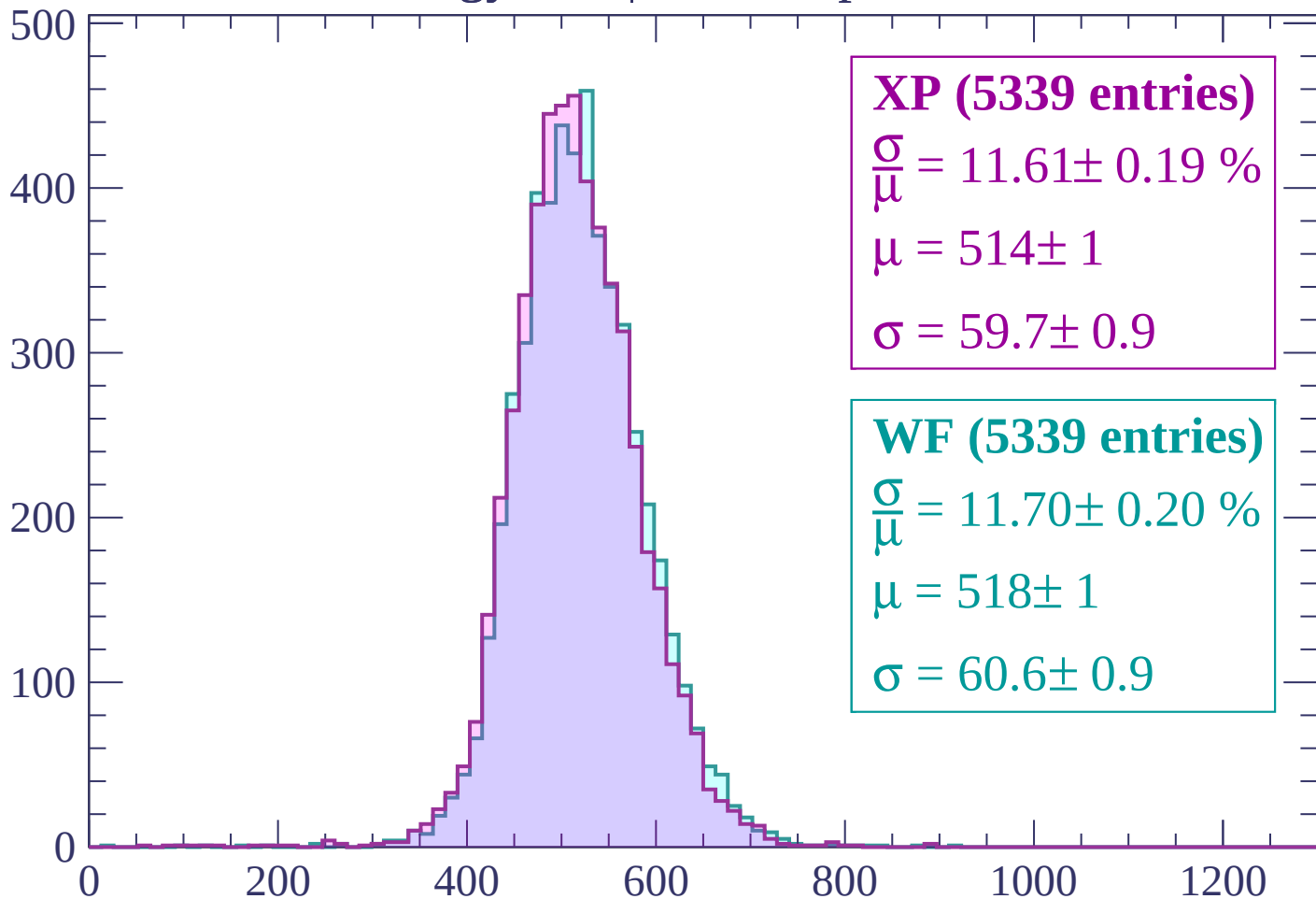
$$\mu = 518 \pm 1$$

$$\sigma = 60.4 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $-2460 < p < -2400$

Count



XP (5339 entries)

$\frac{\sigma}{\mu} = 11.61 \pm 0.19 \%$

$\mu = 514 \pm 1$

$\sigma = 59.7 \pm 0.9$

WF (5339 entries)

$\frac{\sigma}{\mu} = 11.70 \pm 0.20 \%$

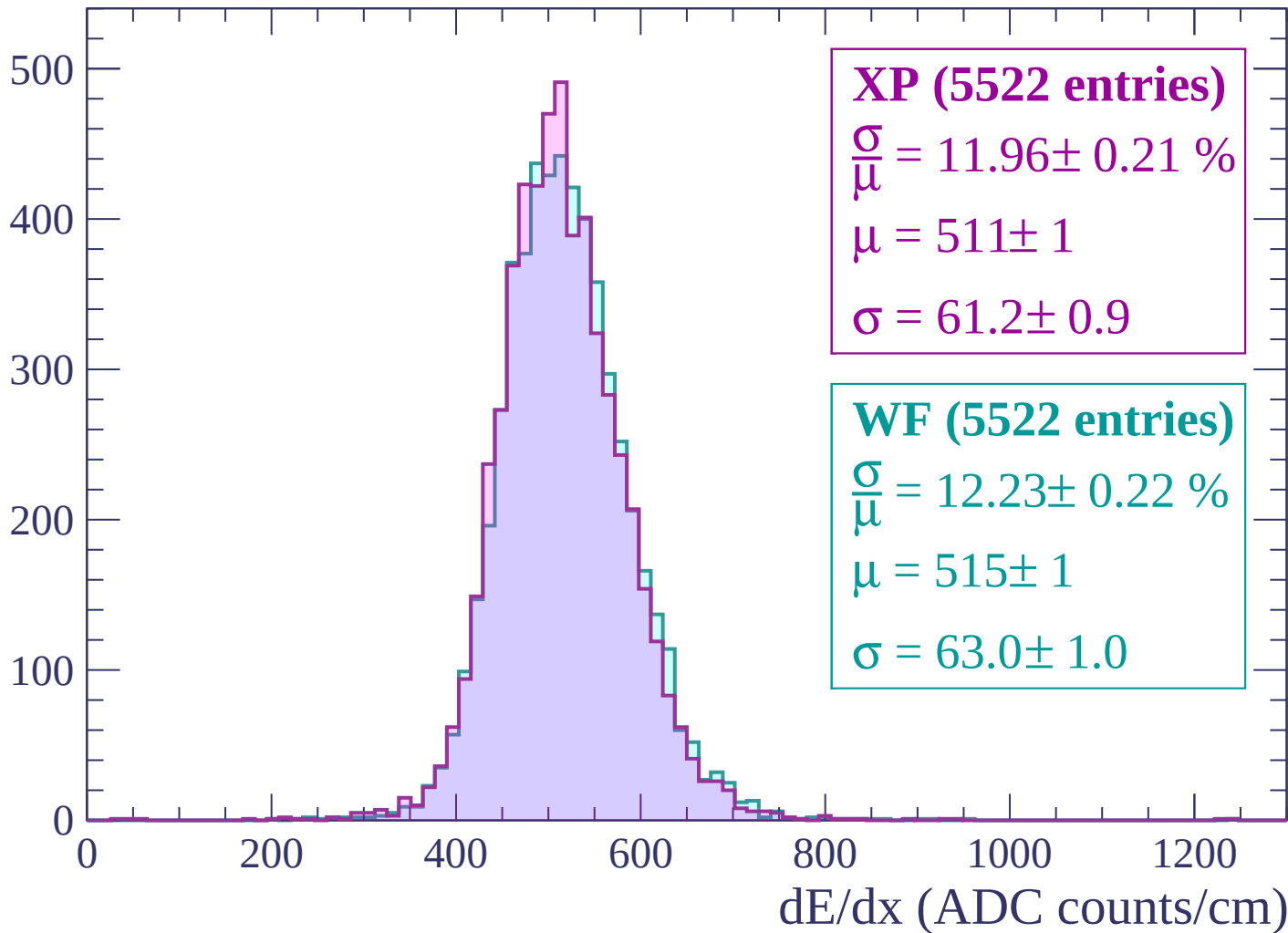
$\mu = 518 \pm 1$

$\sigma = 60.6 \pm 0.9$

dE/dx (ADC counts/cm)

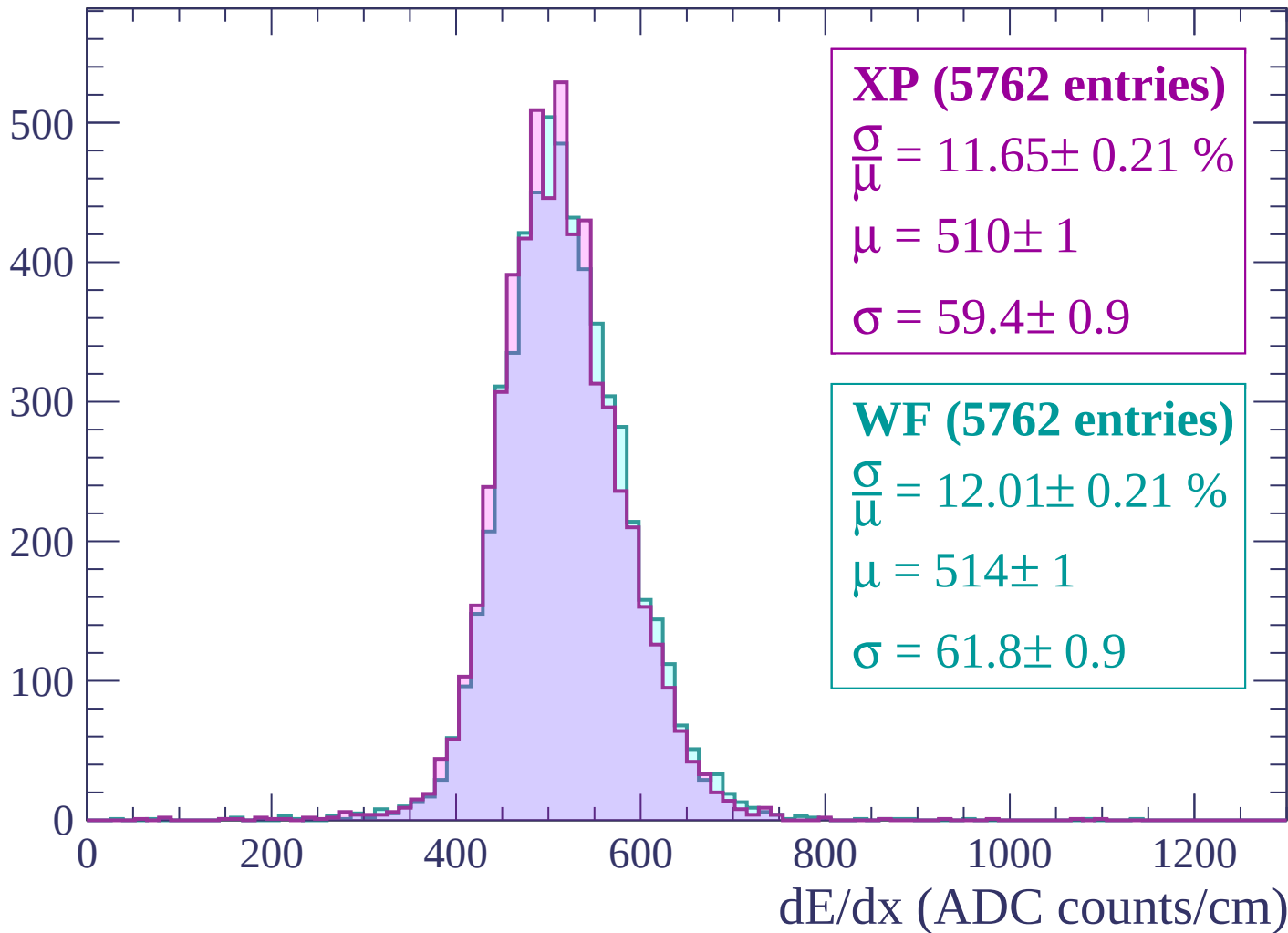
Energy loss | -2400 < p < -2340

Count



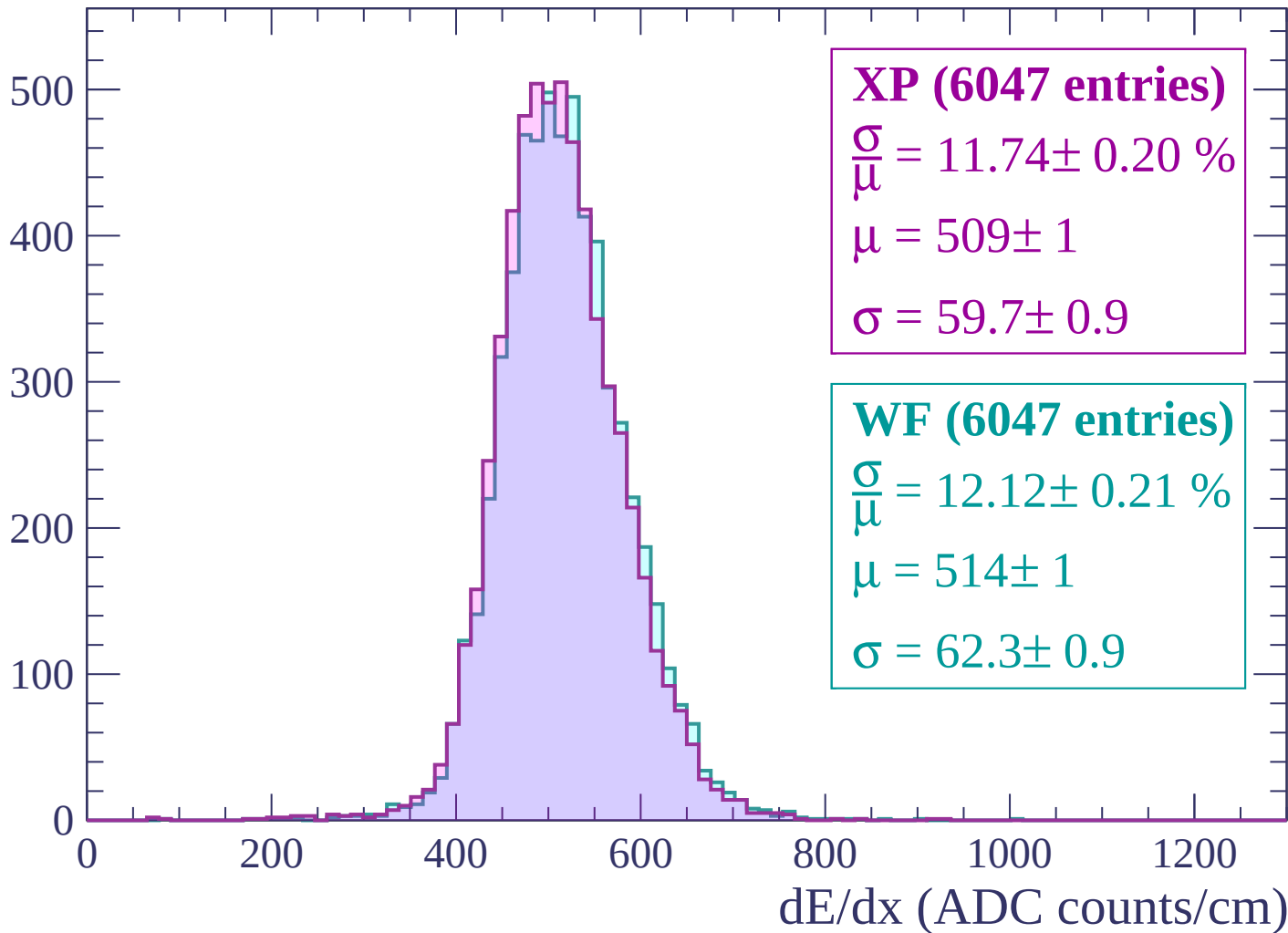
Energy loss | $-2340 < p < -2280$

Count



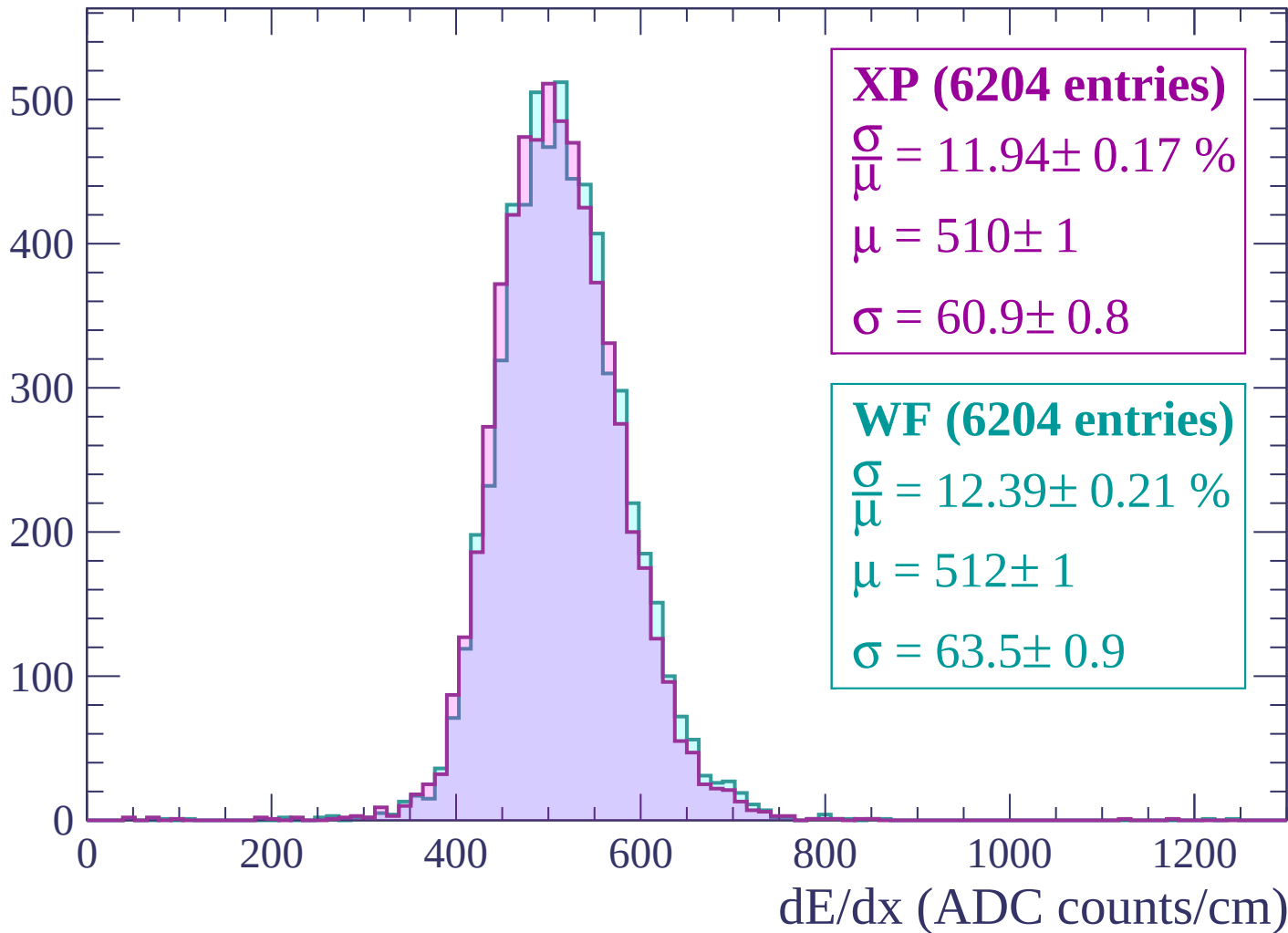
Energy loss | $-2280 < p < -2220$

Count



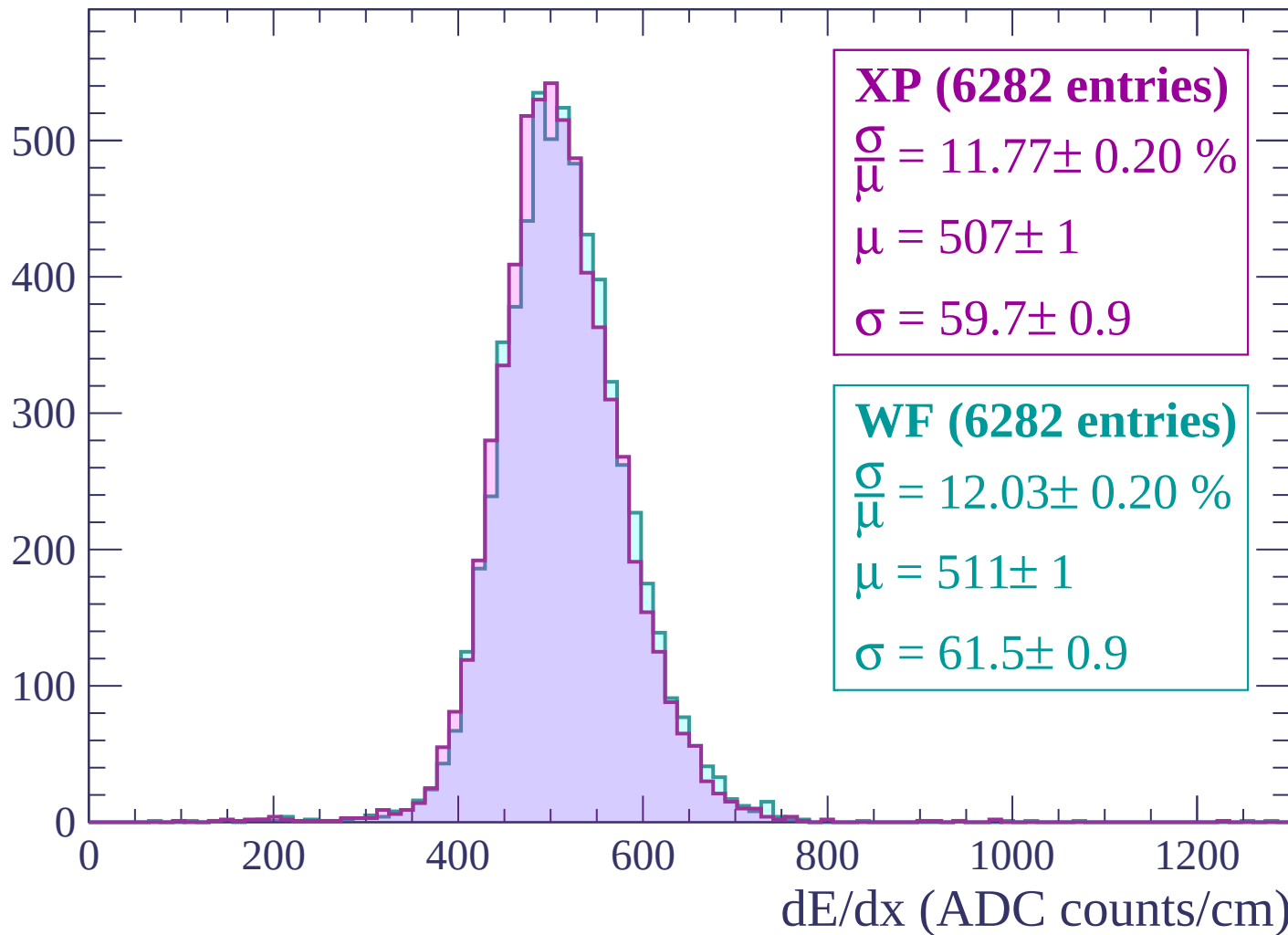
Energy loss | -2220 < p < -2160

Count



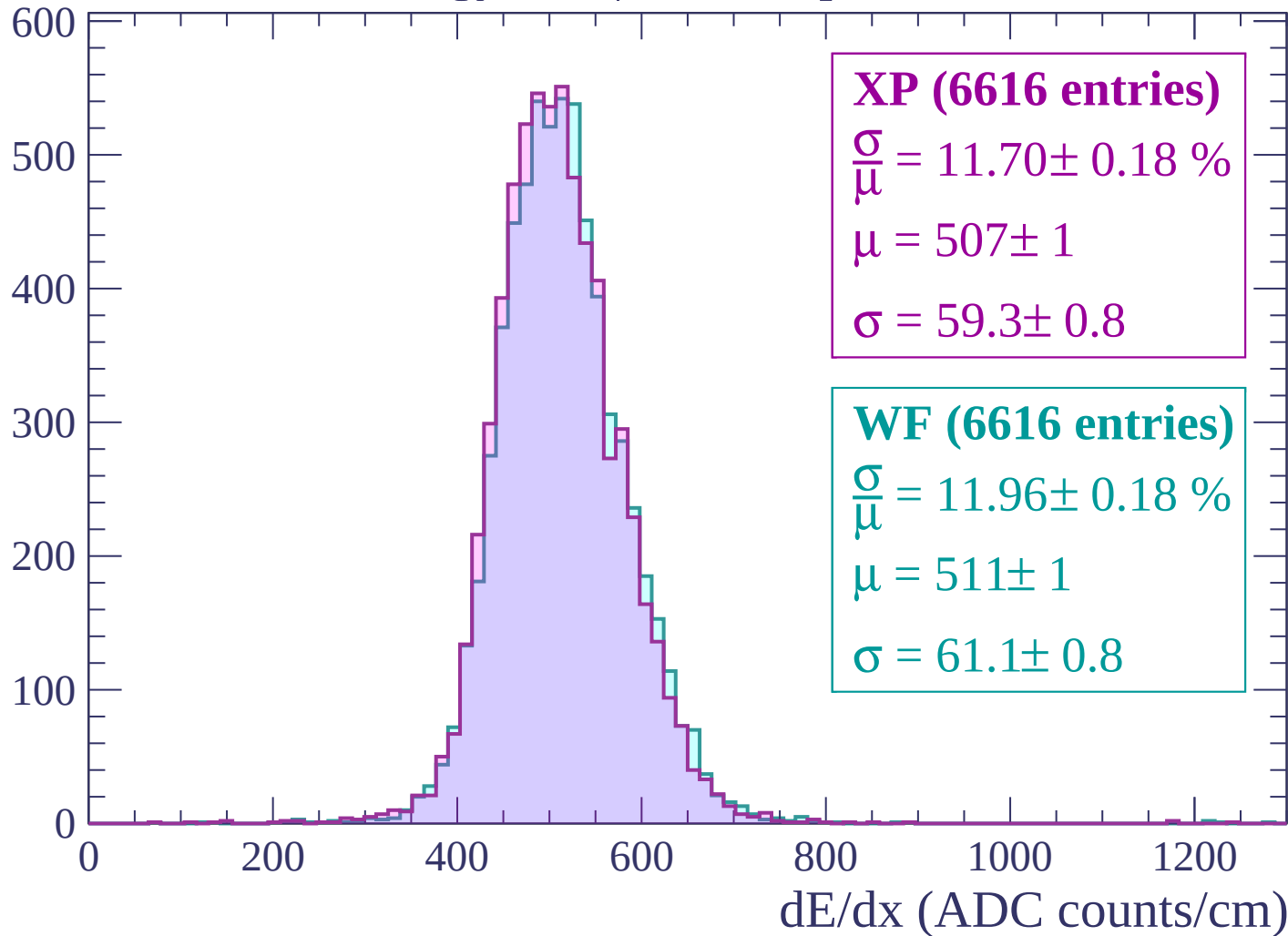
Energy loss | $-2160 < p < -2100$

Count



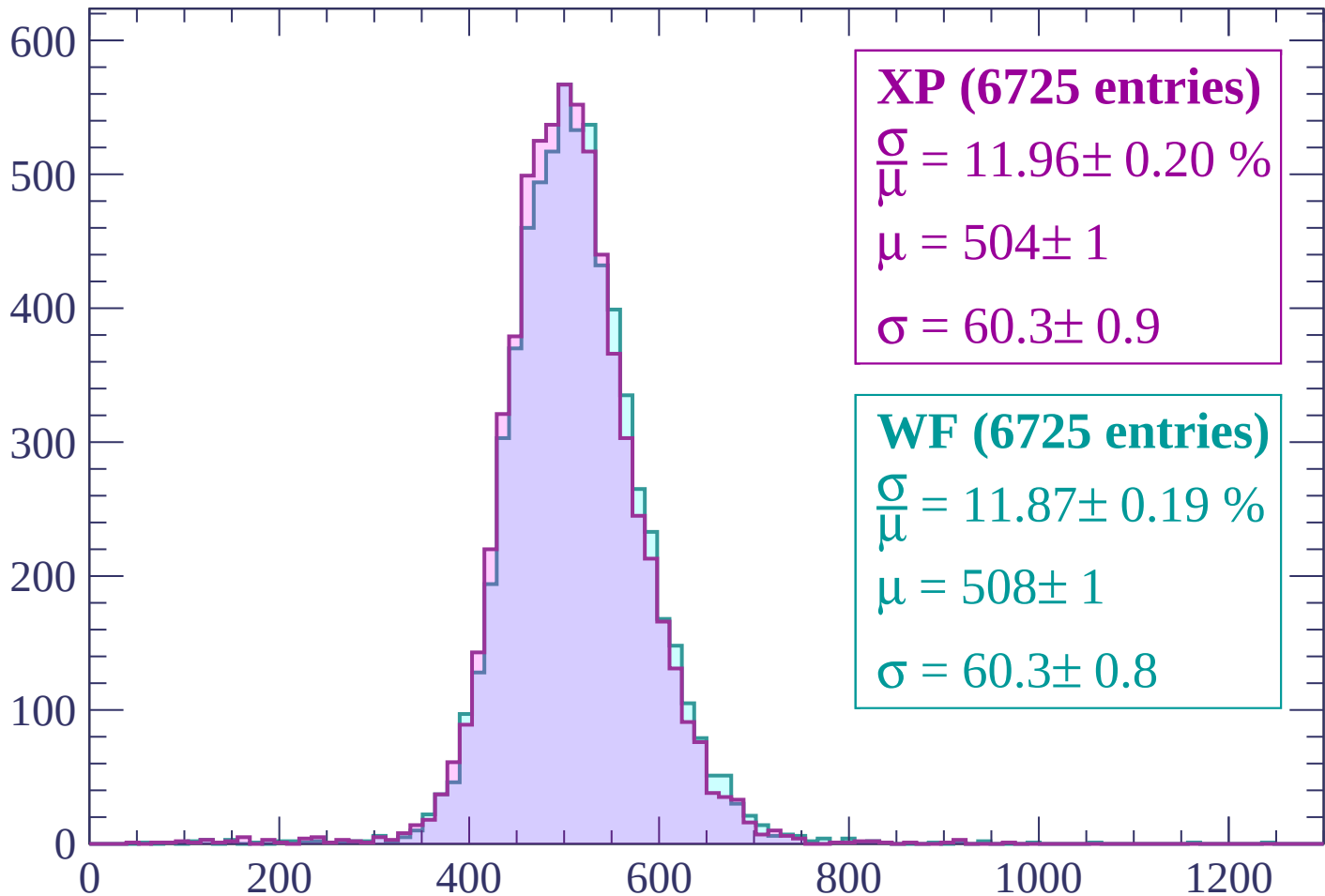
Energy loss | -2100 < p < -2040

Count



Energy loss | -2040 < p < -1980

Count



XP (6725 entries)

$$\frac{\sigma}{\mu} = 11.96 \pm 0.20 \%$$

$$\mu = 504 \pm 1$$

$$\sigma = 60.3 \pm 0.9$$

WF (6725 entries)

$$\frac{\sigma}{\mu} = 11.87 \pm 0.19 \%$$

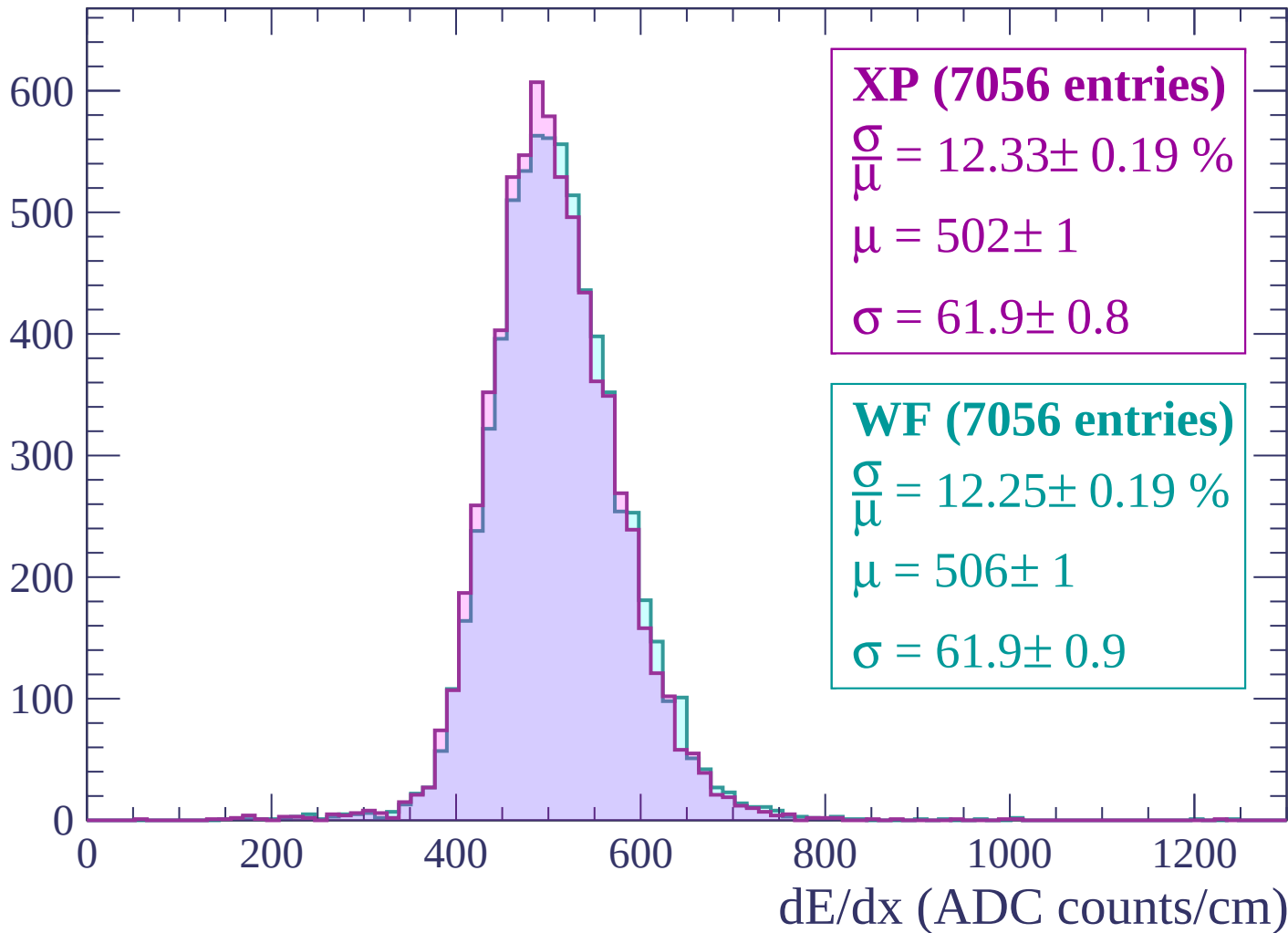
$$\mu = 508 \pm 1$$

$$\sigma = 60.3 \pm 0.8$$

dE/dx (ADC counts/cm)

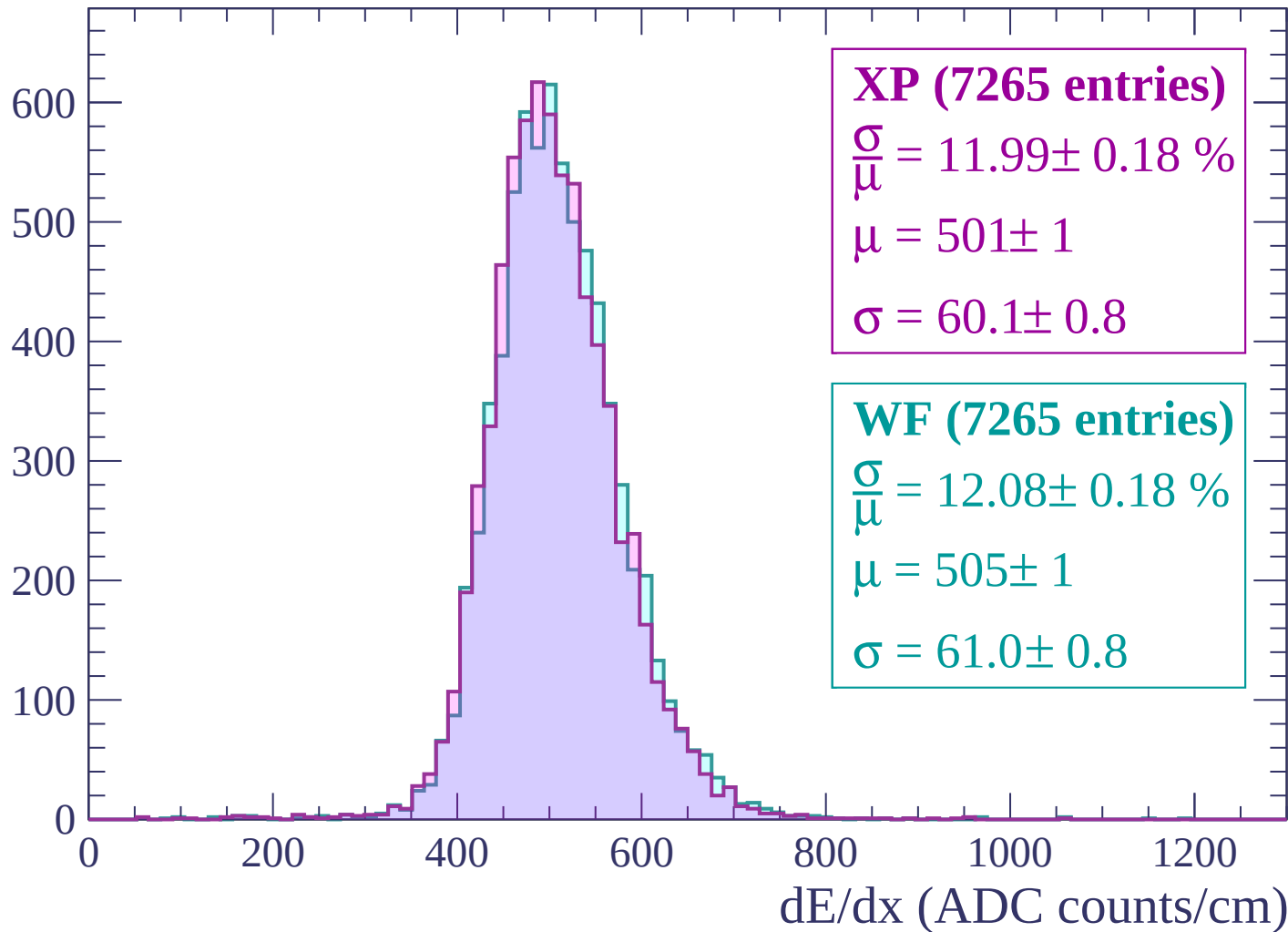
Energy loss | -1980 < p < -1920

Count



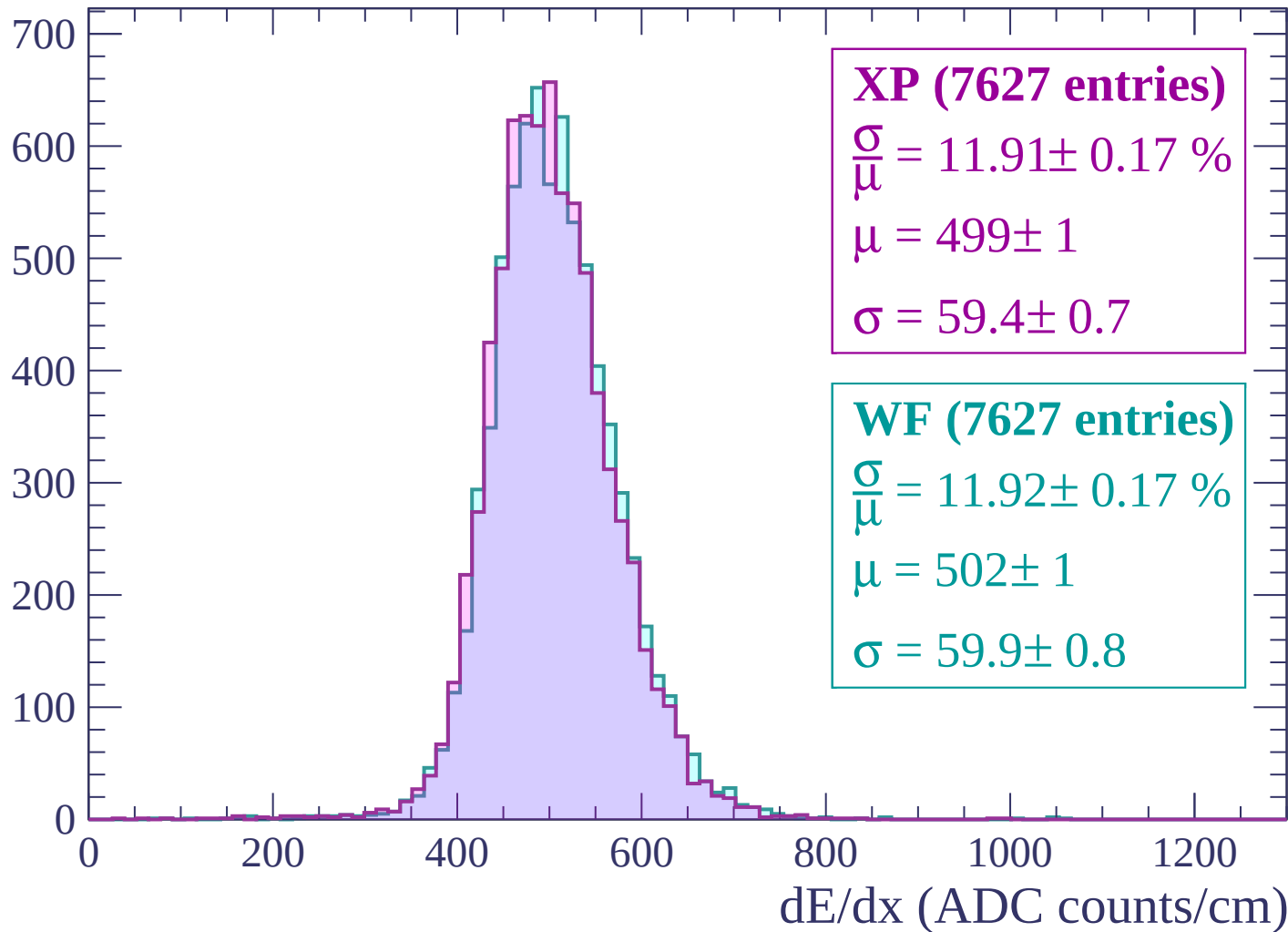
Energy loss | -1920 < p < -1860

Count



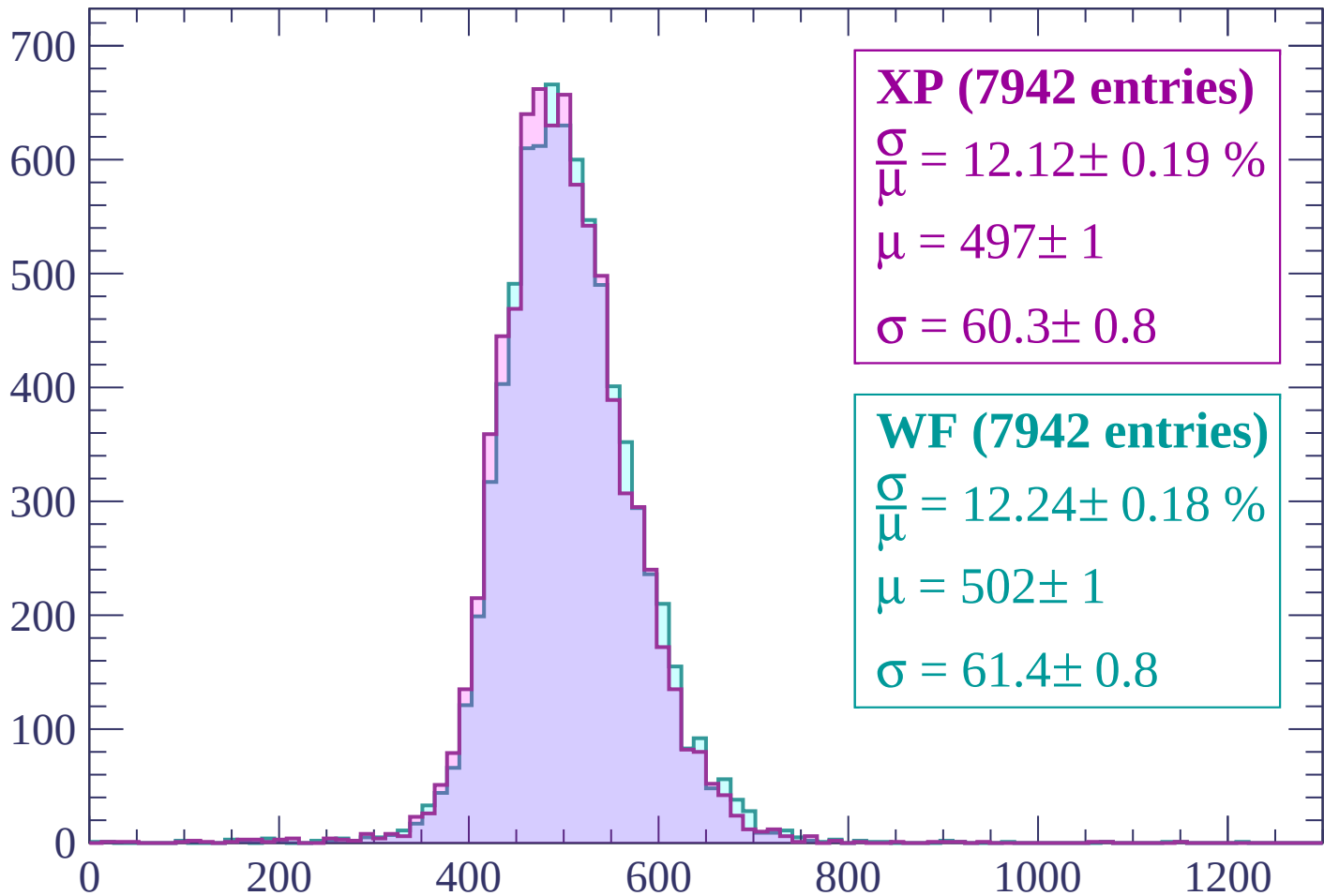
Energy loss | -1860 < p < -1800

Count



Energy loss | -1800 < p < -1740

Count



XP (7942 entries)

$$\frac{\sigma}{\mu} = 12.12 \pm 0.19 \%$$

$$\mu = 497 \pm 1$$

$$\sigma = 60.3 \pm 0.8$$

WF (7942 entries)

$$\frac{\sigma}{\mu} = 12.24 \pm 0.18 \%$$

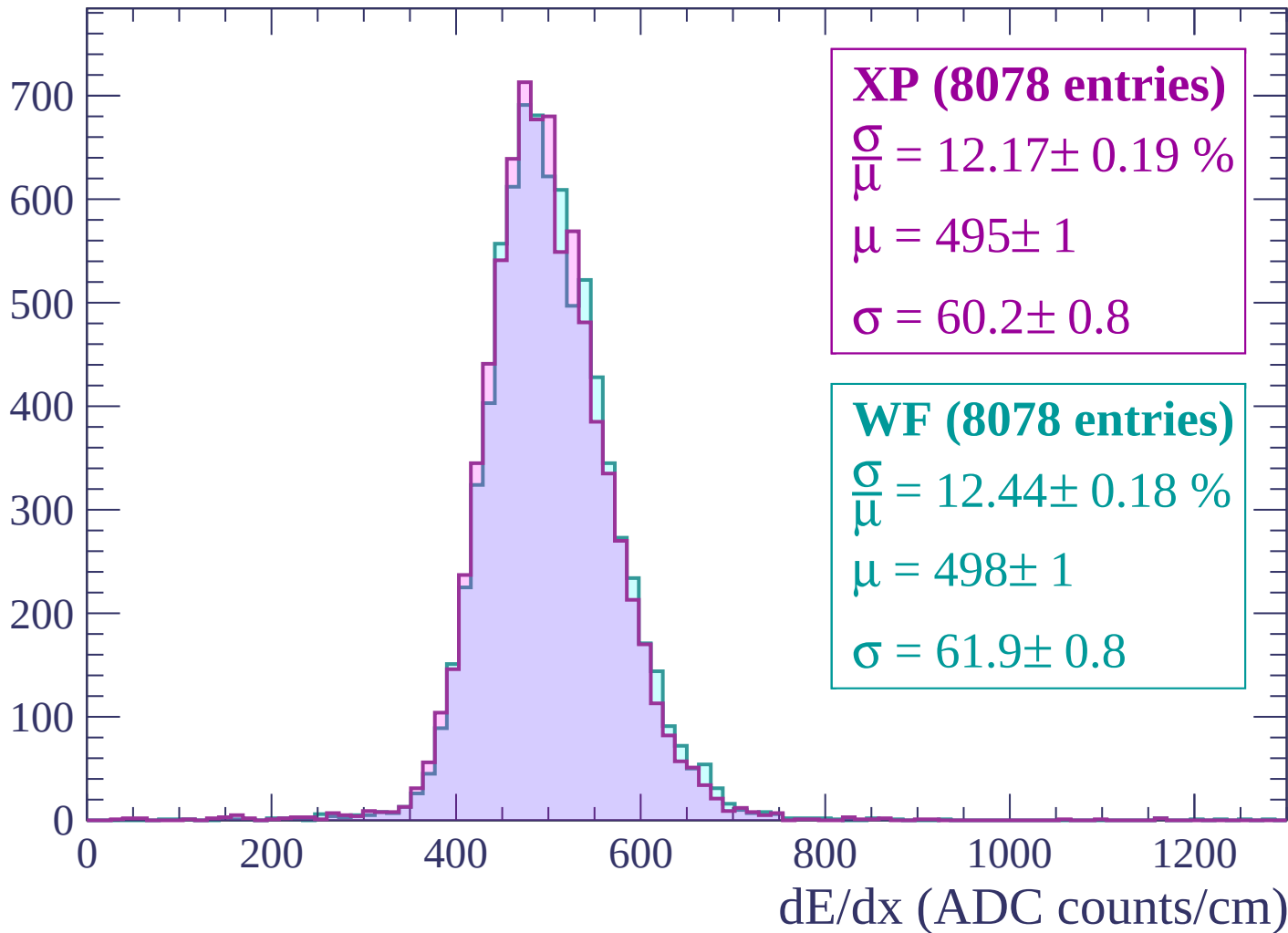
$$\mu = 502 \pm 1$$

$$\sigma = 61.4 \pm 0.8$$

dE/dx (ADC counts/cm)

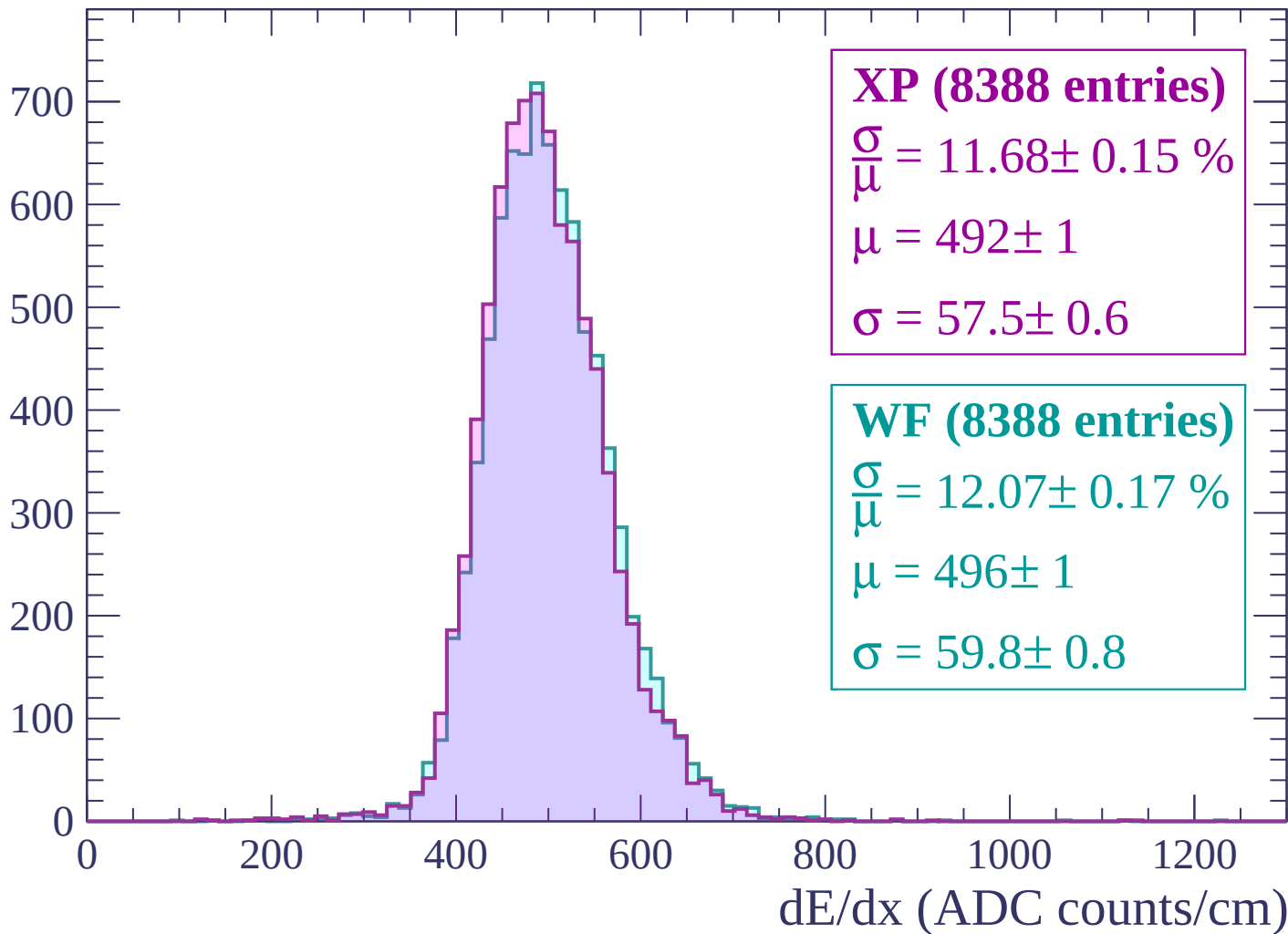
Energy loss | -1740 < p < -1680

Count



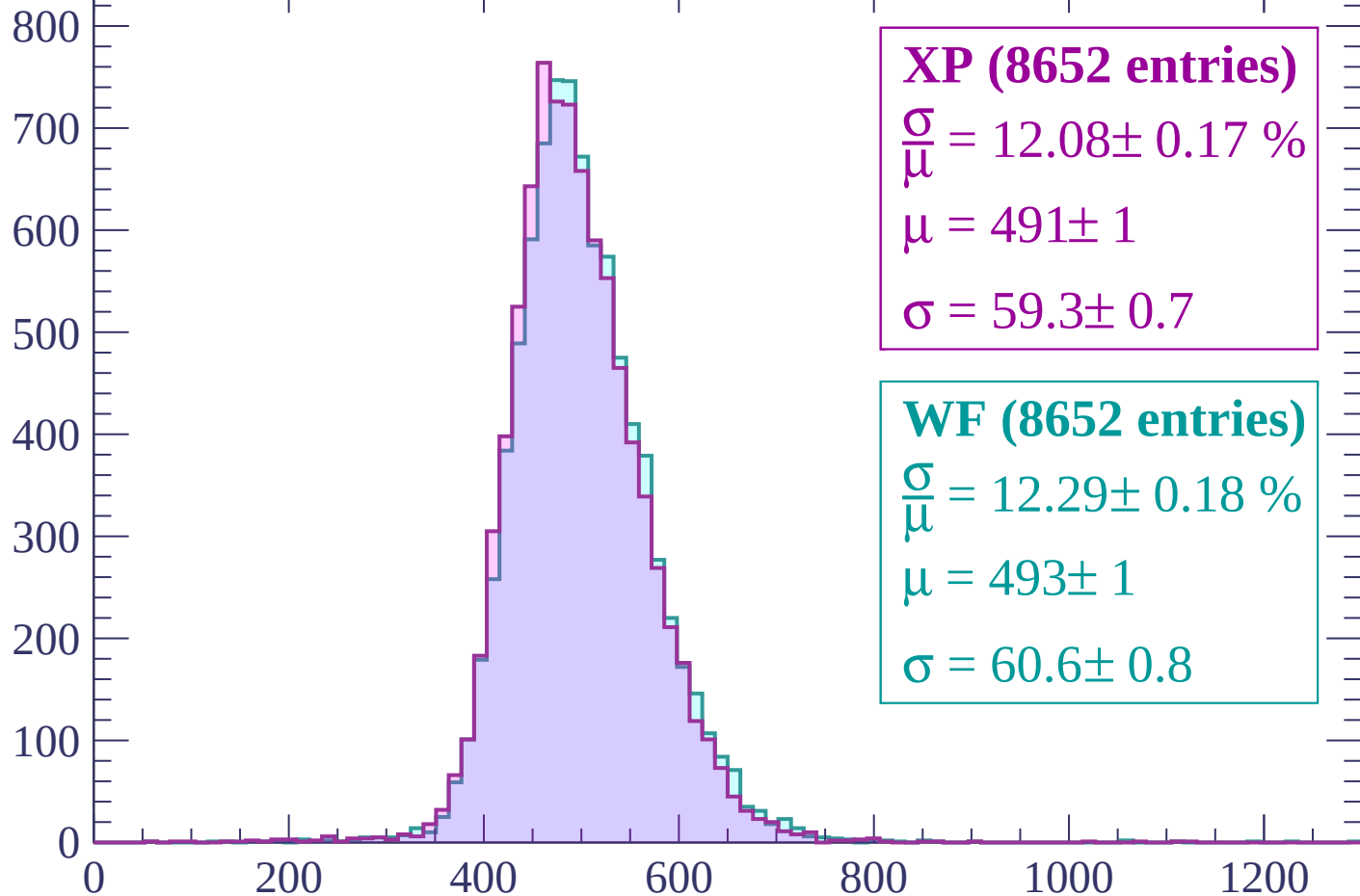
Energy loss | $-1680 < p < -1620$

Count



Energy loss | $-1620 < p < -1560$

Count



XP (8652 entries)

$$\frac{\sigma}{\mu} = 12.08 \pm 0.17 \%$$

$$\mu = 491 \pm 1$$

$$\sigma = 59.3 \pm 0.7$$

WF (8652 entries)

$$\frac{\sigma}{\mu} = 12.29 \pm 0.18 \%$$

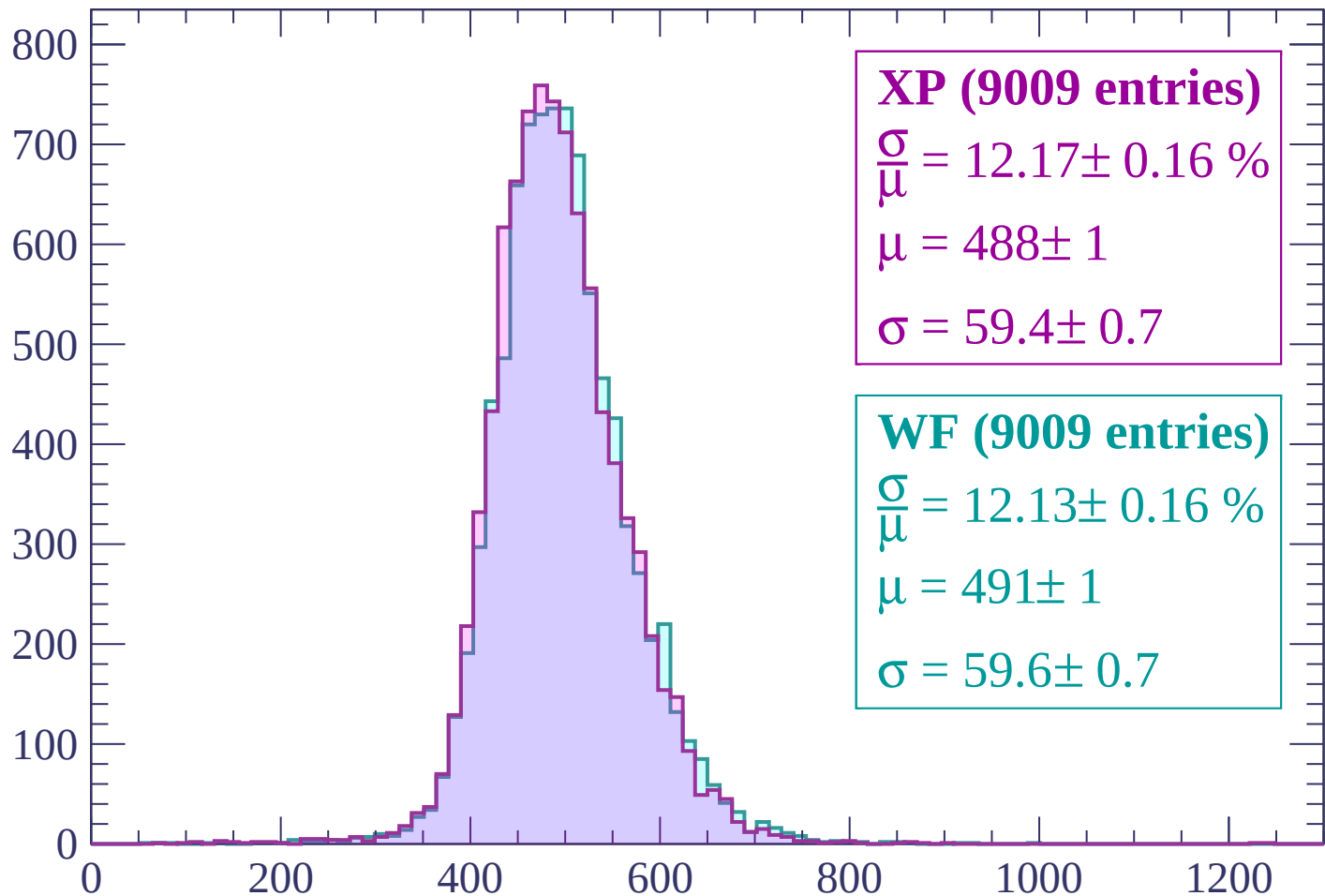
$$\mu = 493 \pm 1$$

$$\sigma = 60.6 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | -1560 < p < -1500

Count



XP (9009 entries)

$$\frac{\sigma}{\mu} = 12.17 \pm 0.16 \%$$

$$\mu = 488 \pm 1$$

$$\sigma = 59.4 \pm 0.7$$

WF (9009 entries)

$$\frac{\sigma}{\mu} = 12.13 \pm 0.16 \%$$

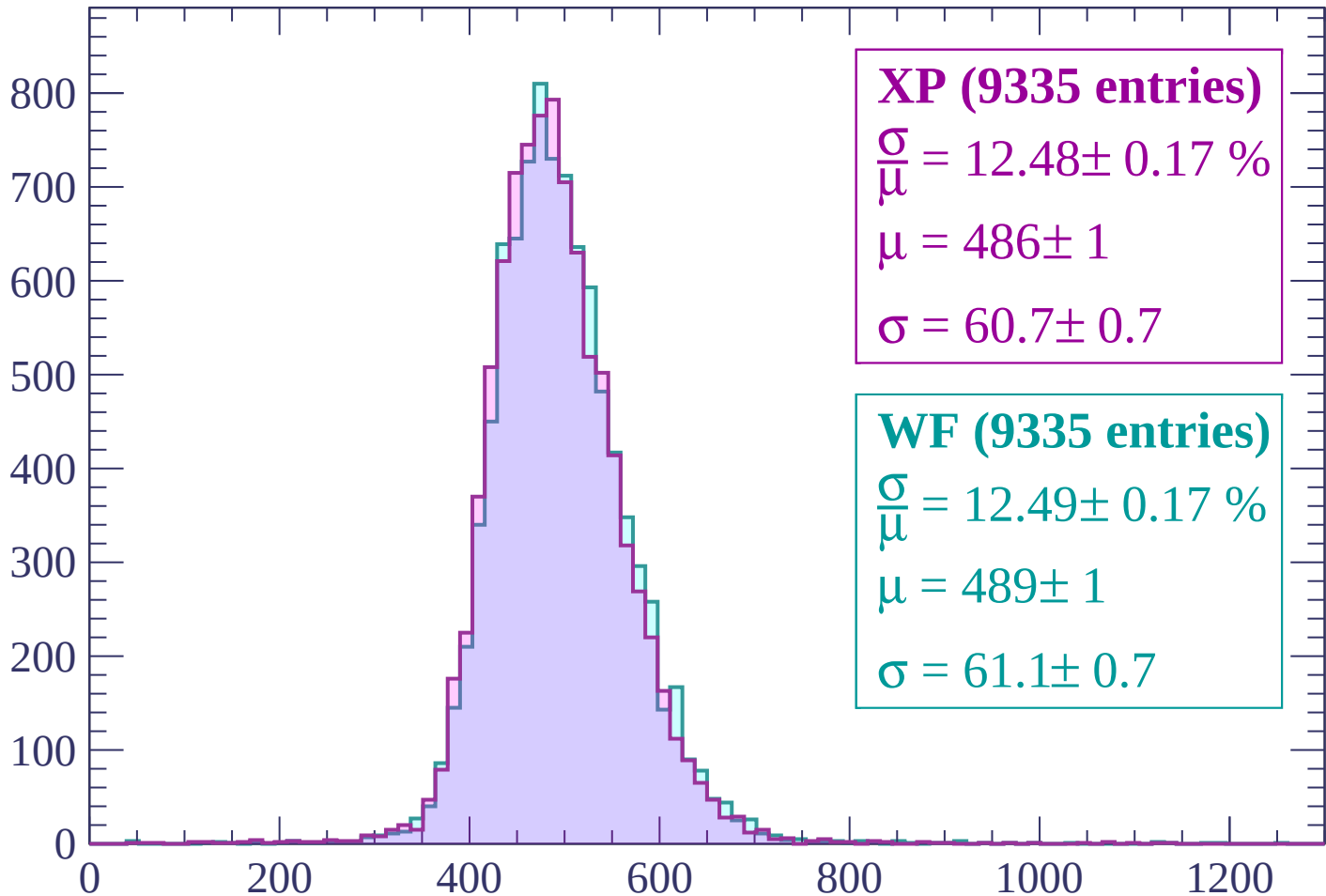
$$\mu = 491 \pm 1$$

$$\sigma = 59.6 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | -1500 < p < -1440

Count



XP (9335 entries)

$$\frac{\sigma}{\mu} = 12.48 \pm 0.17 \%$$

$$\mu = 486 \pm 1$$

$$\sigma = 60.7 \pm 0.7$$

WF (9335 entries)

$$\frac{\sigma}{\mu} = 12.49 \pm 0.17 \%$$

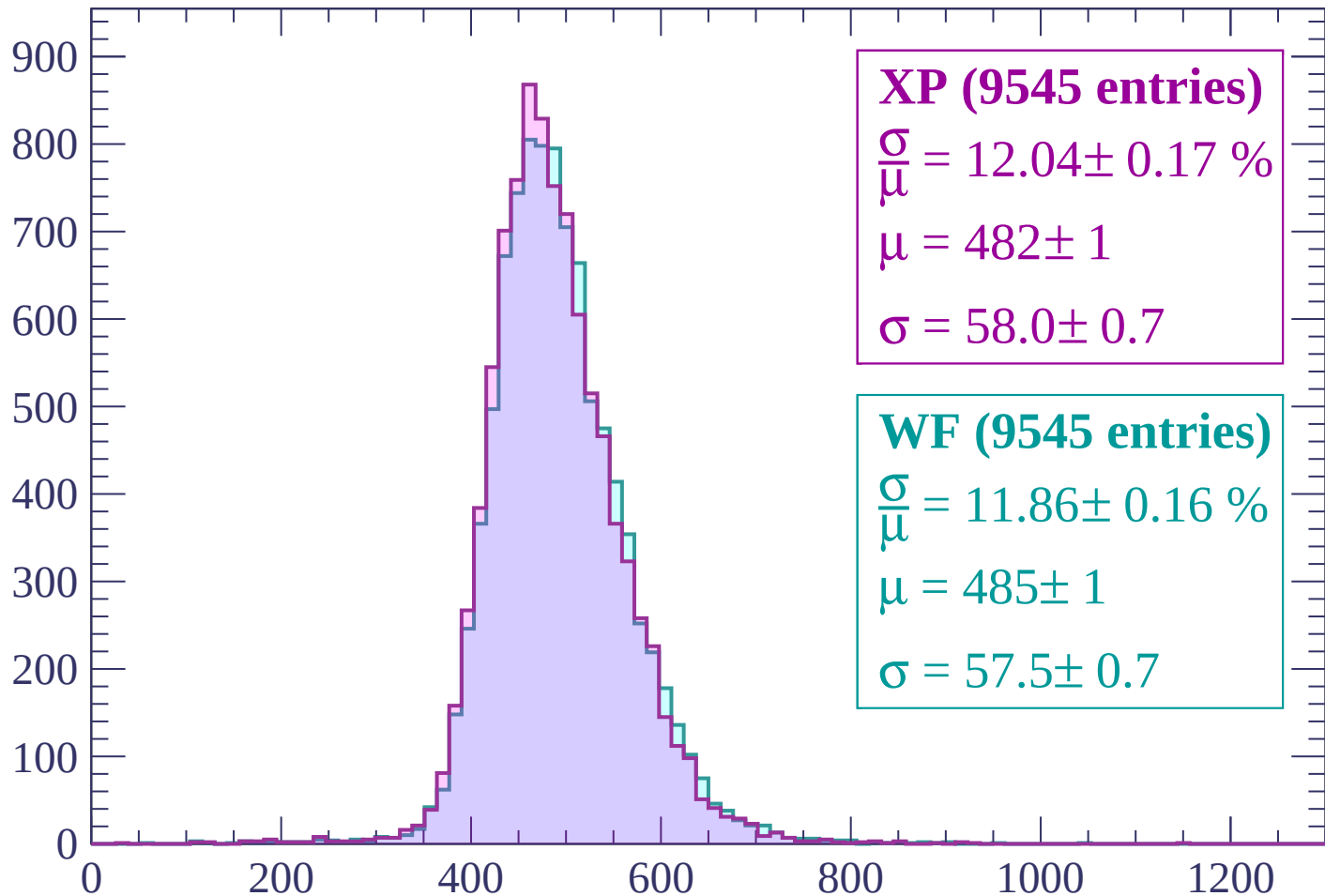
$$\mu = 489 \pm 1$$

$$\sigma = 61.1 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | -1440 < p < -1380

Count



XP (9545 entries)

$$\frac{\sigma}{\mu} = 12.04 \pm 0.17 \%$$

$$\mu = 482 \pm 1$$

$$\sigma = 58.0 \pm 0.7$$

WF (9545 entries)

$$\frac{\sigma}{\mu} = 11.86 \pm 0.16 \%$$

$$\mu = 485 \pm 1$$

$$\sigma = 57.5 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-1380 < p < -1320$

Count

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (10038 entries)

$$\frac{\sigma}{\mu} = 11.60 \pm 0.16 \%$$

$$\mu = 477 \pm 1$$

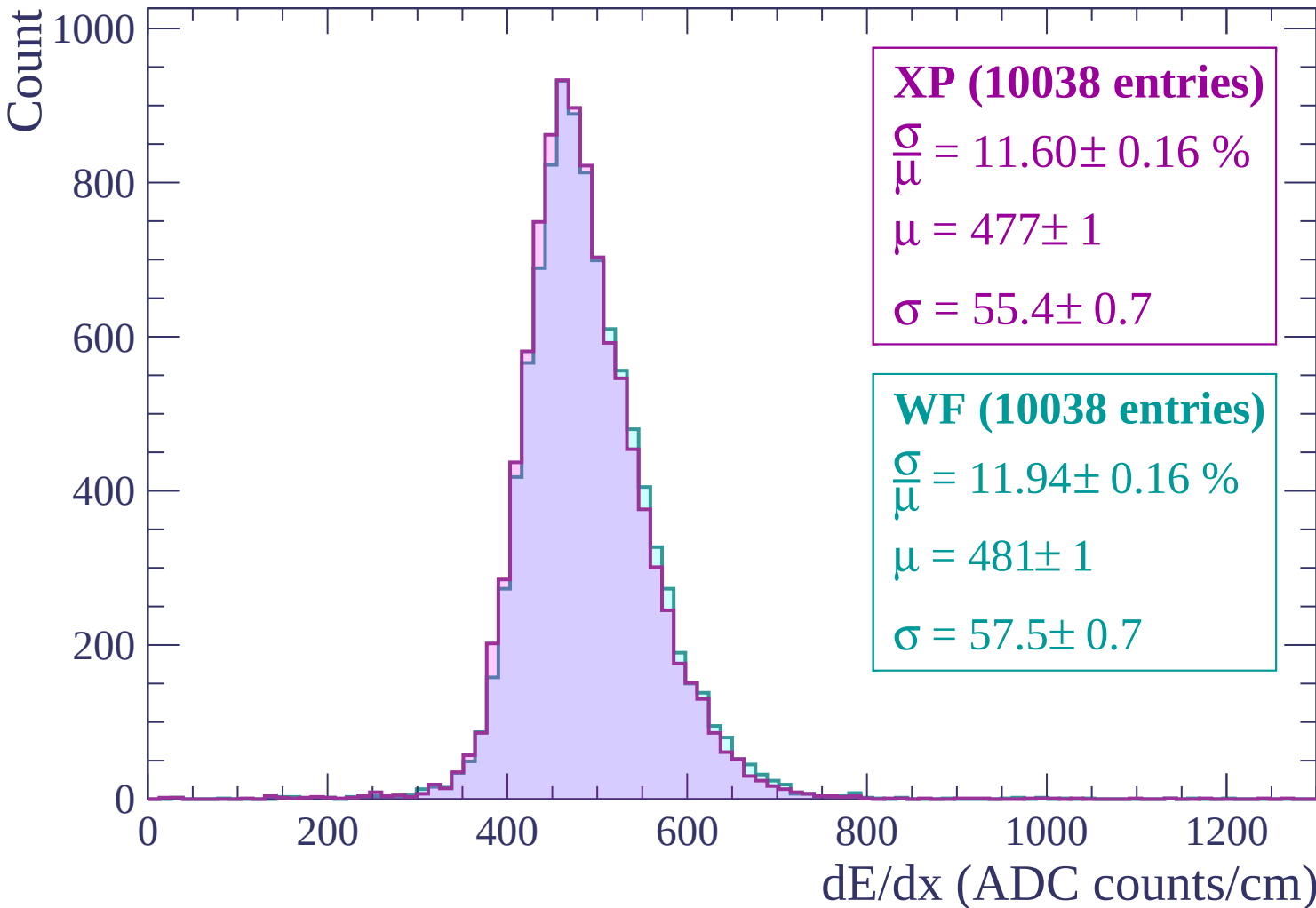
$$\sigma = 55.4 \pm 0.7$$

WF (10038 entries)

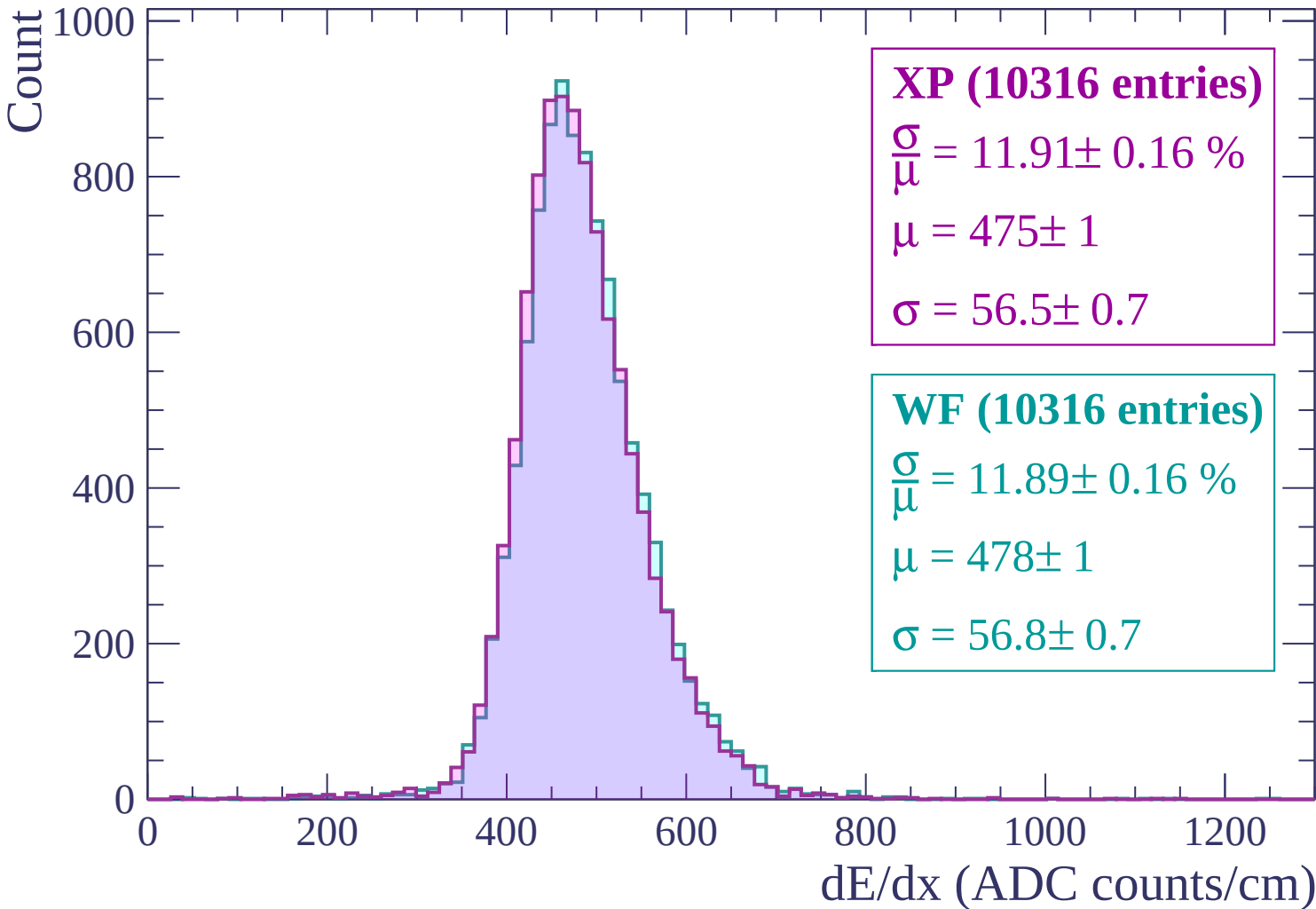
$$\frac{\sigma}{\mu} = 11.94 \pm 0.16 \%$$

$$\mu = 481 \pm 1$$

$$\sigma = 57.5 \pm 0.7$$



Energy loss | -1320 < p < -1260



Energy loss | -1260 < p < -1200

Count

1000

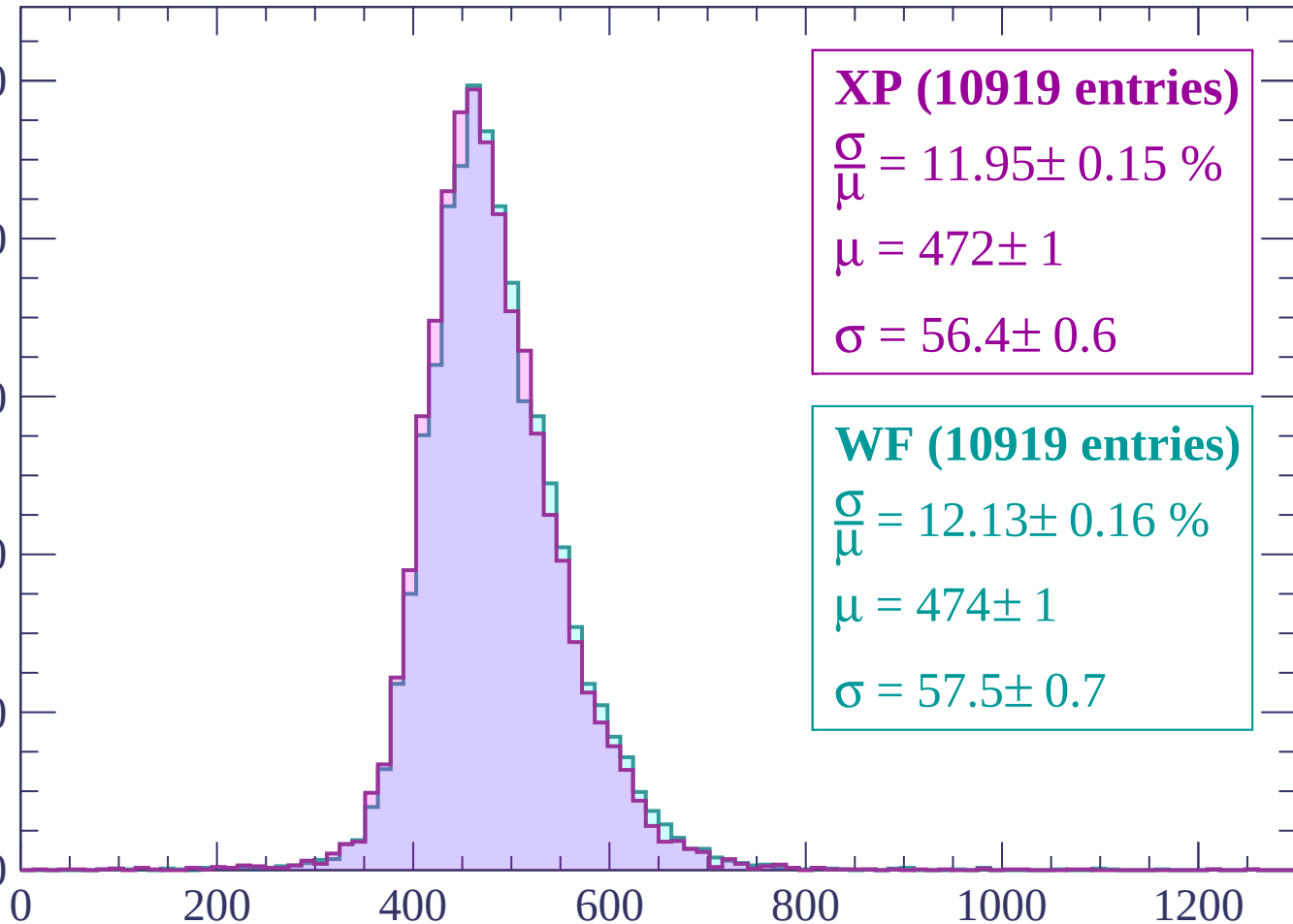
800

600

400

200

0



XP (10919 entries)

$\frac{\sigma}{\mu} = 11.95 \pm 0.15 \%$

$\mu = 472 \pm 1$

$\sigma = 56.4 \pm 0.6$

WF (10919 entries)

$\frac{\sigma}{\mu} = 12.13 \pm 0.16 \%$

$\mu = 474 \pm 1$

$\sigma = 57.5 \pm 0.7$

dE/dx (ADC counts/cm)

Energy loss | -1200 < p < -1140

Count

1000

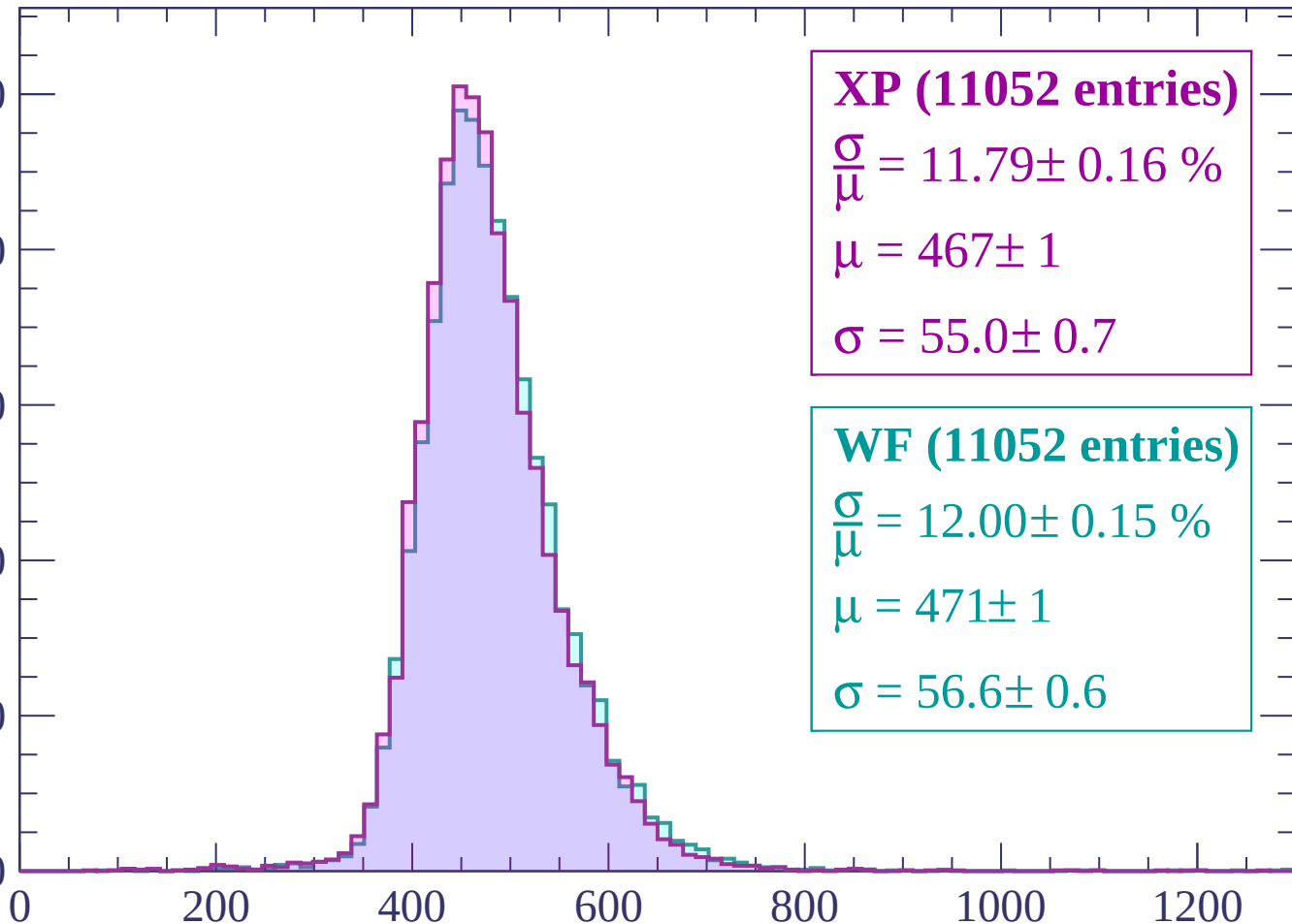
800

600

400

200

0



XP (11052 entries)

$\frac{\sigma}{\mu} = 11.79 \pm 0.16 \%$

$\mu = 467 \pm 1$

$\sigma = 55.0 \pm 0.7$

WF (11052 entries)

$\frac{\sigma}{\mu} = 12.00 \pm 0.15 \%$

$\mu = 471 \pm 1$

$\sigma = 56.6 \pm 0.6$

dE/dx (ADC counts/cm)

Energy loss | -1140 < p < -1080

Count

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (11472 entries)

$$\frac{\sigma}{\mu} = 11.62 \pm 0.15 \%$$

$$\mu = 465 \pm 1$$

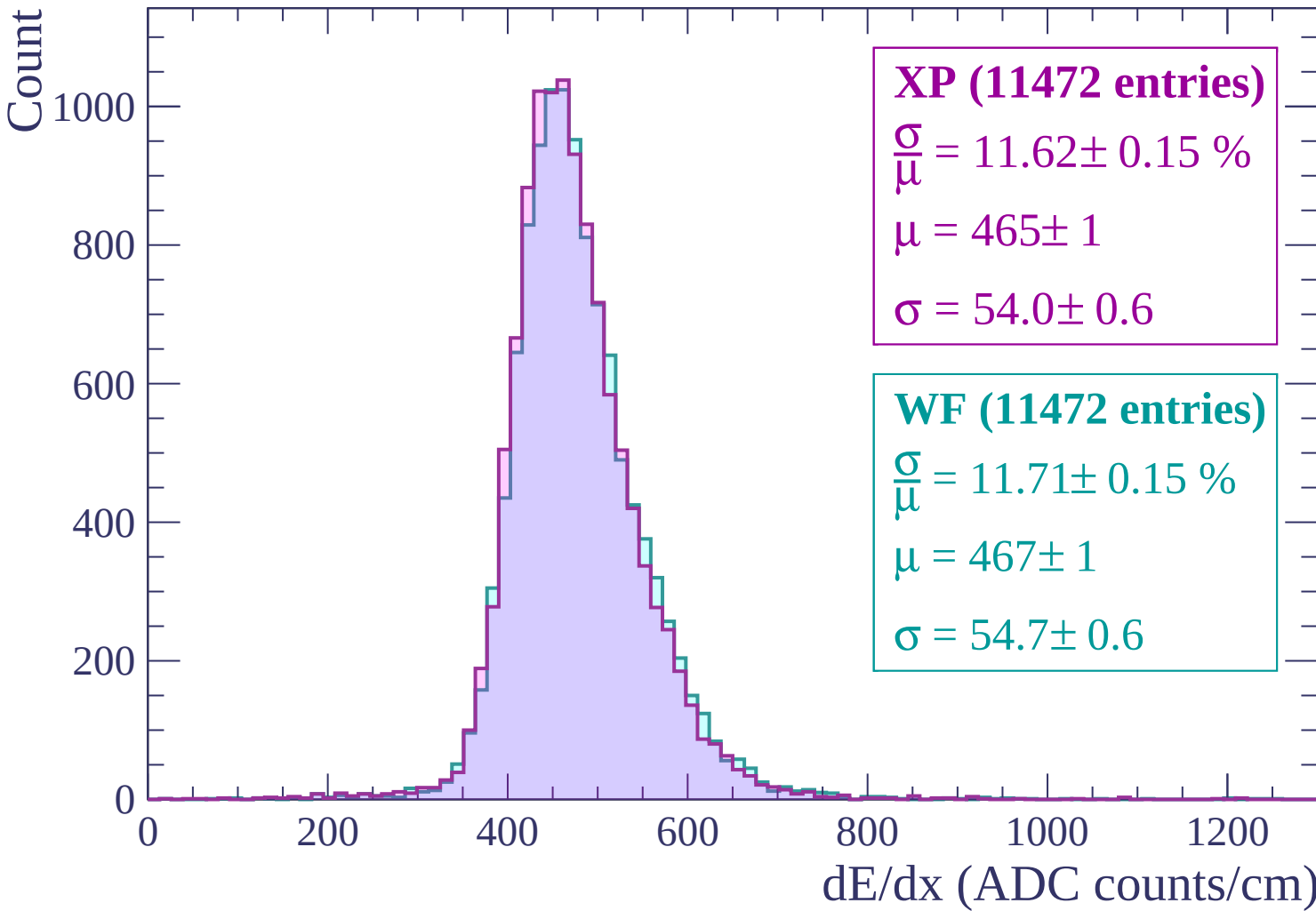
$$\sigma = 54.0 \pm 0.6$$

WF (11472 entries)

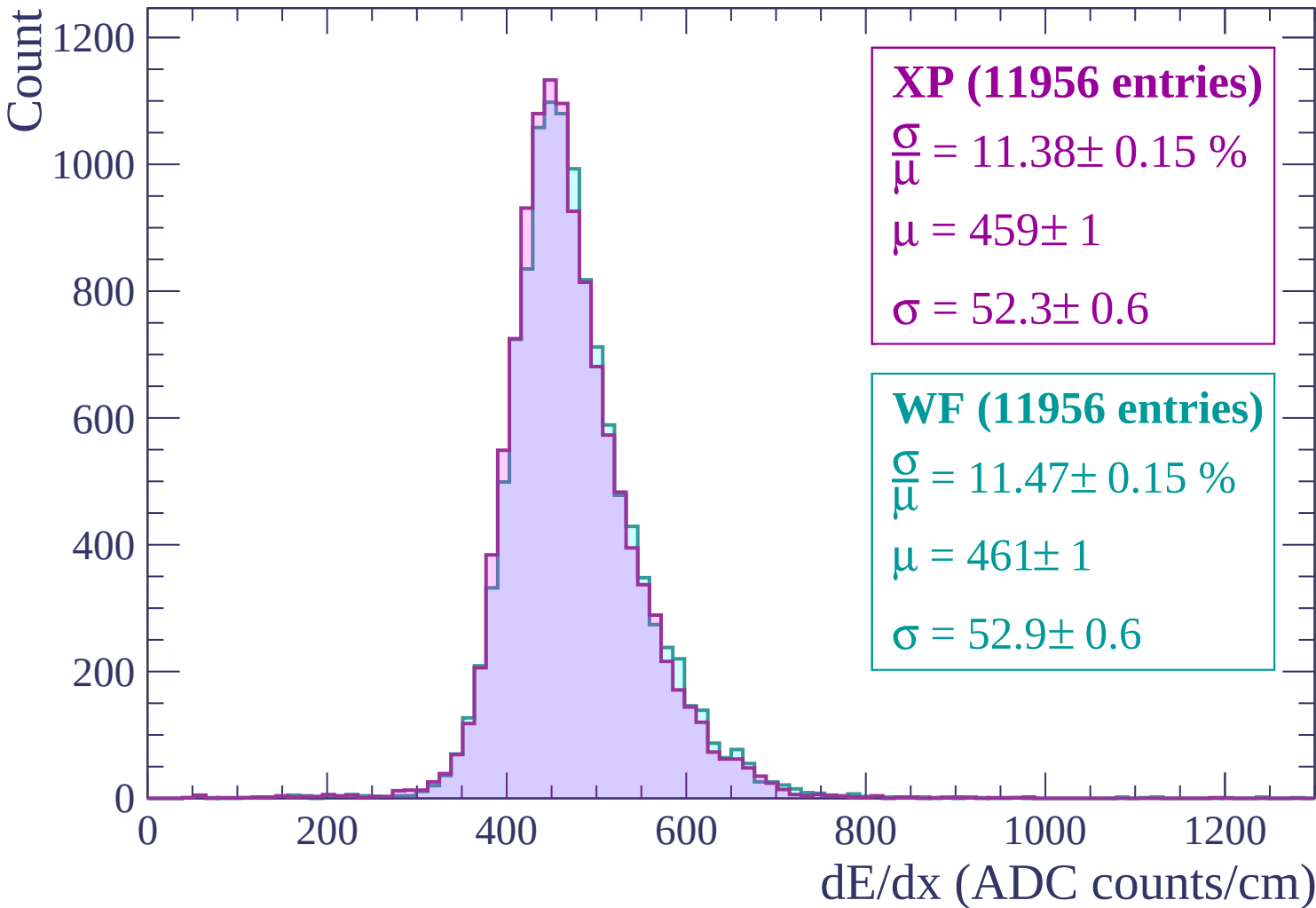
$$\frac{\sigma}{\mu} = 11.71 \pm 0.15 \%$$

$$\mu = 467 \pm 1$$

$$\sigma = 54.7 \pm 0.6$$



Energy loss | -1080 < p < -1020



Energy loss | $-1020 < p < -960$

Count

1200
1000
800
600
400
200
0

XP (12448 entries)

$\frac{\sigma}{\mu} = 11.06 \pm 0.14 \%$

$\mu = 453 \pm 1$

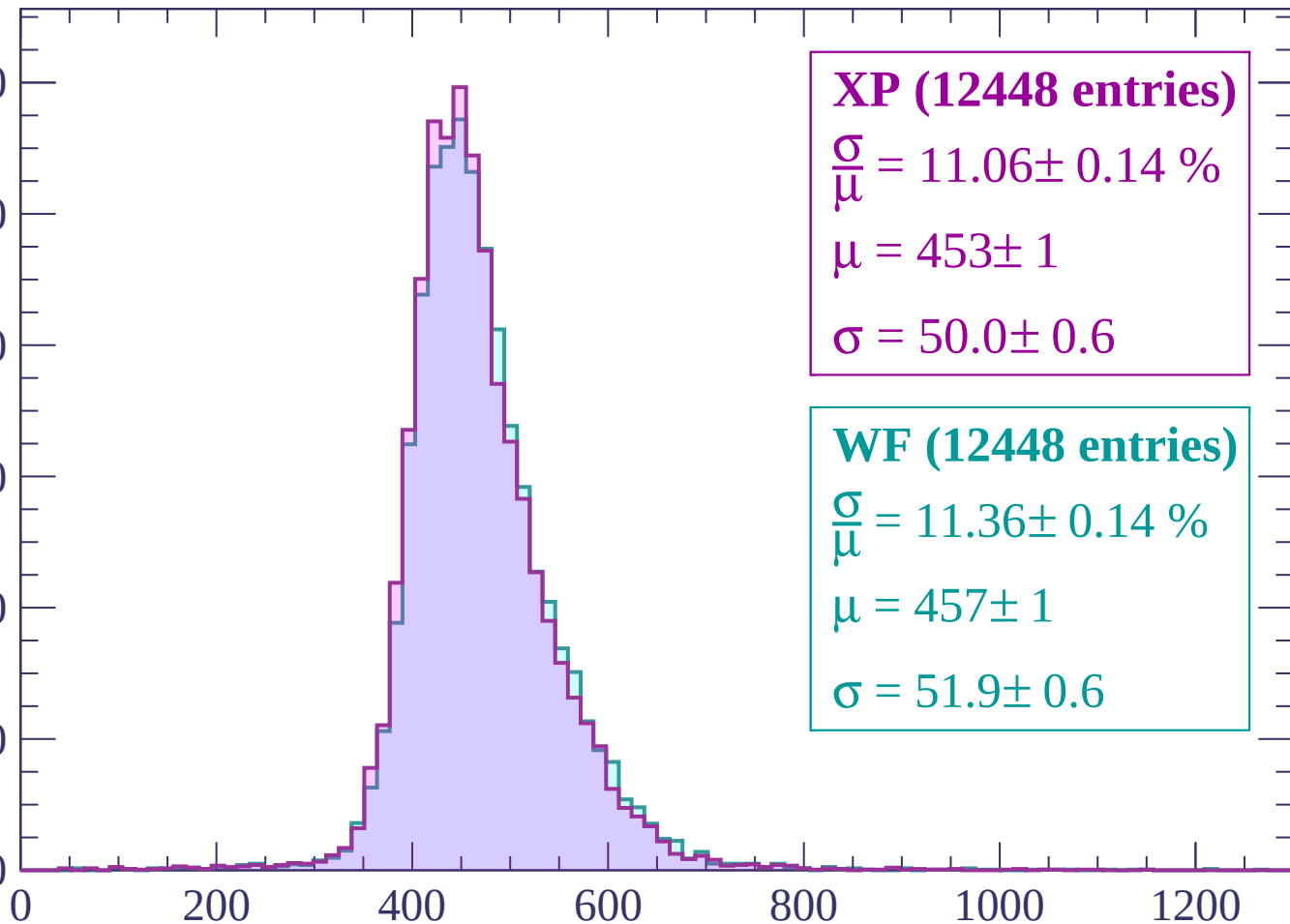
$\sigma = 50.0 \pm 0.6$

WF (12448 entries)

$\frac{\sigma}{\mu} = 11.36 \pm 0.14 \%$

$\mu = 457 \pm 1$

$\sigma = 51.9 \pm 0.6$



dE/dx (ADC counts/cm)

Energy loss | $-960 < p < -900$

Count

1200

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

XP (12638 entries)

$\frac{\sigma}{\mu} = 11.40 \pm 0.15 \%$

$\mu = 450 \pm 1$

$\sigma = 51.3 \pm 0.6$

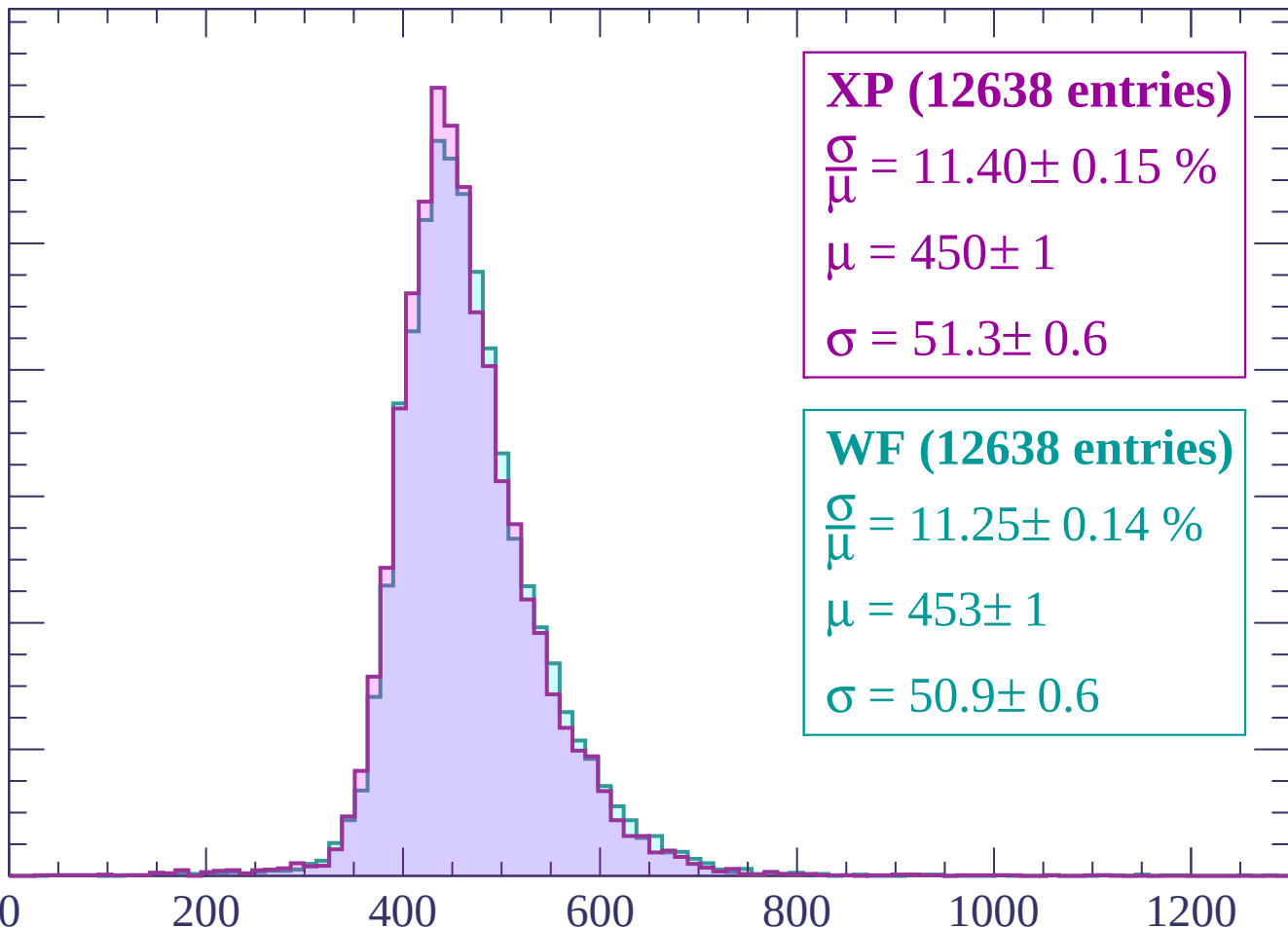
WF (12638 entries)

$\frac{\sigma}{\mu} = 11.25 \pm 0.14 \%$

$\mu = 453 \pm 1$

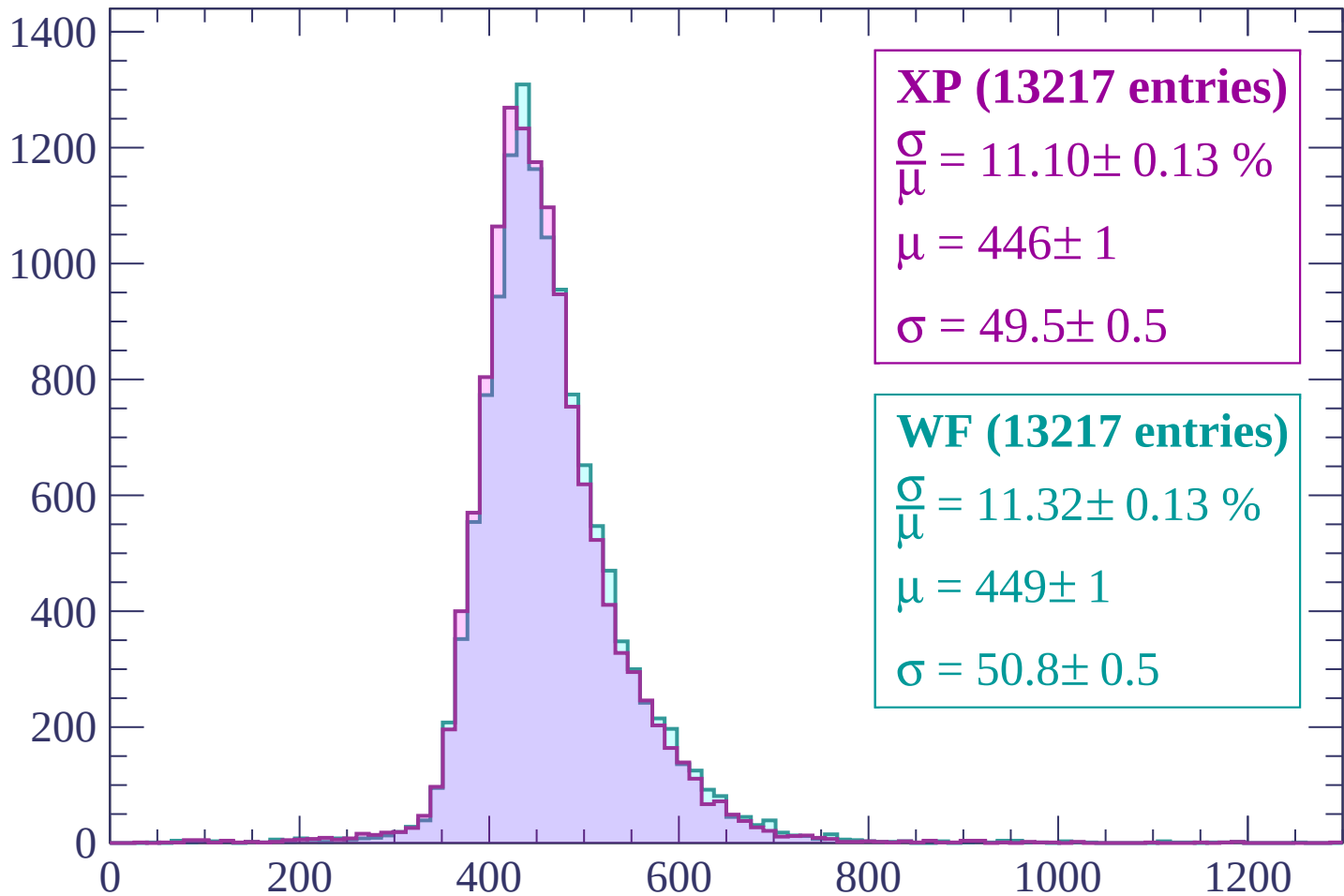
$\sigma = 50.9 \pm 0.6$

dE/dx (ADC counts/cm)



Energy loss | $-900 < p < -840$

Count



XP (13217 entries)

$$\frac{\sigma}{\mu} = 11.10 \pm 0.13 \%$$

$$\mu = 446 \pm 1$$

$$\sigma = 49.5 \pm 0.5$$

WF (13217 entries)

$$\frac{\sigma}{\mu} = 11.32 \pm 0.13 \%$$

$$\mu = 449 \pm 1$$

$$\sigma = 50.8 \pm 0.5$$

dE/dx (ADC counts/cm)

Energy loss | $-840 < p < -780$

Count

1400
1200
1000
800
600
400
200
0

XP (13801 entries)

$\frac{\sigma}{\mu} = 10.84 \pm 0.13 \%$

$\mu = 441 \pm 1$

$\sigma = 47.8 \pm 0.5$

WF (13801 entries)

$\frac{\sigma}{\mu} = 11.00 \pm 0.13 \%$

$\mu = 443 \pm 1$

$\sigma = 48.7 \pm 0.5$

0

200

400

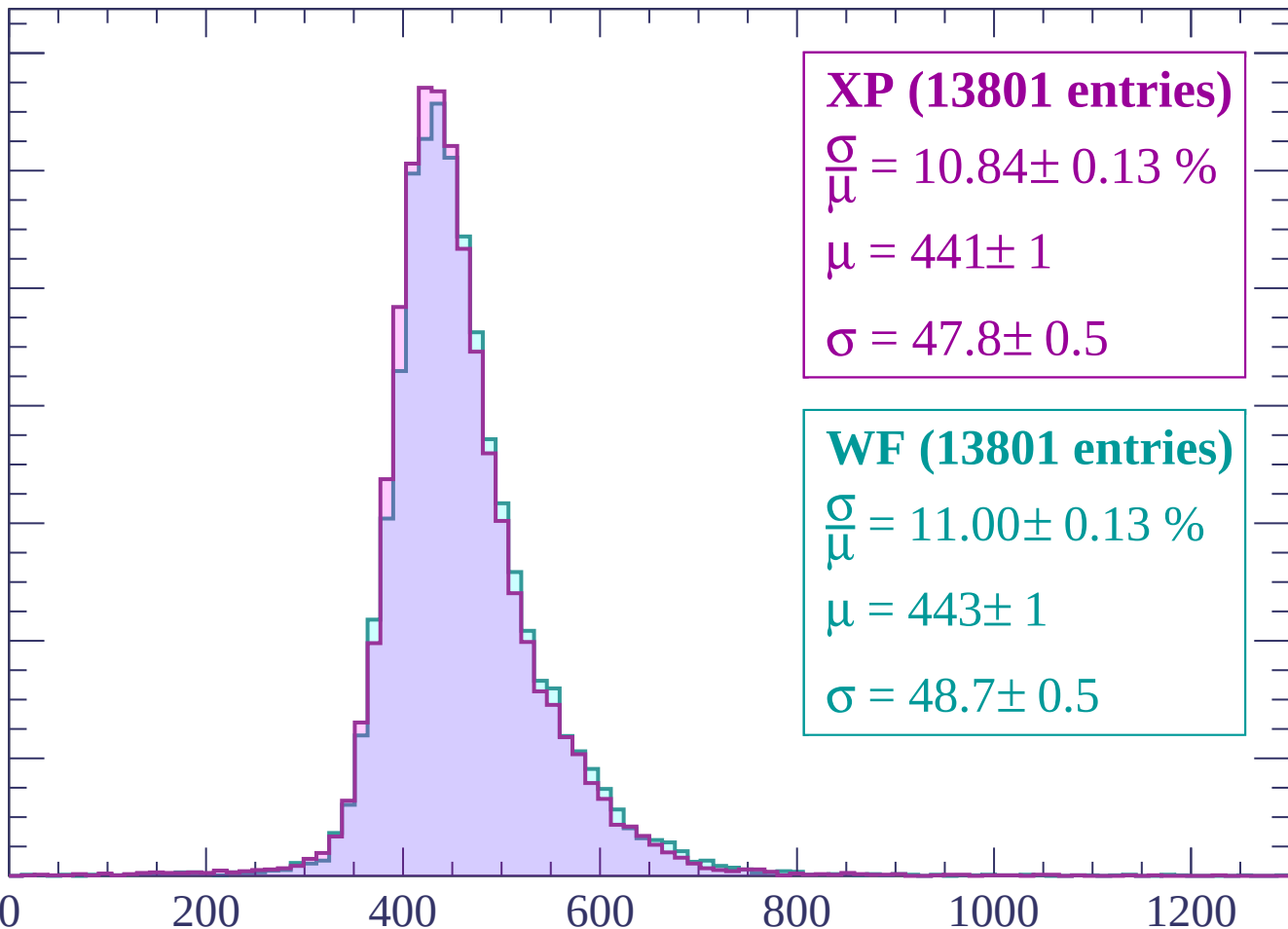
600

800

1000

1200

dE/dx (ADC counts/cm)



Energy loss | $-780 < p < -720$

Count

1400

1200

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

XP (13982 entries)

$\frac{\sigma}{\mu} = 10.68 \pm 0.13 \%$

$\mu = 435 \pm 1$

$\sigma = 46.4 \pm 0.5$

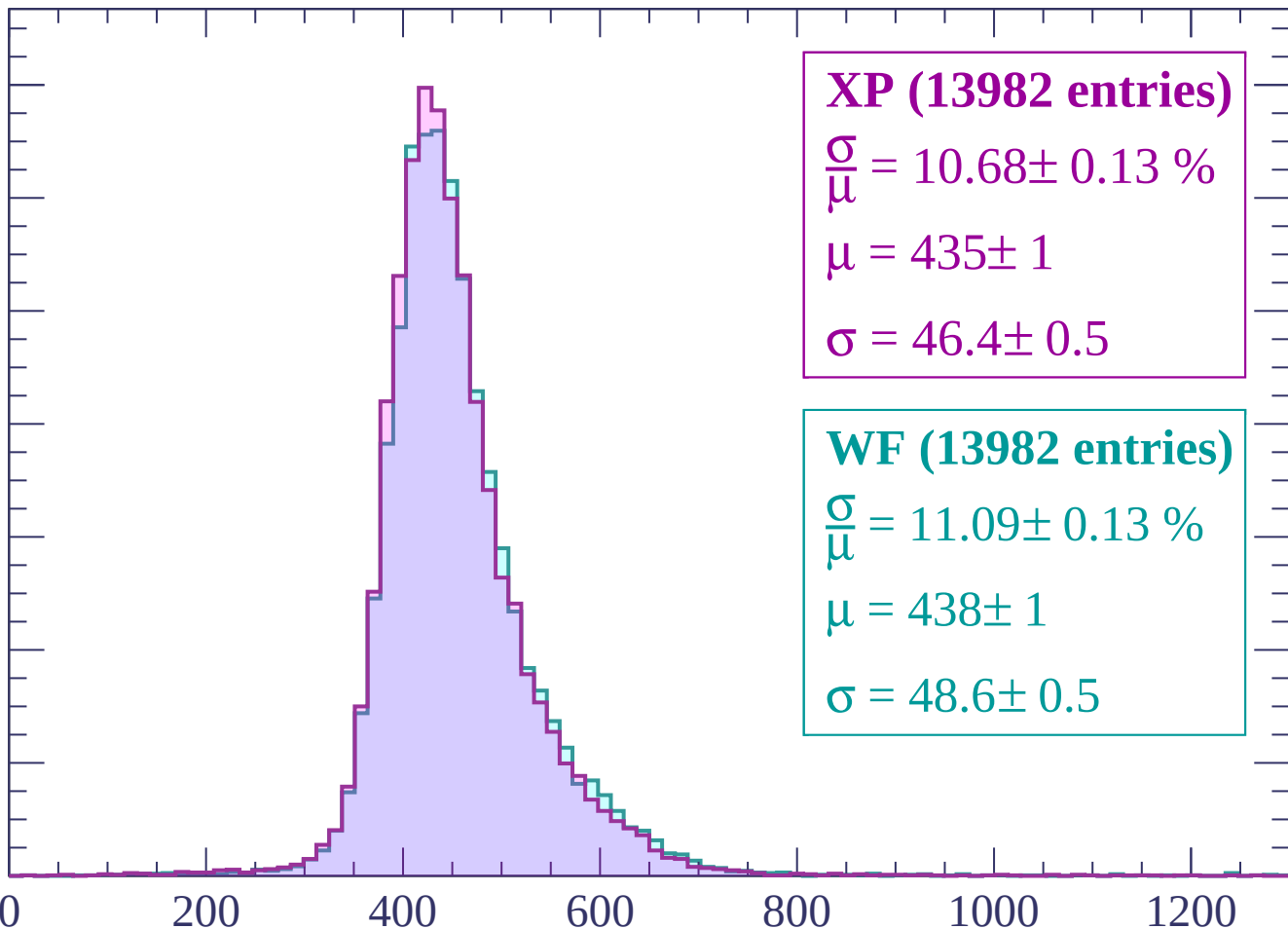
WF (13982 entries)

$\frac{\sigma}{\mu} = 11.09 \pm 0.13 \%$

$\mu = 438 \pm 1$

$\sigma = 48.6 \pm 0.5$

dE/dx (ADC counts/cm)



Energy loss | $-720 < p < -660$

Count

1400
1200
1000
800
600
400
200
0

XP (14264 entries)

$\frac{\sigma}{\mu} = 10.71 \pm 0.13 \%$

$\mu = 430 \pm 1$

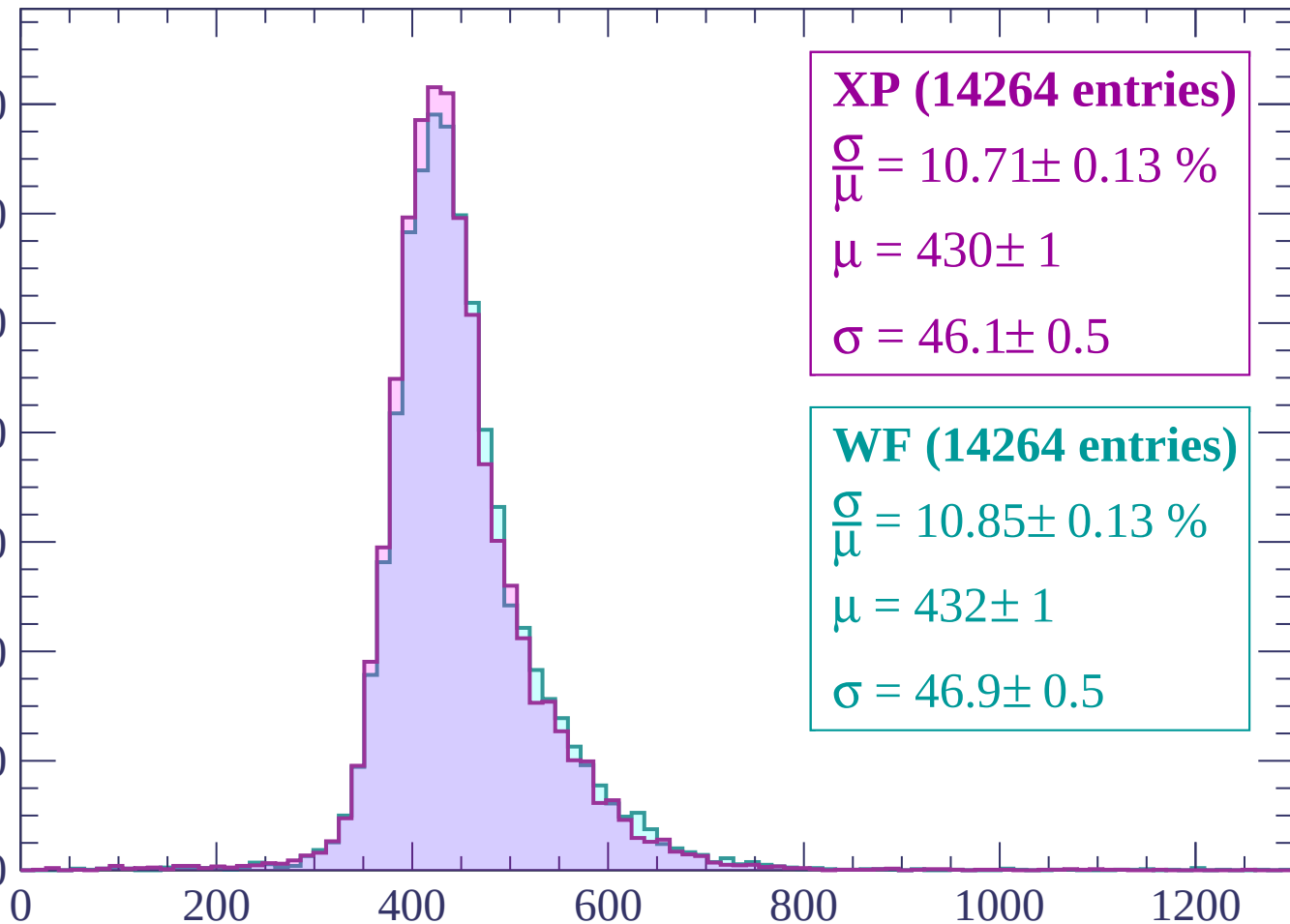
$\sigma = 46.1 \pm 0.5$

WF (14264 entries)

$\frac{\sigma}{\mu} = 10.85 \pm 0.13 \%$

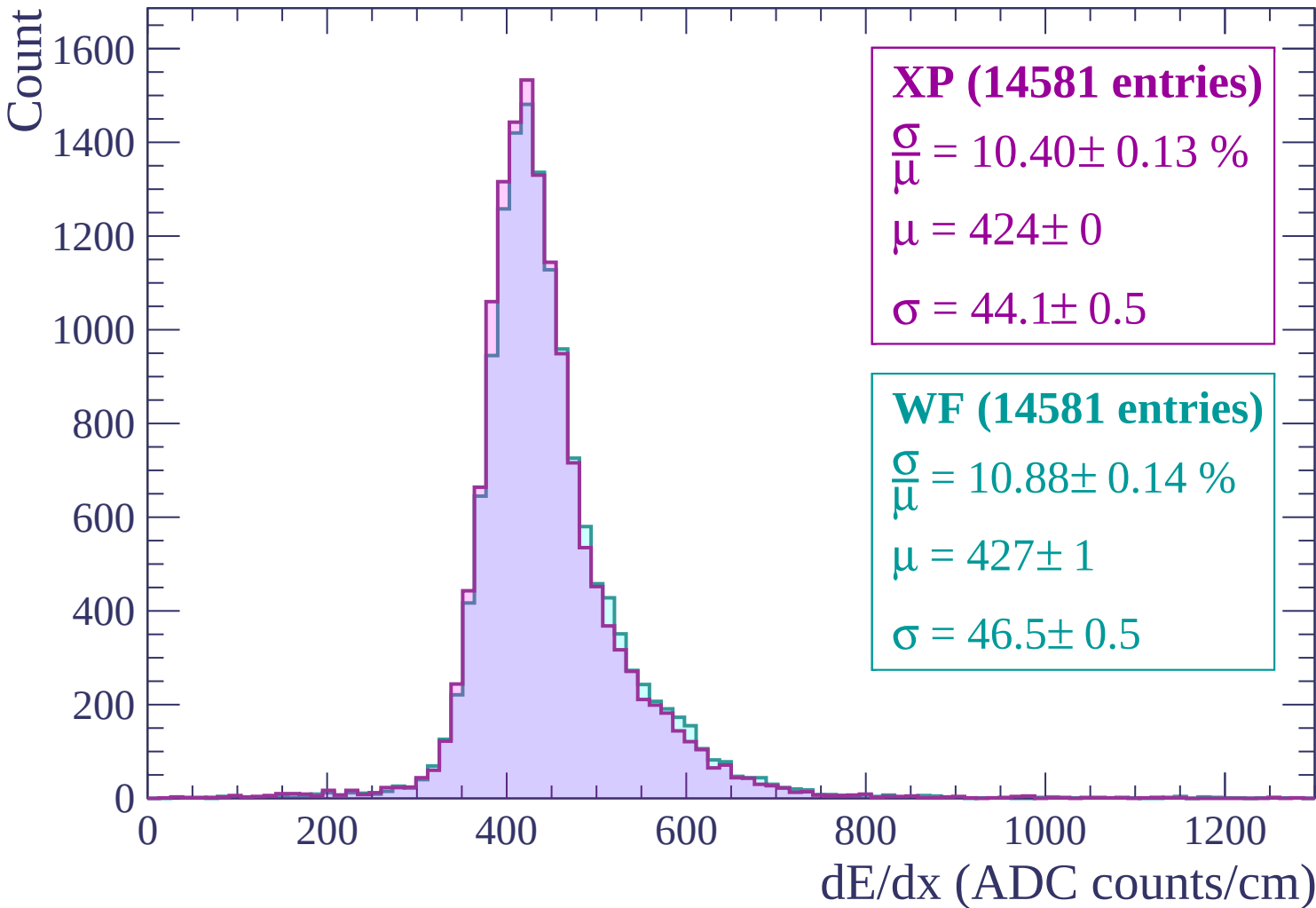
$\mu = 432 \pm 1$

$\sigma = 46.9 \pm 0.5$



dE/dx (ADC counts/cm)

Energy loss | $-660 < p < -600$



Energy loss | $-600 < p < -540$

Count

1600
1400
1200
1000
800
600
400
200
0

XP (14859 entries)

$\frac{\sigma}{\mu} = 10.41 \pm 0.13 \%$

$\mu = 420 \pm 0$

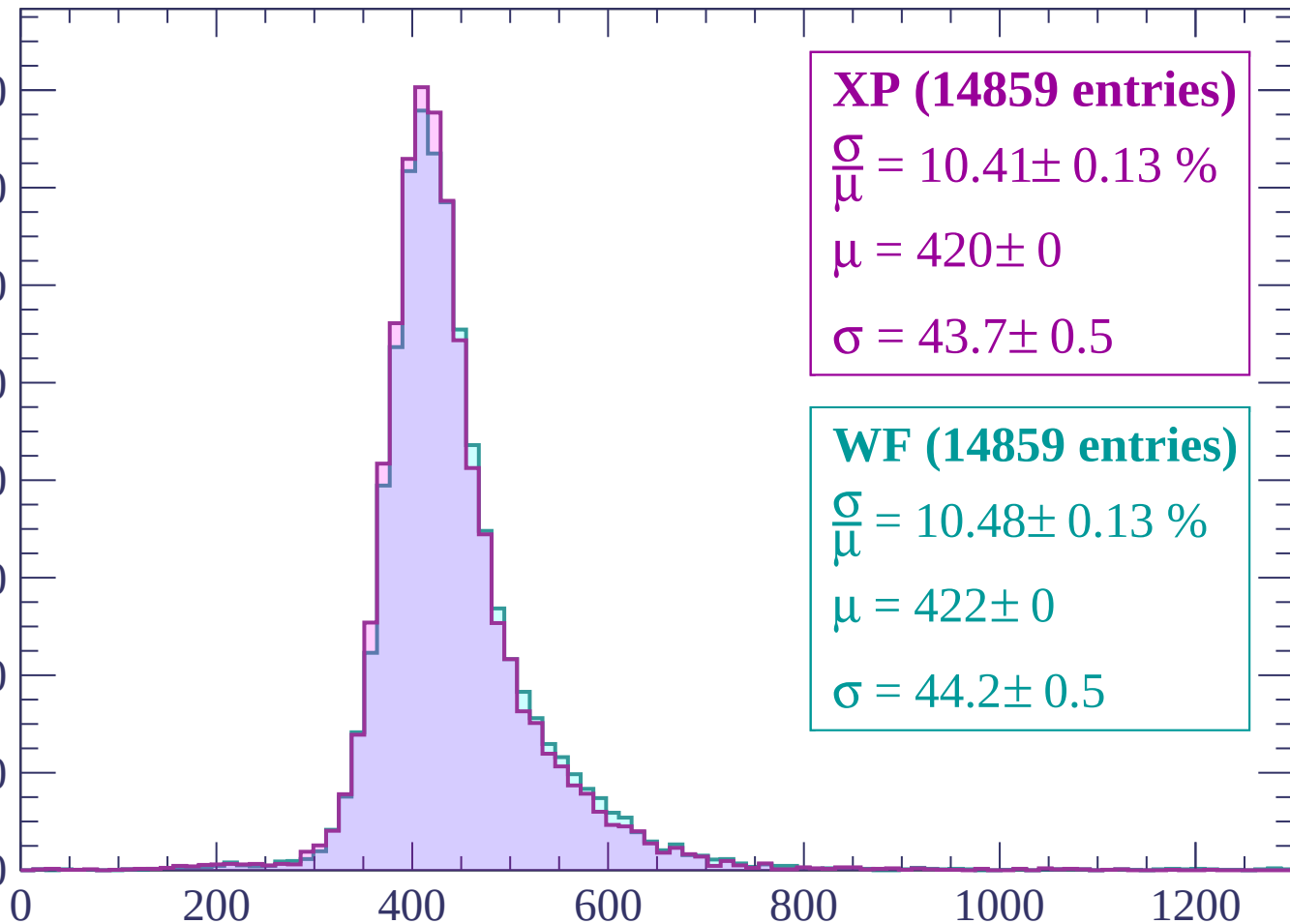
$\sigma = 43.7 \pm 0.5$

WF (14859 entries)

$\frac{\sigma}{\mu} = 10.48 \pm 0.13 \%$

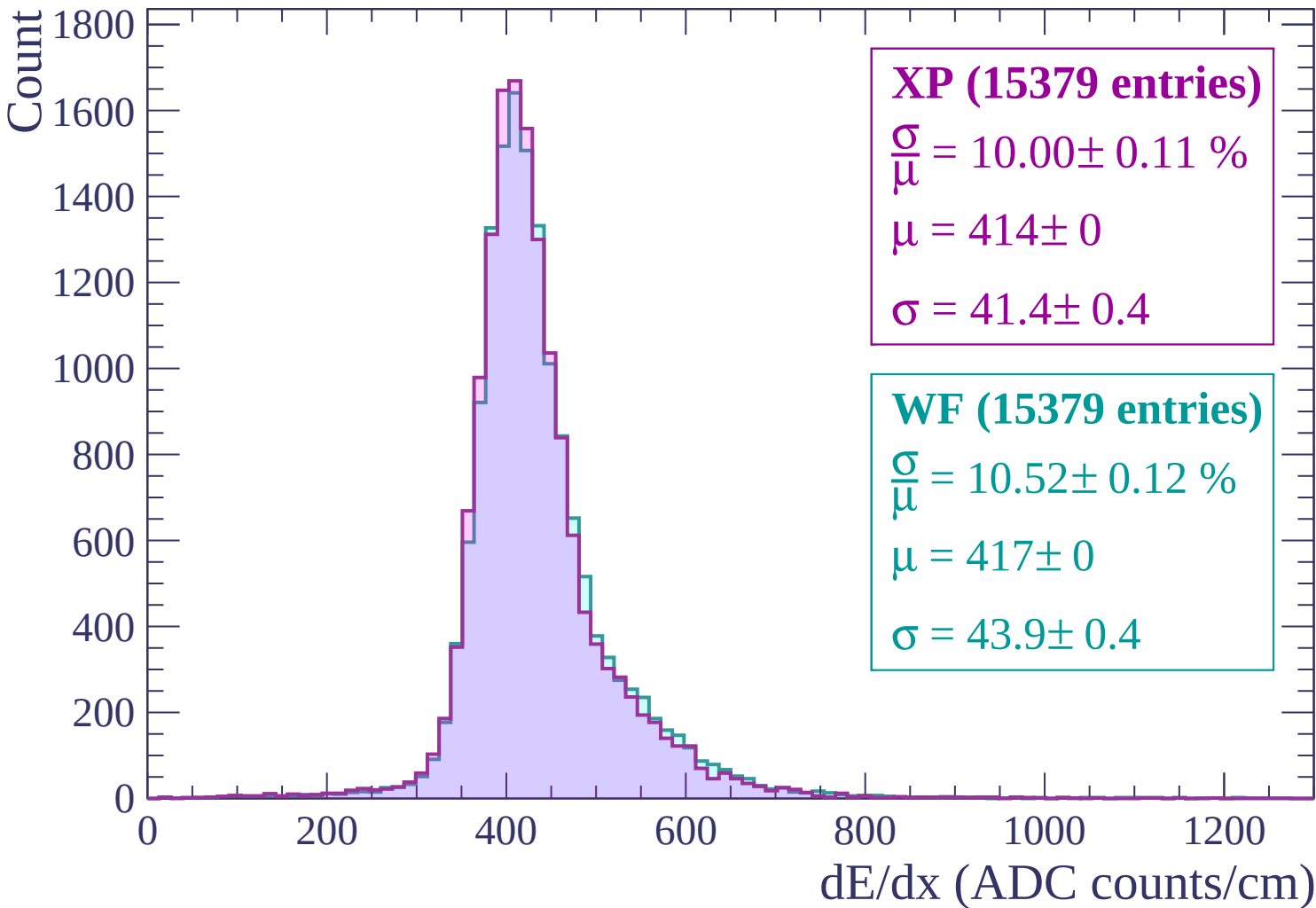
$\mu = 422 \pm 0$

$\sigma = 44.2 \pm 0.5$

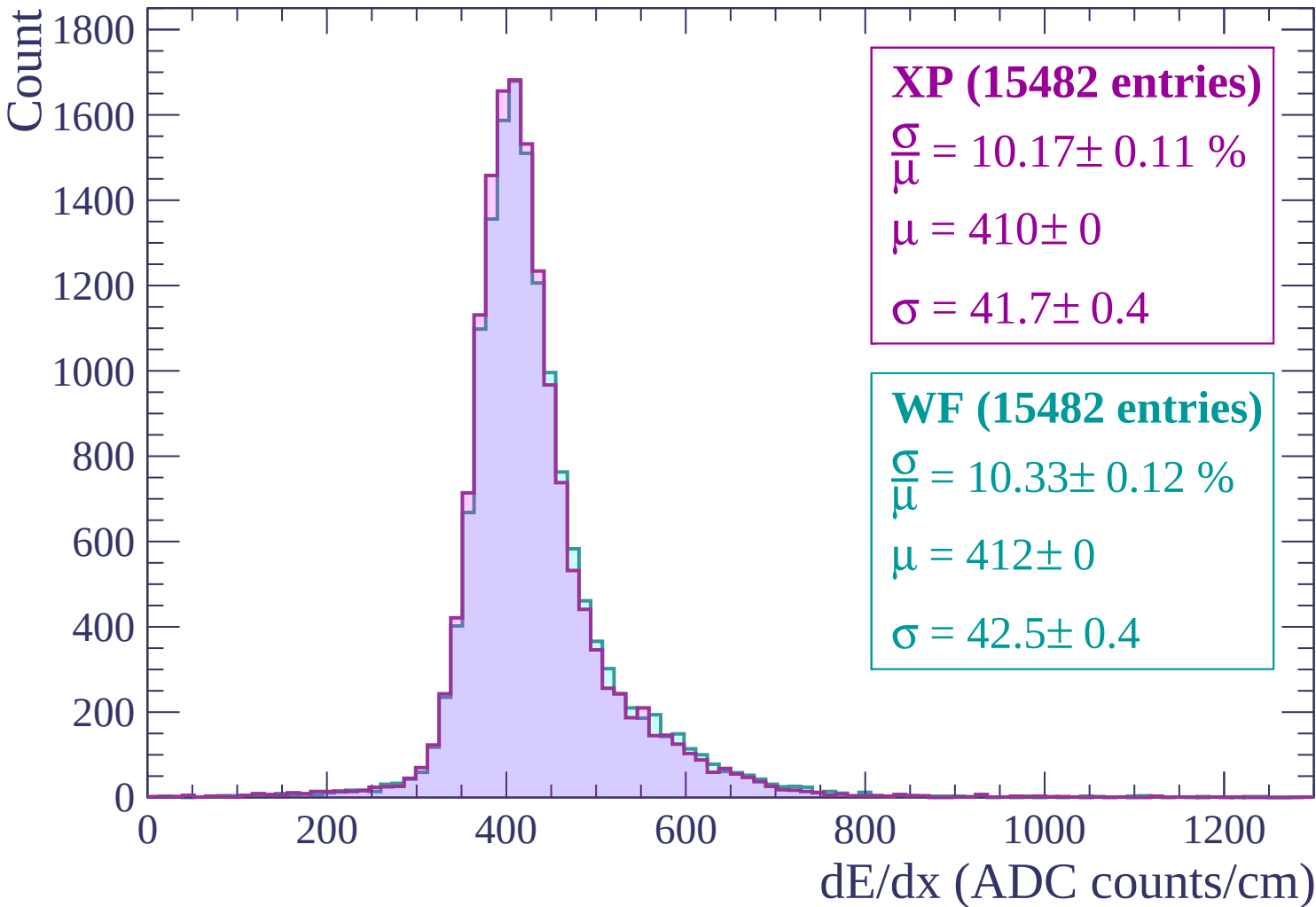


dE/dx (ADC counts/cm)

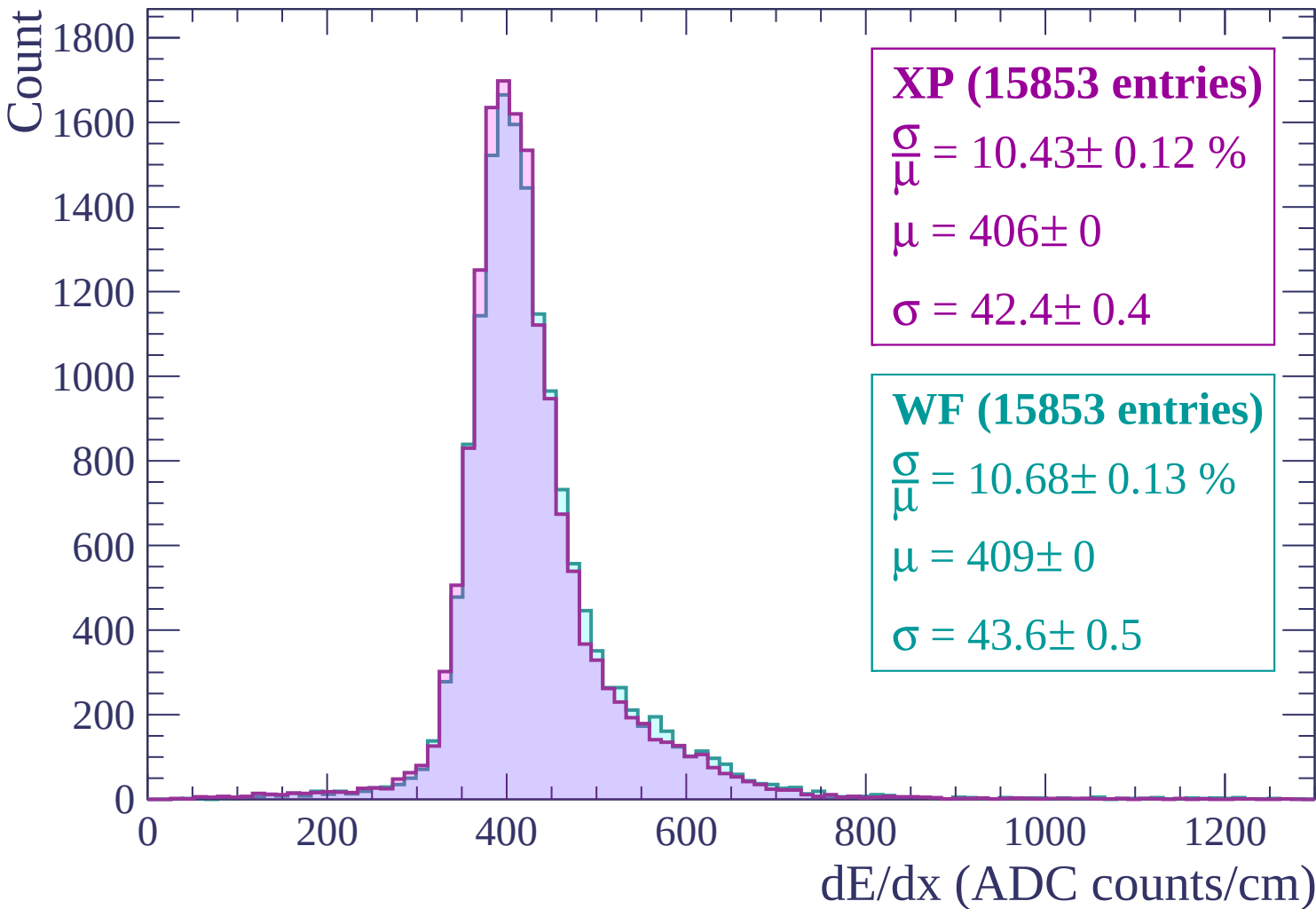
Energy loss | $-540 < p < -480$



Energy loss | $-480 < p < -420$



Energy loss | $-420 < p < -360$



Energy loss | $-360 < p < -300$

Count

1600
1400
1200
1000
800
600
400
200
0

XP (15346 entries)

$$\frac{\sigma}{\mu} = 10.18 \pm 0.12 \%$$

$$\mu = 404 \pm 0$$

$$\sigma = 41.1 \pm 0.5$$

WF (15346 entries)

$$\frac{\sigma}{\mu} = 10.55 \pm 0.13 \%$$

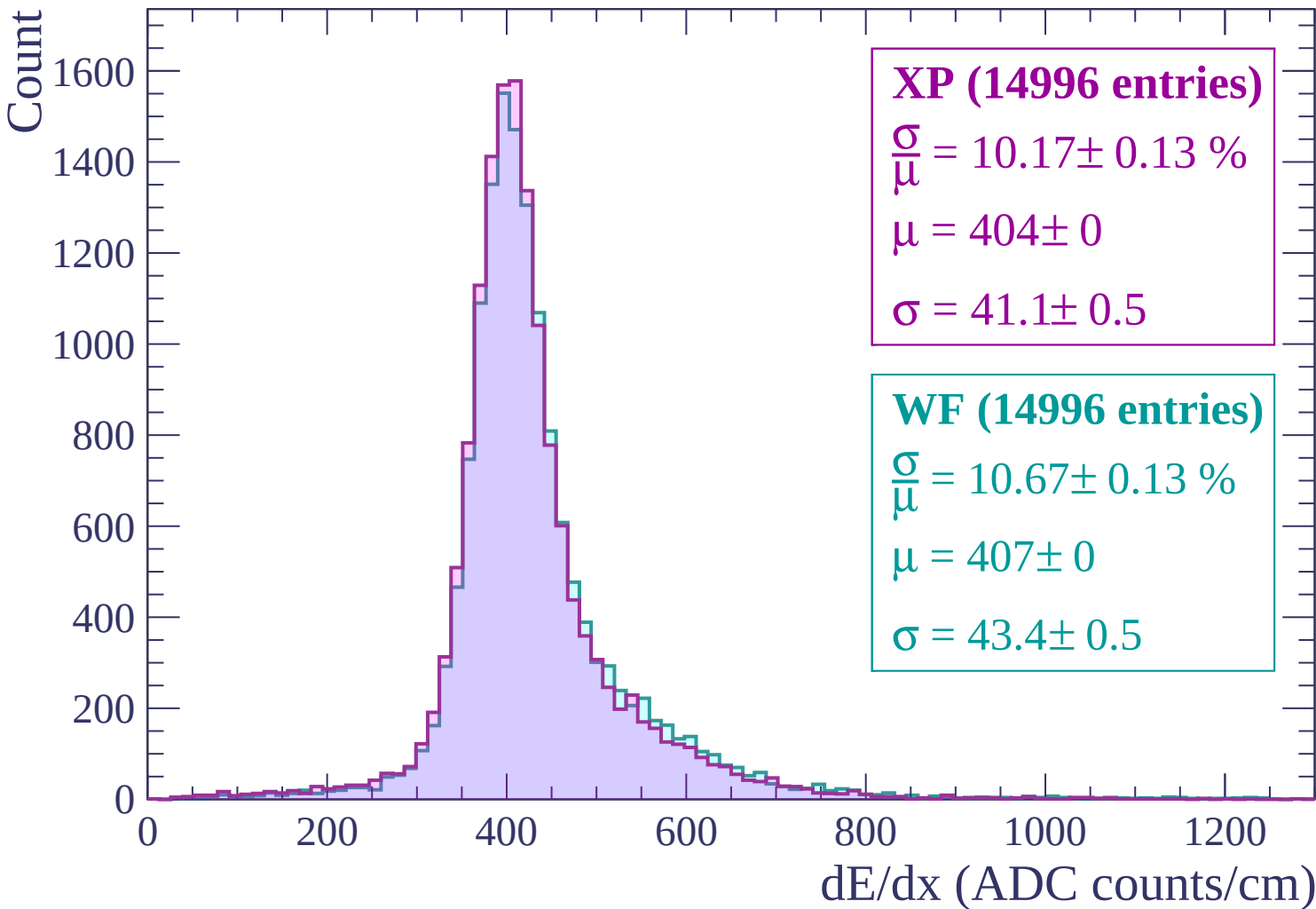
$$\mu = 407 \pm 0$$

$$\sigma = 42.9 \pm 0.5$$

0 200 400 600 800 1000 1200

dE/dx (ADC counts/cm)

Energy loss | $-300 < p < -240$



Energy loss | $-240 < p < -180$

Count

1200
1000
800
600
400
200
0

XP (13385 entries)

$\frac{\sigma}{\mu} = 11.61 \pm 0.16 \%$

$\mu = 416 \pm 1$

$\sigma = 48.3 \pm 0.6$

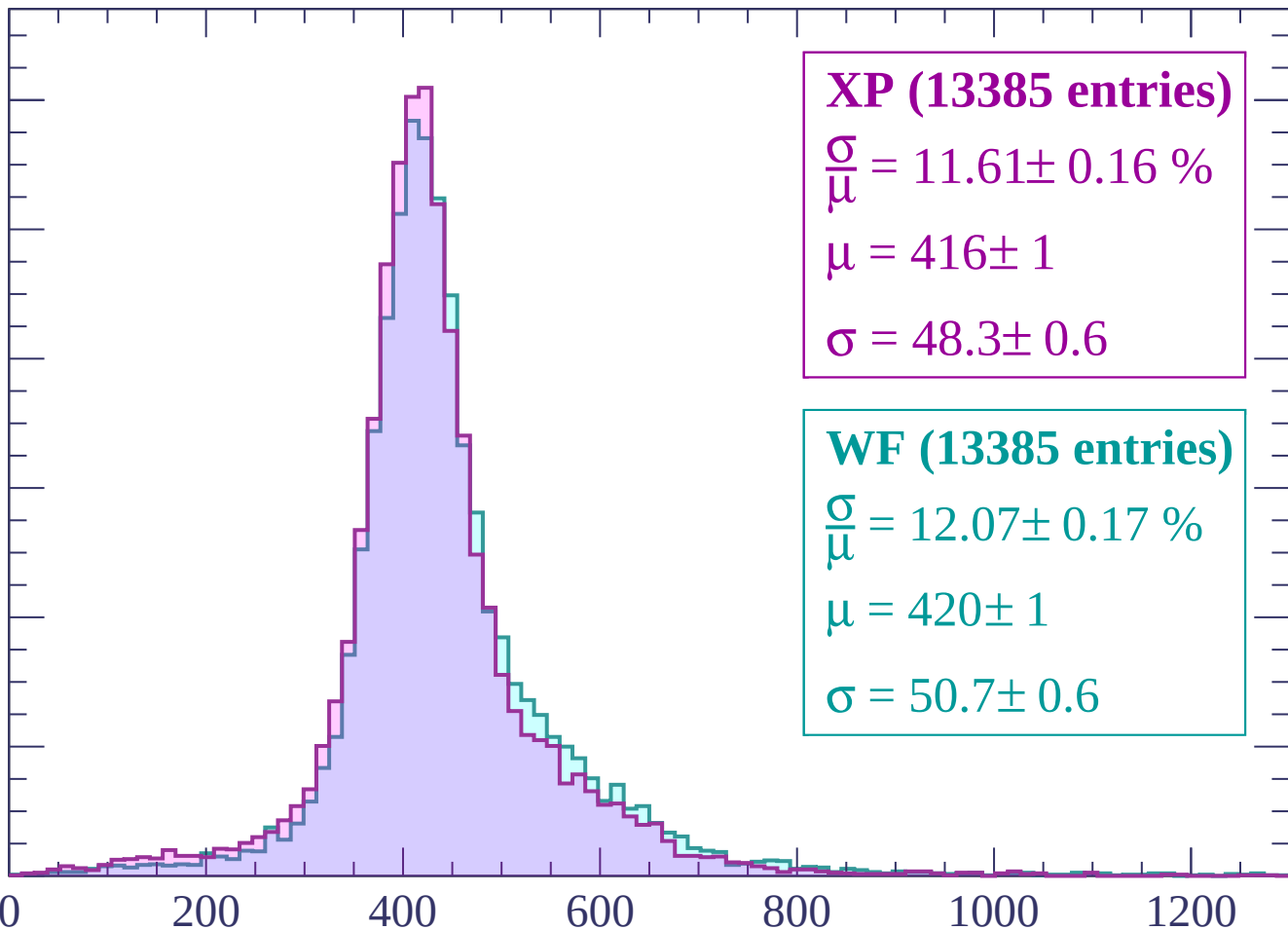
WF (13385 entries)

$\frac{\sigma}{\mu} = 12.07 \pm 0.17 \%$

$\mu = 420 \pm 1$

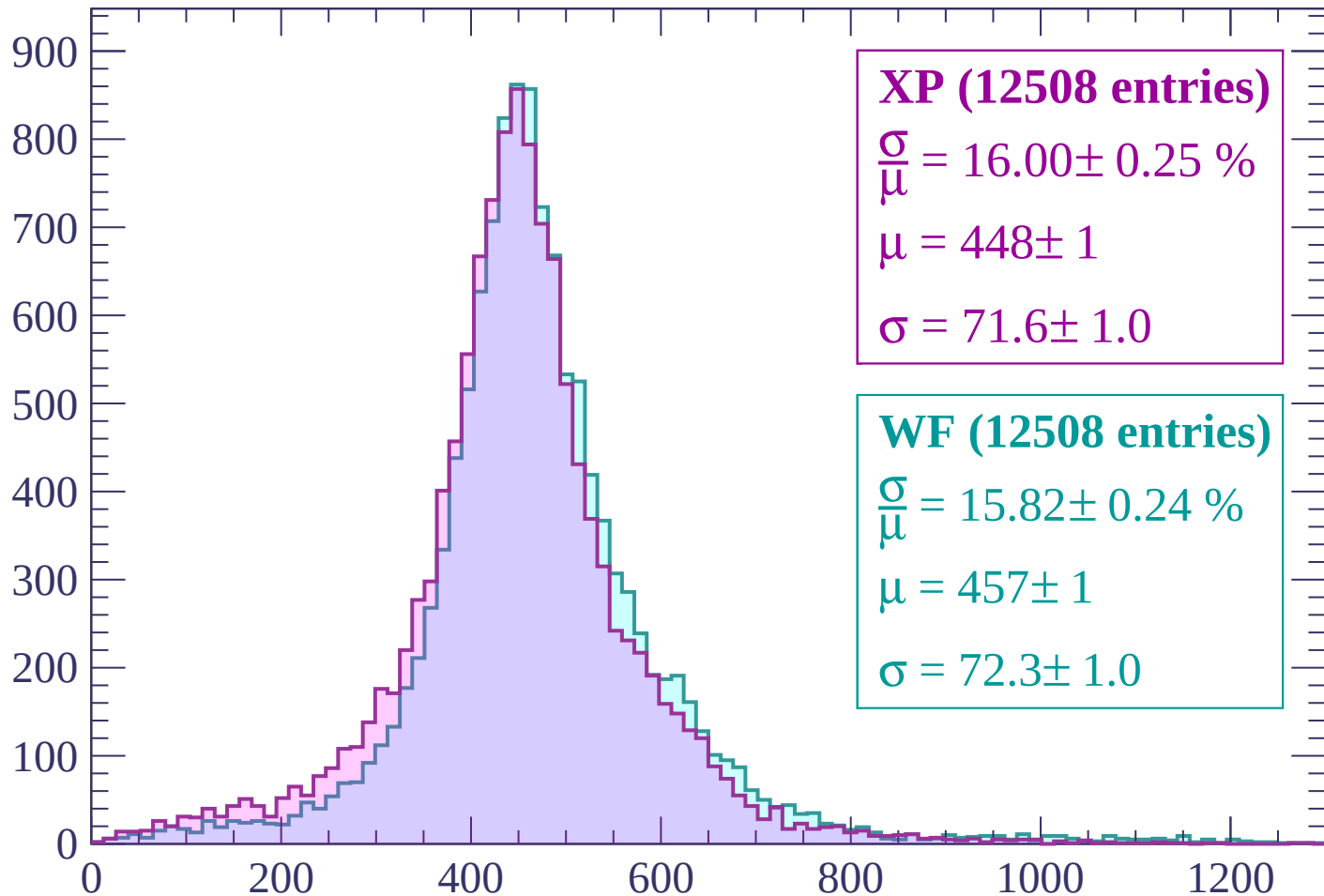
$\sigma = 50.7 \pm 0.6$

dE/dx (ADC counts/cm)



Energy loss | $-180 < p < -120$

Count



XP (12508 entries)

$$\frac{\sigma}{\mu} = 16.00 \pm 0.25 \%$$

$$\mu = 448 \pm 1$$

$$\sigma = 71.6 \pm 1.0$$

WF (12508 entries)

$$\frac{\sigma}{\mu} = 15.82 \pm 0.24 \%$$

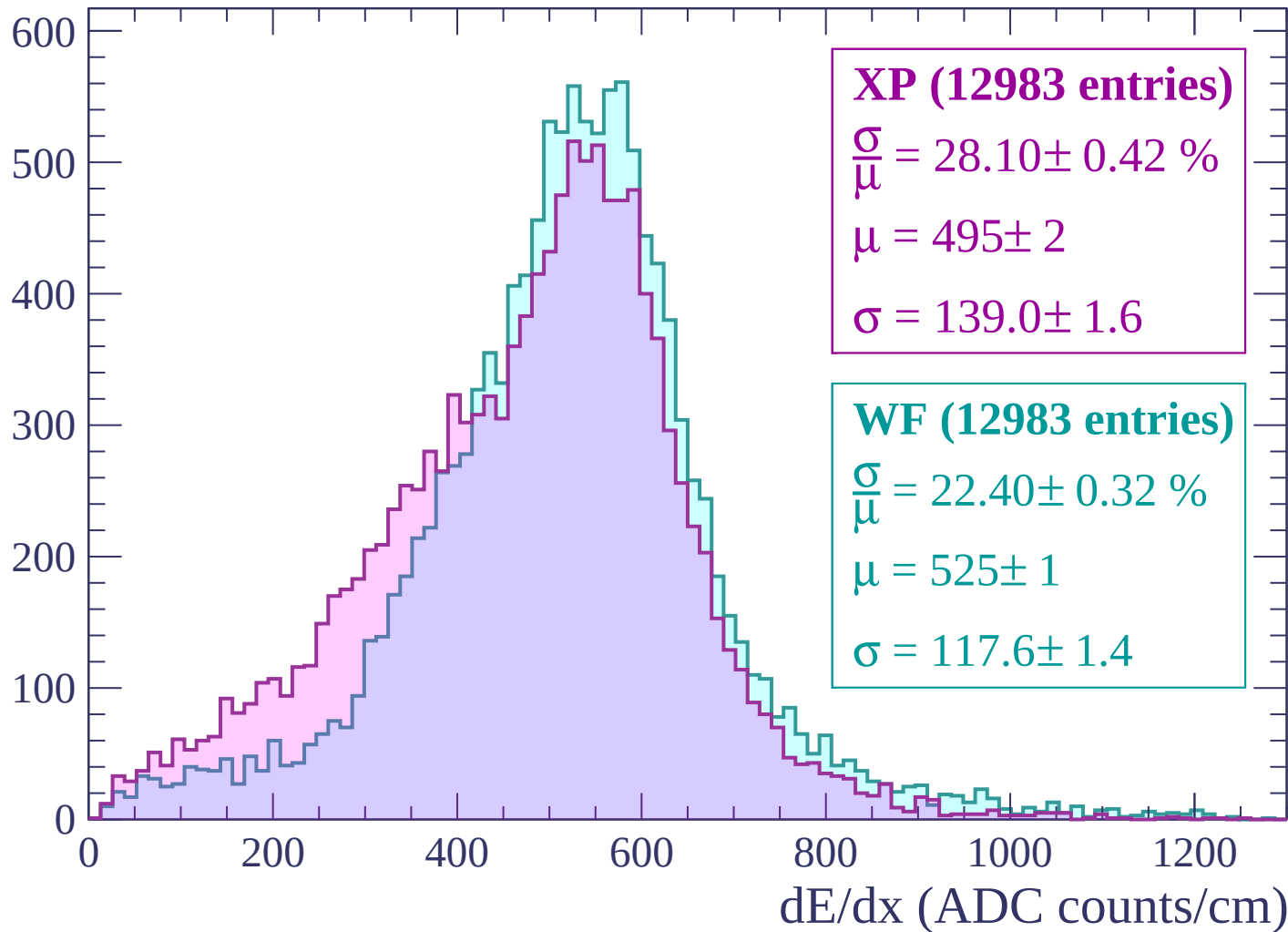
$$\mu = 457 \pm 1$$

$$\sigma = 72.3 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $-120 < p < -60$

Count



Energy loss | $-60 < p < 0$

Count

1600
1400
1200
1000
800
600
400
200
0

XP (26830 entries)

$$\frac{\sigma}{\mu} = 12.91 \pm 0.14 \%$$

$$\mu = 545 \pm 1$$

$$\sigma = 70.4 \pm 0.7$$

WF (26830 entries)

$$\frac{\sigma}{\mu} = 13.46 \pm 0.13 \%$$

$$\mu = 547 \pm 1$$

$$\sigma = 73.6 \pm 0.7$$

0

200

400

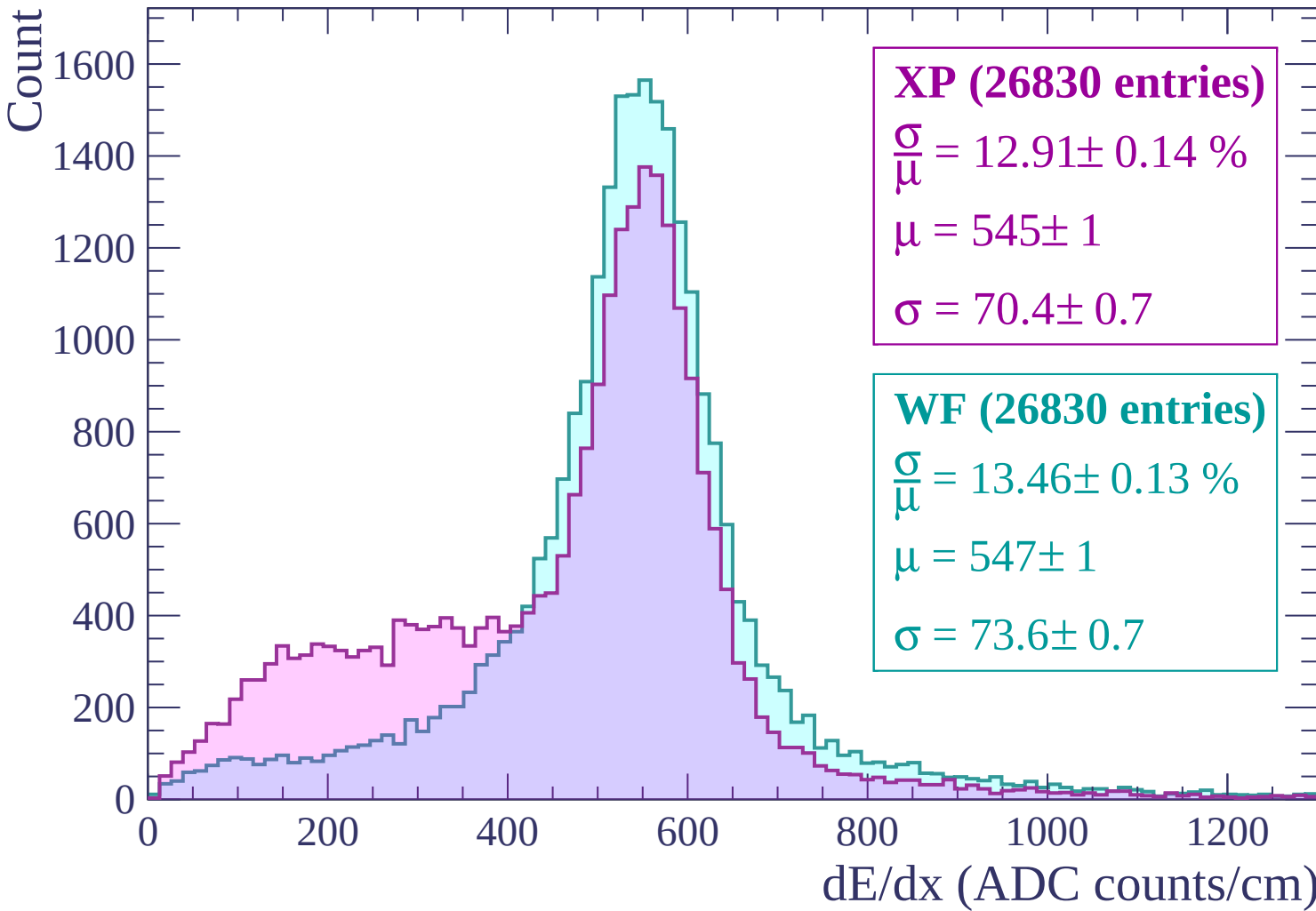
600

800

1000

1200

dE/dx (ADC counts/cm)



Energy loss | $0 < p < 60$

Count

4000
3500
3000
2500
2000
1500
1000
500
0

XP (68308 entries)

$$\frac{\sigma}{\mu} = 18.12 \pm 0.13 \%$$

$$\mu = 486 \pm 0$$

$$\sigma = 88.0 \pm 0.5$$

WF (68308 entries)

$$\frac{\sigma}{\mu} = 15.85 \pm 0.10 \%$$

$$\mu = 501 \pm 0$$

$$\sigma = 79.3 \pm 0.4$$

0

200

400

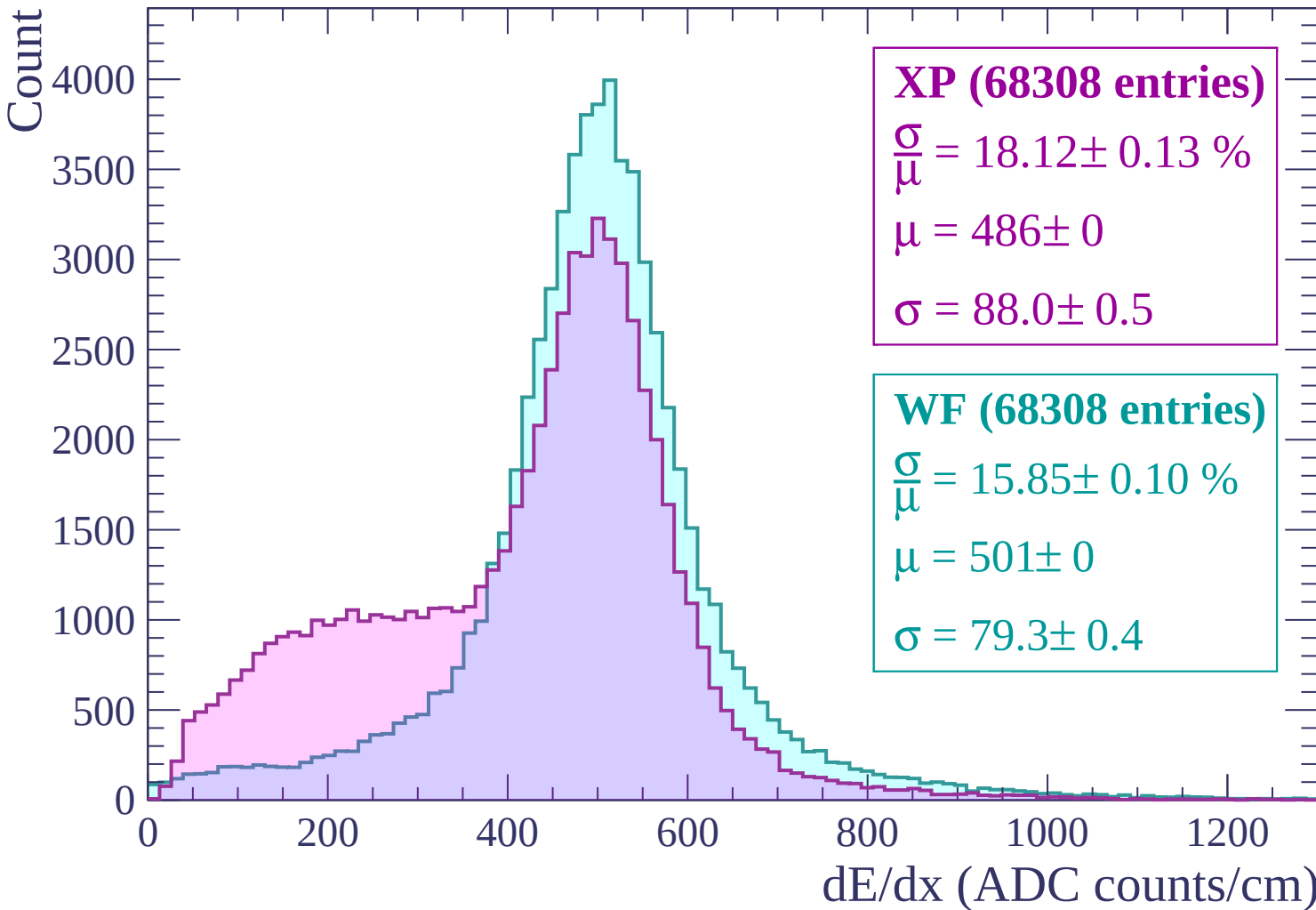
600

800

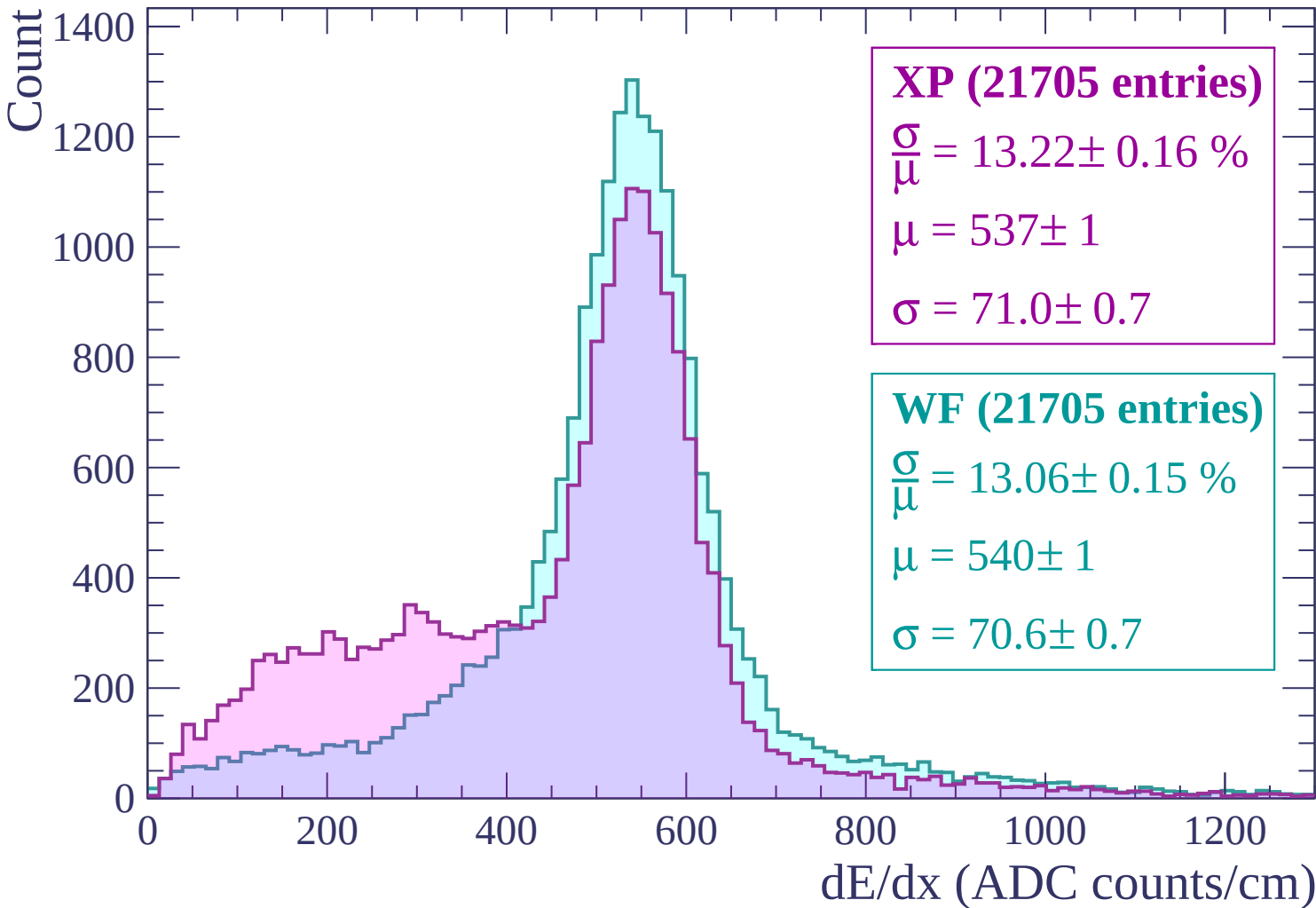
1000

1200

dE/dx (ADC counts/cm)

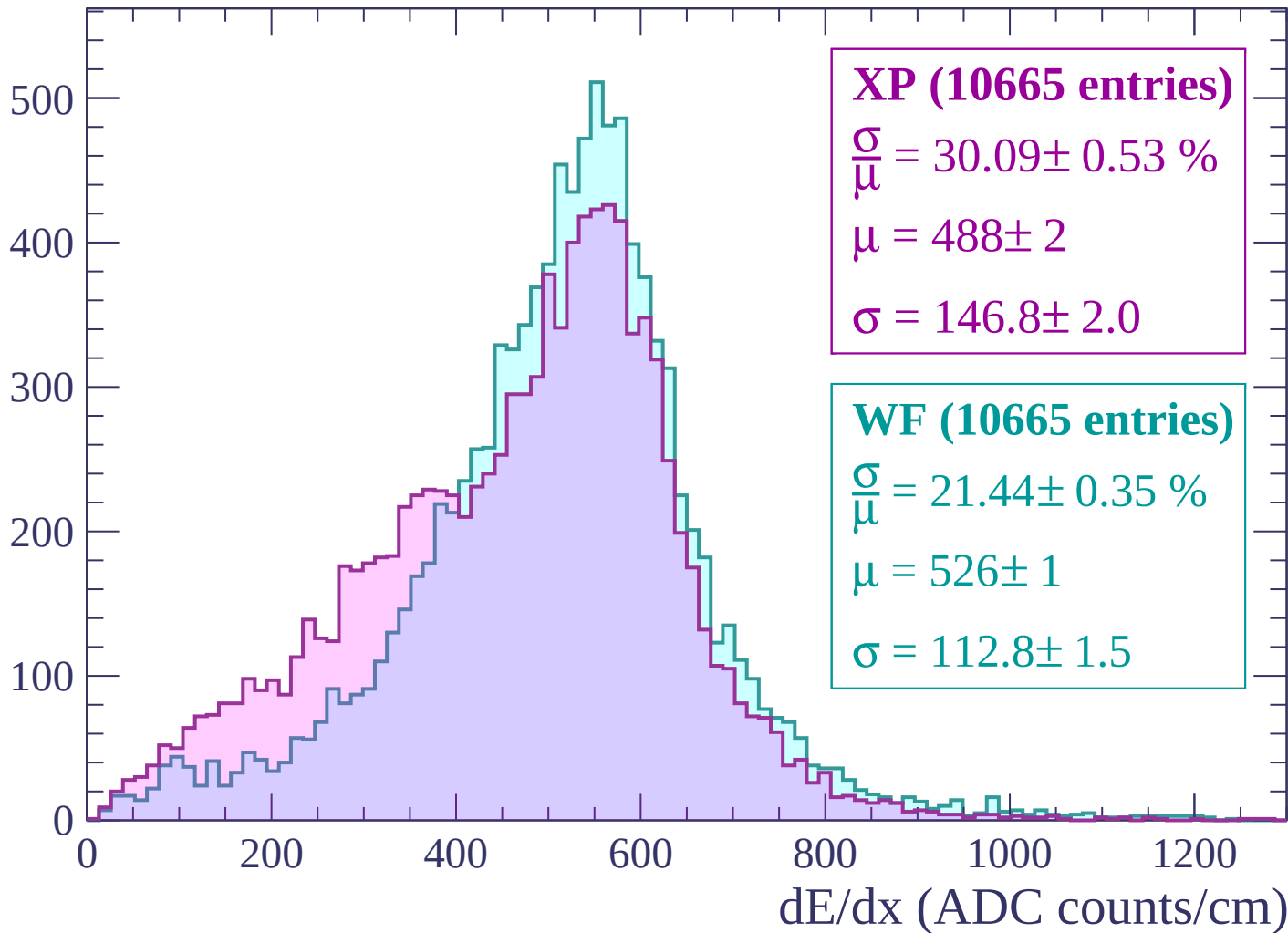


Energy loss | $60 < p < 120$



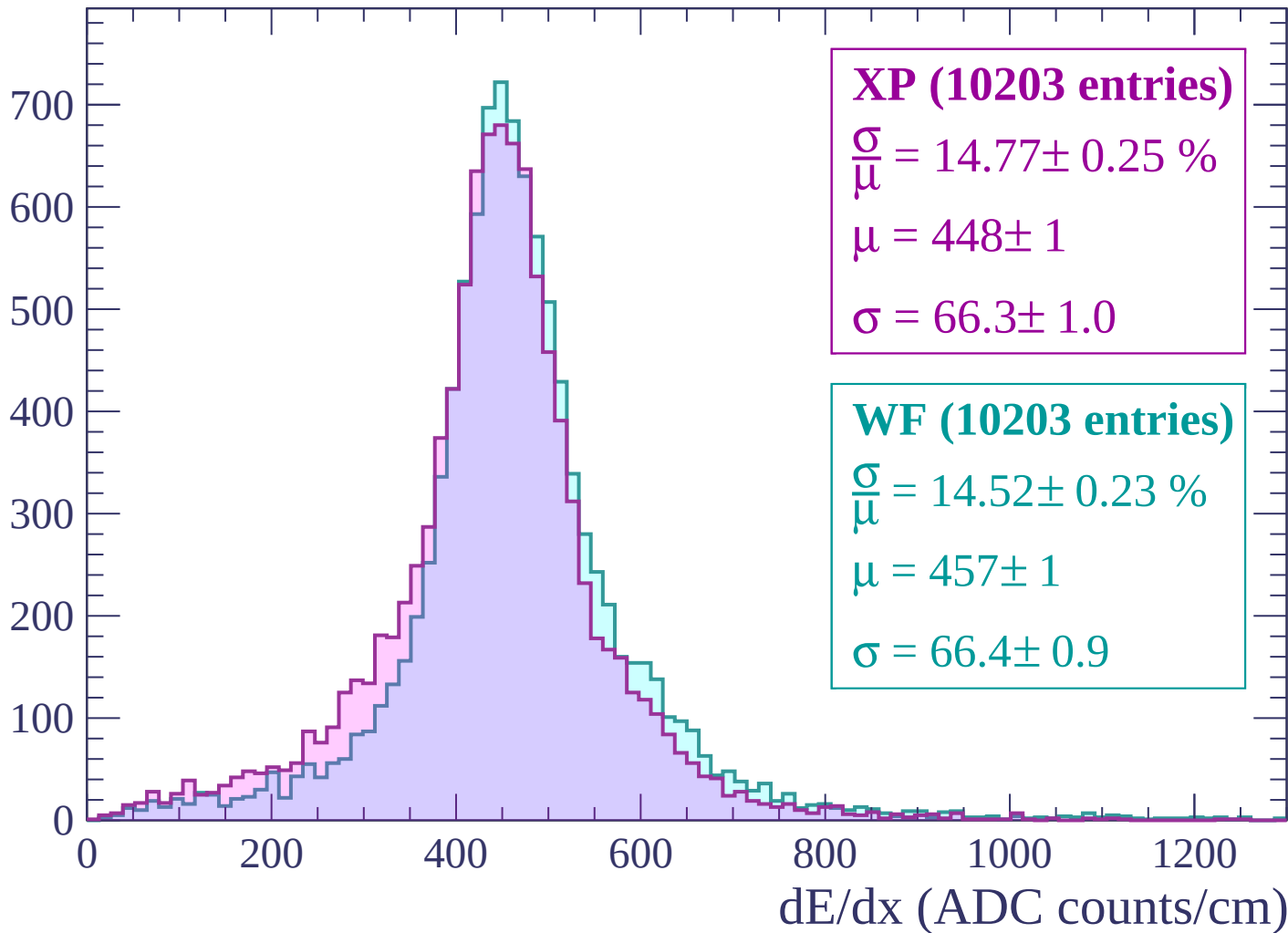
Energy loss | $120 < p < 180$

Count



Energy loss | $180 < p < 240$

Count



Energy loss | $240 < p < 300$

Count

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

XP (11141 entries)

$$\frac{\sigma}{\mu} = 11.60 \pm 0.18 \%$$

$$\mu = 412 \pm 1$$

$$\sigma = 47.8 \pm 0.7$$

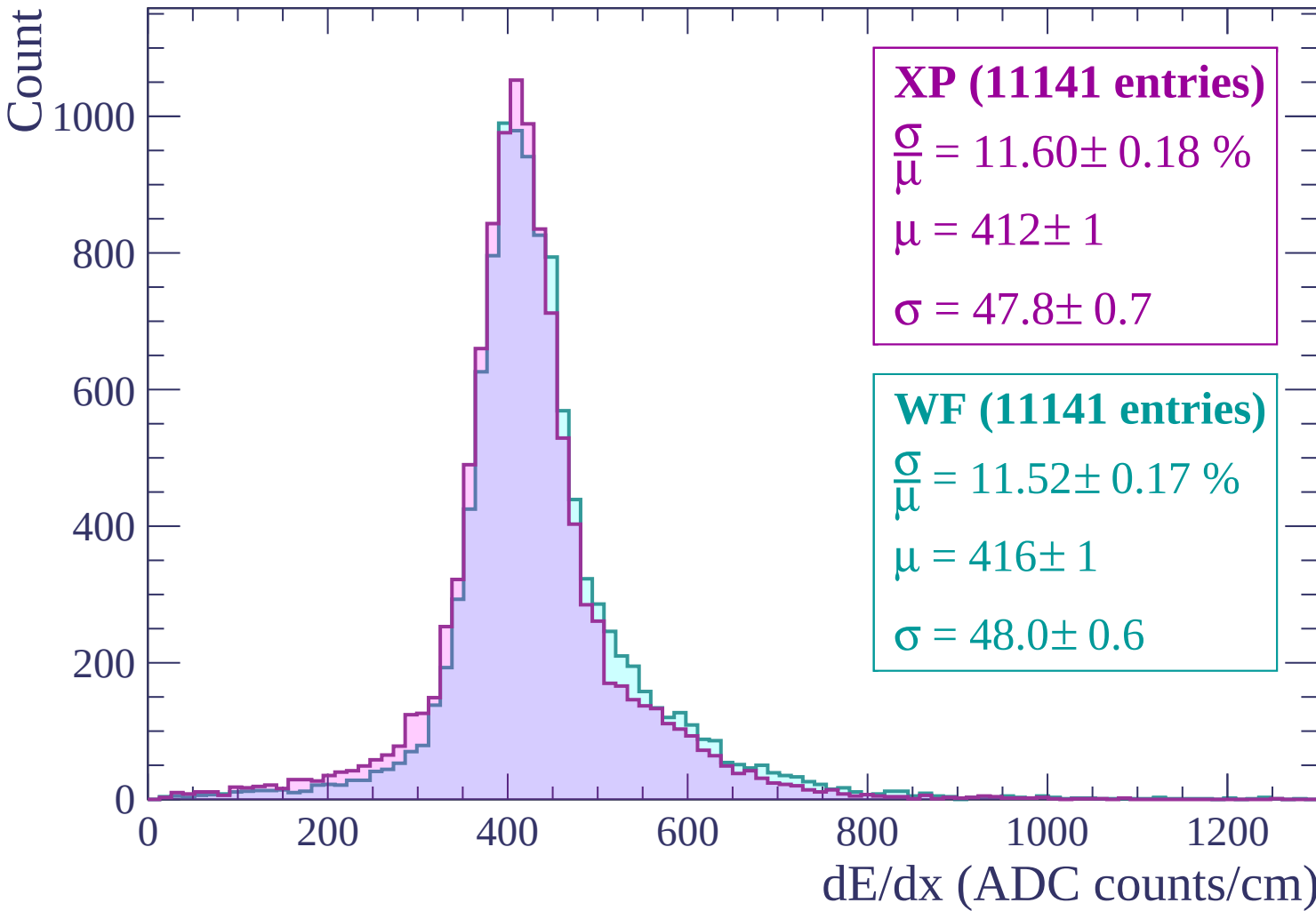
WF (11141 entries)

$$\frac{\sigma}{\mu} = 11.52 \pm 0.17 \%$$

$$\mu = 416 \pm 1$$

$$\sigma = 48.0 \pm 0.6$$

dE/dx (ADC counts/cm)



Energy loss | $300 < p < 360$

Count

1400
1200
1000
800
600
400
200
0

XP (13077 entries)

$$\frac{\sigma}{\mu} = 10.02 \pm 0.13 \%$$

$$\mu = 401 \pm 0$$

$$\sigma = 40.1 \pm 0.5$$

WF (13077 entries)

$$\frac{\sigma}{\mu} = 10.04 \pm 0.13 \%$$

$$\mu = 404 \pm 0$$

$$\sigma = 40.5 \pm 0.5$$

0

200

400

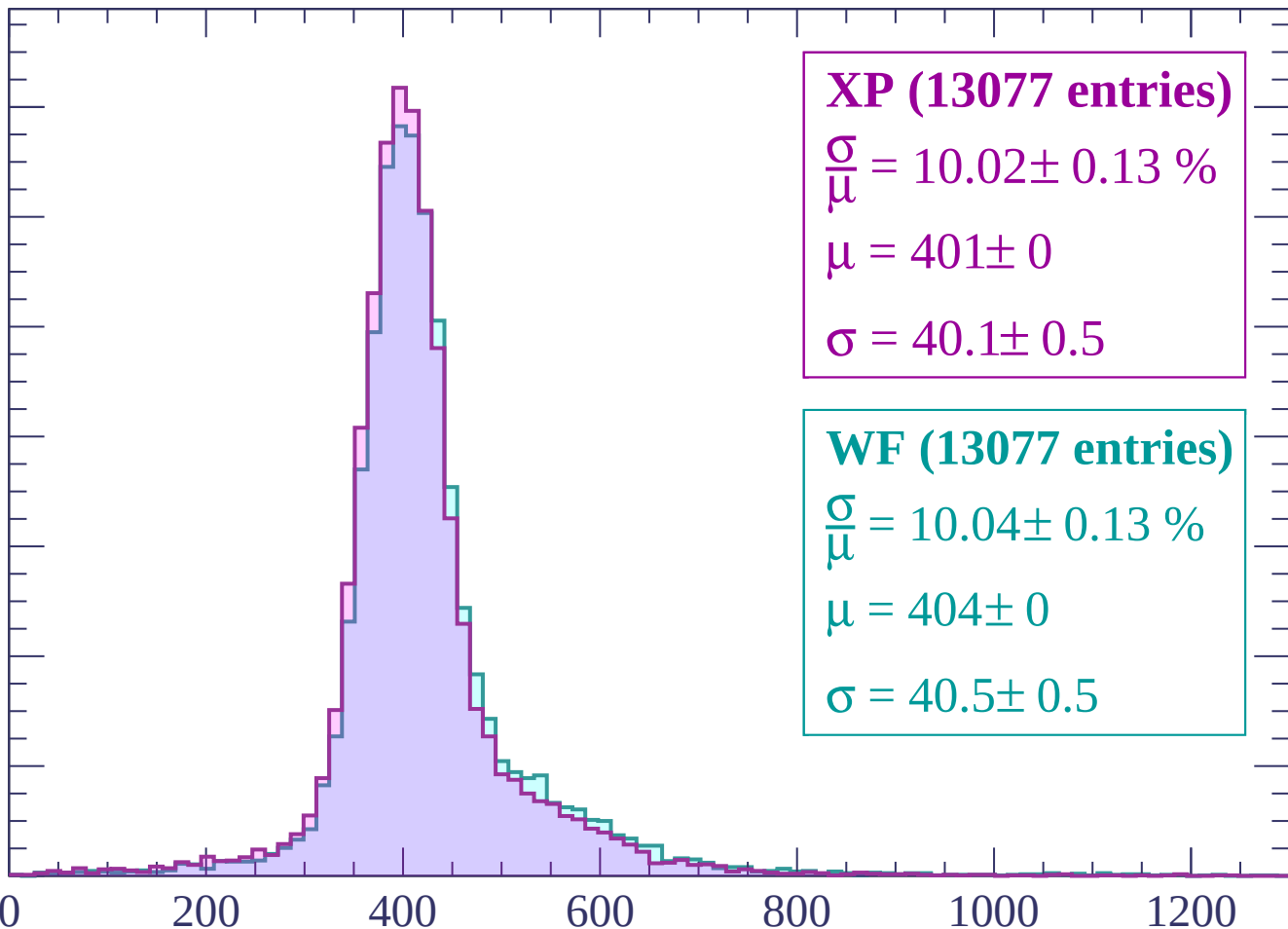
600

800

1000

1200

dE/dx (ADC counts/cm)



Energy loss | $360 < p < 420$

Count

1600
1400
1200
1000
800
600
400
200
0

XP (13532 entries)

$$\frac{\sigma}{\mu} = 10.09 \pm 0.13 \%$$

$$\mu = 398 \pm 0$$

$$\sigma = 40.2 \pm 0.5$$

WF (13532 entries)

$$\frac{\sigma}{\mu} = 10.21 \pm 0.13 \%$$

$$\mu = 401 \pm 0$$

$$\sigma = 41.0 \pm 0.5$$

0

200

400

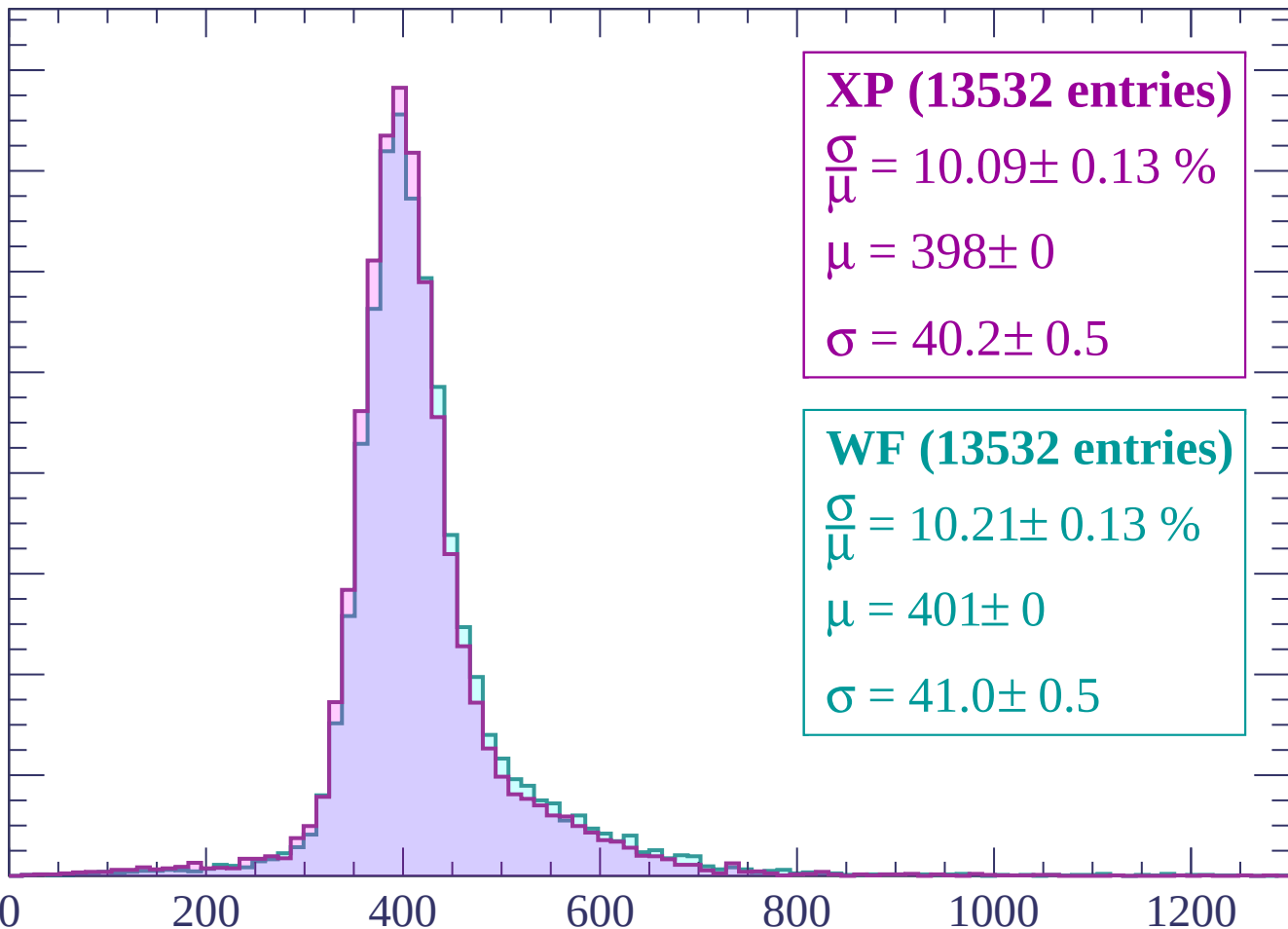
600

800

1000

1200

dE/dx (ADC counts/cm)



Energy loss | $420 < p < 480$

Count

1600
1400
1200
1000
800
600
400
200
0

XP (13112 entries)

$$\frac{\sigma}{\mu} = 9.77 \pm 0.12 \%$$

$$\mu = 400 \pm 0$$

$$\sigma = 39.1 \pm 0.4$$

WF (13112 entries)

$$\frac{\sigma}{\mu} = 9.89 \pm 0.12 \%$$

$$\mu = 403 \pm 0$$

$$\sigma = 39.8 \pm 0.5$$

0

200

400

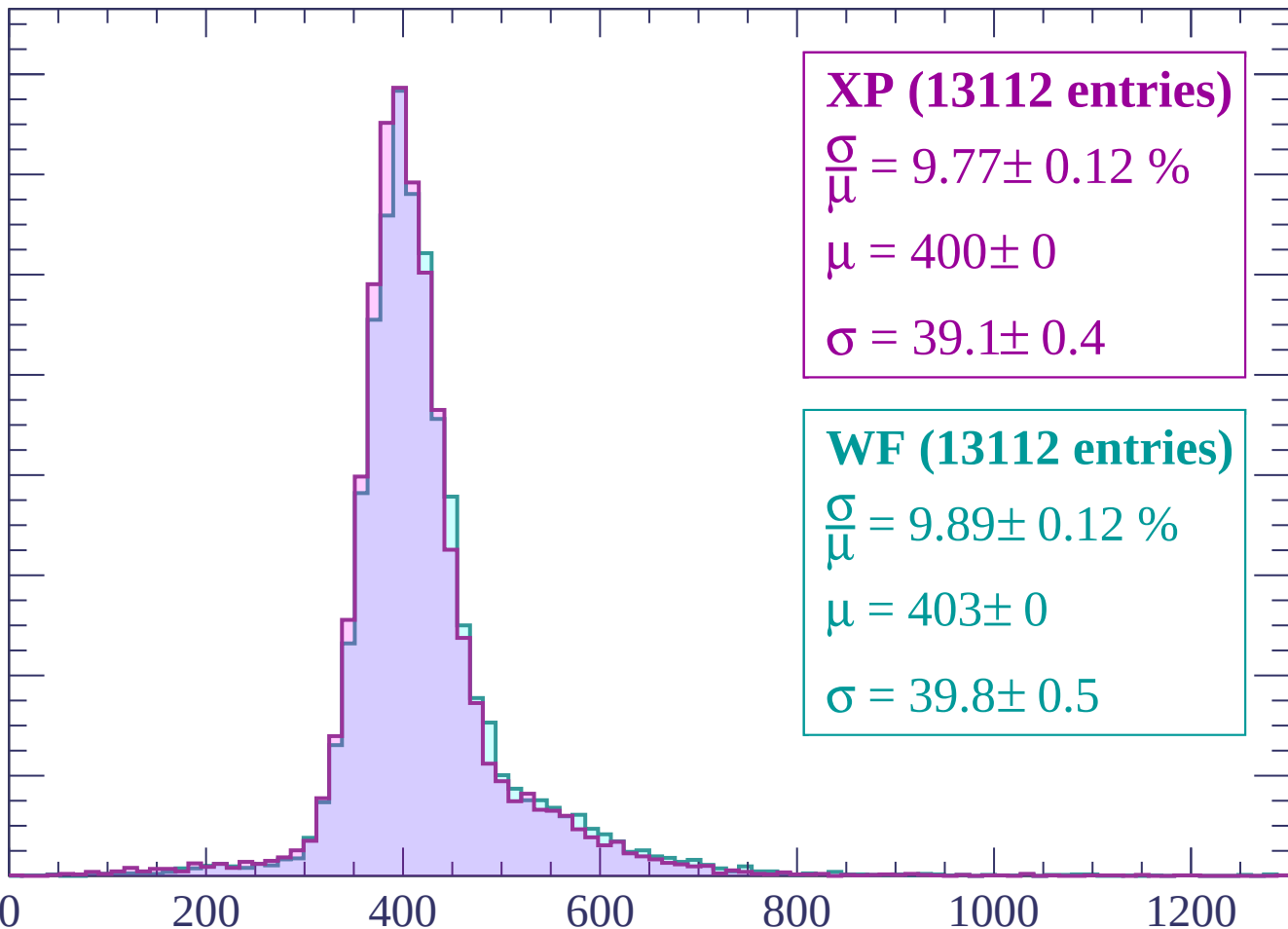
600

800

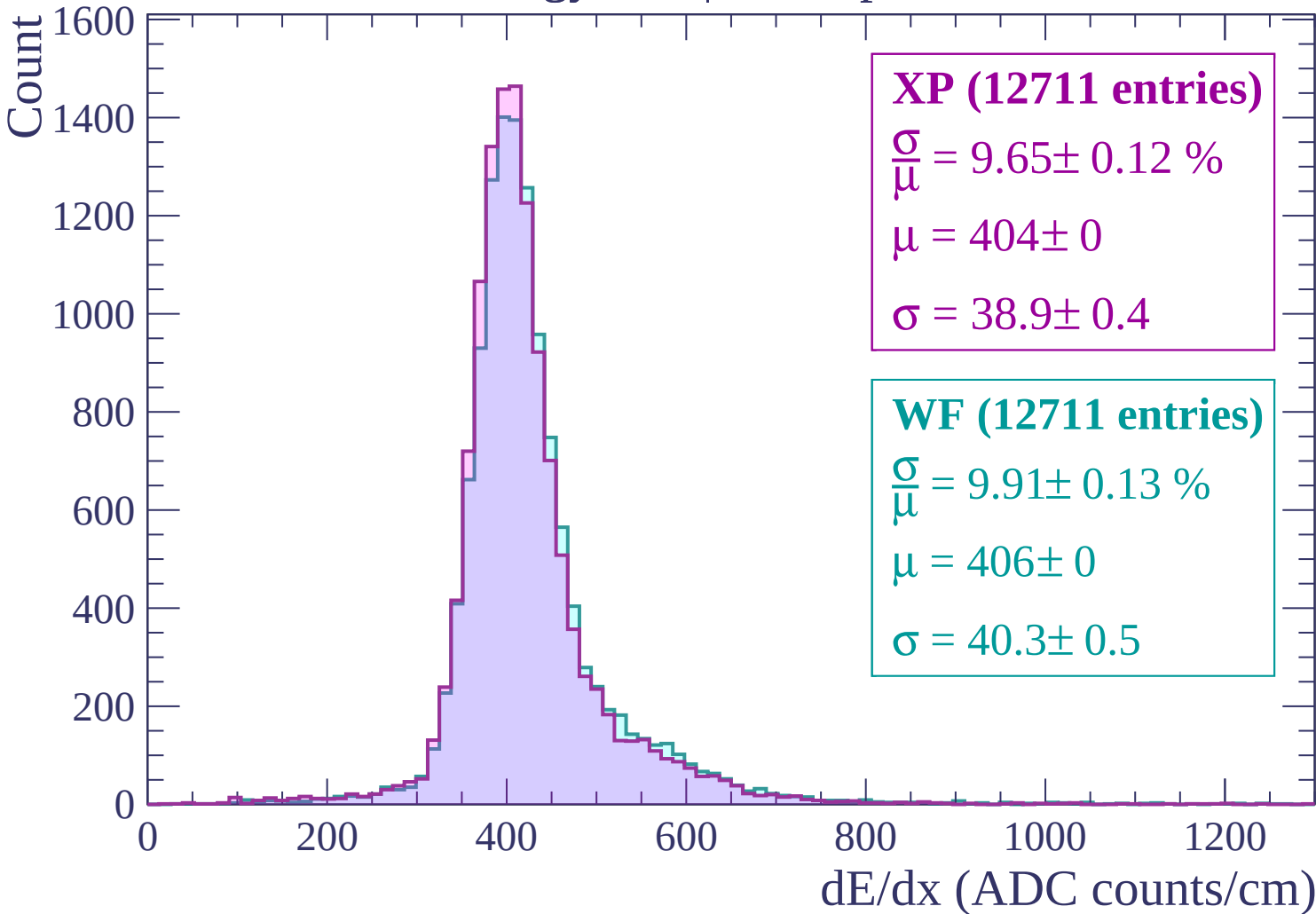
1000

1200

dE/dx (ADC counts/cm)



Energy loss | $480 < p < 540$



Energy loss | $540 < p < 600$

Count

1400
1200
1000
800
600
400
200
0

XP (12189 entries)

$$\frac{\sigma}{\mu} = 9.56 \pm 0.12 \%$$

$$\mu = 407 \pm 0$$

$$\sigma = 38.9 \pm 0.4$$

WF (12189 entries)

$$\frac{\sigma}{\mu} = 10.01 \pm 0.12 \%$$

$$\mu = 410 \pm 0$$

$$\sigma = 41.0 \pm 0.4$$

0

200

400

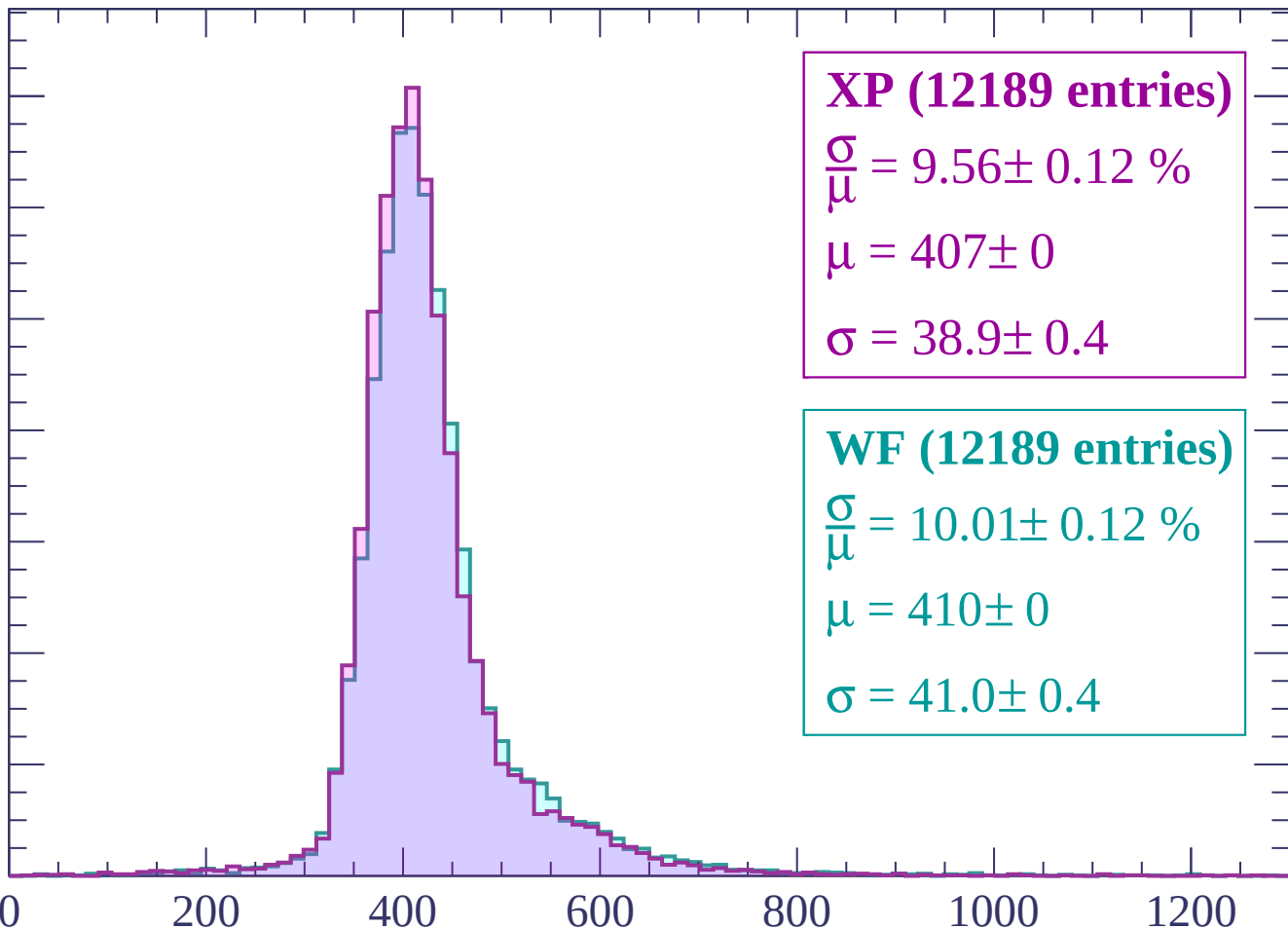
600

800

1000

1200

dE/dx (ADC counts/cm)



Energy loss | $600 < p < 660$

Count

1400
1200
1000
800
600
400
200
0

XP (11612 entries)

$$\frac{\sigma}{\mu} = 9.51 \pm 0.12 \%$$

$$\mu = 410 \pm 0$$

$$\sigma = 39.0 \pm 0.5$$

WF (11612 entries)

$$\frac{\sigma}{\mu} = 10.00 \pm 0.12 \%$$

$$\mu = 414 \pm 0$$

$$\sigma = 41.4 \pm 0.5$$

0

200

400

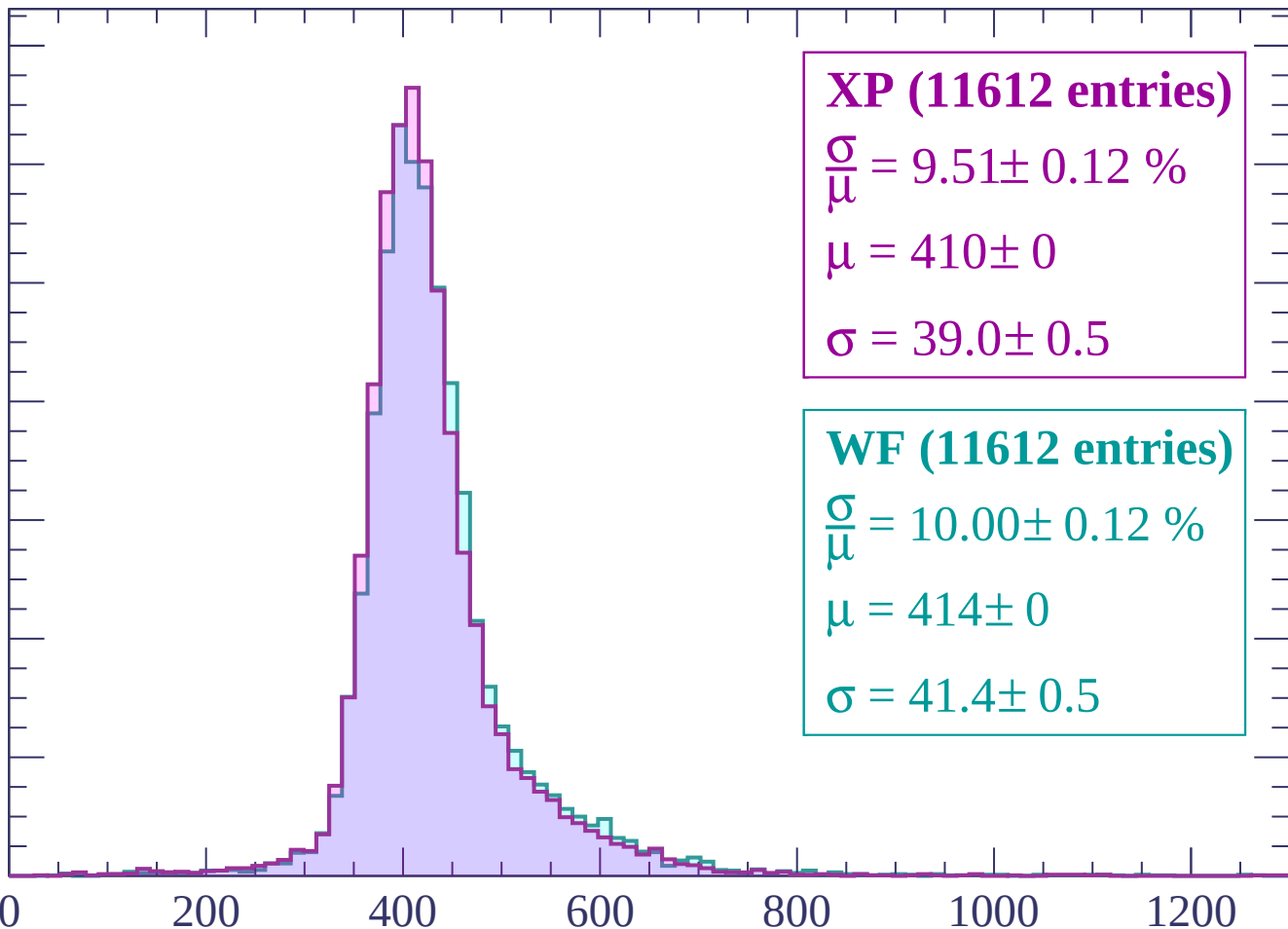
600

800

1000

1200

dE/dx (ADC counts/cm)



Energy loss | $660 < p < 720$

Count

1200

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

XP (11201 entries)

$$\frac{\sigma}{\mu} = 9.79 \pm 0.14 \%$$

$$\mu = 413 \pm 0$$

$$\sigma = 40.4 \pm 0.5$$

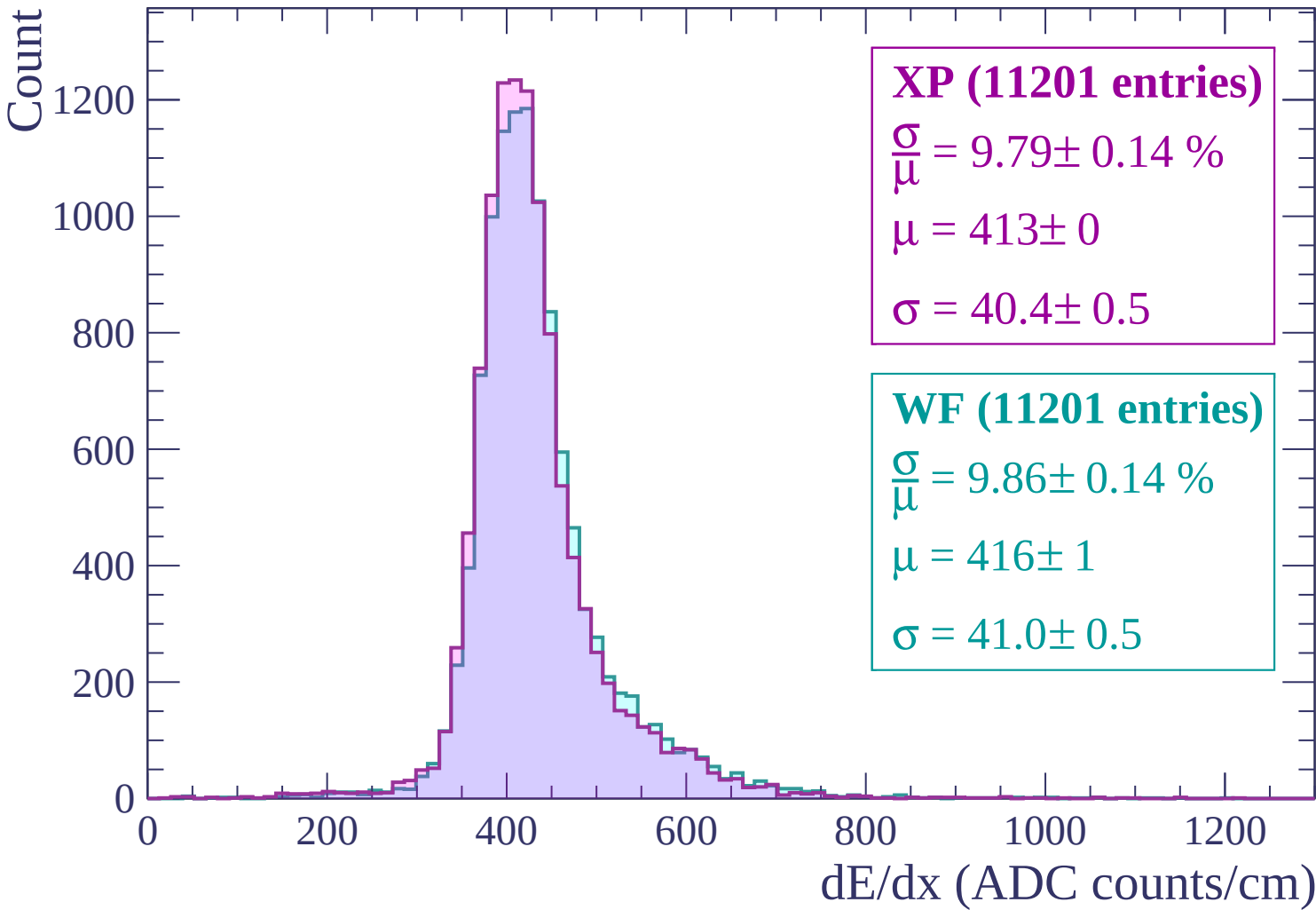
WF (11201 entries)

$$\frac{\sigma}{\mu} = 9.86 \pm 0.14 \%$$

$$\mu = 416 \pm 1$$

$$\sigma = 41.0 \pm 0.5$$

dE/dx (ADC counts/cm)



Energy loss | $720 < p < 780$

Count

1200

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (10988 entries)

$\frac{\sigma}{\mu} = 9.76 \pm 0.14 \%$

$\mu = 417 \pm 1$

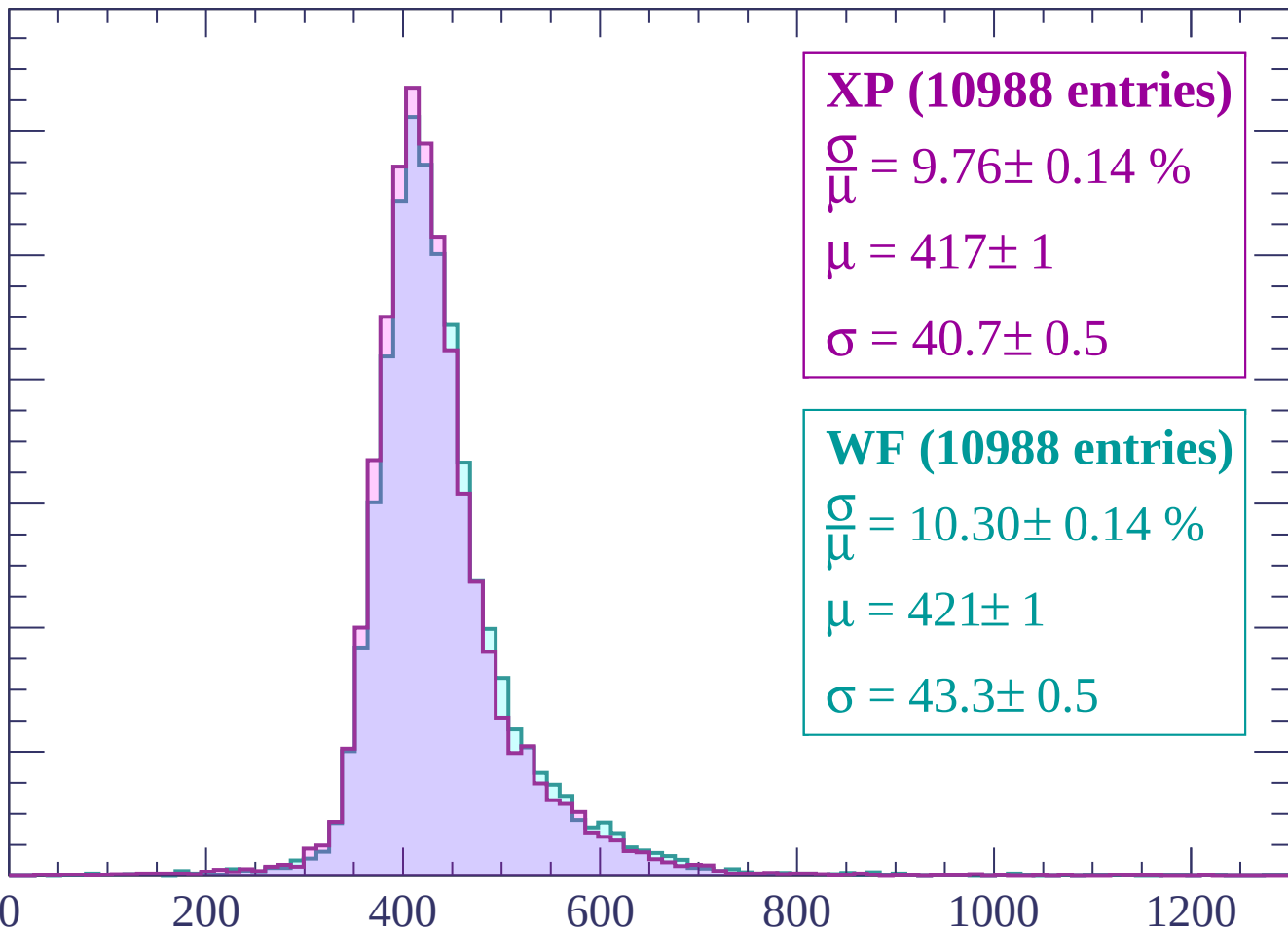
$\sigma = 40.7 \pm 0.5$

WF (10988 entries)

$\frac{\sigma}{\mu} = 10.30 \pm 0.14 \%$

$\mu = 421 \pm 1$

$\sigma = 43.3 \pm 0.5$



Energy loss | $780 < p < 840$

Count

1200

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

XP (10275 entries)

$\frac{\sigma}{\mu} = 9.47 \pm 0.13 \%$

$\mu = 420 \pm 1$

$\sigma = 39.8 \pm 0.5$

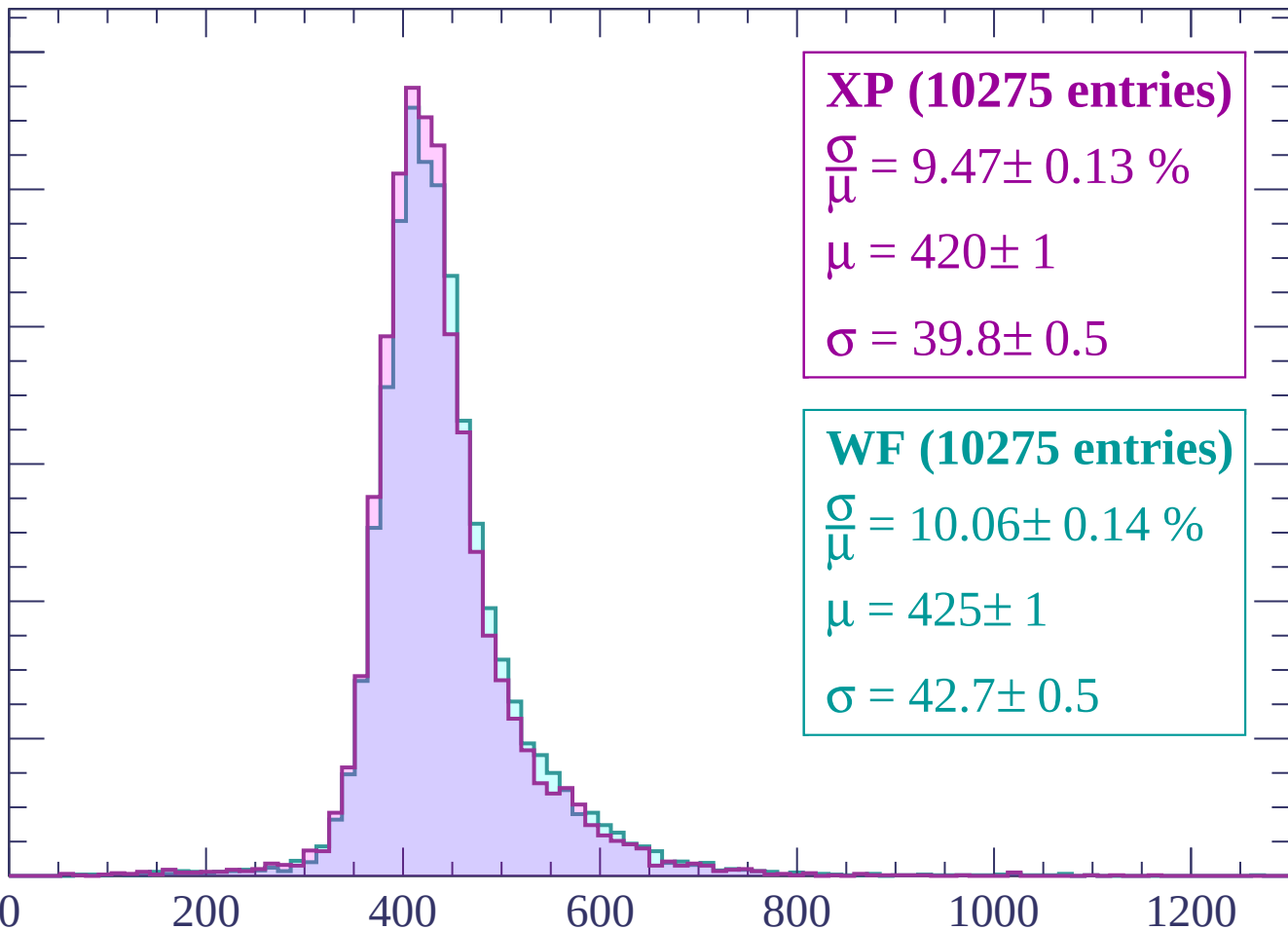
WF (10275 entries)

$\frac{\sigma}{\mu} = 10.06 \pm 0.14 \%$

$\mu = 425 \pm 1$

$\sigma = 42.7 \pm 0.5$

dE/dx (ADC counts/cm)



Energy loss | $840 < p < 900$

Count

1200
1000
800
600
400
200
0

XP (9970 entries)

$$\frac{\sigma}{\mu} = 9.87 \pm 0.13 \%$$

$$\mu = 425 \pm 1$$

$$\sigma = 42.0 \pm 0.5$$

WF (9970 entries)

$$\frac{\sigma}{\mu} = 9.96 \pm 0.14 \%$$

$$\mu = 429 \pm 1$$

$$\sigma = 42.7 \pm 0.5$$

0

200

400

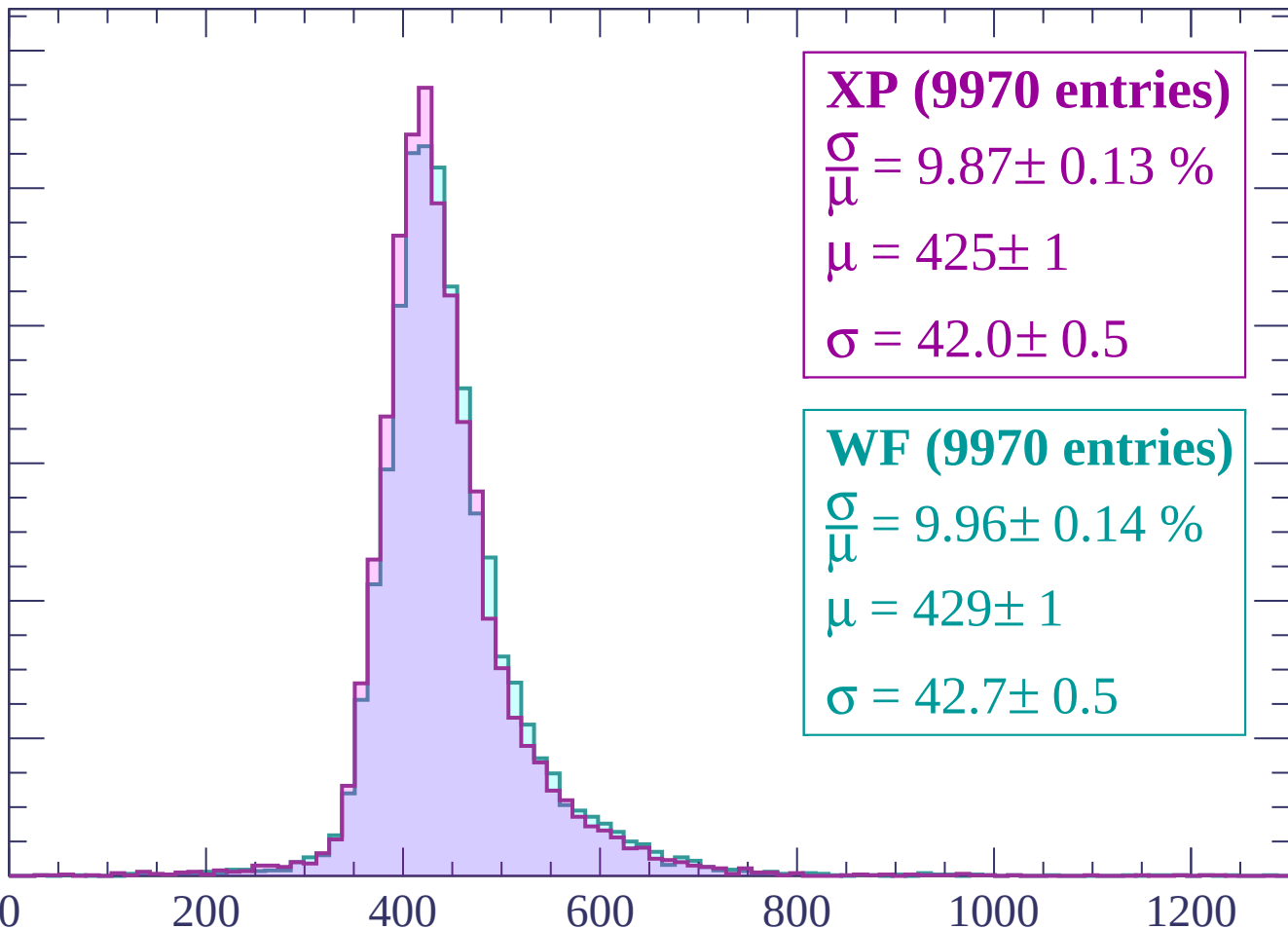
600

800

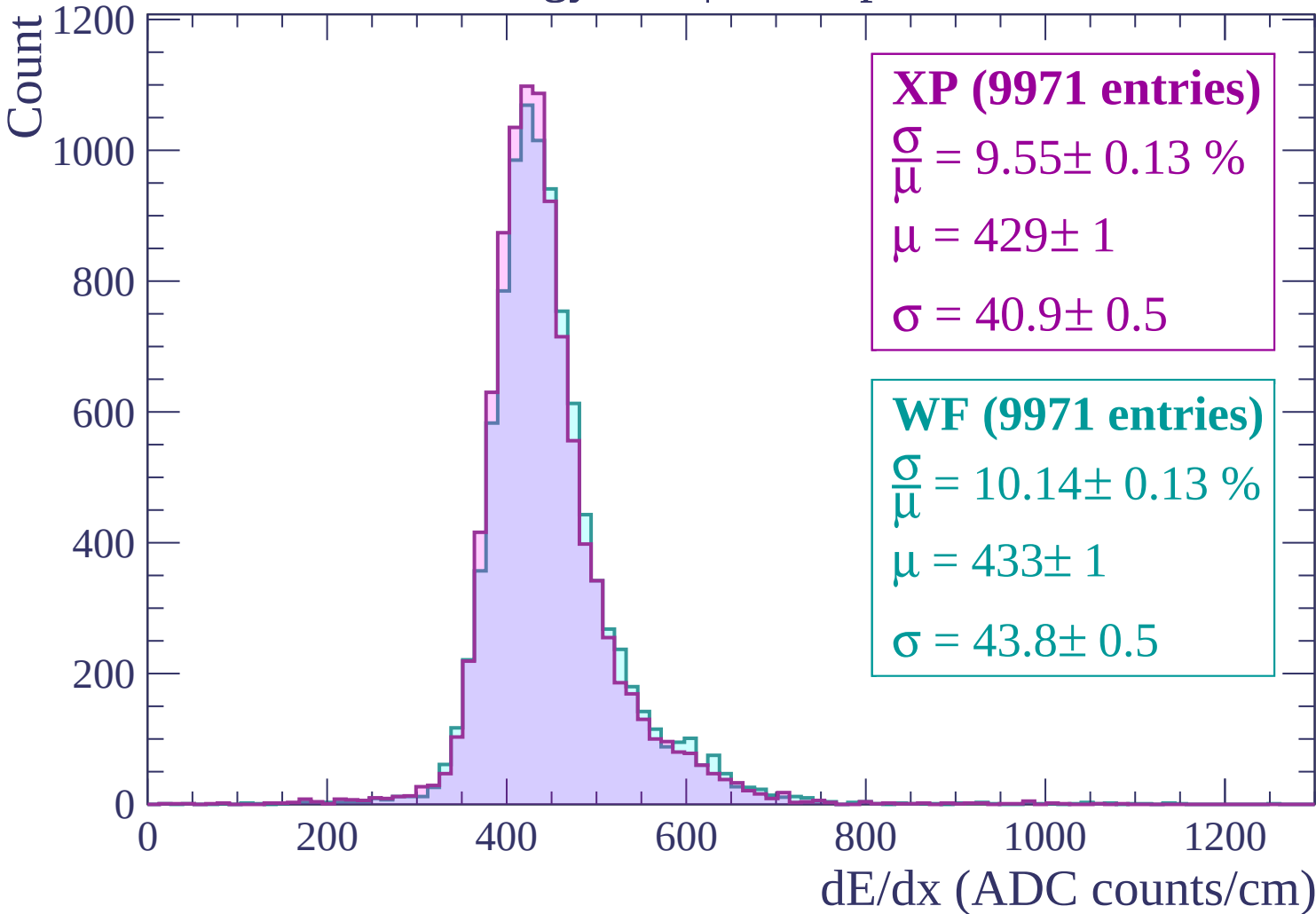
1000

1200

dE/dx (ADC counts/cm)



Energy loss | $900 < p < 960$



Energy loss | $960 < p < 1020$

Count

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

XP (9372 entries)

$$\frac{\sigma}{\mu} = 9.61 \pm 0.15 \%$$

$$\mu = 431 \pm 1$$

$$\sigma = 41.4 \pm 0.6$$

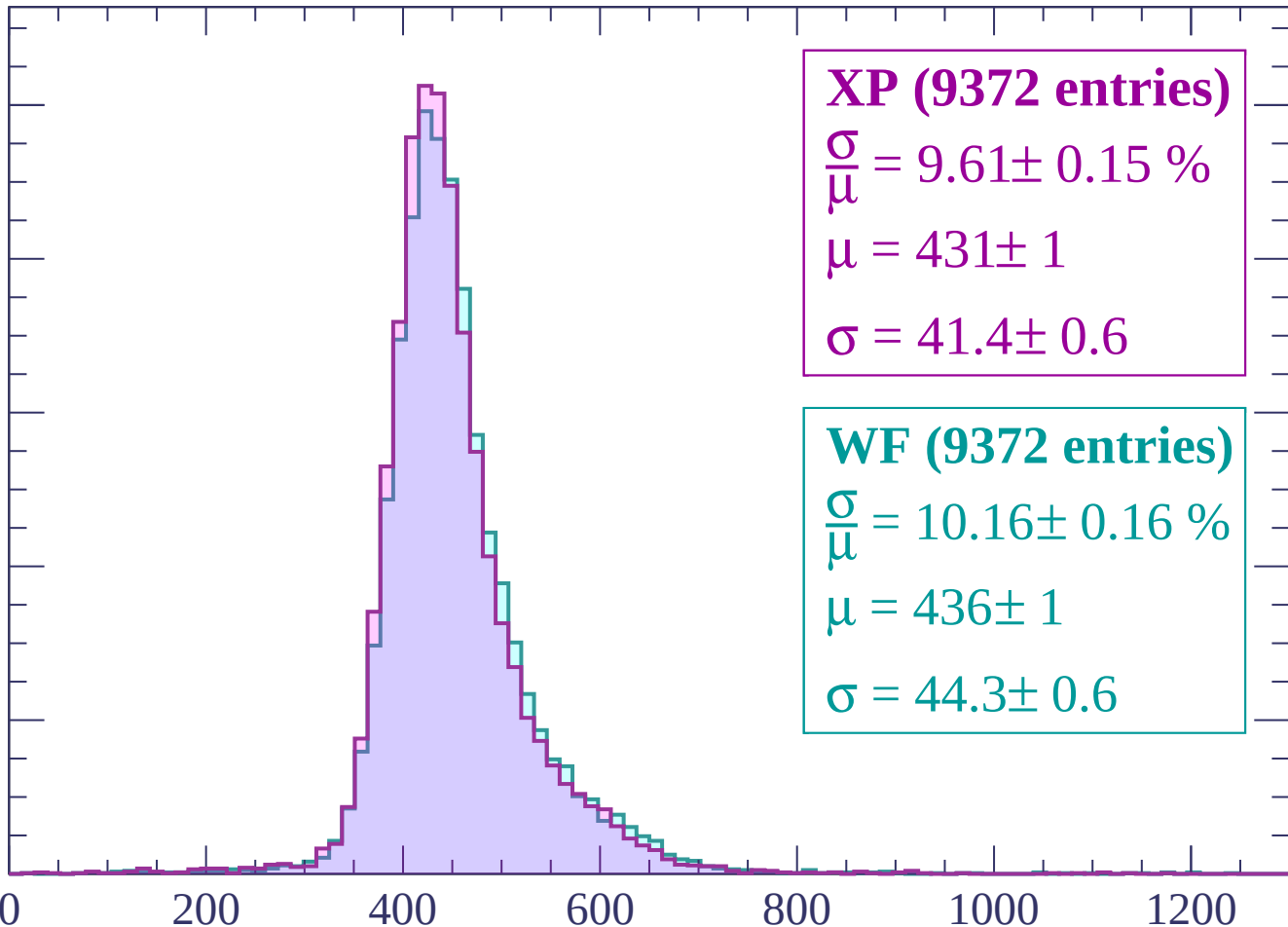
WF (9372 entries)

$$\frac{\sigma}{\mu} = 10.16 \pm 0.16 \%$$

$$\mu = 436 \pm 1$$

$$\sigma = 44.3 \pm 0.6$$

dE/dx (ADC counts/cm)



Energy loss | $1020 < p < 1080$

Count

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

XP (9114 entries)

$\frac{\sigma}{\mu} = 9.84 \pm 0.14 \%$

$\mu = 437 \pm 1$

$\sigma = 43.0 \pm 0.6$

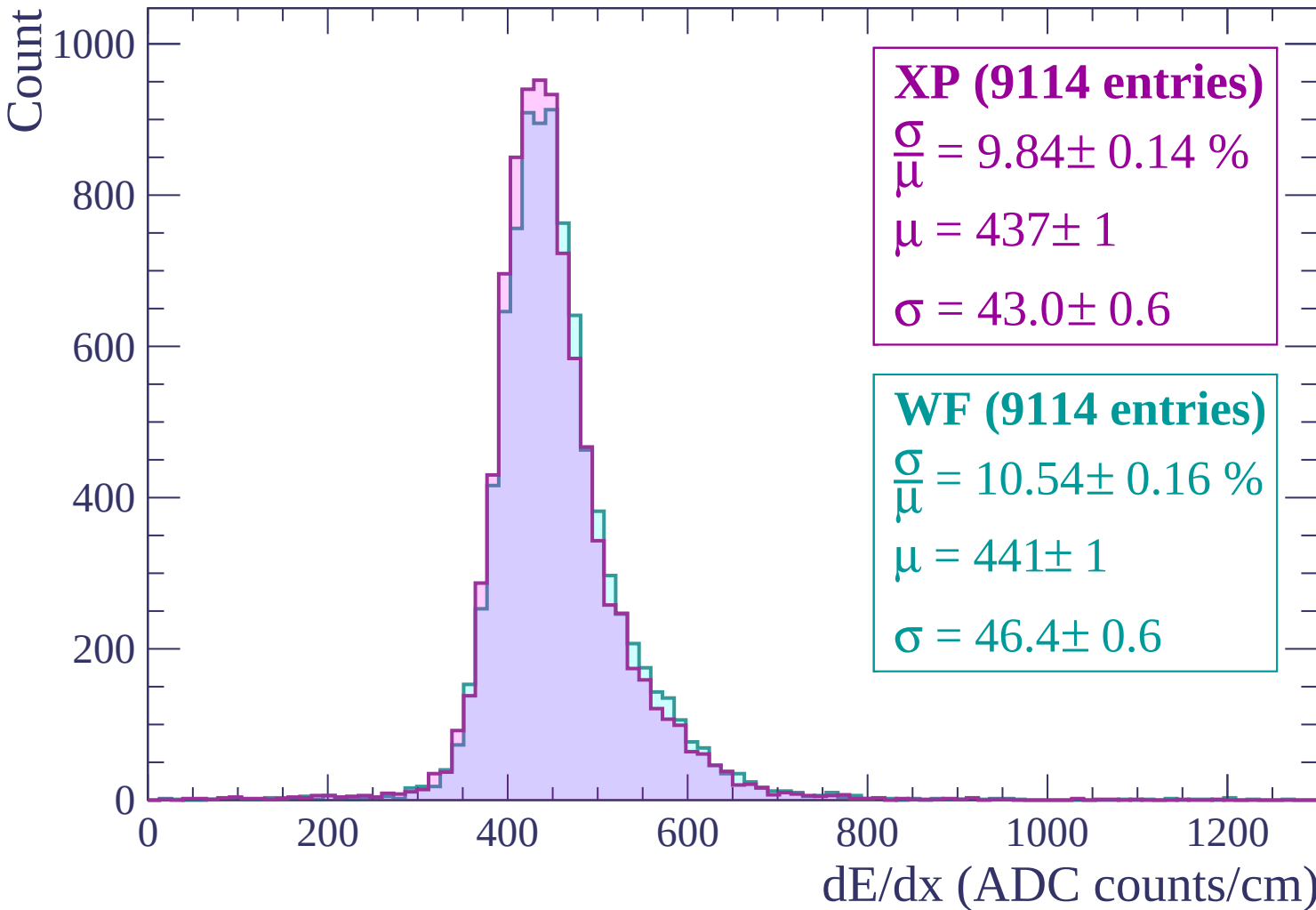
WF (9114 entries)

$\frac{\sigma}{\mu} = 10.54 \pm 0.16 \%$

$\mu = 441 \pm 1$

$\sigma = 46.4 \pm 0.6$

dE/dx (ADC counts/cm)



Energy loss | $1080 < p < 1140$

Count

1000

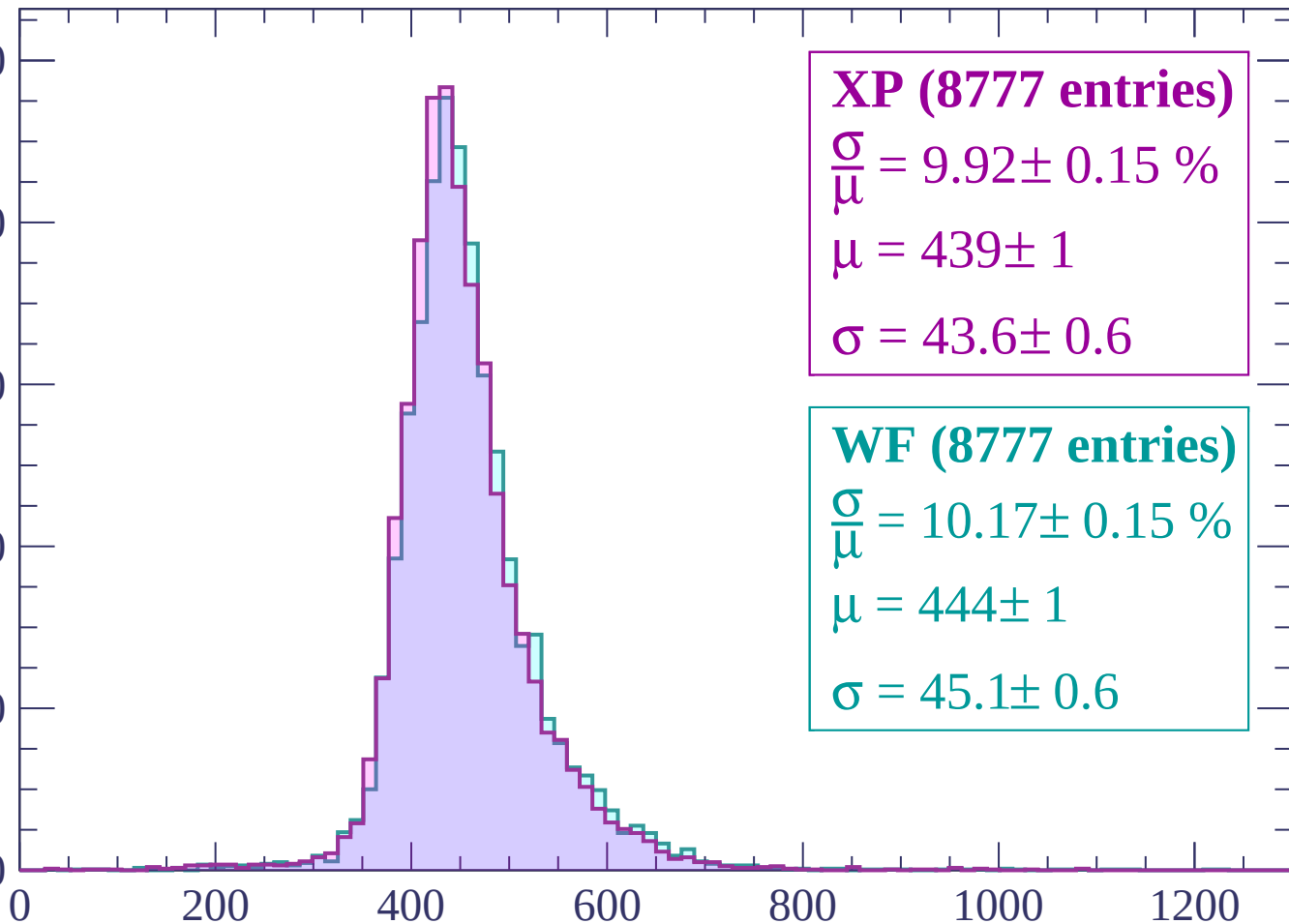
800

600

400

200

0



XP (8777 entries)

$\frac{\sigma}{\mu} = 9.92 \pm 0.15 \%$

$\mu = 439 \pm 1$

$\sigma = 43.6 \pm 0.6$

WF (8777 entries)

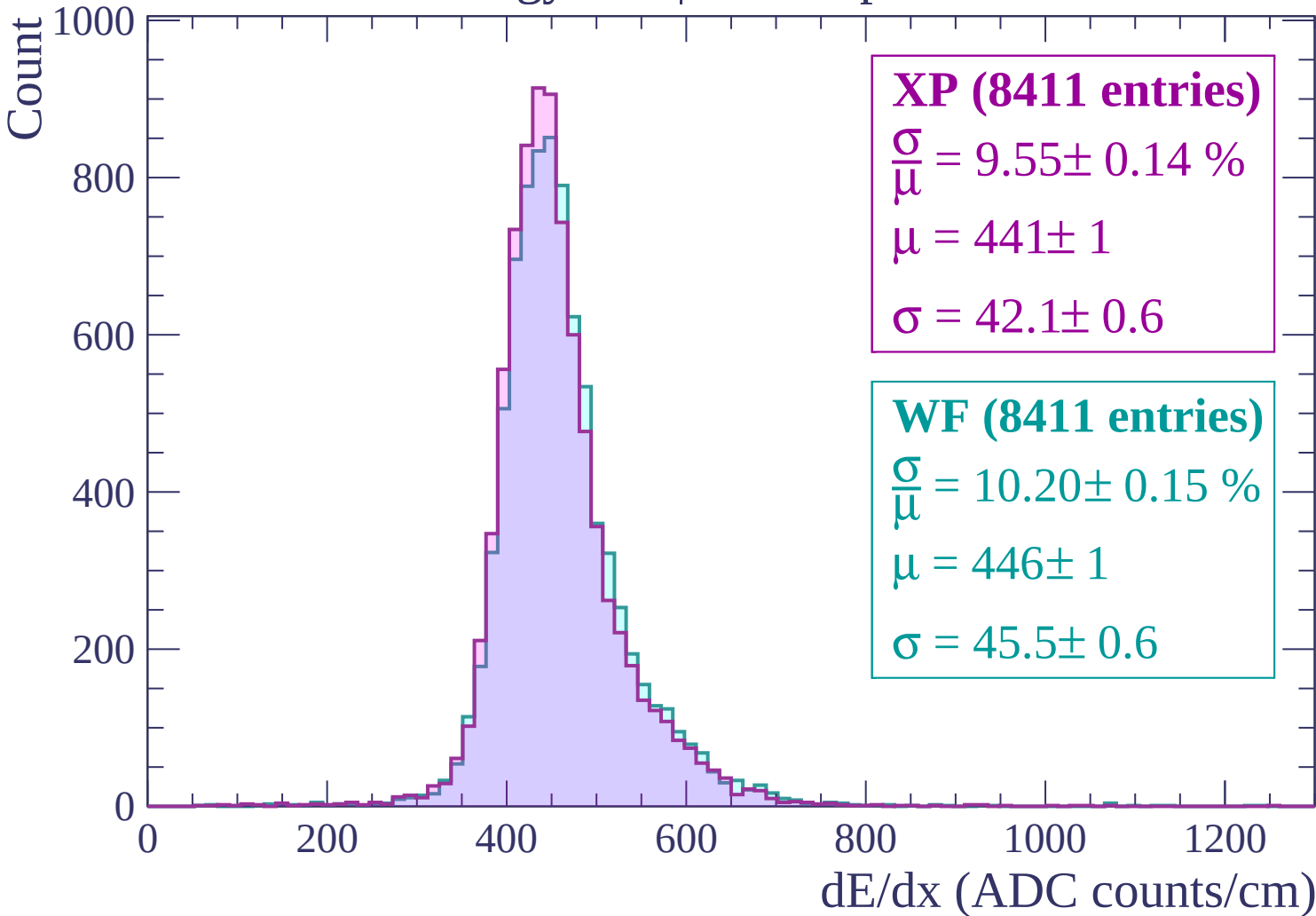
$\frac{\sigma}{\mu} = 10.17 \pm 0.15 \%$

$\mu = 444 \pm 1$

$\sigma = 45.1 \pm 0.6$

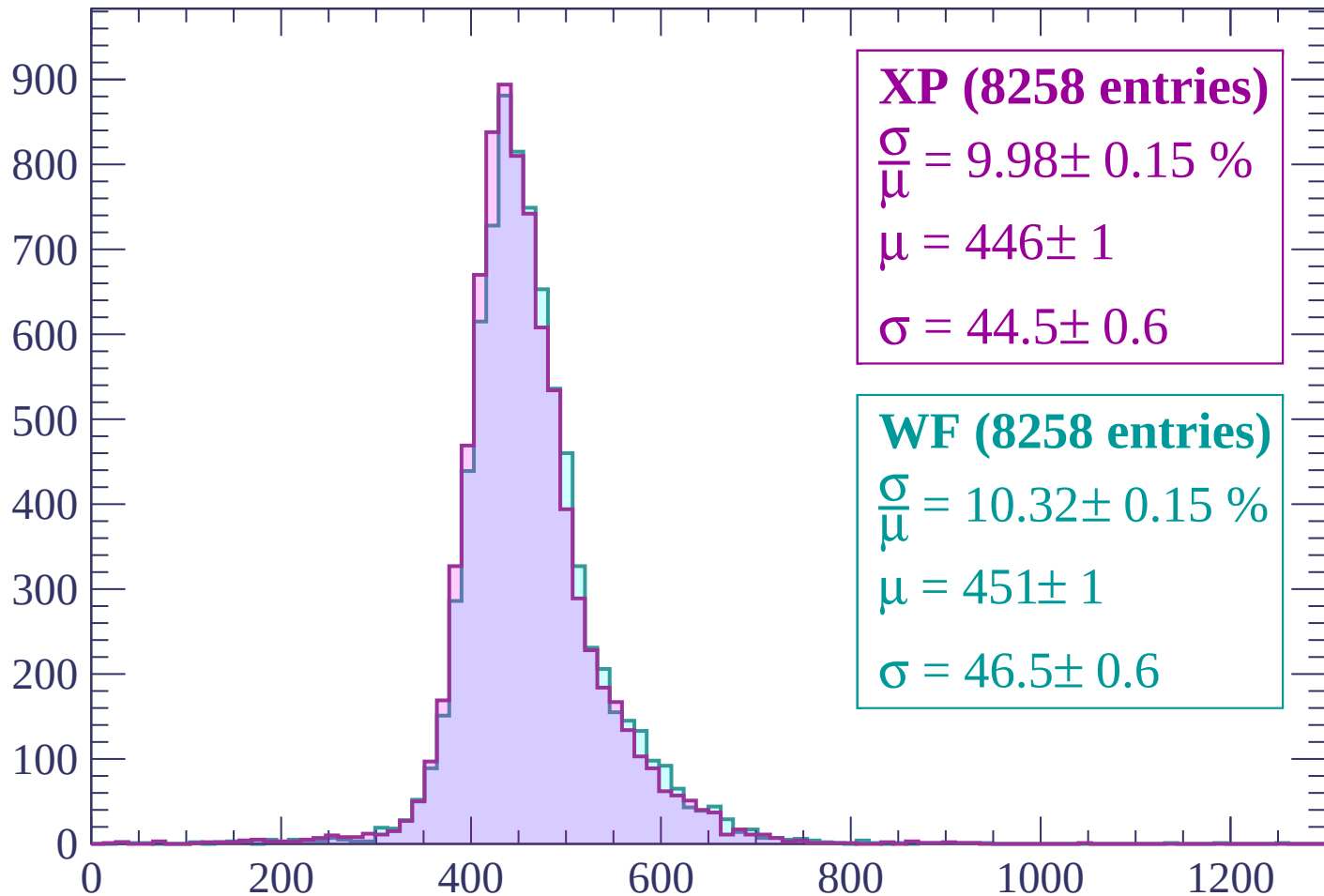
dE/dx (ADC counts/cm)

Energy loss | $1140 < p < 1200$



Energy loss | $1200 < p < 1260$

Count



XP (8258 entries)

$$\frac{\sigma}{\mu} = 9.98 \pm 0.15 \%$$

$$\mu = 446 \pm 1$$

$$\sigma = 44.5 \pm 0.6$$

WF (8258 entries)

$$\frac{\sigma}{\mu} = 10.32 \pm 0.15 \%$$

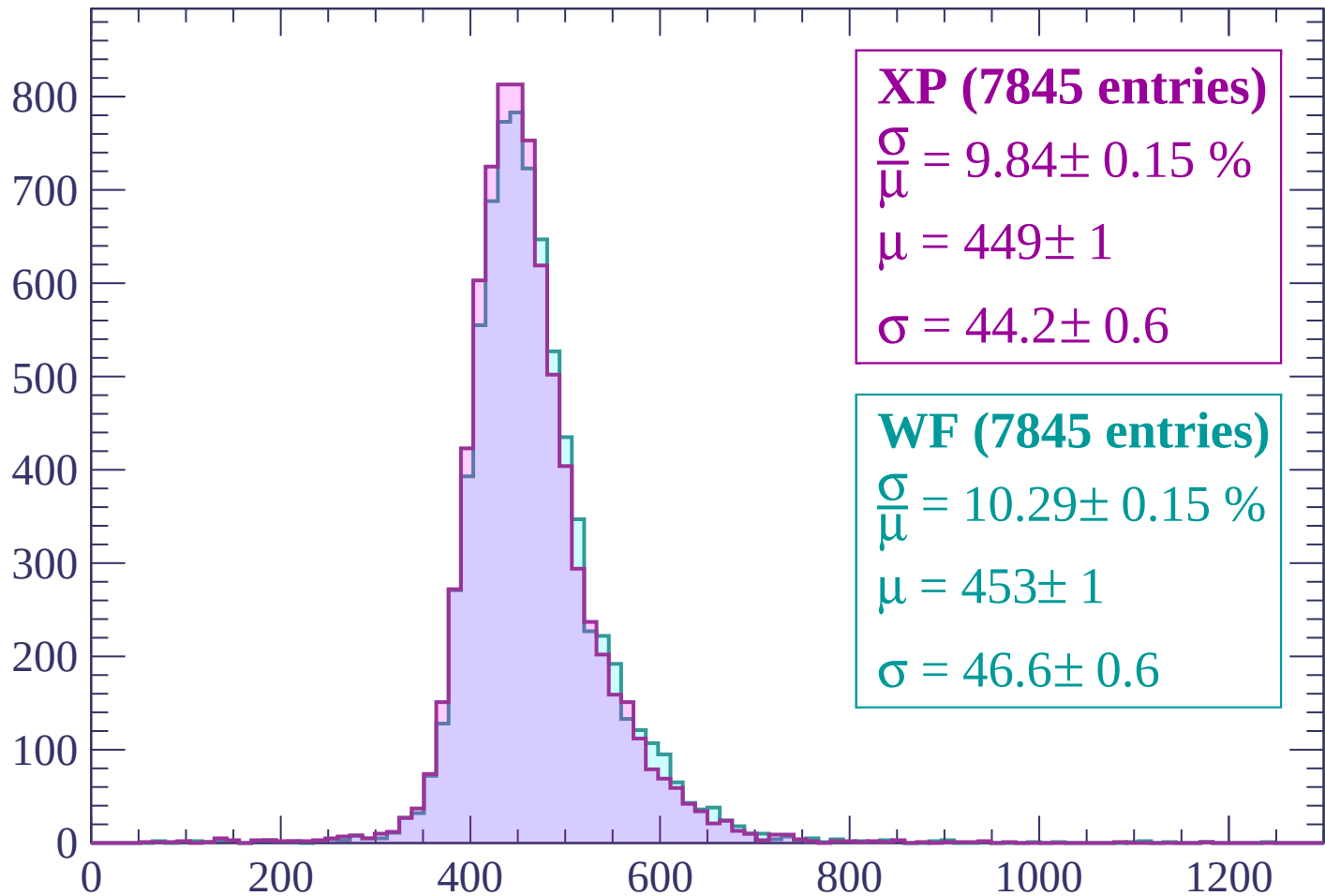
$$\mu = 451 \pm 1$$

$$\sigma = 46.5 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $1260 < p < 1320$

Count



XP (7845 entries)

$$\frac{\sigma}{\mu} = 9.84 \pm 0.15 \%$$

$$\mu = 449 \pm 1$$

$$\sigma = 44.2 \pm 0.6$$

WF (7845 entries)

$$\frac{\sigma}{\mu} = 10.29 \pm 0.15 \%$$

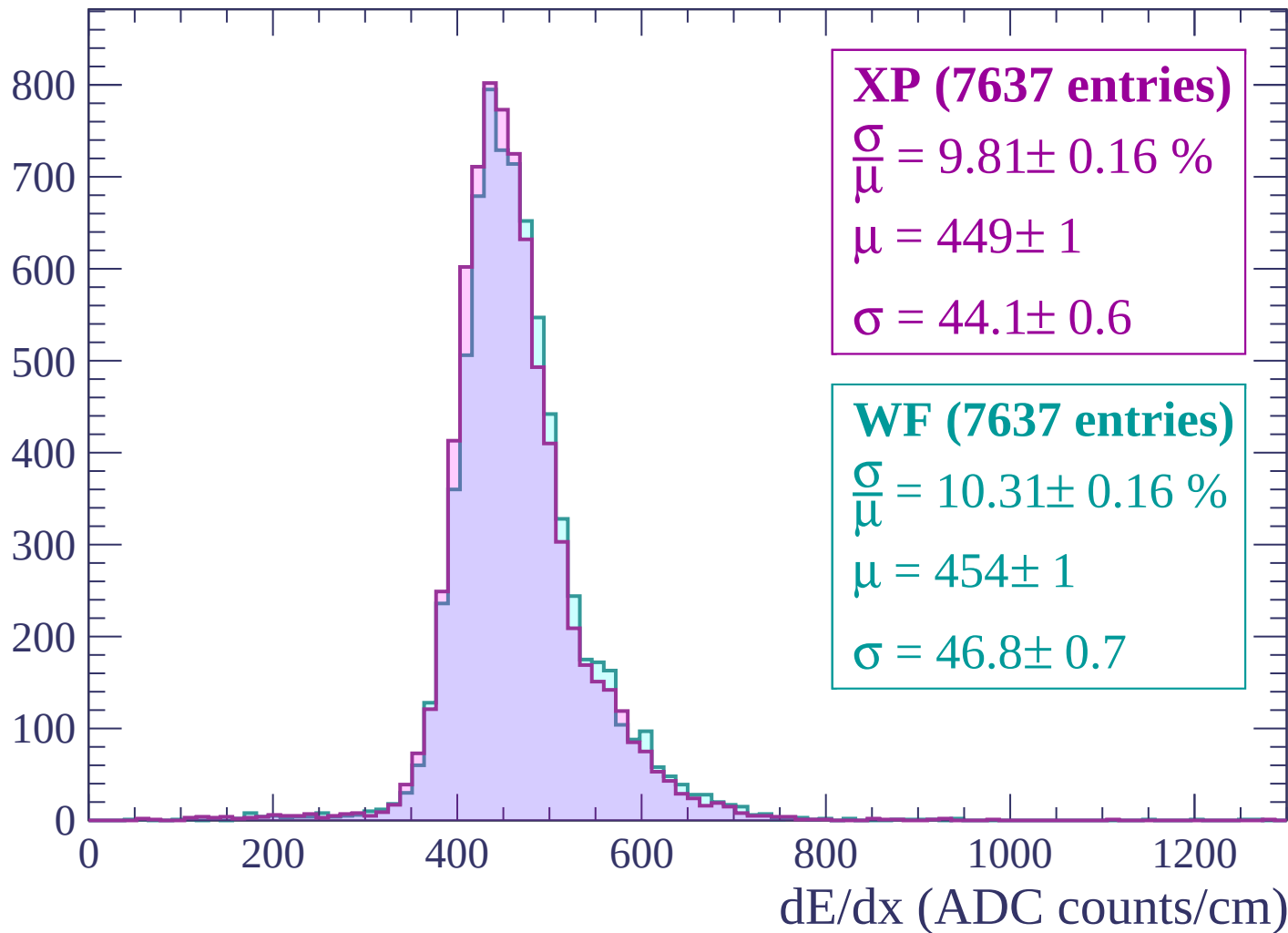
$$\mu = 453 \pm 1$$

$$\sigma = 46.6 \pm 0.6$$

dE/dx (ADC counts/cm)

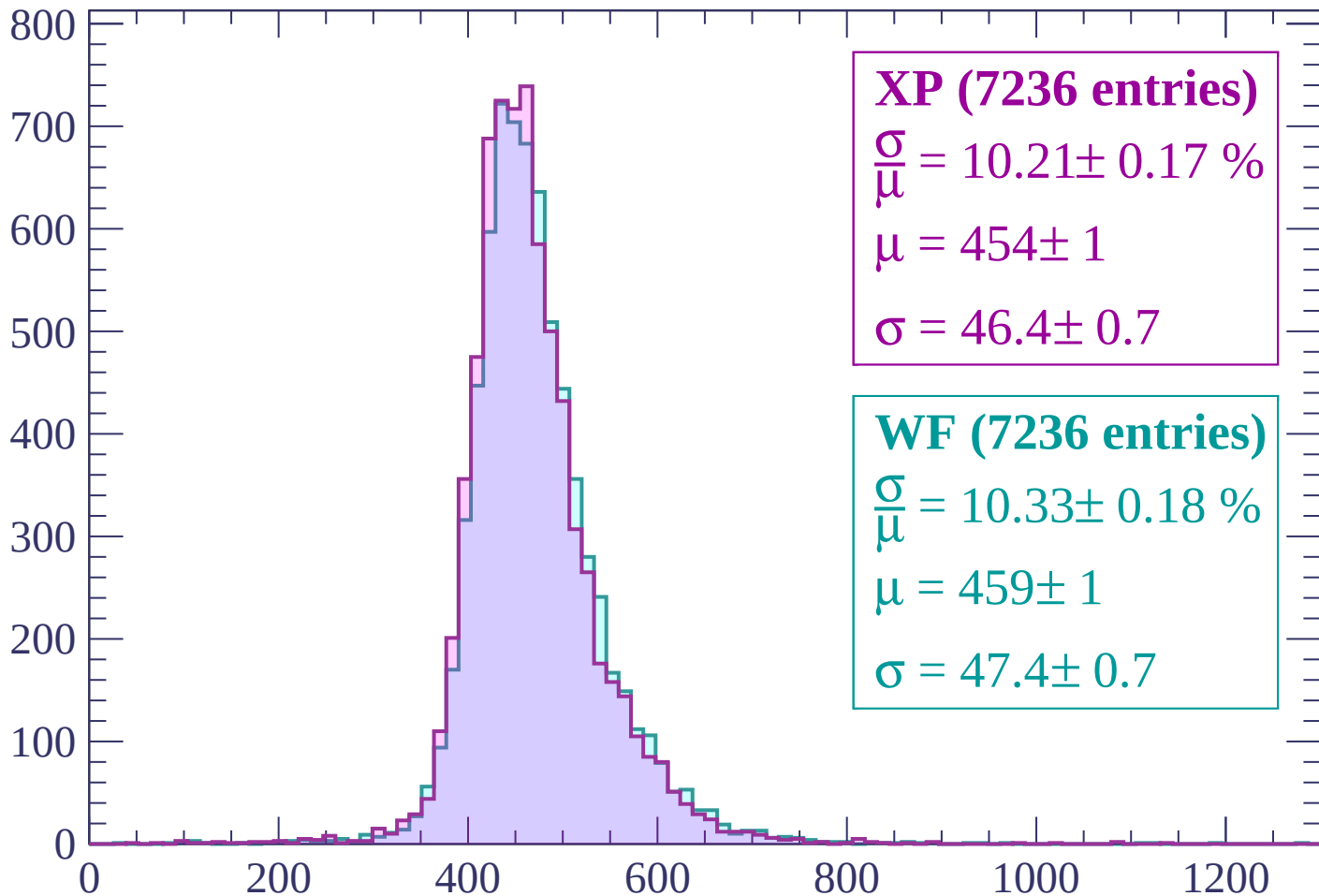
Energy loss | $1320 < p < 1380$

Count



Energy loss | $1380 < p < 1440$

Count



XP (7236 entries)

$$\frac{\sigma}{\mu} = 10.21 \pm 0.17 \%$$

$$\mu = 454 \pm 1$$

$$\sigma = 46.4 \pm 0.7$$

WF (7236 entries)

$$\frac{\sigma}{\mu} = 10.33 \pm 0.18 \%$$

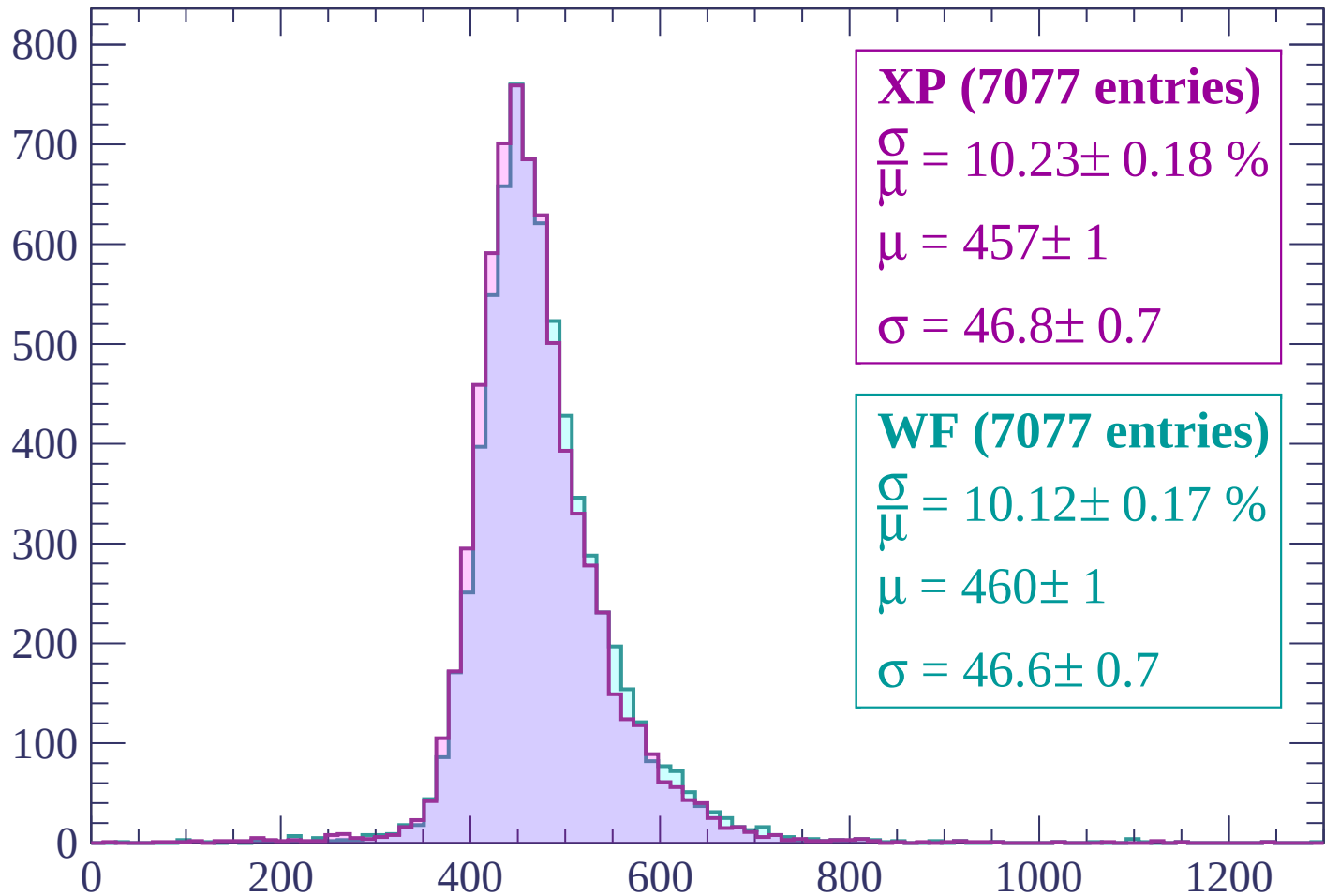
$$\mu = 459 \pm 1$$

$$\sigma = 47.4 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $1440 < p < 1500$

Count



XP (7077 entries)

$\frac{\sigma}{\mu} = 10.23 \pm 0.18 \%$

$\mu = 457 \pm 1$

$\sigma = 46.8 \pm 0.7$

WF (7077 entries)

$\frac{\sigma}{\mu} = 10.12 \pm 0.17 \%$

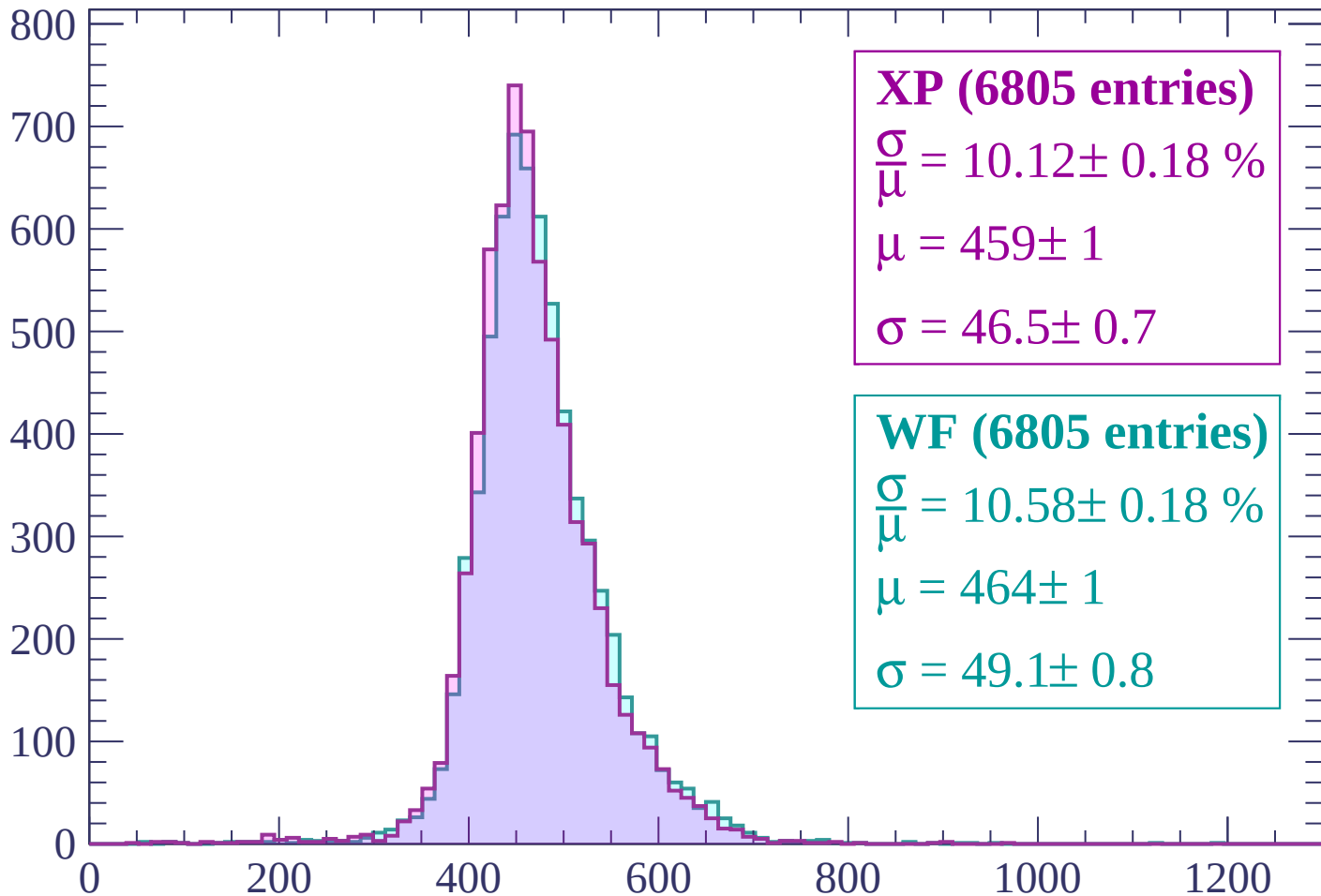
$\mu = 460 \pm 1$

$\sigma = 46.6 \pm 0.7$

dE/dx (ADC counts/cm)

Energy loss | $1500 < p < 1560$

Count



XP (6805 entries)

$\frac{\sigma}{\mu} = 10.12 \pm 0.18 \%$

$\mu = 459 \pm 1$

$\sigma = 46.5 \pm 0.7$

WF (6805 entries)

$\frac{\sigma}{\mu} = 10.58 \pm 0.18 \%$

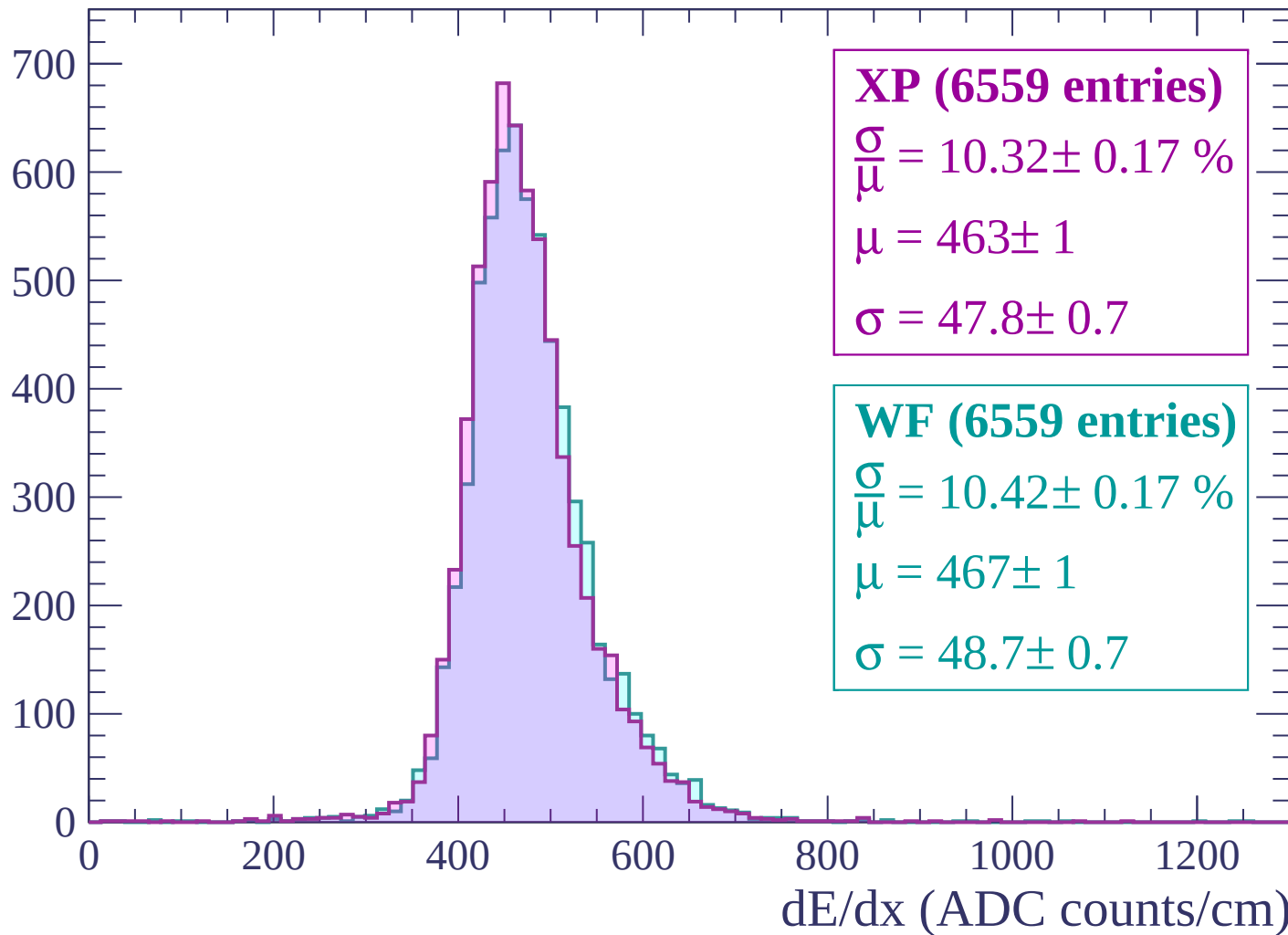
$\mu = 464 \pm 1$

$\sigma = 49.1 \pm 0.8$

dE/dx (ADC counts/cm)

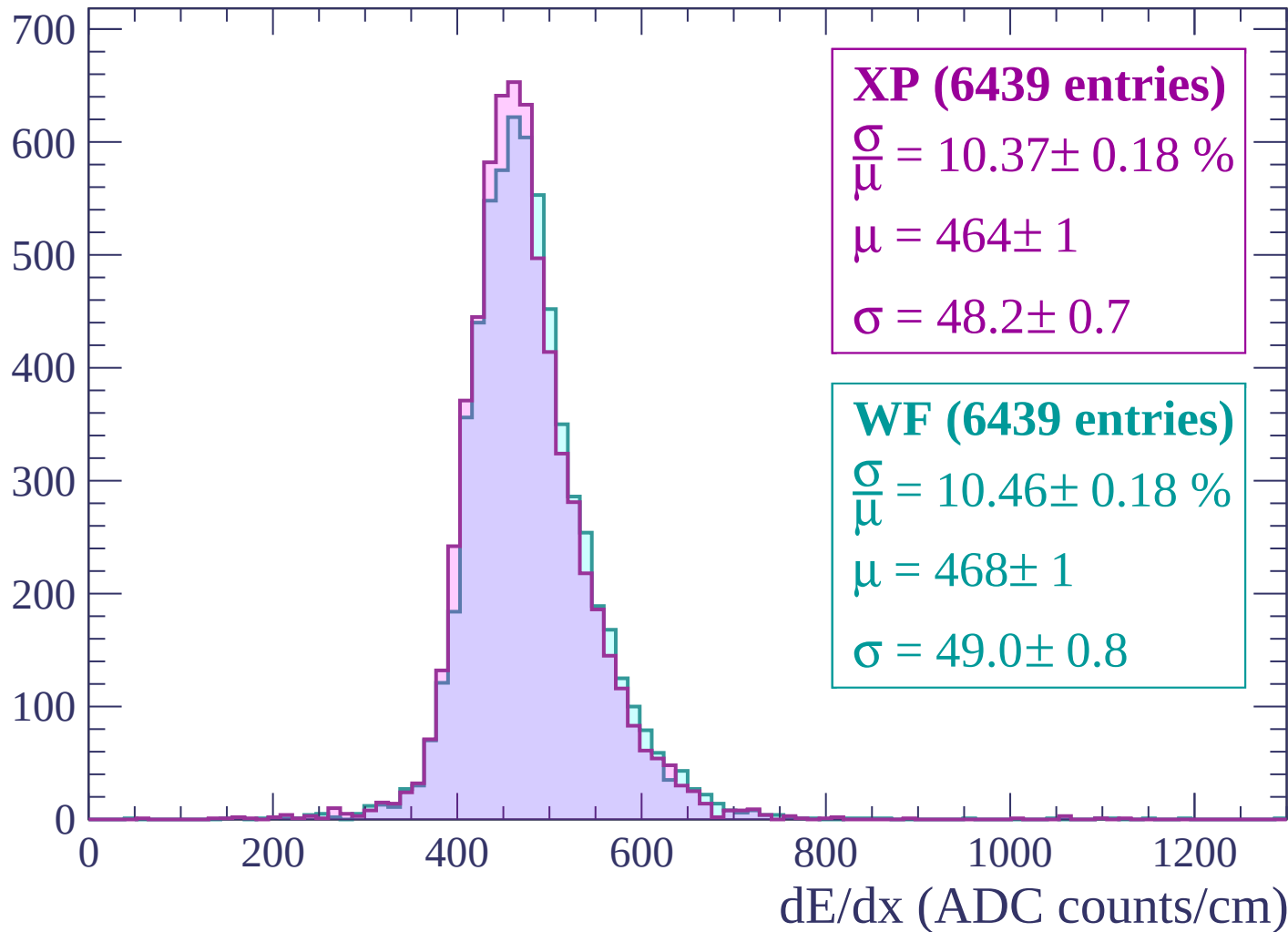
Energy loss | $1560 < p < 1620$

Count



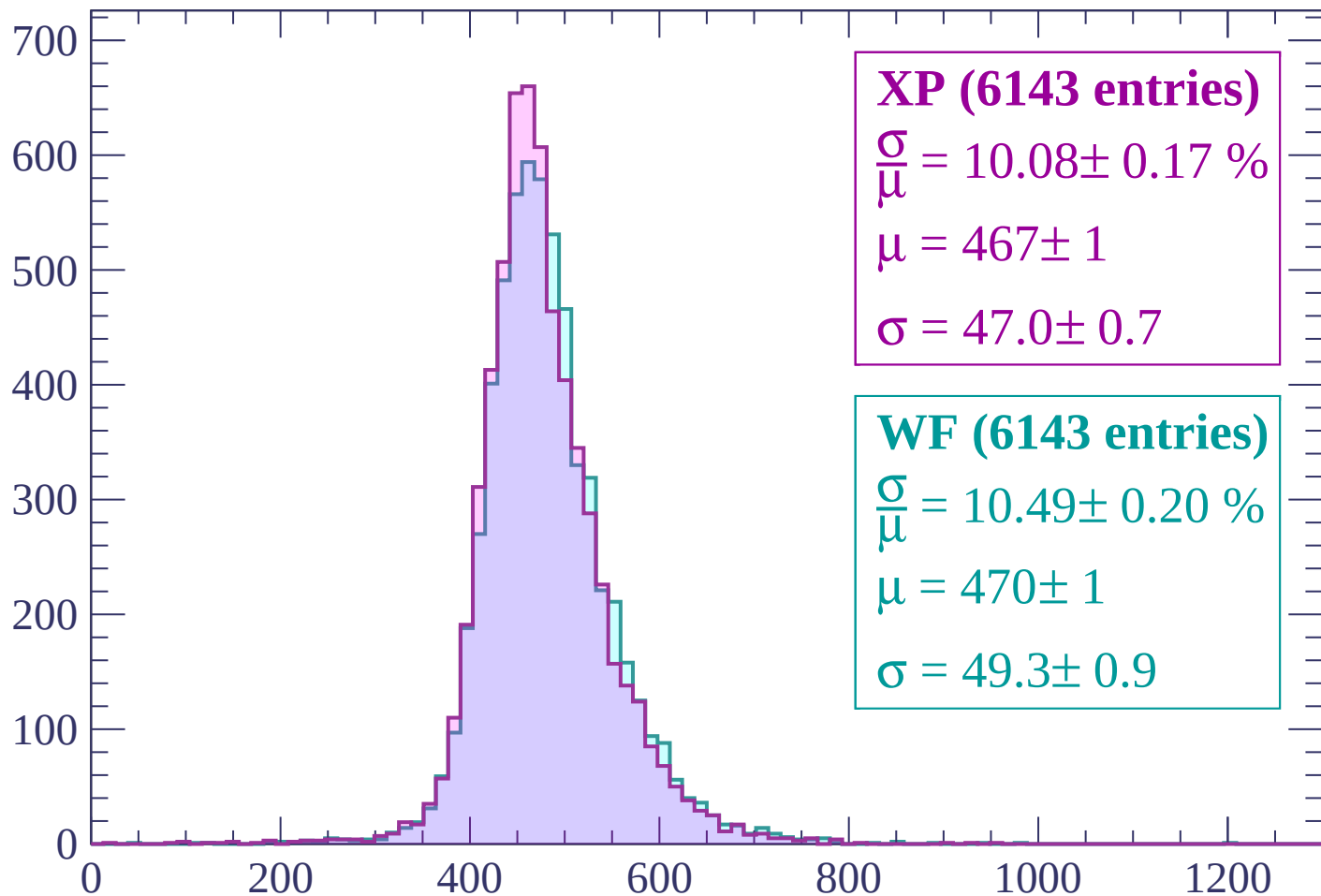
Energy loss | $1620 < p < 1680$

Count



Energy loss | $1680 < p < 1740$

Count



XP (6143 entries)

$\frac{\sigma}{\mu} = 10.08 \pm 0.17 \%$

$\mu = 467 \pm 1$

$\sigma = 47.0 \pm 0.7$

WF (6143 entries)

$\frac{\sigma}{\mu} = 10.49 \pm 0.20 \%$

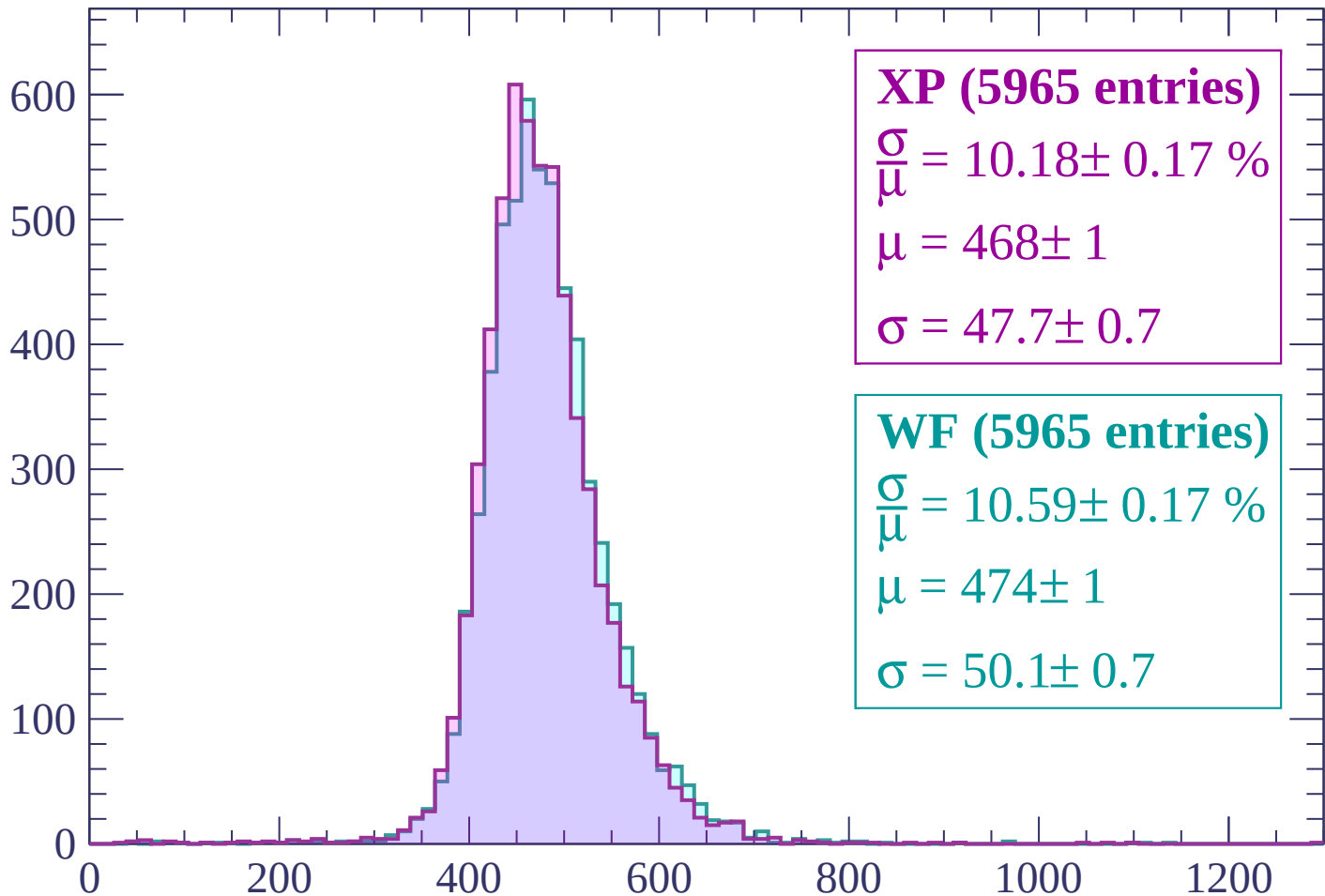
$\mu = 470 \pm 1$

$\sigma = 49.3 \pm 0.9$

dE/dx (ADC counts/cm)

Energy loss | $1740 < p < 1800$

Count



XP (5965 entries)

$\frac{\sigma}{\mu} = 10.18 \pm 0.17 \%$

$\mu = 468 \pm 1$

$\sigma = 47.7 \pm 0.7$

WF (5965 entries)

$\frac{\sigma}{\mu} = 10.59 \pm 0.17 \%$

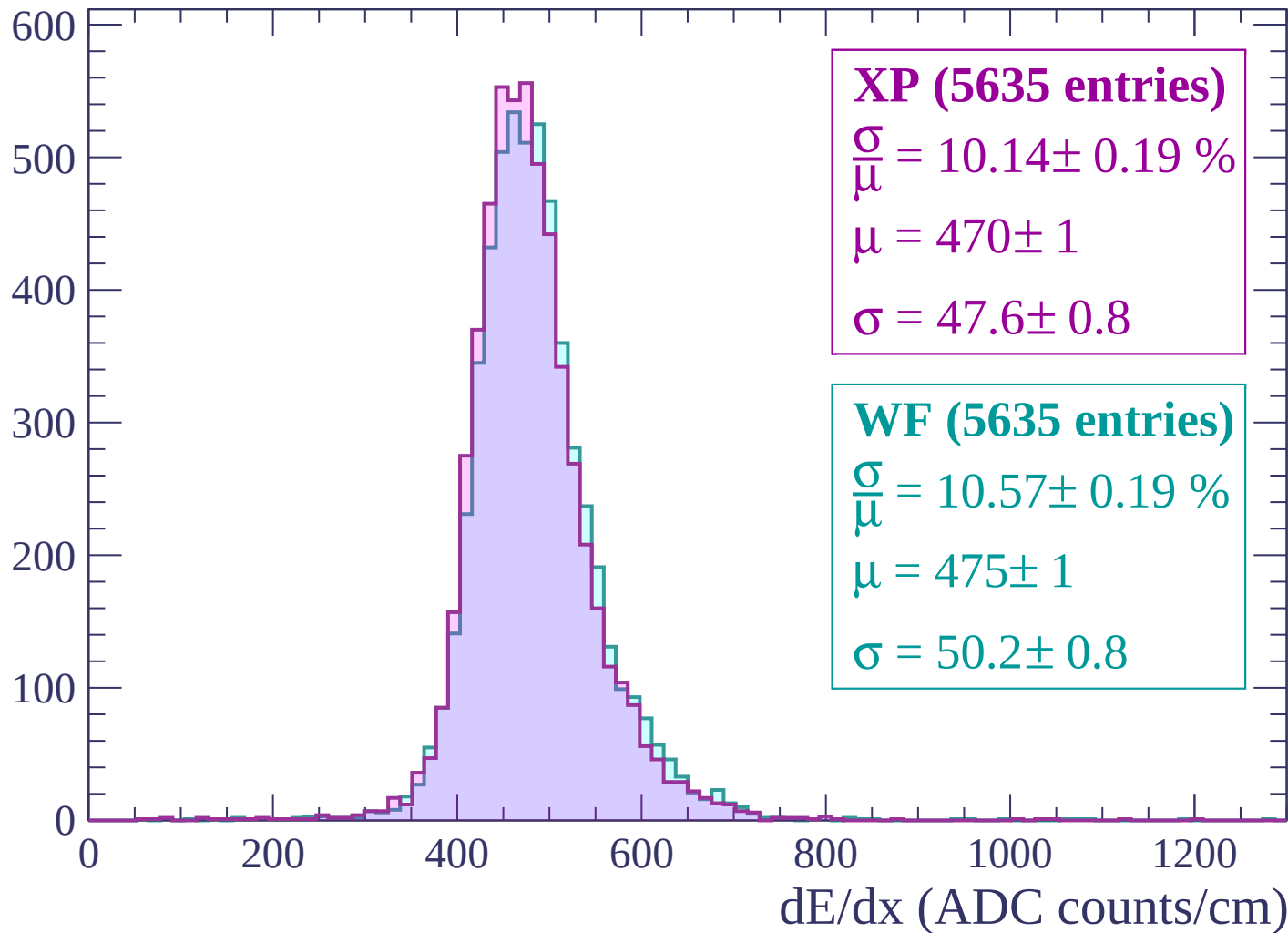
$\mu = 474 \pm 1$

$\sigma = 50.1 \pm 0.7$

dE/dx (ADC counts/cm)

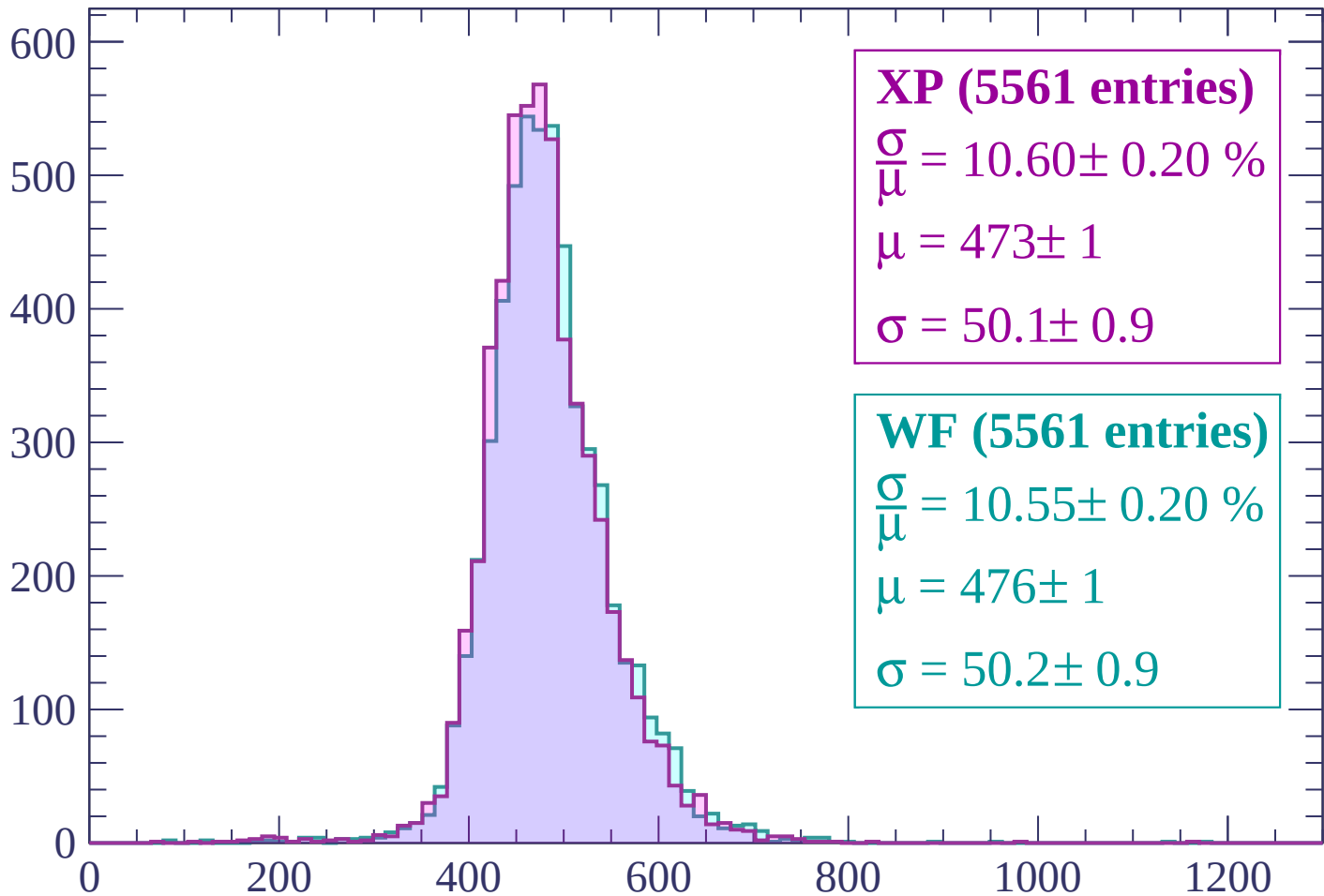
Energy loss | $1800 < p < 1860$

Count



Energy loss | $1860 < p < 1920$

Count



XP (5561 entries)

$\frac{\sigma}{\mu} = 10.60 \pm 0.20 \%$

$\mu = 473 \pm 1$

$\sigma = 50.1 \pm 0.9$

WF (5561 entries)

$\frac{\sigma}{\mu} = 10.55 \pm 0.20 \%$

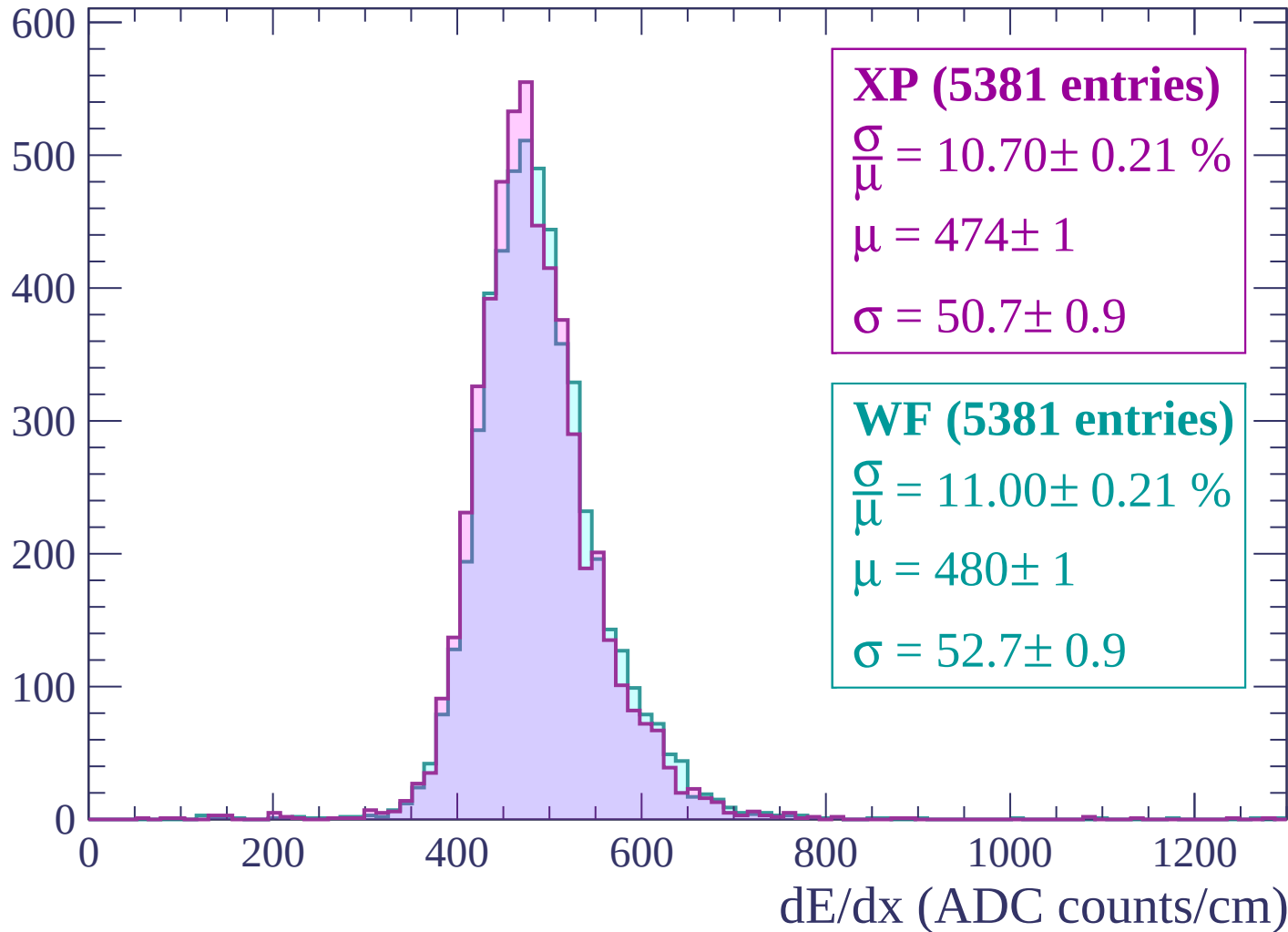
$\mu = 476 \pm 1$

$\sigma = 50.2 \pm 0.9$

dE/dx (ADC counts/cm)

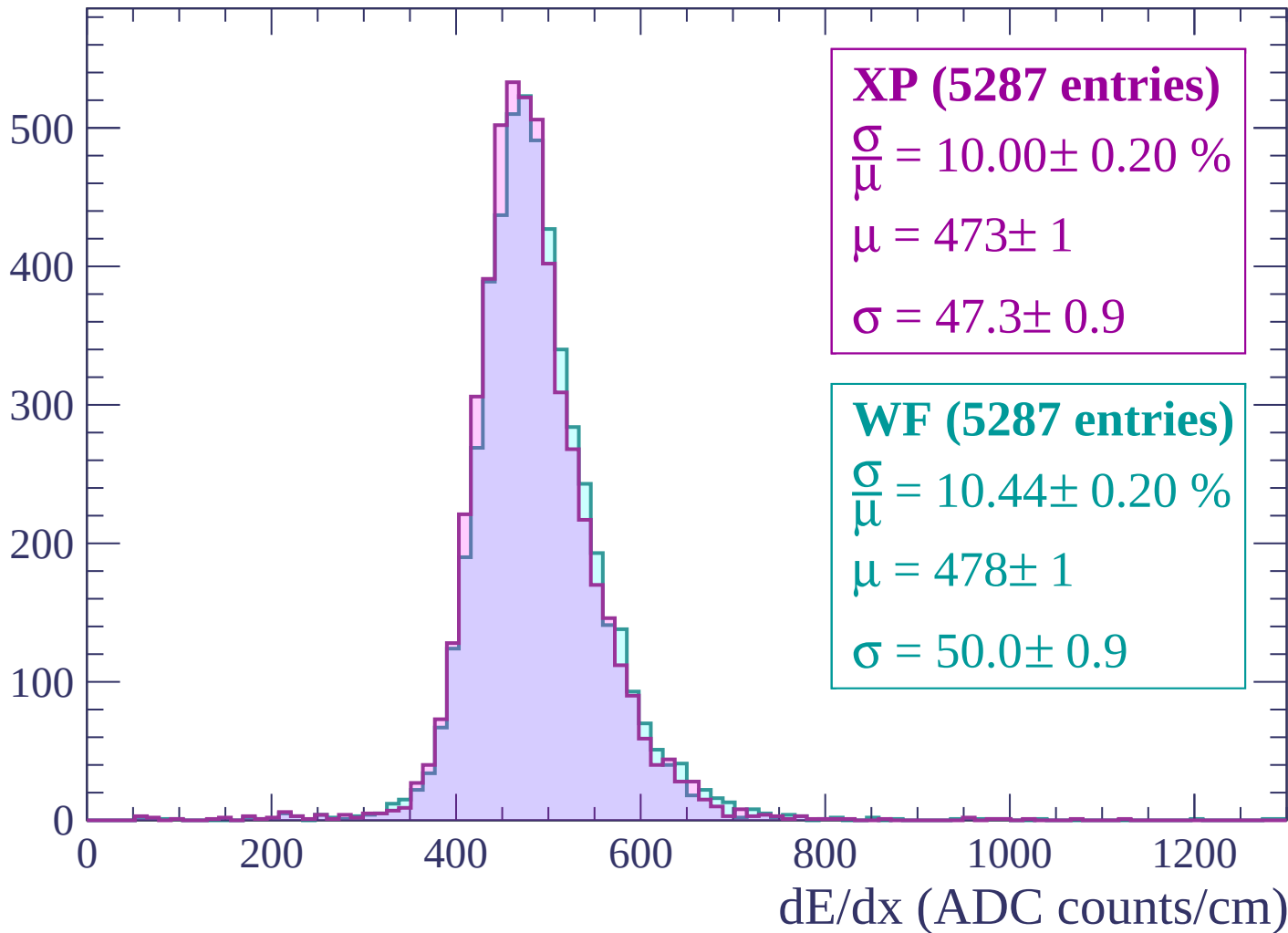
Energy loss | $1920 < p < 1980$

Count



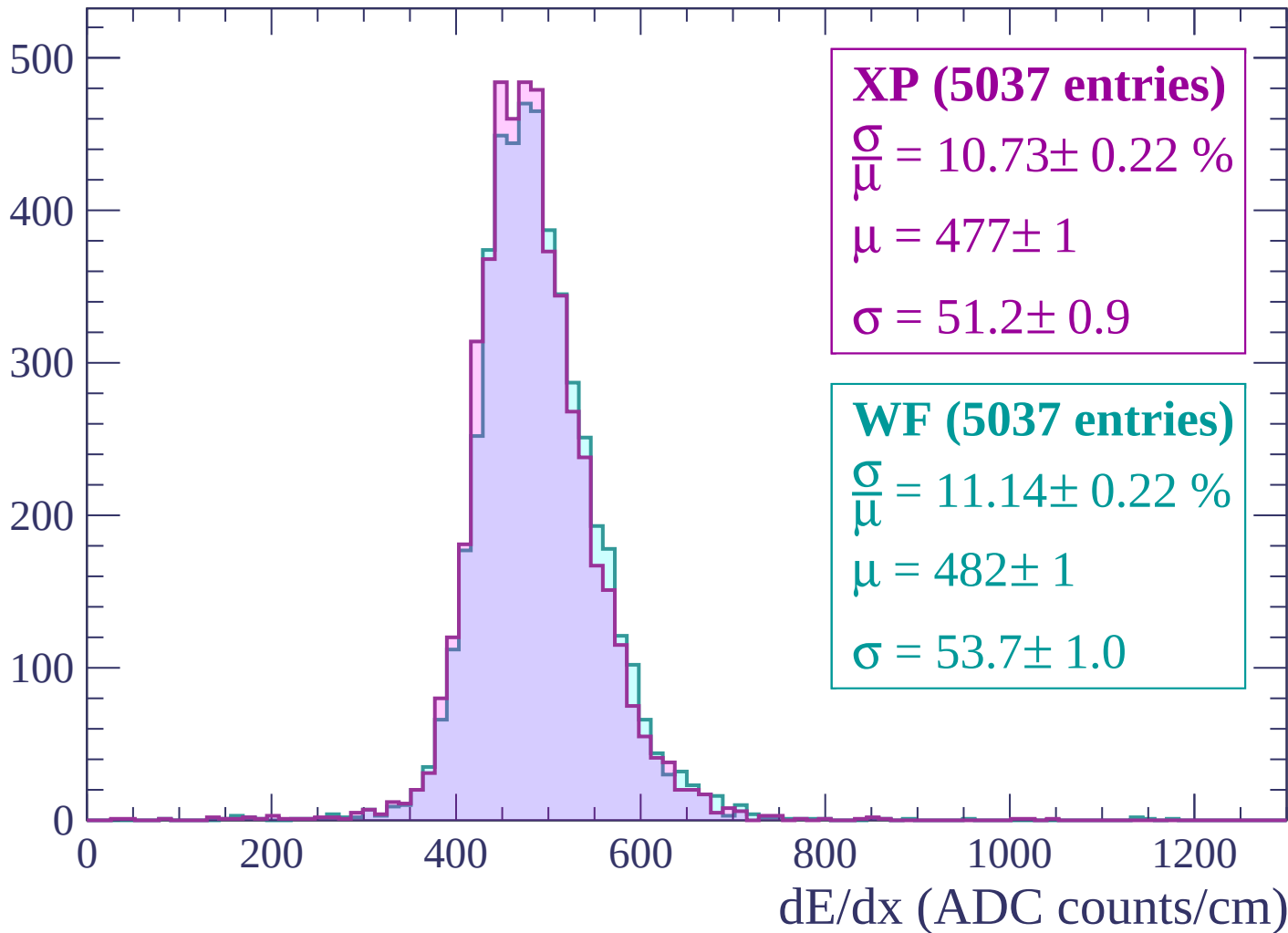
Energy loss | $1980 < p < 2040$

Count



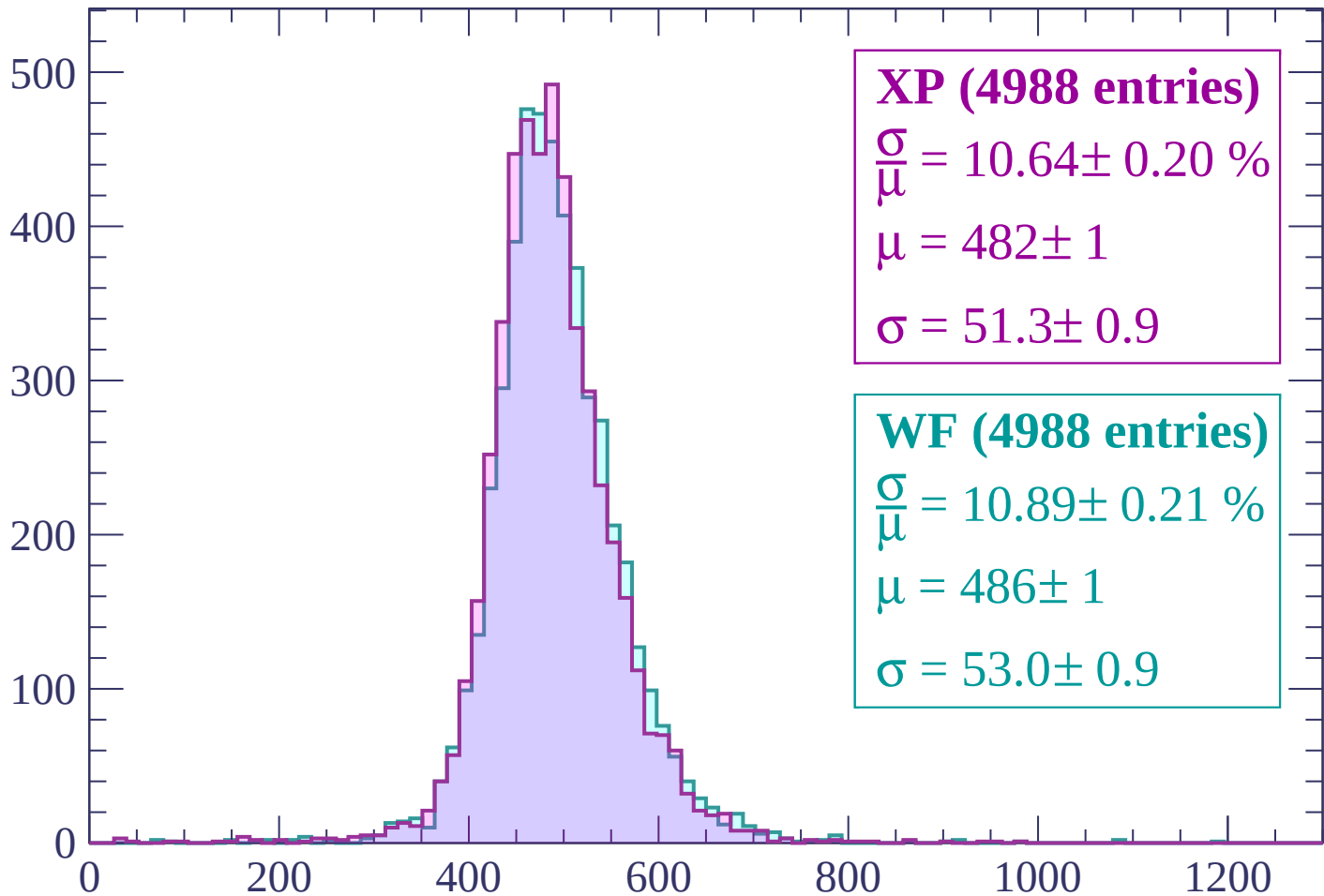
Energy loss | $2040 < p < 2100$

Count



Energy loss | $2100 < p < 2160$

Count



XP (4988 entries)

$$\frac{\sigma}{\mu} = 10.64 \pm 0.20 \%$$

$$\mu = 482 \pm 1$$

$$\sigma = 51.3 \pm 0.9$$

WF (4988 entries)

$$\frac{\sigma}{\mu} = 10.89 \pm 0.21 \%$$

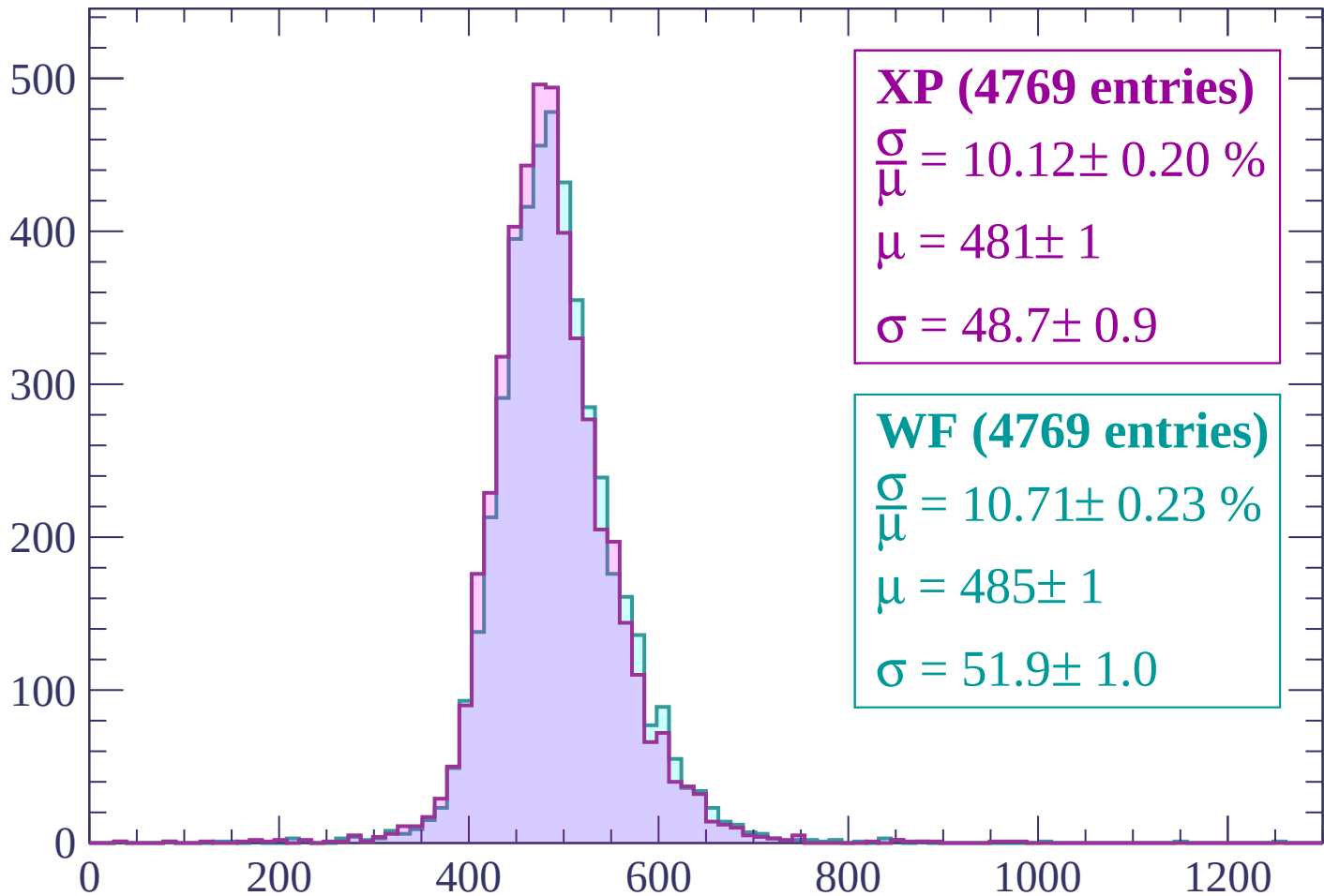
$$\mu = 486 \pm 1$$

$$\sigma = 53.0 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $2160 < p < 2220$

Count



XP (4769 entries)

$\frac{\sigma}{\mu} = 10.12 \pm 0.20 \%$

$\mu = 481 \pm 1$

$\sigma = 48.7 \pm 0.9$

WF (4769 entries)

$\frac{\sigma}{\mu} = 10.71 \pm 0.23 \%$

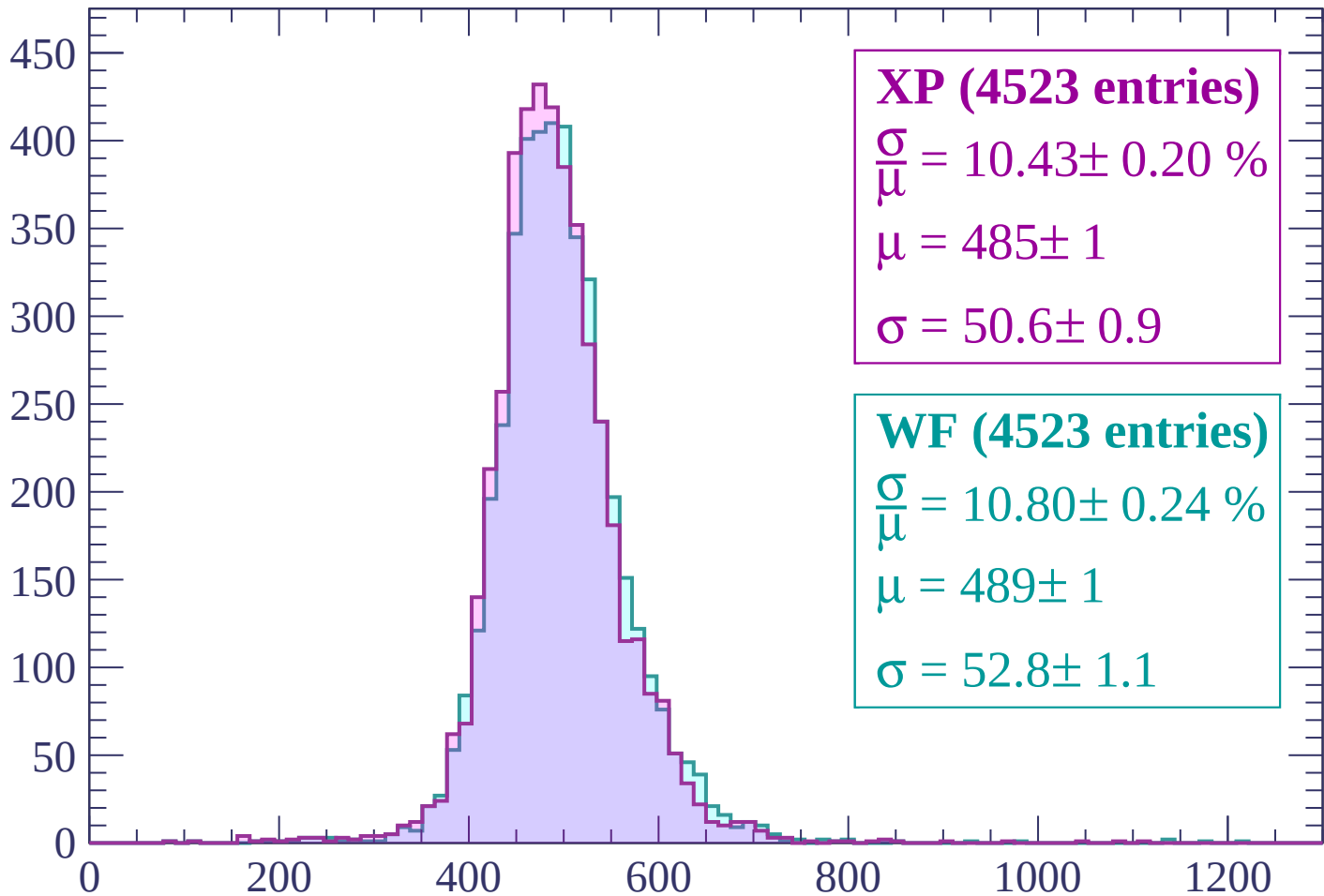
$\mu = 485 \pm 1$

$\sigma = 51.9 \pm 1.0$

dE/dx (ADC counts/cm)

Energy loss | $2220 < p < 2280$

Count



XP (4523 entries)

$\frac{\sigma}{\mu} = 10.43 \pm 0.20 \%$

$\mu = 485 \pm 1$

$\sigma = 50.6 \pm 0.9$

WF (4523 entries)

$\frac{\sigma}{\mu} = 10.80 \pm 0.24 \%$

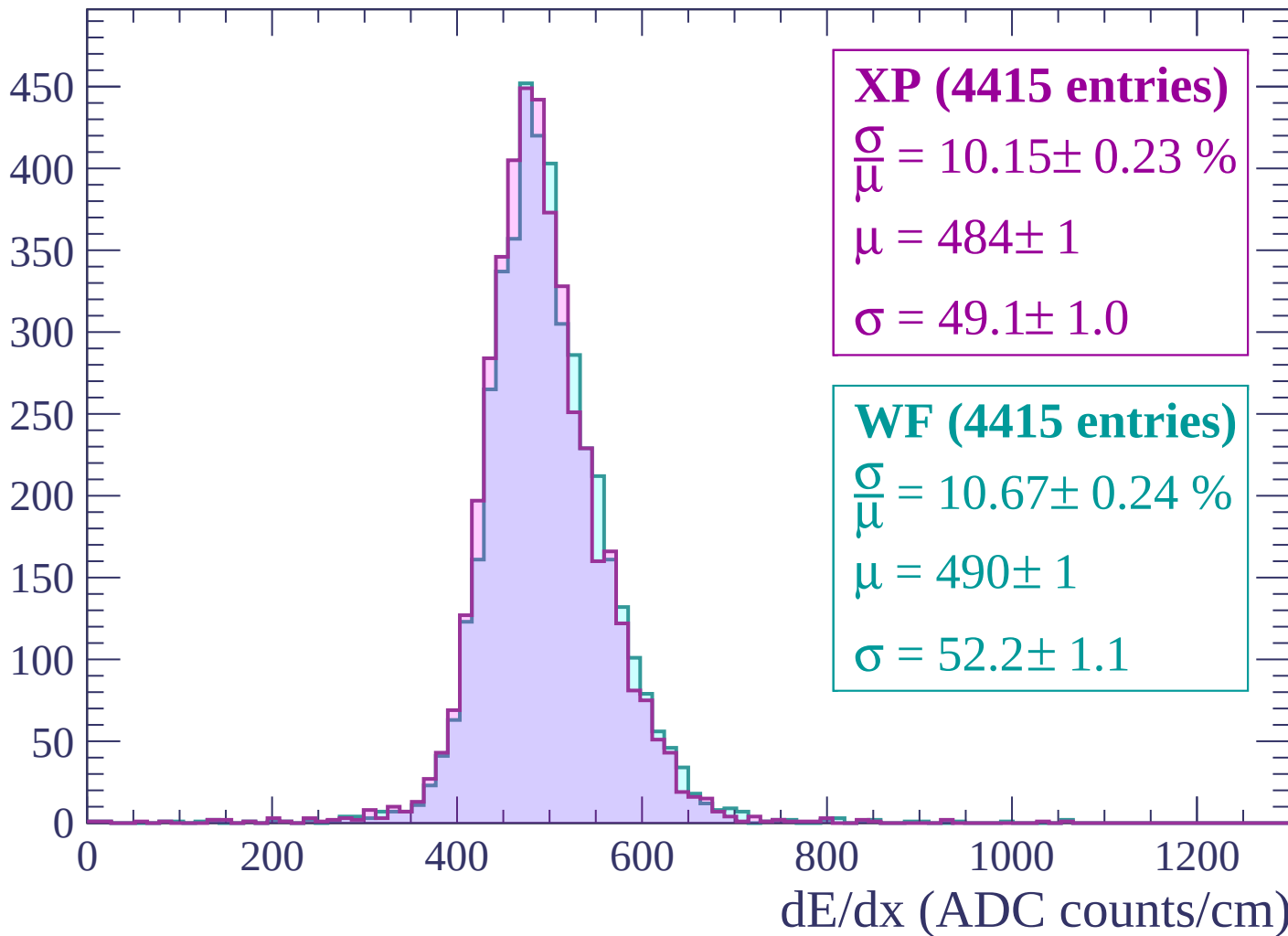
$\mu = 489 \pm 1$

$\sigma = 52.8 \pm 1.1$

dE/dx (ADC counts/cm)

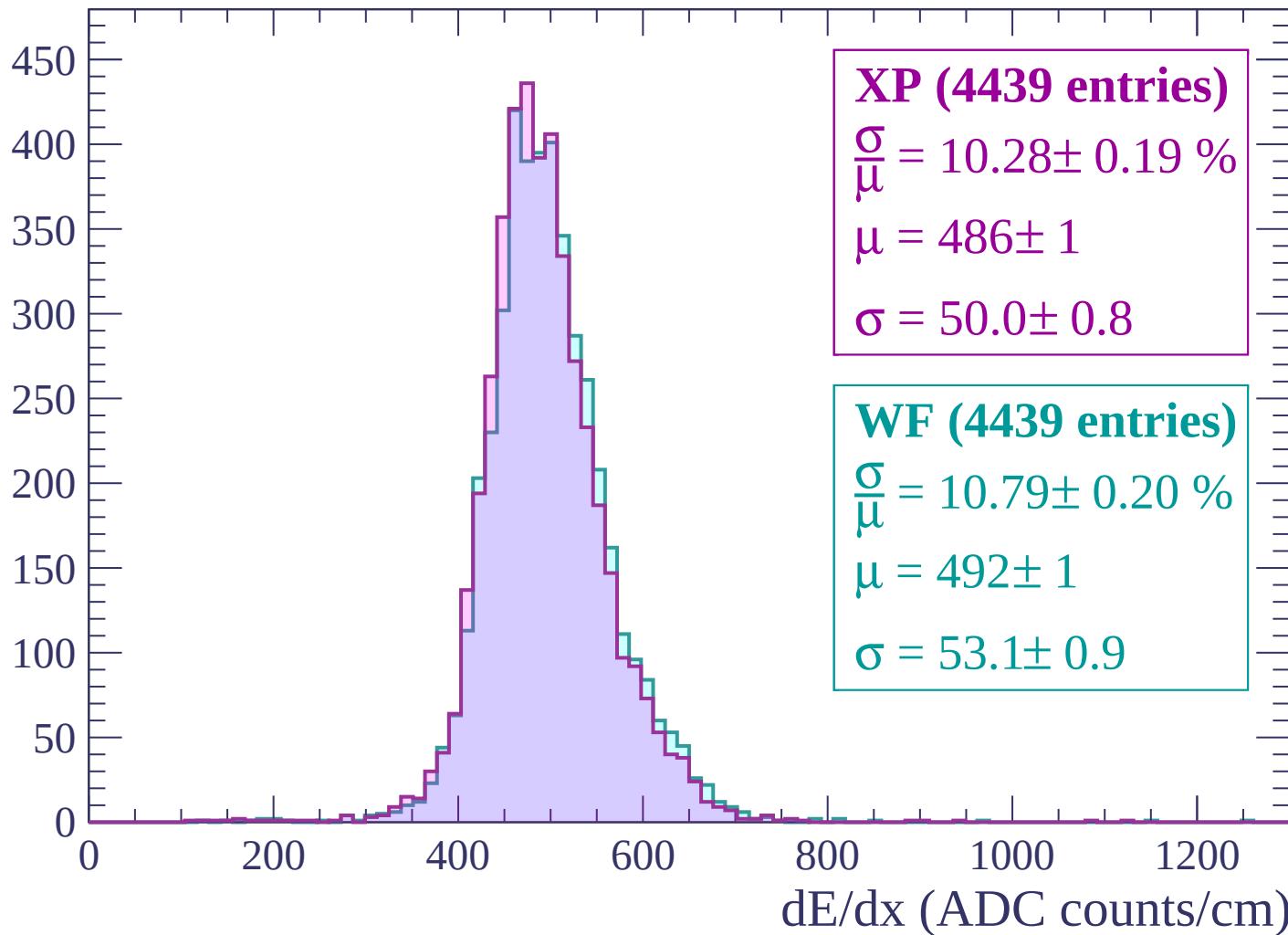
Energy loss | $2280 < p < 2340$

Count



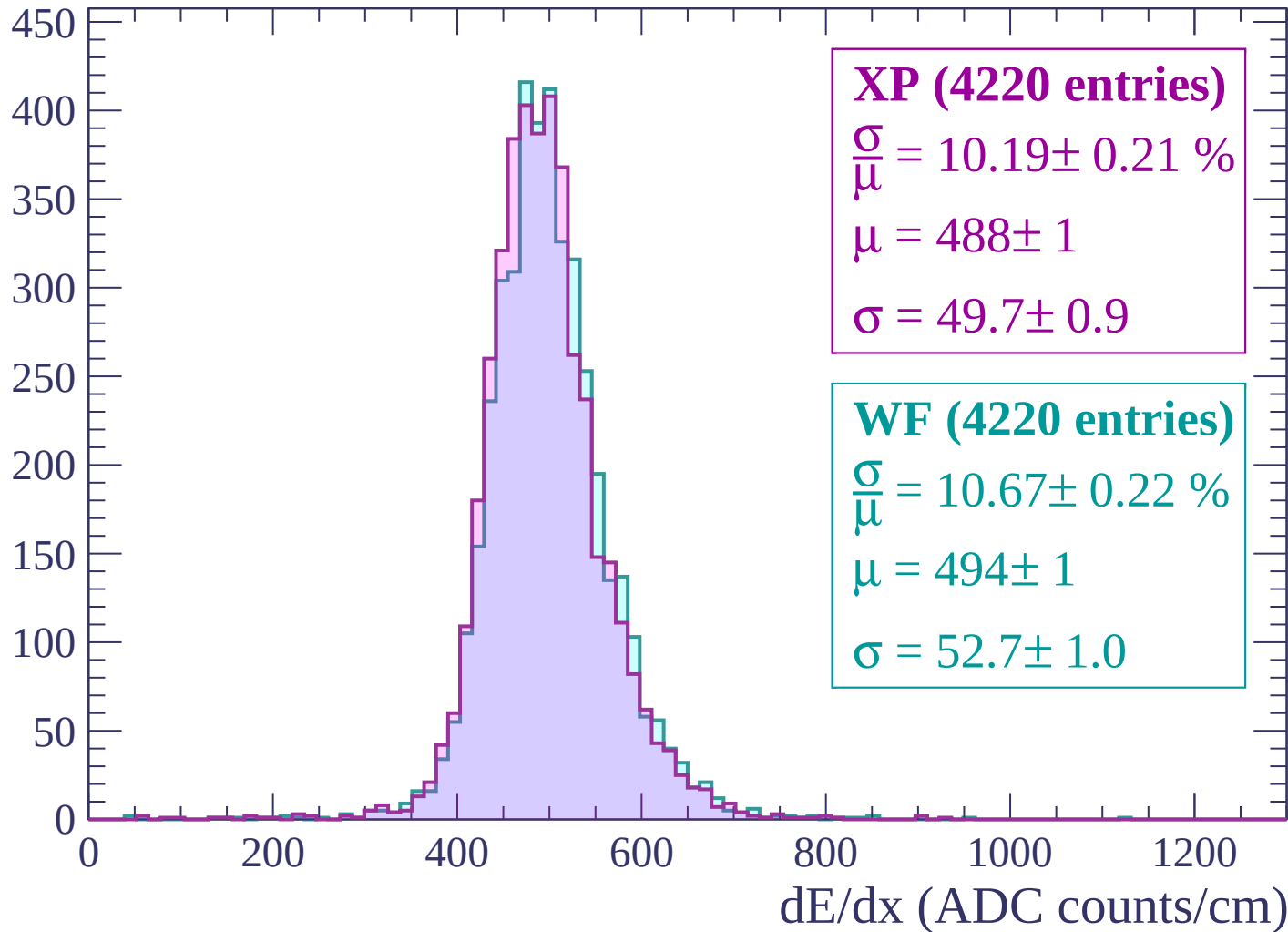
Energy loss | $2340 < p < 2400$

Count



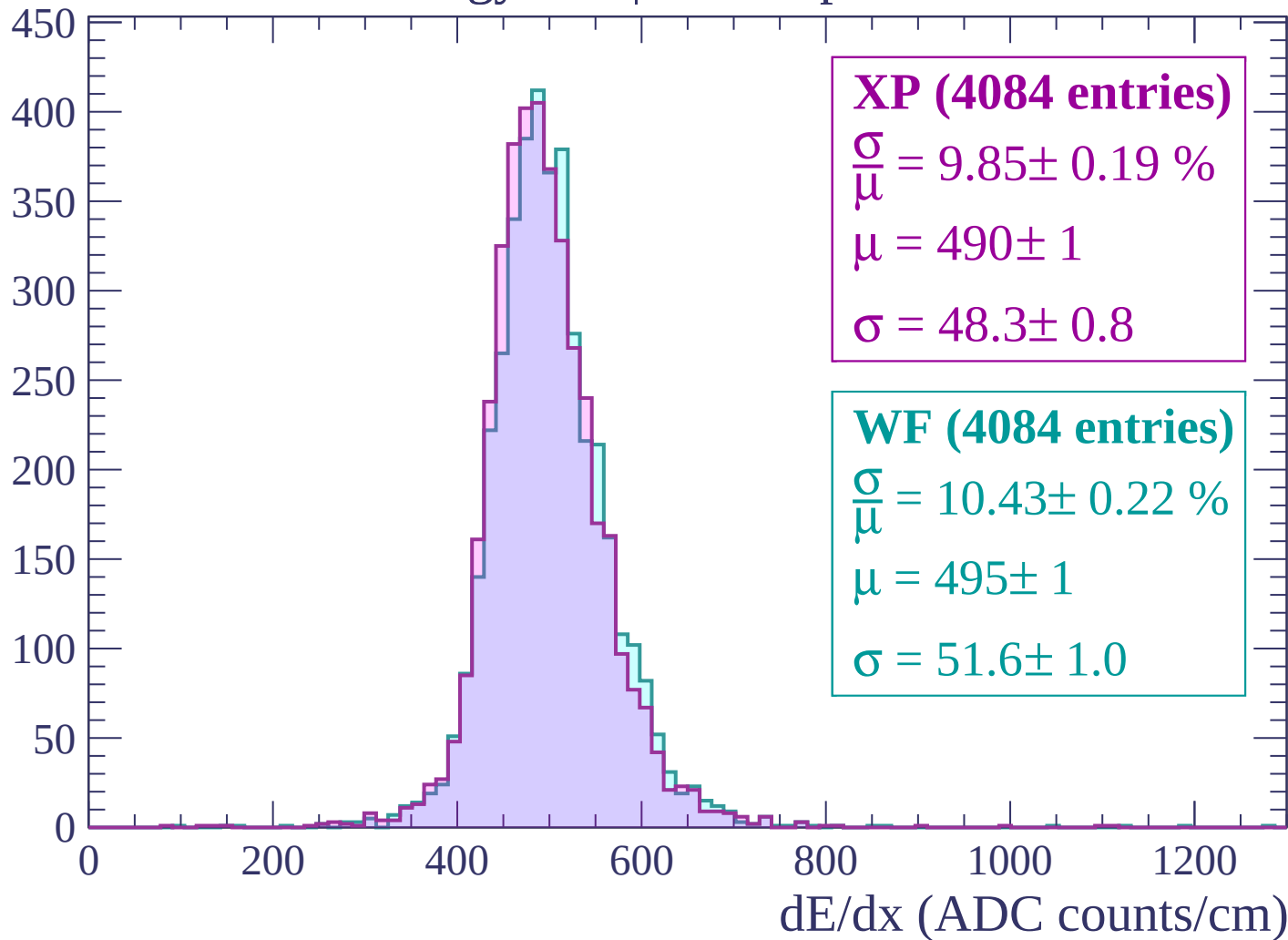
Energy loss | $2400 < p < 2460$

Count



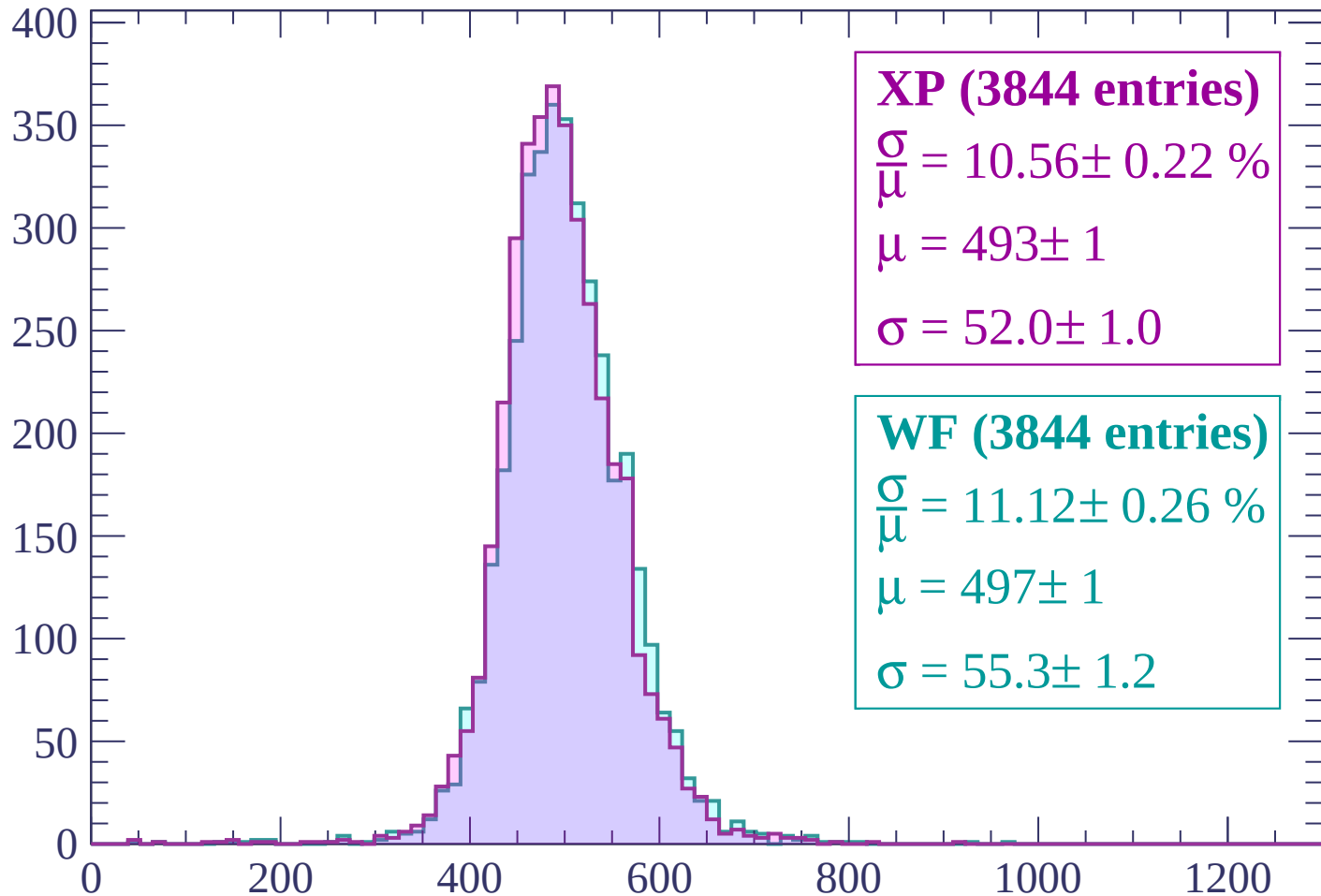
Energy loss | $2460 < p < 2520$

Count



Energy loss | $2520 < p < 2580$

Count



XP (3844 entries)

$\frac{\sigma}{\mu} = 10.56 \pm 0.22 \%$

$\mu = 493 \pm 1$

$\sigma = 52.0 \pm 1.0$

WF (3844 entries)

$\frac{\sigma}{\mu} = 11.12 \pm 0.26 \%$

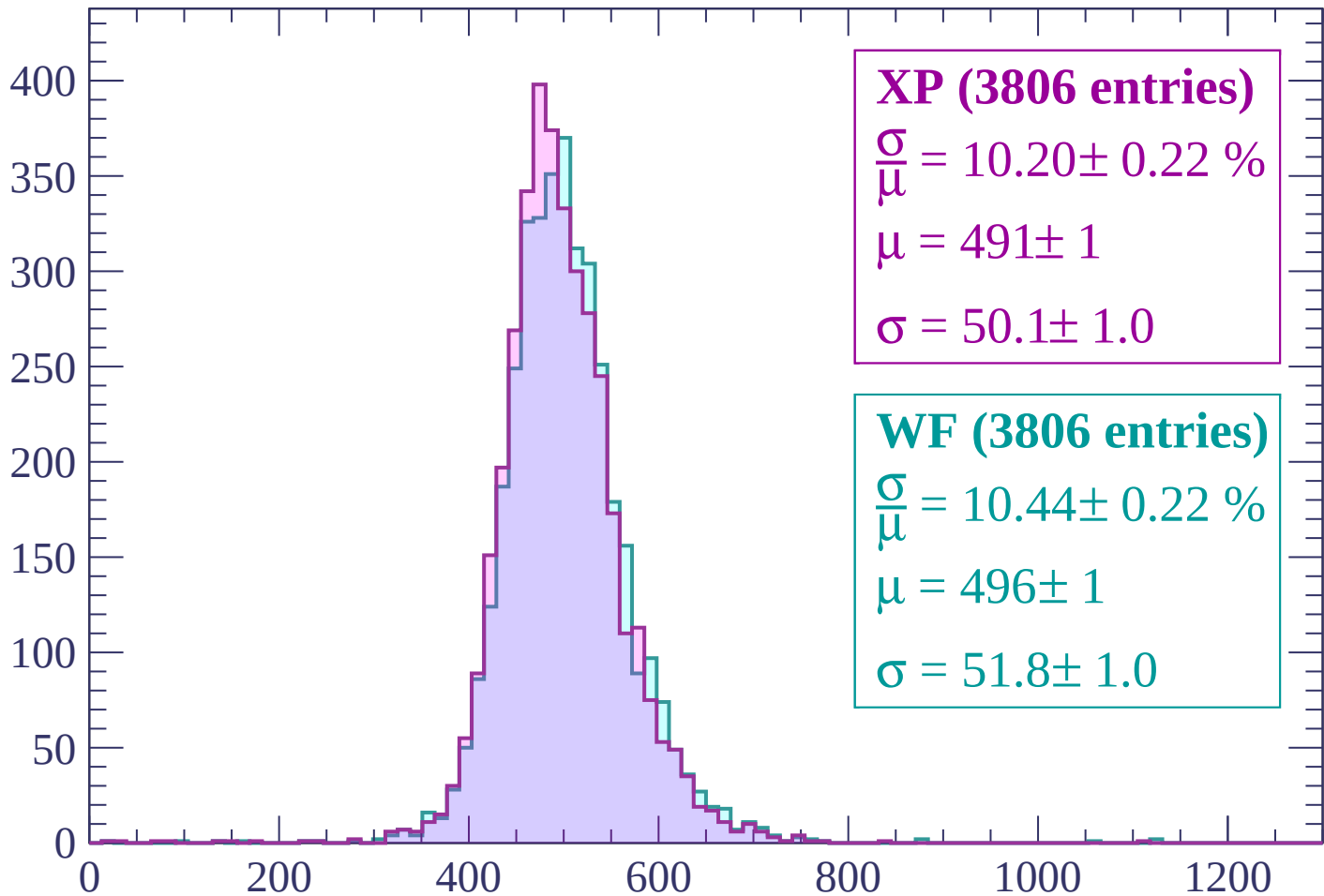
$\mu = 497 \pm 1$

$\sigma = 55.3 \pm 1.2$

dE/dx (ADC counts/cm)

Energy loss | $2580 < p < 2640$

Count



XP (3806 entries)

$$\frac{\sigma}{\mu} = 10.20 \pm 0.22 \%$$

$$\mu = 491 \pm 1$$

$$\sigma = 50.1 \pm 1.0$$

WF (3806 entries)

$$\frac{\sigma}{\mu} = 10.44 \pm 0.22 \%$$

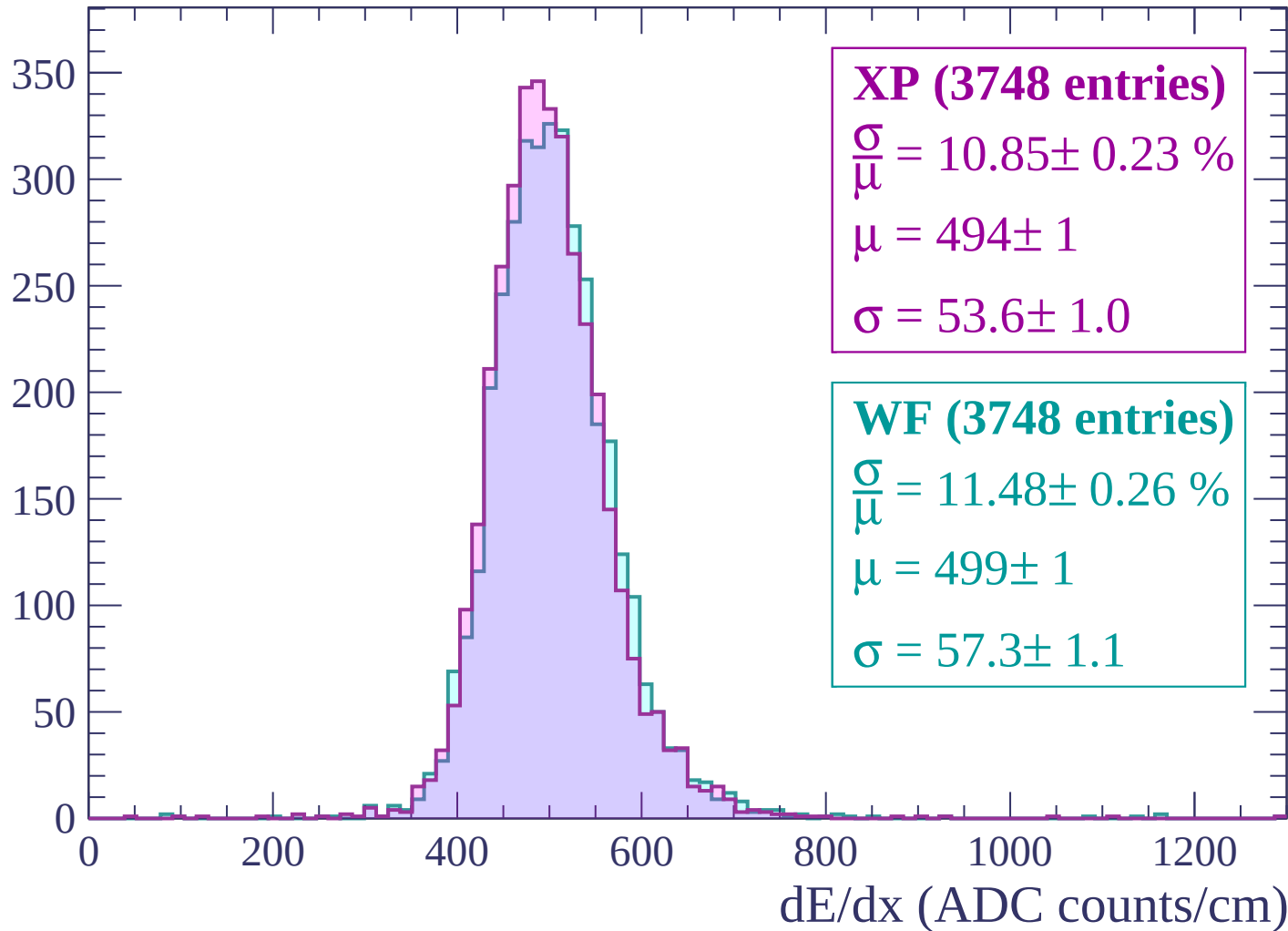
$$\mu = 496 \pm 1$$

$$\sigma = 51.8 \pm 1.0$$

dE/dx (ADC counts/cm)

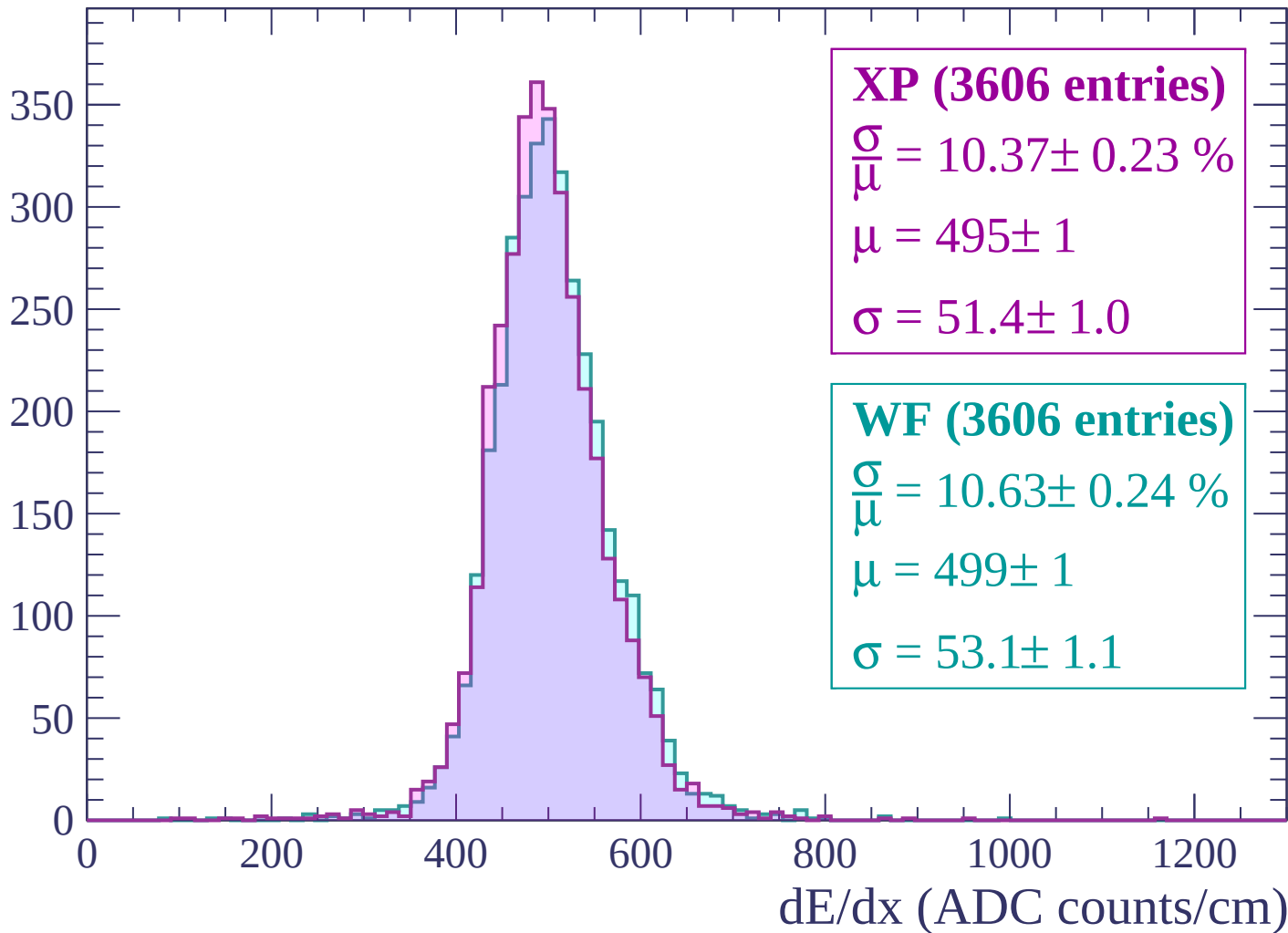
Energy loss | $2640 < p < 2700$

Count



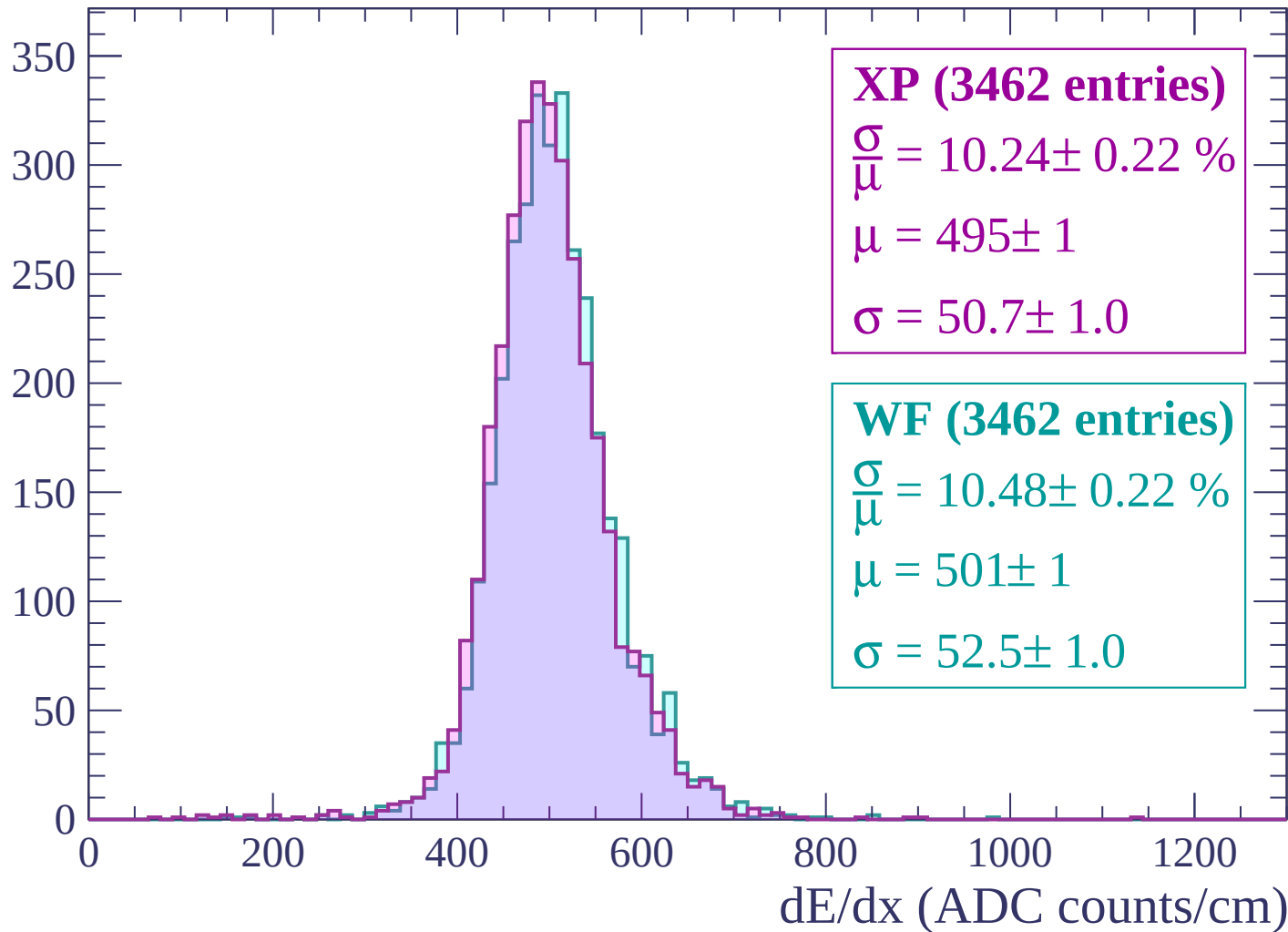
Energy loss | $2700 < p < 2760$

Count



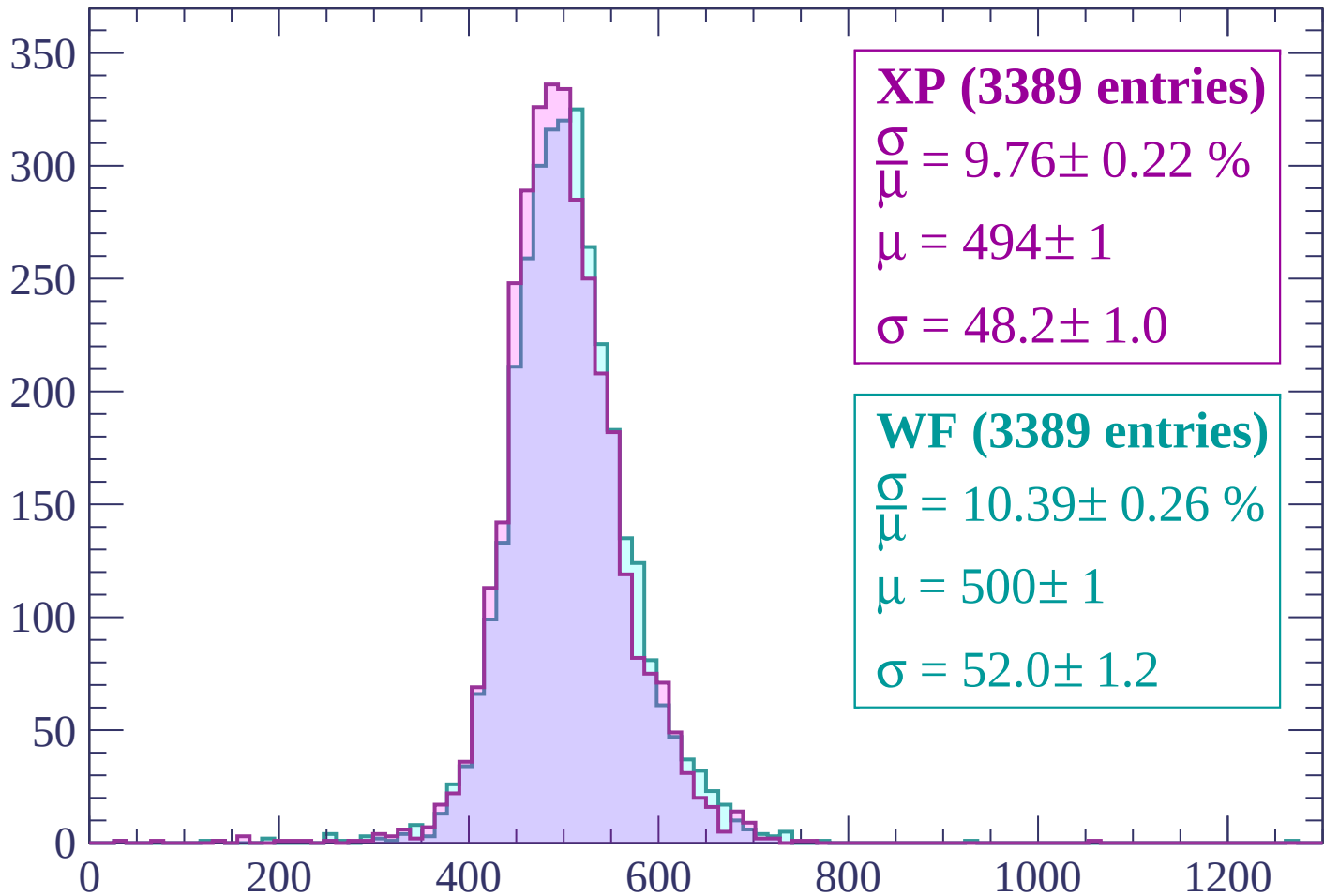
Energy loss | $2760 < p < 2820$

Count



Energy loss | $2820 < p < 2880$

Count



XP (3389 entries)

$$\frac{\sigma}{\mu} = 9.76 \pm 0.22 \%$$

$$\mu = 494 \pm 1$$

$$\sigma = 48.2 \pm 1.0$$

WF (3389 entries)

$$\frac{\sigma}{\mu} = 10.39 \pm 0.26 \%$$

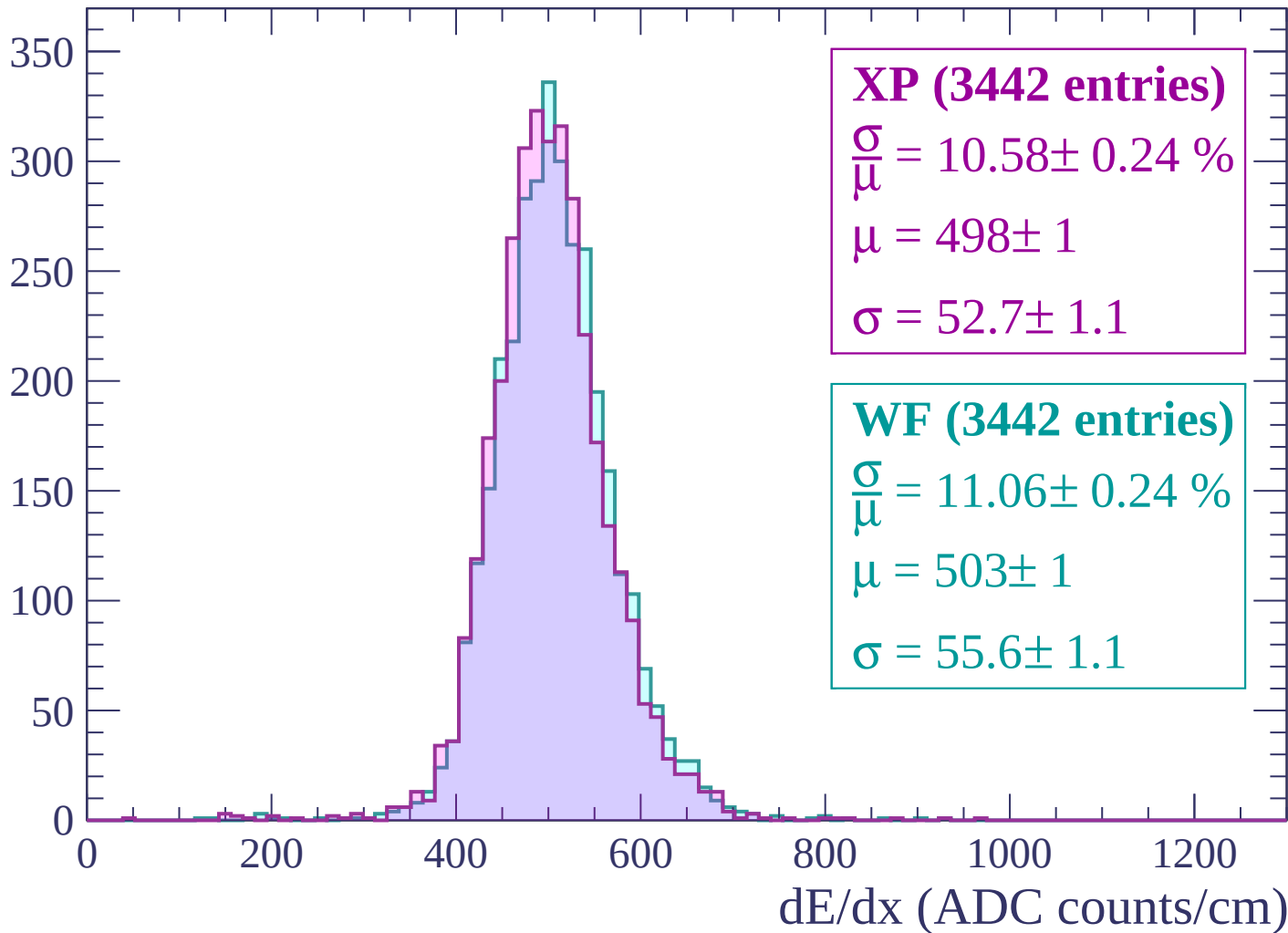
$$\mu = 500 \pm 1$$

$$\sigma = 52.0 \pm 1.2$$

dE/dx (ADC counts/cm)

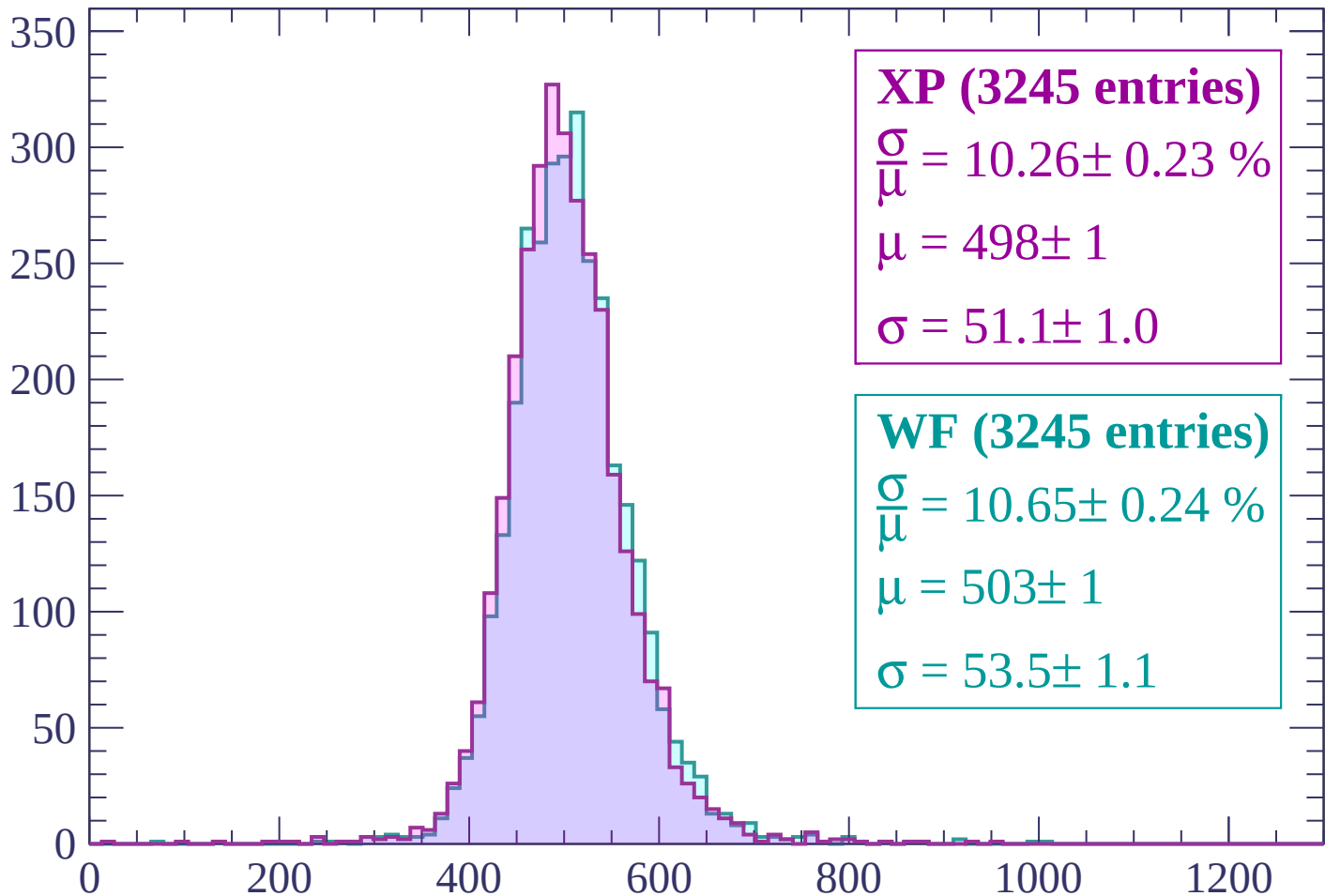
Energy loss | $2880 < p < 2940$

Count



Energy loss | $2940 < p < 3000$

Count



XP (3245 entries)

$\frac{\sigma}{\mu} = 10.26 \pm 0.23 \%$

$\mu = 498 \pm 1$

$\sigma = 51.1 \pm 1.0$

WF (3245 entries)

$\frac{\sigma}{\mu} = 10.65 \pm 0.24 \%$

$\mu = 503 \pm 1$

$\sigma = 53.5 \pm 1.1$

dE/dx (ADC counts/cm)

Energy loss | $3000 < p < 3060$

Count

